The Black-Litterman Model In Detail

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Abstract[1]

In this paper we survey the literature on the Black-Litterman model. This survey is provided both as a chronology and a taxonomy as there are many claims on the model in the literature. We provide a complete description of the canonical model including full derivations from the underlying principles using both Theil's Mixed Estimation model and Bayes Theory. The various parameters of the model are considered, along with information on their computation or calibration. Further consideration is given to several of the key papers, with worked examples illustrating the concepts.

1 The author gratefully acknowledges feedback and comments from Attilio Meucci and Boris Gnedenko.

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Introduction

The Black-Litterman Model was first published by Fischer Black and Robert Litterman in 1990. During the twenty plus years since the original papers, many authors have published research referring to their model as Black-Litterman. This has lead to a variety of models being labeled as Black-Litterman even though they may be very different from the original model created by Black and Litterman. In the chronological survey of the literature we will introduce several papers which make significant contributions to the literature on the Black-Litterman model. We will also document the taxonomy of models which have been labeled as Black Litterman. We will further refer to the Black-Litterman model as described in the papers by the original authors as the canonical Black-Litterman model.

The canonical Black-Litterman model makes two significant contributions to the problem of asset allocation. First, it provides an intuitive prior, the equilibrium market portfolio, as a starting point for estimation of asset returns. Previous similar work started either with the uninformative uniform prior distribution or with the global minimum variance portfolio. The latter method, described by Frost and Savarino (1986), and Jorion (1986), took a shrinkage approach to improve the final asset allocation. Neither of these methods has an intuitive connection back to the market,. The idea that one could use 'reverse optimization' to generate a stable distribution of returns from the equilibrium market portfolio as a starting point for shrinkage is a significant improvement to the process of return estimation.

Second, the Black-Litterman model provides a clear way to specify investors views on returns and to blend the investors views with prior information. The investor's views are allowed to be partial or complete, and the views can span arbitrary and overlapping sets of assets. The model outputs estimates of expected excess returns and the estimates corresponding precision Prior to Black and Litterman, (1991a), nothing similar had been published for the portfolio selection problem. Theil's mixing model had been developed, but nobody had applied it to the problem of estimating asset returns. No research linked the process of specifying views to the blending of the prior and the investors views. The Black-Litterman model provides a quantitative framework for specifying the investor's views, and a clear way to combine those investor's views with an intuitive prior to arrive at a new combined distribution.

The state of the art in the portfolio selection problem has changed significantly during the time since the publication of the Black-Litterman model. Because of its rich theoretical basis it can be applied alongside many modern portfolio selection approaches as can be seen in the literature..

The rest of the document is structured in the following manner. First we will address the literature survey and taxonomy, then we will describe each of the reference models, and then the estimation model. Finally we will illustrate the concepts from many of the papers described in the literature survey.

Historical Taxonomy and Literature Survey

This section of the paper will provide survey of the literature and will classify the model used by each of the various authors. The primary dimensions we will use to classify the authors' models will be; does it specify the estimates as distributions or as point estimates, and secondly does it include the parameter τ. In order for an author's model to be canonical Black-Litterman it would need to match

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both these conditions.

We will collect the models into three distinct Reference Models based on our dimensions provided above. We describe the situation where both conditions are met as the Canonical Reference Model. This is the model described by the original authors. The Alternative Reference Model describes models which use point estimates, but for some reason include τ which now becomes a scaling factor with no quantitative basis. These authors treat the model as just a shrinkage/mixing model and lose the connection to Bayesian statistics. As we will see, using the Bayes law formula, but substituting in different variables is not theoretically tenable.. Finally the Beyond Black-Litterman Reference Model uses point estimates and no longer includes τ at all. Here we see a pure shrinkage/mixing model. Each of the Reference Models is described in depth later in the paper.

The Black-Litterman model was first published by Fischer Black and Robert Litterman of Goldman Sachs in an internal Goldman Sachs Fixed Income Research Note, Black and Litterman (1990). This research note was extended into a paper and published in the Journal of Fixed Income in 1991, Black and Litterman (1991a). This paper does not provide all the formulas used in the model, but does provide a good overview of the features of the model. A second internal Goldman Sachs Fixed Income Research Note was published the same year, Black and Litterman (1991b). This second paper was later extended and published in the Financial Analysts Journal (FAJ) as Black and Litterman (1992). The latter article was then republished by FAJ in the mid 1990's. Copies of the FAJ article are widely available on the Internet. It provides the rationale for the methodology, and some information on the derivation, but does not show all the formulas or a full derivation. It also includes a rather complex worked example based on the global equilibrium, see Litterman (2003) for more details on the methods required to solve this problem. Unfortunately, because of the merging of the two problems, their results are difficult to reproduce. He and Litterman (1999) is the last paper by one of the original authors and it does provide more detail on the workings of the model, but not quite a complete set of formulas. They do provide a much simpler to reproduce working example.

Bevan and Winkelmann (1998) provide details on how they use Black-Litterman as part of their broader Asset Allocation process at Goldman Sachs, including some calibrations of the model which they perform. This is useful information for anybody planning on building Black-Litterman into an ongoing asset allocation process.

Satchell and Scowcroft (2000) attempted to demystify the Black-Litterman model, but instead introduced a new non-Bayesian expression of the model. Their model uses point estimates for the prior and the views, and uses τ and Ω only to control the amount of shrinkage of the views onto the prior. Because they use point estimates instead of distributions, their model does not include any information on the precision of the estimate. This allows them to recommend setting $\tau = 1$. They also introduce a stochastic τ, but because they use point estimates this really becomes a model with a stochastic covariance of returns. Th model was used occasionally in the literature after this point, but was largely replaced by Meucci's model in the mid-2000's.

Drobetz (2001) provides further description of the Black-Litterman model including a good discussion of how to interpret the confidence in the estimates including a diagram. He works an example. [Need more text here]

Fusai and Meucci (2003) introduced yet another non-Bayesian variant of the model which removed the

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parameter τ all together. Meucci (2005) followed up on this paper and coined the phrase, "Beyond Black-Litterman". Once the model is viewed as only a shrinkage model, Ω alone provides enough degrees of freedom to the shrinkage and τ is superfluous and confusing. Since the mid-2000's we have seen a mixture of the canonical and "Beyond Black-Litterman" model used in the literature[2]. Since Fuasi and Meucci (2003) there has been little use of the hybrid reference model until Michaud, et al (2013) debunked it.

Firoozye and Blamont (2003), provide a good overview of the Canonical Reference Model and attempt to provide intuition in the setting of the parameter, τ. They illustrate the reduction in variance of the posterior estimate as a result of the mixing.

Herold (2003) provides an alternative view of the problem where he examines optimizing alpha generation for active portfolio management, essentially specifying that the sample distribution has zero mean. He uses the alternative reference model with point estimates and tracking error to determine how much shrinkage to allow. The two significant contributions of his paper are; the application of the model to active portfolio management and some additional measures which can be used to validate that the views are reasonable.

Koch (2004) is a powerpoint presentation on the Black-Litterman model. It includes derivations of the 'master formula' and the alternative form under 100% certainty. He does not mention posterior variance, or show the alternative form of the 'master formula' under uncertainty (general case). He does include a slide on sensitivity of the posterior estimate on τ, but he uses the alternative reference model so this information is not valid for the canonical model.

Idzorek (2005) introduced a technique for specifying Ω such that the impact of the shrinkage was specified in terms of percentage of change in the weights between 0 and the change caused by 100% confidence. His paper uses the Alternative Reference Model, but his technique can be applied to the Canonical Black-Litterman model because it is sensitive to the value of τ specified by the investor.

Krishnan and Mains (2005) provide an extension to the Black-Litterman model for an additional factor which is uncorrelated with the market. They call this the Two-Factor Black-Litterman model and they show an example of extending Black-Litterman with a recession factor. They show how it intuitively impacts the expected returns computed from the model.

Mankert (2006) provides a detailed walk through of the model and provides a detailed transformation between the two specifications of the Black-Litterman 'master formula' for the estimated asset returns. She approaches the problem from the point of view of sampling theory, and provides some new intuition about the value τ in the context of the alternative reference model.

Meucci (2006) provides a method to use non-normal views in Black-Litterman. Meucci (2008) extends this method to any model parameter, and allow for both analysis of the full distribution as well as scenario analysis.

Beach and Orlov (2006) illustrate using GARCH models to generate the views. They use a model of international equity across 20 developed countries. They show how the results change as the investor uses different values for τ. They do not provide exact details on the uncertainty of the views and appear to be using an alternative reference model even though their techniques can be applied to the

2 A more comprehensive literature survey can be found at http://www.blacklitterman.org/methods.html.

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canonical reference model.

Braga and Natale (2007) describes a method of calibrating the uncertainty in the views using Tracking Error Volatility (TEV). They provide sensitivities for the posterior estimates to the various views. [Verify] This metric is a well known for it's use in active portfolio management. They use the Alternative Reference Model but their work could be applied to the Canonical Reference Model a well.

Martellini and Ziemann (2007) describe an approach to active management of a fund of hedge funds. They use VaR as their objective function for the reverse optimization and they use a variant of the CAPM model extended to include skewness and kurtosis in determining their neutral portfolio. They use a factor model to generate ankings and then convert the rankings into their confidence in the views. They do not use the Bayesian features of the model, but rather use point estimates and thus do not have information on the precision of their estimates.

Giacommeti, et al (2007) provide an approach to computing the neutral portfolio using stable paretian distributions rather than the normal distribution described in Black and Litterman (1992). They also use multiple different risk measures for the portfolio selection model; variance VaR, and CVaR,

Cheung (2009) intoduces the concept of an augmented model. This version of the model integrates a factor model and does a joint estimation of the factor returns. Cheung uses the Alternative Reference Model, but his variant of the model could easily be used with the Canonical Reference Model.

Bertsimas, et al (2013) They introduce a way to measure the alignment of the views with the prior estimate by comparing the view portfolio weights to the eigenvalues from the prior covariance matrix.

Michaud, et al (2013) provides a blistering critique of the Alternative Reference Model. It is indeed a mixture of fact and opinion, only statistical methods will yield valid prior estimates. Because of it's focus on the Alternative Reference Model (a significant part of the paper is devoted to problems with point estimates) much of it is not relevant to the Canonical Reference Model. Further, the basic arguments restrict the reader to only basic statistical properties of time series (essentially ignoring non-stationarity and auto-regressive properties) and ignores richer, state of the art econometric models.

There are some additional teaching notes put together through the years, Christodoulakis (2002) and Da and Jagnannathan (2005) which provide very basic ideas for class room discussions. The first describes the Canonical Reference model but does not explain what the posterior variance represents. The second quickly describes the model and then works an example illustrating some ways to arrive at the various inputs to the model. The authors use the Alternative Reference Model as they include tau, but do not compute a updated posterior covariance..

Much of the work presented with any of the reference models can be used with the Canonical Reference Model, even if it has been initially formulated for the Alternative Reference Model. Later in this paper we will show how many of these results can be generalized to work with the Canonical Reference Model.

The Canonical Black-Litterman Reference Model

The reference model for returns is the base upon which the rest of the Black-Litterman model is built. It includes the assumptions about which variables are random, and which are not. It also defines which parameters are modeled, and which are not modeled. Most importantly, some authors of papers on the

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Black-Litterman model use reference models, e.g. Alternative Reference Model or Beyond Black-Litterman, which do not have the same theoretical basis as the canonical one which was initially specified in Black and Litterman (1992).

We start with normally distributed expected returns

(1) $r \sim N(\mu, \Sigma)$

The fundamental goal of the Black-Litterman model is to model these expected returns, which are assumed to be normally distributed with mean μ and variance Σ. Note that we will need at least these values, the expected returns and covariance matrix later as inputs into a portfolio selection model.

We define μ, the unknown mean return, as a random variable itself distributed as

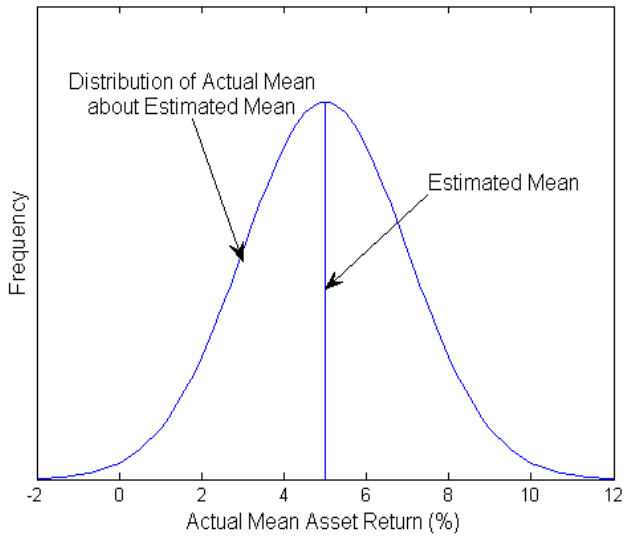
$\mu \sim N(\pi, \Sigma_\pi)$

π is our estimate of the mean and Σ_π is the variance of the unknown mean, μ, about our estimate. Another way to view this simple linear relationship is shown in the formula below.

(2) $\mu = \pi + \epsilon$

The prior returns are normally distributed around π with a disturbance value ϵ. The diagram below shows the distribution of the actual mean about the estimated mean of 5% with a standard deviation of 2% (or a variance of 0.0004). We can see that this is not a very precise estimate from the width of the peak.

Figure 1 - Distribution of Actual Mean about Estimated Mean



ϵ is normally distributed with mean 0 and variance Σ_π and is assumed to be uncorrelated with μ. We can complete the reference model by defining Σ_r as the variance of the returns about our estimate π. From formula (2) and the assumption above that ϵ and μ are not correlated, then the formula to compute Σ_r is

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(3) $\Sigma_r = \Sigma + \Sigma_\pi$

Formula (3) tells us that the proper relationship between the variances is ($\Sigma_r \geq \Sigma, \Sigma_\pi$).

We can check the reference model at the boundary conditions to ensure that it is correct. In the absence of estimation error, e.g. $\varepsilon \equiv 0$, then $\Sigma_r = \Sigma$. As our estimate gets worse, e.g. Σ_π increases, then Σ_r increases as well.

The canonical reference model for the Black-Litterman model expected return is

(4) $r \sim N(\pi, \Sigma_r)$

A common misconception about the canonical Black-Litterman reference model is that formula (1) is the reference model, and that μ is a point estimate. We have previously labeled this approach, the Alternate Reference Model. Several authors approach the problem from this point of view so we cannot neglect it, but it is a fundamentally different model. When considering results from various Black-Litterman implementations it is important to understand which reference model is being used in order to understand how the various parameters will impact the results.

Computing the Equilibrium Returns

The Black-Litterman model starts with a neutral equilibrium portfolio for the prior estimate of returns. The model relies on General Equilibrium theory to state that if the aggregate portfolio is at equilibrium, each sub-portfolio must also be at equilibrium. It can be used with any utility function which makes it very flexible. In practice most practitioners use the Quadratic Utility function and assume a risk free asset, and thus the equilibrium model simplifies to the Capital Asset Pricing Model (CAPM). The neutral portfolio in this situation is the CAPM Market portfolio. Some of the references listed have used other utility functions, most notably Bertsimas (2013), but others have consider CVaR and other measures of portfolio risk without applying the same theoretical basis. In order to preserve the symmetry of the model, the practioner should use the same utility function both to identify the neutral portfolio as well as in the portfolio selection area..

Here we will use the Quadratic Utility function, CAPM and unconsrained mean-variance because it is a well understood model.

Given our previous assumptions, the prior distribution for the Black-Litterman model is the estimated mean excess return from the CAPM market portfolio. The process of computing the CAPM equilibrium excess returns is straight forward.

CAPM is based on the concept that there is a linear relationship between risk (as measured by standard deviation of returns) and return. Further, it requires returns to be normally distributed. This model is of the form

(5) $E(r) = r_f + \beta r_m + \alpha$

Where

r_f The risk free rate.

r_m The excess return of the market portfolio.

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β A regression coefficient computed as $\beta = \rho \frac{\sigma_p}{\sigma_m}$

α The residual, or asset specific (idiosyncratic) excess return.

Under CAPM the idiosyncratic risk associated with an asset is uncorrelated with that from other assets, and this risk can be reduced through diversification. Thus the investor is rewarded for taking systematic risk measured by β, but is not rewarded for taking idiosyncratic risk associated with α.

In the CAPM world, all investors should hold the same risky portfolio, the CAPM market portfolio. Because all investors hold risky assets only in the market portfolio, at equilibrium the market capitalization of the various assets will determine their weights in the market portfolio. The CAPM market portfolio has the maximum Sharpe Ratio[3] of any portfolio on the efficient frontier.

The investor can also invest in a risk free asset. This risk free asset has essentially a fixed positive return for the time period over which the investor is concerned. It is generally some tenor from the sovereign bond yield curve for the investors local currency. Depending on how the asset allocation decision will be framed this risk free asset can range from a 4 week Treasury Bill (1 month horizon) to a 20 year inflation protected bond (20 year horizon).

Note that the CAPM market portfolio contains all investable assets which makes it very hard to actually specify. Because we are in equilibrium, all sub-markets must also be in equilibrium so any sub-market we chose to use is part of the global equilibrium. While this allows us to reverse optimize the excess returns from the market capitalization and the covariance matrix, forward optimization from this point to identify the investors optimal portfolio within CAPM is problematic as we do not have information for the entire market portfolio. In general this is not actually the question an investor is asking, they usually select an investable universe and desire the optimal asset allocation within the universe, so the theoretical problem with the market portfolio can be ignored.

The Capital Market Line is a line through the risk free rate and the CAPM market portfolio. The Two Fund Separation Theorem, closely related to the CAPM, states that all investors should hold portfolios on the Capital Market Line. Any portfolio on the Capital Market Line dominates all portfolios on the Efficient Frontier, the CAPM market portfolio being the only point on the Efficient Frontier and on the Capital Market Line. Depending on their risk aversion an investor will hold arbitrary fractions of their wealth in the risk free asset and/or the CAPM market portfolio.

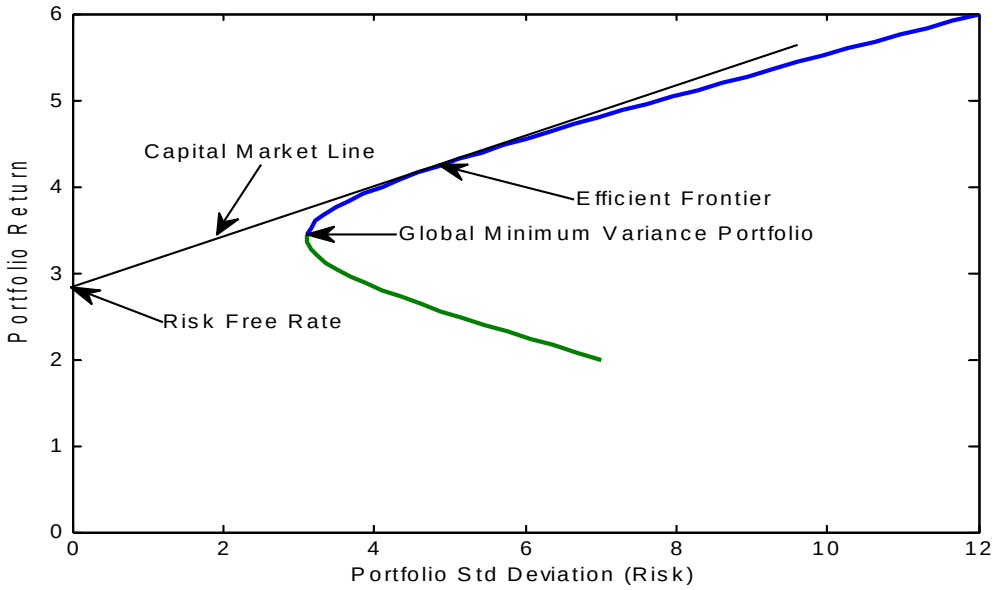
The diagram below illustrates the relationship between the Efficient Frontier and the Capital Market Line.

3 The Sharpe Ratio is the excess return divided by the excess risk, or $(r - r_f) / \sigma$.

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Figure 2 - Efficient Frontier



Since we are starting with the market portfolio, we will be starting with a set of weights which are all greater than zero and naturally sum to one. The market portfolio only includes risky assets, because by definition investors are rewarded only for taking on systematic risk. Thus, in the CAPM, the risk free asset with $\beta = 0$ will not be in the market portfolio. Later on we will see that our Bayesian investor may invest in the risk free asset based on their confidence in their return estimates.

We will constrain the problem by asserting that the covariance matrix of the returns, Σ, is known. In practice, this covariance matrix is estimated from historical return data. It is often computed from higher frequency data and then scaled up to the time frame required for the asset allocation problem. By computing it from actual historical data, we can ensure the covariance matrix is positive definite. Without basing the estimation process on actual data there are significant issues involved in ensuring the covariance matrix is positive definite. We could however apply shrinkage or random matrix theory filters to the covariance matrix in an effort to make it more robust.

For the rest of this section, we will use a common notation, similar to that used in He and Litterman (1999) for all the terms in the formulas. Be aware that this notation is different, and conflicts, with the notation used in the section on Bayesian theory.

Here we derive the equations for 'reverse optimization' starting from the quadratic utility function

(6) $$U = w^T \Pi - (\frac{\delta}{2}) w^T \Sigma w$$

U	Investors utility, this is the objective function during Mean-Variance Optimization.
w	Vector of weights invested in each asset
Π	Vector of equilibrium excess returns for each asset

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δ	Risk aversion parameter
Σ	Covariance matrix of the excess returns for the assets

U is a convex function, so it will have a single global maxima. If we maximize the utility with no constraints, there is a closed form solution. We find the exact solution by taking the first derivative of (6) with respect to the weights (w) and setting it to 0.

$$\frac{dU}{dw} = \Pi - \delta \Sigma w = 0$$

Solving this for Π (the vector of excess returns) yields:

(7) $\Pi = \delta \Sigma w$

In order to use formula (7) to solve for the CAPM market portfolio, we need to have a value for δ, the risk aversion coefficient of the market. One way to find δ is by multiplying both sides of (7) by w^T and replacing vector terms with scalar terms.

$$(r - r_f) = \delta \sigma^2$$

Here our expression at equilibrium is that the excess return to the portfolio is equal to the risk aversion parameter multiplied by the variance of the portfolio.

(8) $\delta = (r - r_f) / \sigma^2$

r	Total return on the market portfolio ($r = w^T \Pi + r_f$)
r_f	Risk free rate
σ^{-2}	Variance of the market portfolio ($\sigma^2 = w^T \Sigma w$)

Most of the authors specify the value of δ that they used. Bevan and Winkelmann (1998) describe their process of calibrating the returns to an average Sharpe Ratio based on their experience. For global fixed income (their area of expertise) they use a Sharpe Ratio of 1.0. Black and Litterman (1992) use a Sharpe Ratio closer to 0.5 in the example in their paper.

Given the Sharpe Ratio we can rewrite formula (8) for δ in terms of the Sharpe Ratio as

(9) $$\delta = \frac{SR}{\sigma_m}$$

Thus we can calibrate our returns in terms of formulas (8) or (9). As part of our analysis we must arrive at the terms on the right hand side of which ever formula we choose to use. For formula (8); this is r, r_f, and $\sigma^{2.}$ in order to calculate a value for δ. For formula (9) this is the Sharpe Ratio and σ.

In order to use formula (8) we need to have an implied return for the market portfolio which may be harder to estimate than the Sharpe Ratio of the market portfolio.

Once we have a value for δ, then we plug w, δ and Σ into formula (7) and generate the set of equilibrium asset returns. Formula (7) is the closed form solution to the reverse optimization problem for computing asset returns given an optimal mean-variance portfolio in the absence of constraints. We can rearrange formula (7) to yield the formula for the closed form calculation of the optimal portfolio weights in the absence of constraints.

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(10) $w = (\delta \Sigma)^{-1} \Pi$

Herold (2005) provides insights into how implied returns can be computed in the presence of simple equality constraints such as the budget or full investment ($\Sigma w = 1$) constraint. He illustrates how errors can be introduced during a reverse optimization process if constraints are assumed to be non-binding when they are actually binding for a given portfolio. Note that because we are dealing with the market portfolio which has only positive weights which sum to 1, we can safely assume that there are no binding constraints on the reverse optimization.

The only missing piece is the variance of our estimate of the mean. Looking back at the reference model, we need Σ_π. Black and Litterman made the simplifying assumption that the structure of the covariance matrix of the estimate is proportional to the covariance of the returns Σ. They created a parameter, τ, as the constant of proportionality. Given that assumption, $\Sigma_\pi = \tau\Sigma$, then the prior distribution P(A) is:

(11) $P(A) \sim N(\Pi, \tau\Sigma), r_A \sim N(P(A), \Sigma)$

This is the prior distribution for the Black-Litterman model. It represents our estimate of the mean, which is expressed as a distribution of the actual unknown mean about our estimate.

Using formula (4) we can rewrite formula (11) in terms of Π as

(12) $r_A \sim N(\Pi, (1 + \tau)\Sigma)$

It is often written that an investor with no views and using an unconstrained mean-variance portfolio selection model will invest 100% in the neutral portfolio, but this is only true if they apply a budget constraint. Because of their uncertainty in the estimates they will invest $1/(1+\tau)$ in the neutral portfolio and $\tau/(1+\tau)$ in the risk free asset. We can see this is the case as follows, starting from (7):

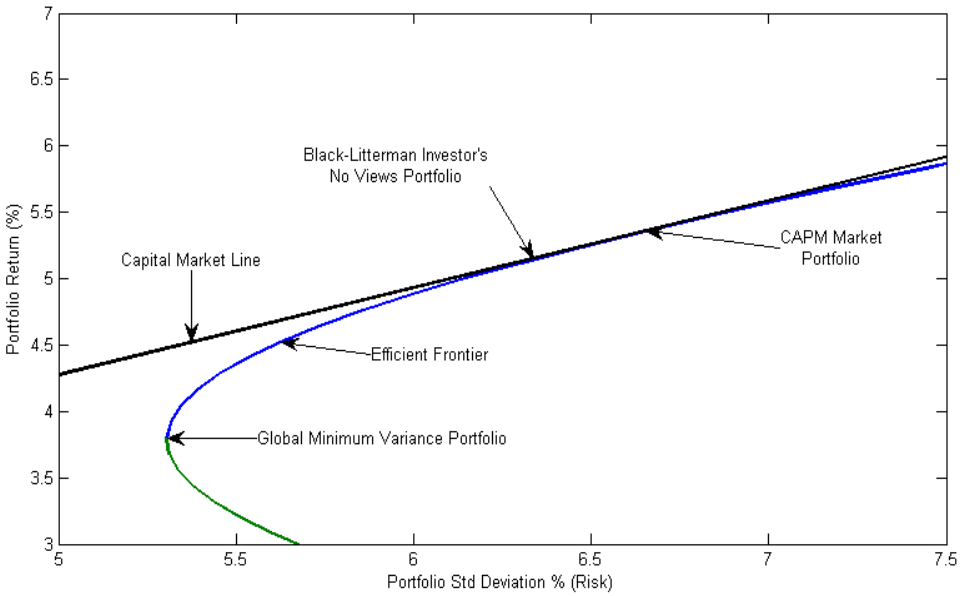
$$\Pi = \delta \Sigma w$$
$$w = (\delta \Sigma)^{-1} \Pi$$
$$\hat{w} = ((1+\tau)\delta \Sigma)^{-1} \Pi$$
$$\hat{w} = (1/1+\tau)(\delta \Sigma)^{-1} \Pi$$
$$\hat{w} = (1/1+\tau) w$$

The diagram below illustrates this concept graphically.

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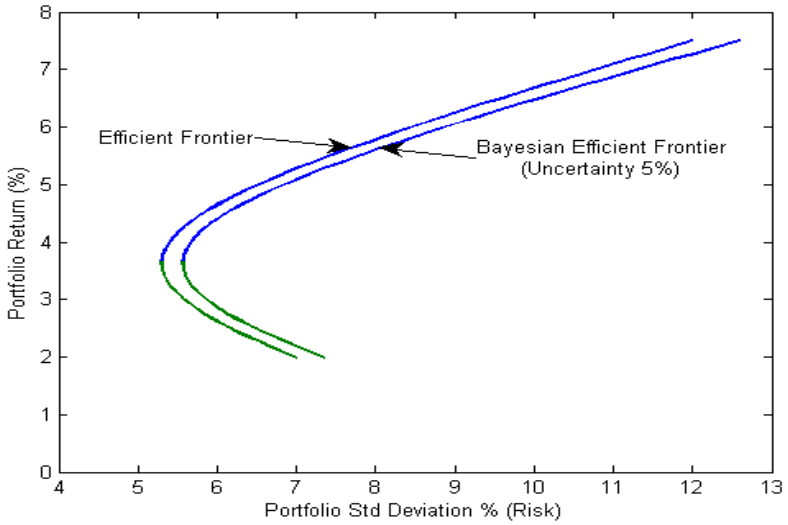
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Figure 3 - Investor's Portfolio in the Absence of Views



We can alternatively view the Bayesian efficient frontier as a shift to the right if we plot the efficient frontier generated with the increased covariance matrix and a budget constraint. In this case the uncertainty just pushes each point further to the right in risk/return space.

Figure 4 - Risk Adjusted Bayesian Efficient Frontier



Specifying the Views

This section will describe the process of specifying the investors views on the estimated mean excess

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returns. We define the combination of the investors views as the conditional distribution. First, by construction we will require each view to be unique and uncorrelated with the other views. This will give the conditional distribution the property that the covariance matrix will be diagonal, with all off-diagonal entries equal to 0. We constrain the problem this way in order to improve the stability of the results and to simplify the problem. Estimating the covariances between views would be even more complicated and error prone than estimating the view variances. Second, we will require views to be fully invested, either the sum of weights in a view is zero (relative view) or is one (an absolute view). We do not require a view on any or all assets. In addition it is actually possible for the views to conflict, the mixing process will merge the views based on the confidence in the views and the confidence in the prior.

We will represent the investors k views on n assets using the following matrices

- P, a k×n matrix of the asset weights within each view. For a relative view the sum of the weights will be 0, for an absolute view the sum of the weights will be 1. Different authors compute the various weights within the view differently, He and Litterman (1999) and Idzorek (2005) use a market capitalization weighed scheme, whereas Satchell and Scowcroft (2000) use an equal weighted scheme in their examples. In practice weights will be a mixture depending on the process used to estimate the view returns.
- Q, a k×1 vector of the returns for each view.
- Ω a k×k matrix of the covariance of the views. Ω is diagonal as the views are required to be independent and uncorrelated. Ω^{-1} is known as the confidence in the investor's views. The i-th diagonal element of Ω is represented as ω_i.

We do not require P to be invertible. Meucci (2006) describes a method of augmenting the matrices to make the P matrix invertible while not changing the net results.

Ω is symmetric and zero on all non-diagonal elements, but may also be zero on the diagonal if the investor is certain of a view. This means that Ω may or may not be invertible. At a practical level we can require that $\omega > 0$ so that Ω is invertible, but we will reformulate the problem so that Ω is not required to be inverted.

As an example of how these matrices would be populated we will examine some investors views. Our example will have four assets and two views. First, a relative view in which the investor believes that Asset 1 will outperform Asset 3 by 2% with confidence. ω_1. Second, an absolute view in which the investor believes that Asset 2 will return 3% with confidence ω_2. Note that the investor has no view on Asset 4, and thus it's return should not be directly adjusted. These views are specified as follows:

$$P=\begin{bmatrix}1 & 0 & -1 & 0\\ 0 & 1 & 0 & 0\end{bmatrix} \;;\; Q=\begin{bmatrix}2\\ 3\end{bmatrix} \;;\; \Omega=\begin{bmatrix}\omega_{11} & 0\\ 0 & \omega_{22}\end{bmatrix}$$

Given this specification of the views we can formulate the conditional distribution mean and variance in view space as

$P(B|A) \sim N(Q, \Omega)$

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We generally cannot convert this into a useful expression into asset space because of the mixture of relative and absolute views, and because the P matrix is not required to be of full rank. If we did express the views in asset space the formula is shown below.

(13) $P(B|A) \sim N(P^{-1}Q, [P^T\Omega^{-1}P]^{-1})$

This representation is not of any practical use. Incomplete views and relative views make the variance non-invertible, and relative views also impact the mean term. Remember that P may not be invertible, and even if P is invertible $[P^T\Omega^{-1}P]$ is probably not invertible, making this expression impossible to evaluate in practice. Luckily, to work with the Black-Litterman model we don't need to evaluate formula (13). It is however, interesting to see how the views are projected into the asset space.

Specifying Ω

Ω, the variance of the views is inversely related to the investors confidence in the views, however the basic Black-Litterman model does not provide an intuitive way to quantify this relationship. It is up to the investor to compute the variance of the views Ω.

There are several ways to calculate Ω.

- Proportional to the variance of the prior
- Use a confidence interval
- Use the variance of residuals in a factor model
- Use Idzorek's method to specify the confidence along the weight dimension

Proportional to the Variance of the Prior

We can just assume that the variance of the views will be proportional to the variance of the asset returns, just as the variance of the prior distribution is. Both He and Litterman (1999), and Meucci (2006) use this method, though they use it differently. He and Litterman (1999) set the variance of the views as follows:

(14)
$$\omega_{ij} = p(\tau\Sigma)p^T \quad \forall i = j$$
$$\omega_{ij} = 0 \quad \forall i \neq j$$

or

$$\Omega = diag(P(\tau\Sigma)P^T)$$

This specification of the variance, or uncertainty, of the views essentially equally weights the investor's views and the market equilibrium weights. By including τ in the expression, the posterior estimate of the returns becomes independent of τ as well. This seems to be the most common method used in the literature.

Meucci (2006) doesn't bother with the diagonalization at all, and just sets

$$\Omega = \frac{1}{c} P \Sigma P^t$$

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He sets c >1, and one obvious choice for c is τ^{-1}.We will see later that this form of the variance of the views lends itself to some simplifications of the Black-Litterman formulas.

Use a Confidence Interval

The investor can specify the variance using a confidence interval around the estimated mean return, e.g. Asset has an estimated 3% mean return with the expectation it is 68% likely to be within the interval (2.0%,4.0%). Knowing that 68% of the normal distribution falls within 1 standard deviation of the mean allows us to translate this into a variance for the view of $(1\%)^2$.

Note that Ω is our uncertainty in the estimate of the mean, we are not specifying the variance of returns about the mean. This formulation of the variance of the view is consistent with the canonical reference model.

Use the Variance of Residuals from a Factor Model

If the investor is using a factor model to compute the views, they can use the variance of the residuals from the model to drive the variance of the return estimates. The general expression for a factor model of returns is:

$$(15) \qquad r = \sum_{i=1}^{n} \beta_i f_i + \epsilon$$

Where

r	is the return of the asset
β_i	is the factor loading for factor (i)
f_i	is the return due to factor (i)
ϵ	is an independent normally distributed residual

And the general expression for the variance of the return from a factor model is:

$$(16) \qquad V(r) = B\,V(F)\,B^T + V(\epsilon)$$

B	is the factor loading matrix
F	is the vector of returns due to the various factors

Given formula (15), and the assumption that ε is independent and normally distributed, then we can compute the variance of ε directly as part of the regression. While the regression might yield a full covariance matrix, the mixing model will be more robust if only the diagonal elements are used.

Beach and Orlov (2006) describe their work using GARCH style factor models to generate their views for use with the Black-Litterman model. They generate the precision of the views using the GARCH models.

Use Idzorek's Method

Idzorek (2005) describes a method for specifying the confidence in the view in terms of a percentage move of the weights on the interval from 0% confidence to 100% confidence. We will look at

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Idzorek's algorithm in the section on extensions.

The Estimation Model

The original Black-Litterman paper references Theil's Mixed Estimation model rather than a Bayesian estimation model, though we can get similar results from both methodologies. We choose to start with Theil's model because it is simpler and cleaner. We also work through the Bayesian version of the derivations for completeness.

With either approach, we will be using the Canonical Black-Litterman Reference Model. In this reference model the estimation model is used to compute the distribution of the estimated returns about the mean return, and then to estimate the distribution of returns about the mean return. This distinction is important in understanding the values used for τ and Ω, and for the computations of the variance of the prior and posterior distributions of returns.

The posterior estimate of the mean generated by the estimation model is more precise than either the prior estimate or the investors views. Note that we would not expect large changes in the variance of the distribution of returns about the mean because our estimate of the mean is more precise. The prototypical example of this would be to blend the distributions, P(A) ~ N(10%, 20%) and P(B|A) ~ N(12%, 20%). If we apply either estimation model in a straightforward fashion, P(A|B) ~ N(11%, 10%). Clearly with financial data we did not really cut the variance of the return distribution about the mean in ½ just because we have a slightly better estimate of the mean. In this case, the mean is the random variable, thus the variance of our posterior corresponds to the variance of the estimated mean around the mean return; not the variance of the distribution of returns about the mean return. In this case, the posterior result of P(A|B) ~ N(11%, 10%) makes sense. By blending these two estimates of the mean, we have an estimate of the mean with much less uncertainty (less variance) than either of the estimates, even though we do not have a better estimate of the the actual distribution of returns around the mean.

Theil's Mixed Estimation Model

Theil's mixed estimation model was created for the purpose of estimating parameters from a mixture of complete prior data and partial conditional data. This is a good fit with our problem as it allows us to express views on only a subset of the asset returns, there is no requirement to express views on all of them. The views can also be expressed on a single asset, or on arbitrary combinations of the assets. The views do not even need be consistent, the estimation model will take each into account based on the investors confidence.

Theil's Mixed Estimation model starts from a linear model for the parameters to be estimated. We can use formula (2) from our reference model as a starting point.

Our simple linear model is shown below:

(17) $x\beta = \pi + u$

Where

π is the n x 1 vector of equilibrium returns for the assets.

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x is the n x n matrix I_n which are the factor loadings for our model.
β is the n x 1 vector of unknown means for the asset return process.
u is a n x n matrix of residuals from the regression where $E(u)=0\,;V(u)=E(u'u)=\Phi$ and Φ is non-singular.

The Black-Litterman model uses a very simple linear model, the expected return for each asset is modeled by a single factor which has a coefficient of 1. Thus, x, is the identity matrix. Given that β and u are independent, and x is constant, then we can model the variance of π as:

$$V(\pi)=x'V(\beta)x+V(u)$$

Which can be simplified to:

(18) $$V(\pi)=\Sigma+\Phi$$

Where

Σ is the historical covariance matrix of the returns as used earlier.
Φ is the covariance of the residuals or of the estimate about the actual mean.

This ties back to formula (3) in the reference model. The total variance of the estimated return is the sum of the variance of the actual return process plus the variance of the estimate of the mean. We will come back to this relation again later in the paper.

Next we consider some additional information which we would like to combine with the prior. This information can be subjective views or can be derived from statistical data. We will also allow it to be incomplete, meaning that we might not have an estimate for each asset return.

(19) $$p\beta=q+v$$

Where

q is the k x 1 vector of returns for the views.
p is the k x n matrix mapping the views onto the assets.
β is the n x 1 vector of unknown means for the asset return process.
v is a k x 1 vector of of residuals from the regression where $E(v)=0\,;V(v)=E(v'v)=\Omega$ and Ω is non-singular.

We can combine the prior and conditional information by writing:

$$\begin{bmatrix} x \\ p \end{bmatrix}\hat{\beta}=\begin{bmatrix} \pi \\ q \end{bmatrix}+\begin{bmatrix} u \\ v \end{bmatrix}$$

Where the expected value of the residual is 0, and the expected value of the variance of the residual is

$$V\left(\begin{bmatrix} u \\ v \end{bmatrix}\right)=E\left(\begin{bmatrix} u \\ v \end{bmatrix}\begin{bmatrix} u' & v' \end{bmatrix}\right)=\begin{bmatrix} \Phi & 0 \\ 0 & \Omega \end{bmatrix}$$

We can then apply the generalized least squares procedure, which leads to estimating $\hat{\beta}$ as

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$$\hat{\beta}=\left[\begin{bmatrix}x' & p'\end{bmatrix}\begin{bmatrix}\Phi & 0\\ 0 & \Omega\end{bmatrix}^{-1}\begin{bmatrix}x\\ p\end{bmatrix}\right]^{-1}\begin{bmatrix}x' & p'\end{bmatrix}\begin{bmatrix}\Phi & 0\\ 0 & \Omega\end{bmatrix}^{-1}\begin{bmatrix}\pi\\ q\end{bmatrix}$$

This can be rewritten without the matrix notation as

(20) $$\hat{\beta}=\left[x'\Phi^{-1}x+p'\Omega^{-1}p\right]^{-1}\left[x'\Phi^{-1}\pi+p'\Omega^{-1}q\right]$$

For the Black-Litterman model which is a single factor per asset, we can drop x as it is the identity matrix. If one wanted to use a multi-factor model for the equilibrium, then x would be the equilibrium factor loading matrix.

(21) $$\hat{\beta}=\left[\Phi^{-1}+p'\Omega^{-1}p\right]^{-1}\left[\Phi^{-1}\pi+p'\Omega^{-1}q\right]$$

This new $\hat{\beta}$ is the weighted average of the estimates, where the weighting factor is the precision of the estimates. The precision is the inverse of the variance. The posterior estimate $\hat{\beta}$ is also the best linear unbiased estimate given the data, and has the property that it minimizes the variance of the residual. Note that given a new $\hat{\beta}$, we also should have an updated expectation for the variance of the residual.

If we were using a factor model for the prior, then we would keep x, the factor weightings, in the formulas. This would give us a multi-factor model, where all the factors will be priced into the equilibrium.

We can reformulate our combined relationship in terms of our estimate of $\hat{\beta}$ and a new residual $\tilde{u}$ as

$$\begin{bmatrix}x\\ p\end{bmatrix}\hat{\beta}=\begin{bmatrix}\pi\\ q\end{bmatrix}+\tilde{u}$$

Once again $E(\tilde{u})=0$, so we can derive the expression for the variance of the new residual as:

(22) $$V(\tilde{u})=E(\tilde{u}'\tilde{u})=\left[\Phi^{-1}+p'\Omega^{-1}p\right]^{-1}$$

and the total variance is

$$V(\begin{bmatrix}y & \pi\end{bmatrix})=V(\hat{\beta})+V(\tilde{u})$$

We began this section by asserting that the variance of the return process is a known quantity. Improved estimation of the quantity $\hat{\beta}$ does not change our estimate of the variance of the return distribution, Σ. Because of our improved estimate, we do expect that the variance of the estimate (residual) has decreased, thus the total variance has changed. We can simplify the variance formula (18) to

(23) $$V(\begin{bmatrix}y & \pi\end{bmatrix})=\Sigma+V(\tilde{u})$$

This is a clearly intuitive result, consistent with the realities of financial time series. We have combined two estimates of the mean of a distribution to arrive at a better estimate of the mean. The variance of this estimate has been reduced, but the actual variance of the underlying process has not

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changed. Given our uncertain estimate of the process, the total variance of our estimated process has also improved incrementally, but it has the asymptotic limit that it cannot be less than the variance of the actual underlying process.

This is the convention for computing the covariance of the posterior distribution of the canonical reference model as shown in He and Litterman (1999).

In the absence of views,formula (23) simplifies to

$$V(\begin{bmatrix} y & \pi \end{bmatrix}) = \Sigma + \Phi$$

Which is the variance of the prior distribution of returns.

Appendix A contains a more detailed derivation of formulas (21) and (22).

Using Bayes Theorem for the Estimation Model

See Appendix B for a brief introduction to Bayes Theorem as it is used in deriving the Black-Litterman model.

In the Black-Litterman model, the prior distribution is based on the equilibrium implied excess returns. One of the major assumptions made by the Black-Litterman model is that the covariance of the prior estimate is proportional to the covariance of the actual returns, but the two quantities are independent. The parameter τ will serve as the constant of proportionality. The prior distribution for the Black-Litterman model was specified in formula (11).

The conditional distribution is based on the investor's views. The investor's views are specified as returns to portfolios of assets, and each view has an uncertainty which will impact the overall mixing process. The conditional distribution from the investor's views was specified in formula (13).

The posterior distribution from Bayes Theorem is the precision weighted average of the prior estimate and the conditional estimate. We can now apply Bayes theory to the problem of blending the prior and conditional distributions to create a new posterior distribution of the asset returns. Given equations (11) and (13), for our prior and conditional distribution respectively, we can apply Bayes Theorem and derive the following formula for the posterior distribution of asset returns.

(24) $$P(A|B) \sim N([(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q][(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}, [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1})$$

This is sometimes referred to as the Black-Litterman master formula. A complete derivation of the formula is shown in Appendix B. An alternate representation of the same formula for the mean returns $\hat{\Pi}$ and covariance (M) is

(25) $$\hat{\Pi} = \Pi + \tau\Sigma P^T[(P\tau\Sigma P^T) + \Omega]^{-1}[Q - P\Pi]$$

(26) $$M = ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}$$

The derivation of formula (25) is shown in Appendix D. Remember that M, the posterior variance, is the variance of the posterior mean estimate about the actual mean. It is the uncertainty in the posterior mean estimate, and is not the variance of the returns.

Computing the posterior covariance of returns requires adding the variance of the estimate about the

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mean, to the variance of the distribution about the estimate the same as in (23). This is mentioned in He and Litterman (1999) but not in any of the other papers.

(27) $\Sigma_p = \Sigma + M$

Substituting the posterior variance from (26) we get

$$\Sigma_p = \Sigma + ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}$$

In the absence of views this reduces to

(28) $\Sigma_p = \Sigma + (\tau\Sigma) = (1 + \tau)\Sigma$

Thus when applying the Black-Litterman model in the absence of views the variance of the estimated returns will be greater than the prior distribution variance. We see the impact of this formula in the results shown in He and Litterman (1999). In their results, the investor's weights sum to less than 1 if they have no views.[4] Idzorek (2005) and most other authors do not compute a new posterior variance, but instead use the known input variance of the returns about the mean.

In the event our investor has only partial views, that is views on a subset of the assets, then by using a posterior estimate of the variance we will tilt the posterior weights towards assets with lower variance (higher precision of the estimated mean) and away from assets with higher variance (lower precision of the estimated mean). Thus the existence of the views and the updated covariance will tilt the optimizer towards using or not using those assets. This tilt will not be very large if we are working with a small value of τ, but it will be measurable.

Since we often build the known covariance matrix of returns, Σ, from historical data we can use methods from basic statistics to compute τ. as $\tau\Sigma$ is analogous to the standard error. We can also estimate τ based on our confidence in the prior distribution. Note that both of these techniques provide some intuition for selecting a value of τ which is closer to 0 than to 1. Black and Litterman (1992), He and Litterman (1999) and Idzorek (2005) all indicate that in their calculations they used small values of τ, on the order of 0.025 – 0.050. Satchell and Scowcroft (2000) state that many investors use a τ around 1, which has no intuitive connection to the data and in fact shows that their paper uses the Alternative Refernce Model.

We can check our results by seeing if the results match our intuition at the boundary conditions.

Given formula (25) it is easy to let $\Omega \rightarrow 0$ showing that the return under 100% certainty of the views is

(29) $$\hat{\Pi} = \Pi + \Sigma P^T [P \Sigma P^T]^{-1} [Q - P \Pi]$$

Thus under 100% certainty of the views, the estimated return is insensitive to the value of τ used.

Furthermore, if P is invertible which means that we have offered a view on every asset, then

$$\hat{\Pi} = P^{-1} Q$$

If the investor is not sure about their views, so $\Omega \rightarrow \infty$, then formula (25) reduces to

$$\hat{\Pi} = \Pi$$

4 This is shown in table 4 and mentioned on page 11 of He and Litterman (1999).

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Finding an analytically tractable way to express and compute the posterior variance under 100% certainty is a challenging problem.

The alternate formula for the posterior variance derived from (26) using the Woodbury Matrix Identity is

$$M = \tau\Sigma - \tau\Sigma P^T (P\tau\Sigma P^T + \Omega)^{-1} P\tau\Sigma \tag{30}$$

If $\Omega \to 0$ (total confidence in views, and every asset is in at least one view) then formula (30) can be reduced to M = 0. If on the other hand the investor is not confident in their views, $\Omega \to \infty$, then formula (30) can be reduced to M = $\tau\Sigma$.

The Alternative Reference Model

This section will discuss the most common alternative reference model used with the Black-Litterman estimation model.

The most common alternative reference model is the one used in Satchell and Scowcroft (2000), and in the work of Meucci prior to his introduction of "Beyond Black-Litterman".

$$E(r) \sim N(\mu, \Sigma)$$

In this reference model, μ is normally distributed with variance Σ. We estimate μ, but μ is not considered a random variable. This is commonly described as having a $\tau = 1$, but more precisely we are making a point estimate and thus have eliminated τ as a parameter. In this model Ω becomes the covariance of the returns to the views around the unknown mean return, , just as Σ is the covariance of the prior return about it's mean. Gven that we are now using point estimates, the posterior is now a point estimate and we no longer worry about posterior covariance of the estimate. In this model we do not have a posterior precision to use downstream in our portfolio selection model.

We rewrite formula (25), noting that we can move around the τ term.

$$\hat{\Pi} = \Pi + \Sigma P^T [(P\Sigma P^T) + \frac{\Omega}{\tau}]^{-1} [Q - P\Pi]$$

Now we see that τ only appears in one term in this formula. Because the Alternative Reference Model does not include updating the Covariance of the estimates this is the only formula. Given that the investor is selecting both Ω and τ to control the blending of the prior and their views, we can eliminate one of the terms. Since τ is a single term for all views and Ω has a separate element for each view, we will keep Ω. We can then rewrite the posterior estimate of the mean, as follows.

$$\hat{\Pi} = \Pi + \Sigma P^T [(P\Sigma P^T) + \Omega]^{-1} [Q - P\Pi] \tag{31}$$

The primary artifacts of this new reference model are, first τ is gone, and second, the investor's portfolio weights in the absence of views equal the equilibrium portfolio weights. Finally at implementation time there is no need or use of formulas (26) or (27).

Note that none of the authors prior to Meucci (2008) except for Black and Litterman (1992), and He and Litterman (1999) make any mention of the details of the Canonical Reference Model, or of the fact that different authors actually use quite different reference models.

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In the Canonical Reference Model, the updated posterior covariance of the unknown mean about the estimate will be smaller than the covariance of either the prior or conditional estimates, indicating that the addition of more information will reduce the uncertainty of the model. The variance of the returns from formula (27) will never be less than the prior variance of returns. This matches our intuition as adding more information should reduce the uncertainty of the estimates. Given that there is some uncertainty in this value (M), then formula (27) provides a better estimator of the variance of returns than the prior variance of returns.

The Impact of τ

The meaning and impact of the parameter τ causes a great deal of confusion for many users of the Black-Litterman model. We know that investors using the Canonical Reference Model use τ and that it has a very precise meaning in the model. An author who selects an essentially random value for τ is likely not using the Canonical Reference Model, but is instead using the Alternative Reference Model.

Given the Canonical Reference Model we can still perform an exercise to understand the impact of τ on the results. We will start with the expression for Ω similar to the one used by He and Litterman (1999). Rather than using only the diagonal, we will retain the entire structure of the covariance matrix to make the math simpler.

(32) $\Omega = P(\tau\,\Sigma)\,P^T$

We can substitute this into formula (25) as

$$\begin{aligned}
\hat{\Pi} &= \Pi + \tau\,\Sigma\,P^T[(P\tau\,\Sigma\,P^T) + \Omega]^{-1}[Q - P\,\Pi^T] \\
&= \Pi + \tau\,\Sigma\,P^T[(P\tau\,\Sigma\,P^T) + (P\tau\,\Sigma\,P^T)]^{-1}[Q - P\,\Pi^T] \\
&= \Pi + \tau\,\Sigma\,P^T[2(P\tau\,\Sigma\,P^T)]^{-1}[Q - P\,\Pi^T] \\
&= \Pi + (\frac{1}{2})\tau\,\Sigma\,P^T(P^T)^{-1}[P\tau\,\Sigma]^{-1}[Q - P\,\Pi^T] \\
&= \Pi + (\frac{1}{2})\tau\,\Sigma\,(\tau\,\Sigma)^{-1}\,P^{-1}[Q - P\,\Pi^T] \\
&= \Pi + (\frac{1}{2})\,P^{-1}[Q - P\,\Pi^T]
\end{aligned}$$

(33) $$\hat{\Pi} = \Pi + (\frac{1}{2})[P^{-1}Q - \Pi^T]$$

Clearly using formula (32) is just a simplification and does not do justice to investors views, but we can still see that setting Ω proportional to τ, will eliminate τ from the final formula for $\hat{\Pi}$. In the Canonical Reference Model it does not eliminate τ from the formulas for posterior covariance (27). In the general form if we formulate Ω as

(34) $\Omega = P(\alpha\tau\,\Sigma)\,P^T$

Then we can rewrite formula (33) as

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(35) $$\hat{\Pi} = \Pi + (\frac{1}{1+\alpha})[P^{-1}Q - \Pi]$$

We can see a similar result if we substitute formula (32) into formula (30).

$$\begin{aligned} M &= \tau\Sigma - \tau\Sigma P^T[P\tau\Sigma P^T + \Omega]^{-1} P\tau\Sigma \\ &= \tau\Sigma - \tau\Sigma P^T[P\tau\Sigma P^T + P\tau\Sigma P^T]^{-1} P\tau\Sigma \\ &= \tau\Sigma - \tau\Sigma P^T[2(P\tau\Sigma P^T)]^{-1} P\tau\Sigma \\ &= \tau\Sigma - (\frac{1}{2})(\tau\Sigma)(P^T)(P^T)^{-1}(\tau\Sigma)^{-1}(P)^{-1}(P)(\tau\Sigma) \\ &= \tau\Sigma - (\frac{1}{2})\tau\Sigma \end{aligned}$$

(36) $$M = (\frac{1}{2})\tau\Sigma$$

Note that τ is not eliminated from formula (36). We can also observe that if τ is on the order of 1 and we were to use formula (27) that the uncertainty in the estimate of the mean would be a significant fraction of the variance of the returns. With the Alternative Reference Model, no posterior variance calculations are performed and the mixing is weighted by the variance of returns.

In both cases, our choice for Ω has evenly weighted the prior and conditional distributions in the estimation of the posterior distribution. This matches our intuition when we consider we have blended two inputs, for both of which we have the same level of uncertainty. The posterior distribution will be the average of the two distributions.

If we instead solve for the more useful general case of $\Omega = \alpha P(\tau\Sigma)P^T$ where $\alpha \geq 0$, substituting into (25) and following the same logic as used to derive (36) we get

(37) $$\hat{\Pi} = \Pi + \frac{1}{(1+\alpha)}\left[P^{-1}Q - \Pi\right]$$

This parametrization of the uncertainty is specified in Meucci (2005) and it allows us an option between using the same uncertainty for the prior and views, and having to specify a separate and unique uncertainty for each view. Given that we are essentially multiplying the prior covariance matrix by a constant this parametrization of the uncertainty of the views does not have a negative impact on the stability of the results.

Note that this specification of the uncertainty in the views changes the assumption from the views being uncorrelated, to the views having the same correlations as the prior returns.

In summary, if the investor uses the Alternative Reference Model and makes Ω proportional to Σ, then they need only calibrate the constant of proportionality, α, which indicates their relative confidence in their views versus the equilibrium. If they use the Canonical Referemce Model and set Ω proportional to $\tau\Sigma$, then the return estimate will not depend on the value of τ, but the posterior covariance of returns will depend on the proper calibration of τ.

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Calibrating τ

This section will discuss some empirical ways to select and calibrate the value of τ.

The first method to calibrate τ relies on falling back to basic statistics. When estimating the mean of a distribution, the uncertainty (variance) of the mean estimate will be proportional to the inverse of the number of samples. Given that we are estimating the covariance matrix from historical data, then

$\tau = \frac{1}{T}$ The maximum likelihood estimator

$\tau = \frac{1}{T-k}$ The best quadratic unbiased estimator

T The number of samples

k The number of assets

There are other estimators, but usually, the first definition above is used. Given that we usually aim for a number of samples around 60 (5 years of monthly samples) then τ is on the order of 0.02. This is consistent with several of the papers which indicate they used values of τ on the range (0.025, 0.05).

The most intuitively easy way to calibrate τ is as part of a confidence interval for the prior mean estimates. We will illustrate this concept with a simple example. Consider the scenario where τ = 0.05 and we consider only a single asset with a prior estimate of 7% as the excess return and 15% as the known standard deviation of returns about the mean. For our view we use a confidence interval of (1%, 5%) with 68% confidence.

We keep the ratio of View Precision to Prior Precision fixed between the two scenarios, one with τ = 0.05 and one with τ = 1. In the first scenario even though we use a seemingly small τ = 0.05, our prior estimate has relatively low precision based on the width of the confidence interval, and thus the posterior estimate will be heavily weighted towards the view. In the second scenario with τ = 1, the prior confidence interval is so wide as to make the prior estimate close to worthless. In order to keep the posterior estimate the same across scenarios, the view estimate also has a wide confidence interval indicating the investor is really not confident in any of their estimates.

τ	Prior @ 68% Confidence	Prior Precision	View σ	View Precision	View @ 68% Confidence	View/Prior Precision
0.05	(4.6%, 9.4%)	888	2.00%	2500	(1%, 5%)	2.81
1.00	(-8%, 22%)	44.4	8.90%	125	(-5.9%, 11.9%)	2.81

Given such wide intervals for the τ = 1 scenario, 16% confidence that our asset has a mean return less than -8%, it is hard to imagine having much conviction in using the final asset allocation. Understanding the interplay between the selection of τ and the specification of the variance of the views is critical.

Note that this example illustrates the difference between the parameters for the Canonical Reference

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Model and the Alternative Reference Model. Specifying $\tau = 1$ is the Alternative Reference Model, but just generates garbage outputs from the Canonical Reference Model.

Finally, we could instead calibrate τ to the amount invested in the risk free asset given the prior distribution. Here we see that the portfolio invested in risky assets given the prior views will be

$$w = \Pi \left[\delta \left(1 + \tau \right) \Sigma \right]^{-1}$$

Thus the weights allocated to the assets are smaller by $[1/(1+\tau)]$ than the CAPM market weights. This is because our Bayesian investor is uncertain in their estimate of the prior, and they do not want to be 100% invested in risky assets.

Results

This section of the document will step through a comparison of the results of the various authors. The java programs used to compute these results are all available as part of the akutan open source finance project at sourceforge.net. All of the mathematical functions were built using the Colt open source numerics library for Java. Selected formulas are also available as MATLAB and/or SciLab scripts on the website blacklitterman.org. Any small differences between my results and the authors reported results are most likely the result of rounding of inputs and/or results.

When reporting results most authors have just reported the portfolio weights from an unconstrained optimization using the posterior mean and variance. Given that the vector Π is the excess return vector, then we do not need a budget constraint ($\Sigma w_i = 1$) as we can safely assume any 'missing' weight is invested in the risk free asset which has expected return 0 and variance 0. This calculation comes from formula (10).

$$w = \Pi(\delta\Sigma_p)^{-1}$$

As a first test of our algorithm we verify that when the investor has no views that the weights are correct, substituting formula (28) into (10) we get

$$w_{nv} = \Pi(\delta(1+\tau)\Sigma)^{-1}$$

$$w_{nv} = w/(1+\tau) \quad (38)$$

Given this result, it is clear that the output weights with no views will be impacted by the choice of τ when the Black-Litterman reference model is used. He and Litterman (1999) indicate that if our investor is a Bayesian, then they will not be certain of the prior distribution and thus would not be fully invested in the risky portfolio at the start. This is consistent with formula (38).

Matching the Results of He and Litterman

First we will consider the results shown in He and Litterman (1999). These results are the easiest to reproduce as they clearly implement the Canonical Reference Model and they provide all the data required to reproduce their results in the paper.

He and Litterman (1999) set

$$\Omega = \text{diag}(P^T(\tau\Sigma)P) \quad (39)$$

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This essentially makes the uncertainty of the views equivalent to the uncertainty of the equilibrium estimates. They select a small value for τ (0.05), and they use the Canonical Reference Model and the updated posterior variance of returns as calculated in formulas (30) and (27).

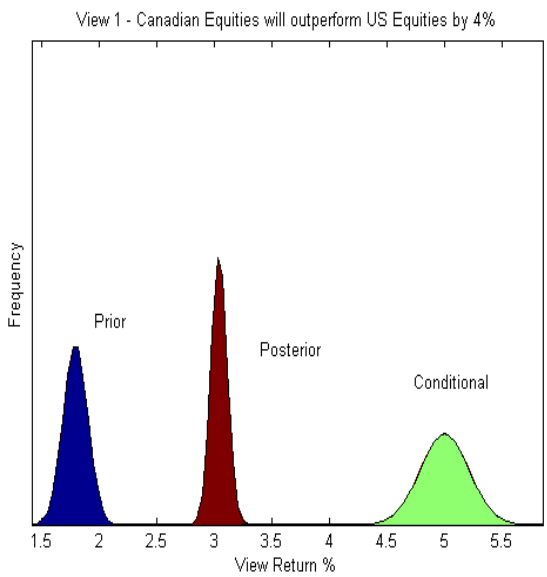
Table 1 – These results correspond to Table 7 in [He and Litterman, 1999].

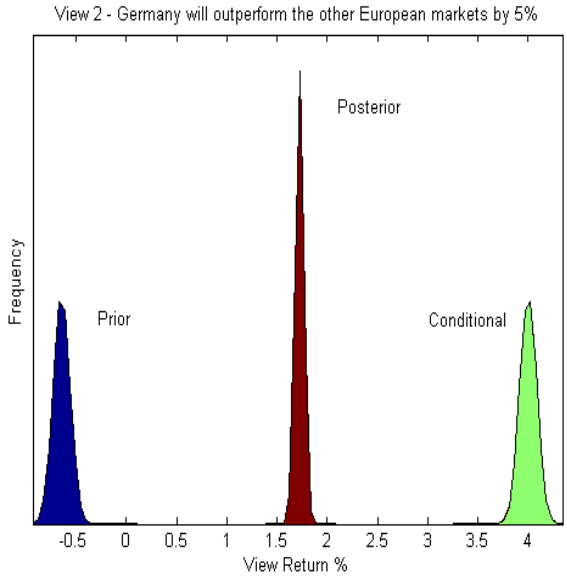
Asset	P_0	P_1	μ	$w_{eq}/(1+\tau)$	w*	w* - $w_{eq}/(1+\tau)$
Australia	0.0	0.0	4.3	16.4	1.5%	.0%
Canada	0.0	1.0	8.9	2.1%	53.9%	51.8%
France	-0.295	0.0	9.3	5.0%	-.5%	-5.4%
Germany	1.0	0.0	10.6	5.2%	23.6%	18.4%
Japan	0.0	0.0	4.6	11.0%	11.0%	.0%
UK	-0.705	0.0	6.9	11.8%	-1.1%	-13.0%
USA	0.0	-1.0	7.1	58.6%	6.8%	-51.8%
q	5.0	4.0				
ω/τ	.043	.017				
λ	.193	.544				

Table 1 contains results computed using the akutan implementation of Black-Litterman and the input data for the equilibrium case and the investor's views from He and Litterman (1999). The values shown for w* exactly match the values shown in their paper.Figure 5 - Distributions of Actual Means about Estimates Means

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The preceding diagrams show the pdf for the prior, view and posterior for each View defined in the problem. The y axis uses the same scale in each graph. Note how in view 1 the view (conditional distribution of the estimated mean) is much more diffuse because the variance of the estimate is larger (precision of the estimate is smaller). Also note how the precision of the prior and views impacts the precision (width of the peak) on the pdf. In view 1 with the less precise view, the posterior is also less precise.

Matching the Results of Idzorek

This section of the document describes the efforts to reproduce the results of Idzorek (2005). In trying to match Idzorek's results we found that he used the Alternative Reference Model. which leaves Σ, the known variance of the returns from the prior distribution, as the variance of the posterior returns. This is a significant difference from the Canonical Reference Model,, but in the end the differences amounted to only 50 basis points per asset. Tables 2 and 3 below illustrate computed results with the data from his paper and how the results differ between the two versions of the model.

Table 2 contains results generated using the data from Idzorek (2005) and the Canonical Reference Model.. Table 3 shows the same results as generated by the Alternative Reference Model.

Table 2 – Canonical Reference Model with Idzorek data (Corresponds with data in Idzorek's Table 6).

Asset Class	μ	w_{eq}	w*	Black-Litterman Reference Model	Idzorek's Results
US Bonds	.07	18.87%	28.96%	10.09%	10.54
Intl Bonds	.50	25.49%	15.41%	-10.09%	-10.54

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Asset Class	μ	w_{eq}	w*	Black-Litterman Reference Model	Idzorek's Results
US LG	6.50	11.80%	9.27%	-2.52%	-2.73
US LV	4.33	11.80%	14.32%	2.52%	-2.73
US SG	7.55	1.31%	1.03%	-.28%	-0.30
US SV	3.94	1.31%	1.59%	.28%	0.30
Intl Dev	4.94	23.59%	27.74%	4.15%	3.63
Intl Emg	6.84	3.40%	3.40%	.0%	0

Note that the results in Table 2 are close, but for several of the assets the difference is about 50 basis points. The values shown in Table 3 are within 4 basis points, essentially matching the results reported by Idzorek.

Table 3 – Alternative Reference Model with Idzorek data (Corresponds with data in Idzorek's Table 6).

Country	μ	w_{eq}	w	Alternative Reference Model	Idzorek's Results
US Bonds	.07	19.34%	29.89%	10.55%	10.54
Intl Bonds	.50	26.13%	15.58%	-10.55%	-10.54
US LG	6.50	12.09%	9.37%	-2.72%	-2.73
US LV	4.33	12.09%	14.81%	2.72%	-2.73
US SG	7.55	1.34%	1.04%	-.30%	-0.30
US SV	3.94	1.34%	1.64%	.30%	0.30
Intl Dev	4.94	24.18%	27.77%	3.59%	3.63
Intl Emg	6.84	3.49%	3.49%	.0%	0

Additional Work

This section provides a brief discussion of efforts to reproduce results from some of the major research papers on the Black-Litterman model.

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Of the major papers on the Black-Litterman model, there are two which would be very useful to reproduce, Satchell and Scowcroft (2000) and Black and Litterman (1992). Satchell and Scowcroft (2000) does not provide enough data in their paper to reproduce their results. They have several examples, one with 11 countries equity returns plus currency returns, and one with fifteen countries. They don't provide the covariance matrix for either example, and so their analysis cannot be reproduced. It would be interesting to confirm that they use the Alernative Reference Model by reproducing their results.

Black and Litterman (1992) do provide what seems to be all the inputs to their analysis, however they chose a non-trivial example including partially hedged equity and bond returns. This requires the application of some constraints to the reverse optimization process which we have been unable to formulate as of this time. We plan on continuing this work with the goal of verifying the details of the Black-Litterman implementation used in Black and Litterman (1992).

Extensions to the Black-Litterman Model

In this section we will cover the extensions to the Black-Litterman model proposed in Idzorek (2005), Fusai and Meucci (2003), Krishnan and Mains (2006) and Qian and Gorman (2001).

Idzorek (2005) presents a means to calibrate the confidence or variance of the investors views in a simple and straightforward method.

Next is a section on measures of extremity or quality of views. Fusai and Meucci (2003) propose a way to measure how consistent a posterior estimate of the mean is with regards to the prior, or some other estimate. Braga and Natale (2007) describe how to use Tracking Error to measure the distance from the equilibirum to the posterior portfolio. I also include additional original work on using relative entropy to measure quality of the views.

Finally, larger extensions to the model such as Krishnan and Mains (2006) present a method to incorporate additional factors into the model. Qian and Gorman (2001) present a method to integrate views on the covariance matrix as well as views on the returns.

Idzorek's Extension

Idzorek's apparent goal was to reduce the complexity of the Black-Litterman model for non-quantitative investors. He achieves this by allowing the investor to specify the investors confidence in the views as a percentage (0-100%) where the confidence measures the change in weight of the posterior from the prior estimate (0%) to the conditional estimate (100%). This linear relation is shown below

(40) $$confidence = (\hat{w} - w_{mkt}) / (w_{100} - w_{mkt})$$

w_{100} is the weight of the asset under 100% certainty in the view

w_{mkt} is the weight of the asset under no views

w is the weight of the asset under the specified view.

He provides a method to back out the value of ω required to generate the proper tilt (change in weights

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from prior to posterior) for each view. These values are then combined to form Ω, and the model is used to compute posterior estimates.

Idzorek includes τ in his formulas, but because of his use of the Alternative Reference Model and his formula (40) there is no need to use τ with the Idzorek method.

In his paper he discusses solving for ω using a least squares method. We can actually solve this analytically[5]. The next section will provide a derivation of the formulas required for this solution.

First we will use the following form of the uncertainty of the views. Idzorek includes τ in this formula, but he uses the Alternative Reference Model so we can drop τ from the formulation of his method.

(41) $$\Omega = \alpha P \Sigma P^T$$

α, the coefficient of uncertainty, is a scalar quantity in the interval [0, ∞]. When the investor is 100% confident in their views, then α will be 0, and when they are totally uncertain then α will be ∞. Note that formula (41) is exact, it is identical to formula (39) the Ω used by He and Litterman (1999) because it is a 1x1 matrix. This allows us to find a closed form solution to the problem of Ω for Idzorek's confidence.

First we substitute formula (37)

$$\hat{\Pi} = \Pi + \frac{1}{(1+\alpha)}\left[P^{-1}Q - \Pi\right]$$

into formula (10).

$$\hat{w} = \hat{\Pi}(\delta \Sigma)^{-1}$$

Which yields

(42) $$\hat{w} = \left[\Pi + \frac{1}{(1+\alpha)}\left[P^{-1}Q - \Pi\right]\right](\delta \Sigma)^{-1}$$

Now we can solve formula (42) at the boundary conditions for α.

$$\lim_{\alpha \to \infty}, w_{mk} = \Pi(\delta \Sigma)^{-1}$$

$$\lim_{\alpha \to 0}, w_{100} = P^{-1}Q(\delta \Sigma)^{-1}$$

And recombining some of the terms in (42) we arrive at

(43) $$\hat{w} = \Pi(\delta \Sigma)^{-1} + \left[\frac{1}{(1+\alpha)}\right]\left[P^{-1}Q(\delta \Sigma)^{-1} - \Pi(\delta \Sigma)^{-1}\right]$$

Substituting w_{mk} and w_{100} back into (43) we get

$$\hat{w} = w_{mk} + \left[\frac{1}{(1+\alpha)}\right]\left[w_{100} - w_{mk}\right]$$

5 Thanks to Boris Gnedenko for pointing this fact out to me.

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And comparing the above with formula (40)

$$confidence = (\hat{w} - w_{mk}) / (w_{100} - w_{mk})$$

We see that

$$confidence = \frac{1}{(1+\alpha)}$$

And if we solve for α

(44) $$\alpha = (1 - confidence) / confidence$$

Using formulas (44) and (41) the investor can easily calculate the value of ω for each view, and then roll them up into a single Ω matrix. To check the results for each view, we then solve for the posterior estimated returns using formula (25) and plug them back into formula (40). Note that when the investor applies all their views at once, the interaction amongst the views may pull the posterior estimate for individual assets away from the results generated when the views were taken one at a time.

Idzorek's method greatly simplifies the investor's process of specifying the uncertainty in the views when the investor does not have a quantitative model driving the process. In addition, this model does not add meaningful complexity to the process.

An Example of Idzorek's Extension

Idzorek describes the steps required to implement his extension in his paper, but does not provide a complete worked example. In this section I will work through his example from where he leaves off in the paper.

Idzorek's example includes 3 views:

- International Dev Equity will have absolute excess return of 5.25%, Confidence 25.0%
- International Bonds will outperform US bonds by 25bps, Confidence 50.0%
- US Growth Equity will outperform US Value Equity by 2%, Confidence 65.0%

In his paper Idzorek defines the steps in his method which include calculations of w_{100} and then the calculation of ω for each view given the desired change in the weights. From the previous section, we can see that we only need to take the investor's confidence for each view, plug it into formula (44) and compute the value of alpha. Then we plug α, P and Σ into formula (41) and compute the value of ω for each view. At this point we can assemble our Ω matrix and proceed to solve for the posterior returns using formula (24) or (25).

In working the example, I will show the results for each view including the w_{mkt} and w_{100} in order to make the workings of the extension more transparent. Tables 4, 5 and 6 below each show the results for a single view.

Table 4 – Calibrated Results for View 1

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Asset	ω	w_{mkt}	w*	$w_{100\%}$	Implied Confidence
Intl Dev Equity	.002126625	24.18%	25.46%	29.28%	25.00%

Table 5 – Calibrated Results for View 2

Asset	ω	w_{mkt}	w*	$w_{100\%}$	Implied Confidence
US Bonds	.000140650	19.34%	29.06%	38.78%	50.00%
Intl Bonds	.000140650	26.13%	16.41%	6.69%	50.00%

Table 6 – Calibrated Results for View 3

Asset	ω	w_{mkt}	w*	$w_{100\%}$	Implied Confidence
US LG	.000466108	12.09%	9.49%	8.09%	65.00%
US LV	.000466108	12.09%	14.69%	16.09%	65.00%
US SG	.000466108	1.34%	1.05%	.90%	65.00%
US SV	.000466108	1.34%	1.63%	1.78%	65.00%

Table 7 – Final Results for Idzorek's Confidence Extension Example

Asset	View 1	View 2	View 3	μ	σ	w_{mkt}	Posterior Weight	change
US Bonds	0.0	-1.0	0.0	.1	3.2	19.3%	29.6%	10.3%
Intl Bonds	0.0	1.0	0.0	.5	8.5	26.1%	15.8%	10.3%
US LG	0.0	0.0	0.9	6.3	24.5	12.1%	8.9%	3.2%
US LV	0.0	0.0	-0.9	4.2	17.2	12.1%	15.2%	3.2%
US SG	0.0	0.0	0.1	7.3	32.0	1.3%	1.0%	-.4%
US SV	0.0	0.0	-0.1	3.8	17.9	1.3%	1.7%	.4%

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Asset	View 1	View 2	View 3	μ	σ	w_{mkt}	Posterior Weight	change
Intl Dev	1.0	0.0	0.0	4.8	16.8	24.2%	26.0%	1.8%
Intl Emg	0.0	0.0	0.0	6.6	28.3	3.5%	3.5%	-.0%
Total							101.8%	
Return	5.2	.2	2.0					
Omega/ tau	.08507	.00563	.01864					
Lambda	.002	-.006	-.002					

Then we use the freshly computed values for the Ω matrix with all views specified together and arrive at the final result shown above in Table 7 blending all 3 views together.

Measuring the Impact of the Views

This section will discuss several methods used in the literature to measure the impact of the views on the posterior distribution. In general we can divide these measures into two groups. The first group allows us to test the hypothesis that the views or posterior contradict the prior. The second group allows us to measure a distance or information content between the prior and posterior.

Theil (1971), and Fusai and Meucci (2003) describe measures that are designed to allow a hypothesis test to ensure the views or the posterior does not contradict the prior estimates. Theil (1971) describes a method of performing a hypothesis test to verify that the views are compatible with the prior. We will extend that work to measure compatibility of the posterior and the prior. Fusai and Meucci (2003) describe a method for testing the compatibility of the posterior and prior when using the alternative reference model.

He and Litterman (1999), and Braga and Natale (2007) describe measures which can be used to measure the distance between two distributions, or the amount of tilt between the prior and the posterior. These measures don't lend themselves to hypothesis testing, but they can potentially be used as constraints on the optimization process. He and Litterman (1999) define a metric, Λ, which measures the tilt induced in the posterior by each view. Braga and Natale (2007) use Tracking Error Volatility (TEV) to measure the distance from the prior to the posterior.

Theil's Measure of Compatibility Between the Views and the Prior

Theil (1971) describes this as testing the compatibility of the views with the prior information. Given the linear mixed estimation model, we have the prior (17) and the conditional (19).

(17) $x\beta = \pi + u$

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(19) $p\beta = q + v$

The mixed estimation model defines *u* as a random vector with mean 0 and covariance $\tau\Sigma$, and v as a random vector with mean 0 and covariance Ω.

The approach we will take is very similar to the approach taken when analyzing a prediction from a linear regression. We have two estimates of the views, the conditional estimate of the view and the posterior estimate of the returns.

We can define the posterior estimate as $\hat{\beta}$. We can measure the estimation error between the prior and the views as:

(45) $\zeta = (x\hat{\beta} - \mu) = x(\hat{\beta} - \beta) + u$

The vector ζ has mean 0 and variance $V(\zeta)$. We will form our hypothesis test using the formulation

(46) $\xi = E(\zeta)V(\zeta)^{-1}E(\zeta)$

The quantity, ξ, is known as the Mahalanobis distance (multi-dimensional analog of the z-score) and is distributed as $\chi^2(n)$. In order to use this form we need to solve for the $E(\zeta)$ and $V(\zeta)$.

If we consider only the information in the views, the estimator of β is:

$$\hat{\beta} = (P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}Q$$

Note that since P is not required to be a matrix of full rank that we might not be able to evaluate this formula as written. We work in return space here (as opposed to view space) as it seems more natural. Later on we will transform the formula into view space to make it computable.

We then substitute the new estimator into the formula (45) and eliminate x as it is the identity matrix in the Black-Litterman application of mixed-estimation.

(47) $\zeta = (P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}Q - \beta + u$

Next we substitute formula (19) for Q.

$$\zeta = (P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}(P\beta + v) - \beta + u$$
$$\zeta = (P^T\Omega^{-1}P)^{-1}(P^T\Omega^{-1}P)\beta + (P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}v - \beta + u$$
$$\zeta = (P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}v + u$$

Give our estimator, we want to find the variance of the estimator.

$$V(\zeta) = E(\zeta\zeta^T)$$
$$V(\zeta) = E((P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}vv^T\Omega^{-1}P(P^T\Omega^{-1}P)^{-1} - 2(P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}vu^T + uu^T)$$

But $E(vu) = 0$ so we can eliminate the cross term, and simplify the formula.

$$V(\zeta) = E[(P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}vv^T\Omega^{-1}P(P^T\Omega^{-1}P)^{-1} + uu^T]$$
$$V(\zeta) = E[(P^{-1}vv^T(P^T)^{-1}) + uu^T]$$
$$V(\zeta) = (P^T\Omega^{-1}P)^{-1} + \tau\Sigma$$

The last step is to take the expectation of formula (47). At the same time we will substitute the

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posterior estimate (μ) for β.

$$E(\zeta)=(P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}Q-\Pi$$

Now substitute the various values into (46) as follows

$$\xi=((P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}Q-\Pi)\left[(P^T\Omega^{-1}P)^{-1}+\tau\Sigma\right]^{-1}((P^T\Omega^{-1}P)^{-1}P^T\Omega^{-1}Q-\Pi)^T$$

Unfortunately, under the usual circumstances we cannot compute ξ. Because P does not need to contain a view on every asset, several of the terms are not always computable as written. However, we can easily convert it to view space by multiplying by P and P^T.

(48) $$\hat{\xi}=(Q-P\Pi)\left[\Omega+P\tau\Sigma P^T\right]^{-1}(Q-P\Pi)^T$$

This new test statistic statistic, ξ, in formula (48) is distributed as $\chi^2(q)$ where q is the number of views. It is the square of the Mahalanobis distance of the posterior return estimate versus the posterior covariance of the estimate. We can use this test statistic to determine if our views are consistent with the prior by means of a standard confidence test.

$$P(q)=1-F(\xi(q))$$

Where F(ξ) is the CDF of the $\chi^2(q)$ distribution.

We can also compute the sensitivities of this measure to the views using the chain rule.

$$\frac{\partial P}{\partial q}=\frac{\partial P}{\partial \xi}\frac{\partial \xi}{\partial q}$$

Substituting the various terms

(49) $$\frac{\partial P}{\partial q}=-f(\xi)\left[2((\Omega+P\tau\Sigma P^T)^{-1}(Q-P\Pi)^T)\right]$$

Where f(ξ) is the PDF of the $\chi^2(q)$ distribution.

Theil's Measure of the Source of Posterior Information

Theil (1963) describes a measure which can be used to determine the contribution to the posterior precision of the prior and the views. This measure which he calls θ sums to 1 across all sources, and conveniently also sums across the views if we measure the contribution of each view.

The measure for the contribution to posterior precision from the prior is

(50) $$\theta_{prior}=\frac{1}{n}tr((\tau\Sigma)^{-1}[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P]^{-1})$$

where n is the number of assets. The formula below can be used for all views by using the full matrices P and Ω. For a single view (i) just use the relevant slices of P and Ω.

(51) $$\theta_i=\frac{1}{n}tr((P_i^T\Omega_{i,i}^{-1}P_i)[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P]^{-1})$$

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Formulas (50) and (51) provide the formulas we can use to compute the contribution towards the posterior precision from the prior and from the views. We can use this diagnostic to identify if the relative contributions match our intuitive view on this proportion.

Fusai and Meucci's Measure of Consistency

Next we will look at the work of Fusai and Meucci (2003). In their paper they present a way to quantify the statistical difference between the posterior return estimates and the prior estimates. This provides a way to calibrate the uncertainty of the views and ensure that the posterior estimates are not extreme when viewed in the context of the prior equilibrium estimates.

In their paper they use the Alternative Reference Model. Their measure is analagous to Theil's Measure of Compatibility, but because the alternative reference model uses the prior variance of returns for the posterior they do not need any derivation of the variance. We can apply a variant of their measure to the Canonical Reference Model as well.

They propose the use of the squared Mahalanobis distance of the posterior returns from the prior returns. We include τ here to match the Canonical Reference Model, but their work does not include τ as they use the Alternative Reference Model.

$$M(q)=(\mu_{BL}-\mu)(\tau\Sigma)^{-1}(\mu_{BL}-\mu) \tag{52}$$

It is essentially measuring the distance from the prior, μ, to the estimated returns, μ_{BL}, normalized by the uncertainty in the estimate. We use the covariance matrix of the prior distribution as the uncertainty. The squared Mahalanobis distance is distributed as $\chi^2(q)$ where q is the number of assets.. This can easily be used in a hypothesis test. Thus the probability of this event occurring can be computed as:

$$P(q)=1-F(M(q)) \tag{53}$$

Where F(M(q)) is the CDF of the chi square distribution of M(q) with n degrees of freedom.

Finally, in order to identify which views contribute most highly to the distance away from the equilibrium, we can also compute sensitivities of the probability to each view. We use the chain rule to compute the partial derivatives

$$\frac{\partial P(q)}{\partial q}=\frac{\partial P}{\partial M}\frac{\partial M}{\partial \mu_{BL}}\frac{\partial \mu_{BL}}{\partial q}$$

$$\frac{\partial P(q)}{\partial q}=-f(M)\left[2(\mu_{BL}-\mu)\right]\left[(P(\tau\Sigma)P+\Omega)^{-1}P\right] \tag{54}$$

Where f(M) is the PDF of the chi square distribution with n degrees of freedom for M(q). Note that this measure is very similar to (49) Theil's measure of compatibility between the prior and the views.

They work an example in their paper which results in an initial probability of 94% that the posterior is consistent with the prior. They specify that their investor desires this probability to be no less than 95% (a commonly used confidence level in hypothesis testing), and thus they would adjust their views to bring the probability in line. Given that they also compute sensitivities, their investor can identify

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which views are providing the largest marginal increase in their measure and they investor can then adjust these views. These sensitivities are especially useful since some views may actually be pulling the posterior towards the prior, and the investor could strengthen these views, or weaken views which pull the posterior away from the prior. This last point may seem non-intuitive. Given that the views are indirectly coupled by the covariance matrix, one would expect that the views only push the posterior distribution away from the prior. However, because the views can be conflicting, either directly or via the correlations, any individual view can have a net impact pushing the posterior closer to the prior, or pushing it further away.

They propose to use their measure in an iterative method to ensure that the posterior is consistent with the prior to the specified confidence level.

With the Canonical Reference Model we could rewrite formula (52) using the posterior variance of the return instead of $\tau\Sigma$ yielding:

(55) $$M(q)=(\mu_{BL}-\mu)((\tau\Sigma)^{-1}+P^T\Omega^{-1}P)^{-1}(\mu_{BL}-\mu)$$

Otherwise their Consistency Measure and it's use is the same for both reference models.

He and Litterman Lambda

He and Litterman (1999) use a measure, Λ, to measure the impact of each view on the posterior. They define the Black-Litterman unconstrained posterior portfolio as a blend of the equilibrium portfolio (prior) and a contribution from each view, that contribution is measured by Λ.

Deriving the formula for Λ we will start with formula (10) and substitute in the various values from the posterior distribution.

$$w=(\delta\Sigma)^{-1}\hat{\Pi}$$

We substitute the return from formula (0) for $\hat{\Pi}$

(56) $$\hat{w}=\frac{1}{\delta}\tilde{\Sigma}^{-1}M^{-1}[(\tau\Sigma)^{-1}\Pi+P^T\Omega^{-1}Q]$$

We will first simplify the covariance term

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$$\tilde{\Sigma}^{-1}M^{-1}=(\Sigma+M^{-1})^{-1}M^{-1}$$
$$\tilde{\Sigma}^{-1}M^{-1}=(\Sigma M+I)^{-1}$$
$$\tilde{\Sigma}^{-1}M^{-1}=\Sigma^{-1}(\Sigma^{-1}+M)^{-1}$$
$$\tilde{\Sigma}^{-1}M^{-1}=\Sigma^{-1}(\Sigma^{-1}+(\tau\Sigma)^{-1}+P\Omega^{-1}P^{T})^{-1}$$
$$\tilde{\Sigma}^{-1}M^{-1}=\Sigma^{-1}(\frac{1+\tau}{\tau}\Sigma^{-1}+P\Omega^{-1}P^{T})^{-1}$$
$$\tilde{\Sigma}^{-1}M^{-1}=(P^{T}\tau\Sigma P)^{-1}[(1+\tau)(P^{T}\Sigma P)^{-1}+\tau\Omega^{-1}]^{-1}$$
$$\tilde{\Sigma}^{-1}M^{-1}=(P^{T}\tau\Sigma P)^{-1}[\frac{(P^{T}\Sigma P)}{(1+\tau)}-\frac{(P^{T}\tau\Sigma P)}{(1+\tau)}\frac{(P^{T}\Sigma P)}{(1+\tau)}[\frac{\Omega}{\tau}+\frac{P^{T}\Sigma P}{1+\tau}]^{-1}]$$
$$\tilde{\Sigma}^{-1}M^{-1}=\frac{\tau}{1+\tau}[I-\frac{(P^{T}\Sigma P)}{1+\tau}(\frac{\Omega}{\tau}+\frac{P^{T}\Sigma P}{1+\tau})^{-1}]$$

Then we can define

(57) $$A=\left[\frac{\Omega}{\tau}+\frac{P^{T}\Sigma P}{1+\tau}\right]$$

And finally rewrite as

(58) $$\tilde{\Sigma}^{-1}M^{-1}=\frac{\tau}{1+\tau}[I-(P^{T}A^{-1}P)(\frac{\Sigma}{1+\tau})]$$

Our goal is to simplify the formula to the form

$$\hat{w}=\frac{1}{1+\tau}\left\{w_{eq}+P^{T}\Lambda\right\}$$

We use the multiplier, $\frac{1}{1+\tau}$, because in the Black-Litterman Reference Model the investor is not fully invested in the prior (equilibrium) portfolio.

In order to find Λ we substitute formula (58) into (56) and then gather terms.

$$\hat{w}=\frac{1}{\delta}\frac{\tau}{1+\tau}\left[I-(P^{T}A^{-1}P)(\frac{\Sigma}{1+\tau})\right]\left[(\tau\Sigma)^{-1}\Pi+P^{T}\Omega^{-1}Q\right]$$
$$\hat{w}=\frac{1}{\delta}\frac{\tau}{1+\tau}\left[(\tau\Sigma)^{-1}\Pi-(P^{T}A^{-1}P)(\frac{\tau}{1+\tau})\Pi+P^{T}\Omega^{-1}Q-(P^{T}A^{-1}P)(\frac{\Sigma}{1+\tau})P^{T}\Omega^{-1}Q\right]$$
$$\hat{w}=\frac{1}{1+\tau}\left\{w_{eq}+P^{T}\left[-A^{-1}P\Pi(\frac{1}{\delta(1+\tau)})+\frac{\tau\Omega^{-1}Q}{\delta}-(A^{-1}P)(\frac{\Sigma}{\delta(1+\tau)})P^{T}\Omega^{-1}Q\right]\right\}$$
$$\hat{w}=\frac{1}{1+\tau}\left\{w_{eq}+P^{T}\left[\frac{\tau}{\delta}\Omega^{-1}Q-\frac{A^{-1}P\Sigma w_{eq}}{1+\tau}-A^{-1}\frac{\tau}{1+\tau}(P^{T}\Sigma P)\frac{\Omega^{-1}Q}{\delta}\right]\right\}$$

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So we can see that along with (57), the following formula defines Λ.

$$(59) \qquad \Lambda = \frac{\tau}{\delta}\Omega^{-1}Q - \frac{A^{-1}P\Sigma w_{eq}}{1+\tau} - A^{-1}\frac{\tau}{1+\tau}(P\Sigma P^T)\frac{\Omega^{-1}Q}{\delta}$$

He and Litterman Λ represents the weight on each of the view portfolios on the final posterior weight. As a result, we can use Λ as a measure of the impact of our views.

We can also derive the He and Litterman Λ for the Alternative Reference Model, we will call this Λ_A. We start from the same reverse optimization formula.

$$w = (\delta\Sigma)^{-1}\hat{\Pi}$$

We substitute the return from formula (24) for $\hat{\Pi}$, but we use the prior covariance matrix as we are using the Alternative Reference Model.

$$(60) \qquad \hat{w} = \frac{1}{\delta}\Sigma^{-1}((\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}[\Sigma^{-1}\Pi + P^T\Omega^{-1}Q]$$

We will first simplify the covariance term using the Woodbury Matrix Identity.

$$\begin{aligned} &= \Sigma^{-1}((\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1} \\ &= I^{-1} - I^{-1}P^T(\Omega + P\Sigma I^{-1}P^T)^{-1}P\Sigma I^{-1} \\ &= I - P^T(\Omega + P\Sigma P^T)^{-1}P\Sigma \end{aligned}$$

Then we can define

$$A_A = \Omega + P^T\Sigma P$$

Then we can substitute the above result back into formula (43) and expand the terms.

$$\hat{w} = \frac{1}{\delta}(I - P^T A_A^{-1}P\Sigma)[\Sigma^{-1}\Pi + P^T\Omega^{-1}Q]$$

$$\hat{w} = \frac{1}{\delta}(\Sigma^{-1}\Pi + P^T\Omega^{-1}Q - P^T A_A^{-1}P\Pi - P^T A_A^{-1}P\Sigma P^T\Omega^{-1}Q)$$

$$\hat{w} = \frac{\Sigma^{-1}\Pi}{\delta} + \frac{1}{\delta}P^T\left[(\Omega^{-1}Q - A_A^{-1}P\Pi - A_A^{-1}P\Sigma P^T\Omega^{-1}Q)\right]$$

$$\hat{w} = \frac{\Sigma^{-1}\Pi}{\delta} + \frac{1}{\delta}P^T\left[(\Omega^{-1}Q - A_A^{-1}P\delta\Sigma w_{eq} - A_A^{-1}P\Sigma P^T\Omega^{-1}Q)\right]$$

He and Litterman Λ takes the similar form in the Alternative Reference Model as shown below.

$$\hat{w} = w_{eq} + P^T\Lambda_A$$

Note that in the Alternative Reference Model, the investor's prior portfolio has the exact same weights as the equilibrium portfolio.

We can see that the following formula defines Λ_A.

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(61) $$\Lambda_A = \frac{\Omega^{-1} Q}{\delta} - A_A^{-1} P \Sigma w_{eq} - A_A^{-1} P \Sigma P^T \frac{\Omega^{-1} Q}{\delta}$$

Braga and Natale and Tracking Error Volatility

Braga and Natale (2007) propose the use of tracking error between the posterior and prior portfolios as a measure of distance from the prior. Tracking error is commonly used by investors to measure risk versus a benchmark, and can be used as an investment constraint. As it is so commonly used, most investors have an intuitive understanding and a level of comfort with TEV. Tracking error volatility is defined as

(62) $$TEV = \sqrt{w_{actv}^T \Sigma w_{actv}} \quad where\ w_{actv} = \hat{w} - w_r$$

Where

w_{actv}	Active weights, or active portfolio
$\hat{w}$	Weight in the investor's portfolio
w_r	Weight in the reference portfolio
Σ	Covariance matrix of returns

They also derive the formula for tracking error sensitivities as follows: Given that

$$TEV = f(w_{actv})$$

and we can further refine

$$w_{actv} = g(q) \quad where\ q\ represents\ the\ views$$

Then we can use the chain rule to decompose the sensitivity of TEV to the views

(63) $$\frac{\partial TEV}{\partial q} = \frac{\partial TEV}{\partial w_{actv}} \frac{\partial w_{actv}}{\partial q}$$

We can solve for the first term of formula (63) directly,

$$\frac{\partial TEV}{\partial w_{actv}} = \partial \frac{(\sqrt{w_{actv}^T \Sigma w_{actv}})}{\partial w_{actv}}$$

$$Let\ x = w_{actv}^T \Sigma w_{actv},\ \ then\ apply\ the\ chain\ rule$$

$$\frac{\partial TEV}{\partial w_{actv}} = \frac{\partial TEV}{\partial x} \frac{\partial x}{\partial w_{actv}}$$

$$\frac{\partial TEV}{\partial w_{actv}} = [\frac{1}{2\sqrt{w_{actv}^T \Sigma w_{actv}}}][2 \Sigma w_{actv}]$$

$$\frac{\partial TEV}{\partial w_{actv}} = \frac{\Sigma w_{actv}}{\sqrt{w_{actv}^T \Sigma w_{actv}}}$$

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Solving for the second term of formula (63) is slightly more complicated.

$$\frac{\partial w_{actv}}{\partial q}=\frac{\partial(\hat{w}-w_{ref})}{\partial q}$$
$$\frac{\partial w_{actv}}{\partial q}=\frac{\partial((\delta\Sigma)^{-1}E(r)-(\delta\Sigma)^{-1}\Pi)}{\partial q}$$
$$\frac{\partial w_{actv}}{\partial q}=(\delta\Sigma)^{-1}\frac{\partial(E(r)-\Pi)}{\partial q}$$
$$\frac{\partial w_{actv}}{\partial q}=(\delta\Sigma)^{-1}\partial\frac{\tau\Sigma P^{T}[(P\tau\Sigma P^{T})+\Omega]^{-1}[Q-P\Pi]}{\partial Q}$$
$$\frac{\partial w_{actv}}{\partial q}=(\delta\Sigma)^{-1}\tau\Sigma P^{T}[(P\tau\Sigma P^{T})+\Omega]^{-1}$$
$$\frac{\partial w_{actv}}{\partial q}=\frac{1}{\delta}P^{T}[(P\Sigma P^{T})+\frac{\Omega}{\tau}]^{-1}$$

This result is somewhat different from that found in the Braga and Natale (2007) paper because we use the form of the Black-Litterman model which requires less matrix inversions. The formula for the sensitivities is

$$\frac{\partial TEV}{\partial q}=\frac{\Sigma w_{actv}}{\sqrt{w_{actv}^{T}\Sigma w_{actv}}}\frac{1}{\delta}P^{T}\left[(P\Sigma P^{T})+\frac{\Omega}{\tau}\right]^{-1} \tag{64}$$

We can come up with the equivalent metric for the Canoni6`1cal Reference Model model quite easily. If we just notice that in a tracking error scenario, the covariance matrix should be the most accurate which would be the posterior covariance matrix.

$$\frac{\partial TEV}{\partial q}=\frac{\tilde{\Sigma} w_{actv}}{\sqrt{w_{actv}^{T}\tilde{\Sigma} w_{actv}}}\frac{\partial w_{actv}}{\partial q}$$
$$\frac{\partial TEV}{\partial q}=\frac{\tilde{\Sigma} w_{actv}}{\sqrt{w_{actv}^{T}\tilde{\Sigma} w_{actv}}}\frac{P^{T}}{1+\tau}\frac{\partial\Lambda}{\partial q}$$

Yielding the TEV sensitivities for the Canonical Reference Model.

$$\frac{\partial TEV}{\partial q}=\frac{\tilde{\Sigma} w_{actv}}{\sqrt{w_{actv}^{T}\tilde{\Sigma} w_{actv}}}\frac{P^{T}}{1+\tau}\left[\frac{\tau}{\delta}\Omega^{-1}-A^{-1}\frac{\tau}{1+\tau}(P\Sigma P^{T})\frac{\Omega^{-1}}{\delta}\right] \tag{65}$$

Braga and Natale work through a fairly simple example in their paper, but they do not provide all the raw data required to reproduce their results. I have been unable to reproduce either their equilibrium or their mixing results. Given their posterior distribution as presented in the paper one can easily reproduce their TEV results.

One advantage of the TEV is that most investors are familiar with it, and so they will have some intuition as to what it represents. The consistency metric introduced by Fusai and Meucci (2003) will

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not be as familiar to investors.

Metrics Introduced in Herold (2003)

Herold (2003) discusses the concept of diagnostics which an investor can apply to their model in order to validate the outputs. One of these diagnostics is the correlation between views as one diagnostic which can be used to determine how the updated portfolio may perform. By examining the correlation matrix of the views $P \Sigma P^t$ we can determine how correlated the views are. If the views are highly correlated then we can expect all views to contribute to out performance if they are correct and underperformance if they are incorrect. If the views are not highly correlated then we can expect a diversified contribution to performance.

As an example, if we use He and Litterman's data and their two views we compute this measure as

$$P \Sigma P^t = \begin{bmatrix} 0.0213 & 0.0020 \\ 0.0020 & 0.0170 \end{bmatrix}$$

The off-diagonal elements, 0.0020, are one order of magnitude smaller than the on-diagonal elements, (0.0213 and 0.0170). This difference indicates that the views are not strongly correlated. This is consistent with He and Littermans' specification of the two views on mutually exclusive sets of assets which are loosely correlated.

Herold (2003) also discusses the marginal contribution to tracking error by view. This is the same metric computed by Braga and Natale, though he proposes deriving the formula for it's calculation from the Alternative Reference Model version of He and Litterman's Λ, Λ_A.

Herold (2003) considers the case of Active Management, which means that the Black-Litterman model is being applied to an active management overlay on some portfolio. The prior distribution in this case corresponds to $w_{eq} = 0$. The posterior weights are the overlay weights. He uses the Alternative Reference Model and computes a quantity, Φ, which is the active management version of Λ_A.

$$\Lambda_A = \frac{\Omega^{-1} Q}{\delta} - A_A^{-1} P \Sigma w_{eq} - A_A^{-1} P \Sigma P^T \frac{\Omega^{-1} Q}{\delta}$$

but $w_{eq} = 0$, *so*

$$\Phi = \frac{\Omega^{-1} Q}{\delta} - A_A^{-1} P \Sigma P^T \frac{\Omega^{-1} Q}{\delta}$$

where $w_{actv} = P^T \Phi$

Herold starts from formula (62) just as Braga and Natale, but then uses an alternative formula for tracking error.

$$TEV = \sqrt{w_{actv}^T \Sigma w_{actv}} \quad where\, w_{actv} = \hat{w} - w_r$$

and also

$$w_{actv} = \hat{w} - w_{eq} = P^T \Phi$$

$$TEV = \sqrt{(\Phi^T P \Sigma P^T \Phi)}$$

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Then we can just take $\frac{\delta TEV}{\delta Q}$ to find the marginal contribution to tracking error by views. By the chain rule

$$\frac{\delta TEV}{\delta Q} = \frac{\delta TEV}{\delta \Phi} \times \frac{\delta \Phi}{\delta Q}$$

$$\frac{\delta TEV}{\delta Q} = \frac{P \Sigma P^T \Phi}{\sqrt{(\Phi P \Sigma P^T \Phi)}} \times \frac{1}{\delta} \left[\Omega^{-1} - (P \Sigma P^T + \Omega)^{-1} P \Sigma P^T \Omega^{-1} \right]$$

Note that we could perform the same calculation for the Canonical Reference Model using He and Littermans' Λ, but we would also need to use the posterior covariance of the distribution rather than just Σ in the calculations.

A Demonstration of the Measures

Now we will work a sample problem to illustrate all of the metrics, and to provide some comparison of their features. We will start with the equilibrium from He and Litterman (1999) and for Example 1 use the views from their paper, Germany will outperform other European markets by 5% and Canada will outperform the US by 4%.

Table 8 – Example 1 Returns and Weights, equilibrium from He and Litterman, (1999).

Asset	P_0	P_1	μ	μ_{eq}	$w_{eq}/(1+\tau)$	w*	w* - $w_{eq}/(1+\tau)$
Australia	0.0	0.0	4.45	3.9	1.50%	1.5%	0.0%
Canada	0.0	1.0	9.06	6.9	2.1%	53.3%	51.2%
France	-0.295	0.0	9.53	8.4	5.0%	-3.3%	-8.3%
Germany	1.0	0.0	11.3	9	5.2%	33.1%	27.9%
Japan	0.0	0.0	4.65	4.3	11.0%	11.0%	0.0%
UK	-0.705	0.0	6.98	6.8	11.8%	-7.8%	-19.6%
USA	0.0	-1.0	7.31	7.6	58.6%	7.3%	-51.3%
q	5.0	4.0					
ω/τ	0.02	.017					

Table (8) illustrates the results of applying the views and Table (0) displays the various impact measures.

Table 9 - Impact Measures for Example 1

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Measure	Value (Confidence Level)	Sensitivity (V1)	Sensitivity (V2)
Theil's Measure	1.672 (43.3%)	-5.988	-11.46
Theil's θ	0.858 (prior)	0.0712	0.0712
Fusai and Meucci's Measure	0.8728 (99.7%)	-0.1838	-0.3327
Λ		0.2920	0.5380
TEV	8.28%	0.688	1.294

If we examine the change in the estimated returns vs the equilibrium, we see where the USA returns decreased by only 29 bps, but the allocation decreased 51.25% caused by the optimizer favoring Canada whose returns increased by 216 bps and whose allocation increased by 51.25%. This shows that what appear to be moderate changes in return forecasts can cause very large swings in the weights of the assets, a common issue with unconstrained mean variance optimization. Here we use unconstrained mean variance optimization not because we have to, but because it is well understood and transparent.

Next looking at the impact measures, Theil's measure indicates that we can be confident only at the 43% level that the views are consistent with the prior. If we examine the diagram (5) we can see that indeed these views are significantly different from the prior. Fusai and Meucci's measure of compatibility of the prior and posterior on the other hand comes with a confidence level of 99.8% so they are much more confident that the posterior is consistent with the prior. A major difference in their approaches is that Theil is working in view space, which for our example has order = 2 whilst Fusai and Meucci are working in asset space which has order = 7.

All of the metrics' sensitivities to the views indicate that the second view has a relative weight just about twice as large as the first view, so the second view contributes much more to the change in the weights.

The TEV of the posterior portfolio is 8.28% which is significant in terms of how closely the posterior portfolio will track the equilibrium portfolio. This seems to be a very large TEV value given the scenario we're working with

Next we change our confidence in the views by multiplying the variance by 2, this will increase the change from the prior to the posterior and allow us to make some judgements based on the impact measures.

Table 10 – Example 2 Returns and Weights, equilibrium from He and Litterman, (1999).

Asset	P_0	P_1	μ	$w_{eq}/(1+\tau)$	w*	w* - $w_{eq}/(1+\tau)$
Australia	0.0	0.0	4.72	1.5%	1.5%	0.0%
Canada	0.0	1.0	10.3	2.1%	22.7%	20.6%

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Asset	P_0	P_1	μ	$w_{eq}/(1+\tau)$	w*	w* - $w_{eq}/(1+\tau)$
France	-0.295	0.0	10.2	5.0%	1.6%	-3.4%
Germany	1.0	0.0	12.4	5.2%	16.8%	11.6%
Japan	0.0	0.0	4.84	11.0%	11.0%	0.0%
UK	-0.705	0.0	7.09	11.8%	3.7%	-8.1%
USA	0.0	-1.0	7.14	58.6%	38.0%	-20.6%
q	5.0	4.0				
ω/τ	0.09	0.07				

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Table 11 - Impact Measures for Example 2

Measure	Value (Confidence Level)	Sensitivity (V1)	Sensitivity (V2)
Theil's Measure	1.537 (46.4%)	-4.27	-12.39
Theils θ	0.88	0.05	0.07
Fusai and Meucci's Measure	0.7479 (99.8%)	-0.06	-0.24
Λ		0.193	0.544
TEV	7.67%	0.332	1.406

Examining the updated results in Table (10) we see that the changes to the forecast returns have decreased consistent with our increased uncertainty in the views. We now have an 20% increase in the allocation to Canada and an 20% decrease in the allocation to the USA. From Table (11) we can see that Theil's measure has increased, but only a fraction and we continue to have little confidence that the views our consistent with the prior. Fusai and Meucci's measure now is 99.8% confident that the posterior is consistent with the prior. It is unclear in practice what bound we would want to use, but their measure usually presents which a much higher confidence than Theil's measure. The TEV has decreased, and is now 7.67% which is not significantly smaller.

Once again all the sensitivities show the second view having more of an impact on the final weights.

Across both scenarios we can draw some conclusions about these various measures. Theil's consistency measure test ranged from a high of 46% confident to a low of 44% confident that the views were consistent with the prior estimates. This measure is very sensitive and it is unclear what would be a good confidence threshold. 50% does seem intuitively appealing.

Fusai and Meucci's Consistency measure ranged from 98.38% to 99.98% confident, indicating that the posterior estimates was generally highly consistent with the prior. Fusai and Meucci present that an investor may have a requirement that the confidence level be 5%. In light of these results that would seem to be a fairly large value. The sensitivities of the Consistency measure scale with the measure, and for low values of the measure the sensitivities are very low.

Theil's θ changed with the confidence of the views and generally indicated that much of the information in the posterior originated with the prior. It moved intuitively with the changes in confidence level.

Across the two scenarios the TEV decreased by 61bps, but against a starting point of 8.28% it still indcates large active weights. It is not clear what a realistic threshold for the TEV is in this case, but these values are likely towards the upper limit that would be tolerated. Note that between the two scenarios, the sensitivity to the first view dropped by 50% which is consistent with the change in confidence that we applied.

In analyzing these various measures of the tilt caused by the views, the TEV of the weights measures the impact of the views and the optimization process, which we can consider as the final outputs. If the

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investor is concerned about limits on TEV, they could be easily added as constraints on the optimization process.

He and Litterman's Lambda measures the weight of the view on the posterior weights, but only in the case of an unconstrained optimization. This makes it suitable for measuring impact and being a part of the process, but it cannot be used as a constraint in the optimization process.

Theil's compatibility measure and Fusai and Meucci's consistency measure measure the posterior distribution, including the returns and the covariance matrix. The former in view space the latteri in asset space.

Active Management and the Black-Litterman Model

We will consider Active Management to be the case when our investor is managing an overlay portfolio on top of a 100% invested passive benchmark portfolio. In this case we are only interested in the overlay, and not in the benchmark so we start with a prior distribution with 0 active weights and 0 expected excess returns over the benchmark. All returns to views input to the model are relative to the equilibrium benchmark returns rather than to the risk free rate. The weights are for the active portfolio, so the weights should always sum to 0.

Herold (2003) discusses the application of the Black-Litterman model to the problem of Active Management. He introduces a measure Φ which is He and Litterman's Λ_A modified for Active Management. When we use the Black-Litterman model for Active Management versus a passive benchmark portfolio, then the equilibrium weights (w_{eq}) are 0, and the thus the equilibrium returns (Π) are also 0.

Because the value $\Pi = 0$ the middle term in He and Litterman's Λ_A disappears.

$$\Phi = \frac{1}{\delta}\left[\Omega^{-1}Q - (P\Sigma P^T + \Omega)^{-1} P\Sigma P^T \Omega^{-1} Q\right] \tag{66}$$

Two-Factor Black-Litterman

Krishnan and Mains (2005) developed an extension to the alternate reference model which allows the incorporation of additional uncorrelated market factors. The main point they make is that the Black-Litterman model measures risk, like all MVO approaches, as the covariance of the assets. They advocate for a richer measure of risk. They specifically focus on a recession indicator, given the thesis that many investors want assets which perform well during recessions and thus there is a positive risk premium associated with holding assets which do poorly during recessions. Their approach is general and can be applied to one or more additional market factors given that the market has zero beta to the factor and the factor has a non-zero risk premium.

They start from the standard quadratic utility function (6), but add an additional term for the new market factor(s).

$$U = w^T \Pi - (\frac{\delta_0}{2}) w^T \Sigma w - \sum_{j=1}^{n} \delta_j w^T \beta_j \tag{67}$$

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U is the investors utility, this is the objective function during portfolio optimization.
w is the vector of weights invested in each asset
Π is the vector of equilibrium excess returns for each asset
Σ is the covariance matrix for the assets
δ_0 is the risk aversion parameter of the market
δ_j is the risk aversion parameter for the j-th additional risk factor
β_j is the vector of exposures to the j-th additional risk factor

Given their utility function as shown in formula (67) we can take the first derivative with respect to w in order to solve for the equilibrium asset returns.

$$\Pi = \delta_0 \Sigma w + \sum_{j=1}^{n} \delta_j \beta_j \tag{68}$$

Comparing this to formula (7), the simple reverse optimization formula, we see that the equilibrium excess return vector (Π) is a linear composition of (7) and a term linear in the β_j values. This matches our intuition as we expect assets exposed to this extra factor to have additional return above the equilibrium return.

We will further define the following quantities:

r_m as the return of the market portfolio.
f_j as the time series of returns for the factor
r_j as the return of the replicating portfolio for risk factor j.

In order to compute the values of δ we will need to perform a little more algebra. Given that the market has no exposure to the factor, then we can find a weight vector, v_j, such that $v_j^T \beta_j = 0$. In order to find v_j we perform a least squares fit of $\| f_j - v_j^T \Pi \|$ subject to the above constraint. v_0 will be the market portfolio, and $v_0 \beta_j = 0 \; \forall$ j by construction. We can solve for the various values of δ by multiplying formula (68) by v and solving for δ_0.

$$v_0^T \Pi = \delta_0 v_0^T \Sigma v_0 + \sum_{j=1}^{n} \delta_j v_0^T \beta_j$$

By construction $v_0 \beta_j = 0$, and $v_0 \Pi = r_m$, so

$$\delta_0 = \frac{r_m}{(v_0^T \Sigma v_0)}$$

For any $j \geq 1$ we can multiply formula (68) by v_j and substitute δ_0 to get

$$v_j^T \Pi = \delta_0 v_j^T \Sigma v_j + \sum_{i=1}^{n} \delta_i v_j^T \beta_i$$

Because these factors must all be independent and uncorrelated, then $v_i \beta_j = 0 \; \forall \; i \neq j$ so we can solve for each δ_j.

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$$\delta_j = \frac{(r_j - \delta_0 v_j^T \Sigma v_j)}{(v_j^T \beta_j)}$$

The authors raise the point that this is only an approximation because the quantity $\| f_j - v_j^T \Pi \|$ may not be identical to 0. The assertion that $v_i \beta_j = 0 \; \forall \; i \neq j$ may also not be satisfied for all i and j. For the case of a single additional factor, we can ignore the latter issue.

In order to transform these formulas so we can directly use the Black-Litterman model, Krishnan and Mains change variables, letting

$$\hat{\Pi} = \Pi - \sum_{j=1}^{n} \delta_j \beta_j$$

Substituting back into (67) we are back to the standard utility function

$$U = w^T \hat{\Pi} - (\frac{\delta_0}{2}) w^T \Sigma w$$

and from formula (13)

$$P \hat{\Pi} = P(\Pi - \sum_{j=1}^{n} \delta_j \beta_j)$$

$$P \hat{\Pi} = P \Pi - \sum_{j=1}^{n} \delta_j P \beta_j$$

thus

$$\hat{Q} = Q - \sum_{j=1}^{n} \delta_j P \beta_j$$

We can directly substitute $\hat{\Pi}$ and $\hat{Q}$ into formula (25) for the posterior returns in the Black-Litterman model in order to compute returns given the additional factors. Note that these additional factor(s) do not impact the posterior variance in any way.

Krishnan and Mains work an example of their model for world equity models with an additional recession factor. This factor is comprised of the Altman Distressed Debt index and a short position in the S&P 500 index to ensure the market has a zero beta to the factor. They work through the problem for the case of 100% certainty in the views. They provide all of the data needed to reproduce their results given the set of formulas in this section. In order to perform all the regressions, one would need to have access to the Altman Distressed Debt index along with the other indices used in their paper.

Qian and Gorman

Qian and Gorman (2001) discuss a method to provide both a conditional mean estimate and a conditional covariance estimate. They use a Bayesian framework and reference the Black-Litterman model. They use the alternative reference model as τ does not appear in their paper and they neglect the conditional (or posterior) covariance estimate.

In this section we will compare the Qian and Gorman approach with the approach taken in the Black-

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Litterman Reference Model.

We can match the variance portion of Qian and Gorman's formula (15) our label (QG 15) with our formula (30) if we set $\Omega = 0$, and remove τ (this is the alternative reference model). This describes the scenario where the investor has 100% confidence in their views. For those assets where the investor has absolute views then the variance of the posterior estimate will be zero. For all other assets, the posterior variance will be non-zero.

(QG 15) $$\epsilon \sim N\left[0, \Sigma - (\Sigma P^T)(\Sigma_v)^{-1}(P\Sigma)\right]$$

where $$\Sigma_v = P\Sigma P^T$$

$$Var(\epsilon) = \Sigma - (\Sigma P^T)(P\Sigma P^T)^{-1}(P\Sigma)$$

(30) $$M = \tau\Sigma - \tau\Sigma P^T(P\tau\Sigma P^T + \Omega)^{-1} P\tau\Sigma$$

setting $\Omega = 0$ and removing τ

$$M = \Sigma - \Sigma P^T(P\Sigma P^T)^{-1} P\Sigma$$

In order to get to Qian and Gorman's formula 17 (QG 17), we need to re-introduce the covariance of the views, Ω, but rather than mix the covariances as is done in the Black-Litterman model, we rely on their lack of correlation and just add the two variance terms. We take the variance of the conditional from (13).

$$Var(\epsilon) = \Sigma - (\Sigma P^T)(P\Sigma P^T)^{-1}(\Sigma P^T) + (P^T\Omega^{-1}P)^{-1}$$

$$Var(\epsilon) = \Sigma + (\Sigma P^T)(P\Sigma P^T)^{-1}(\Sigma P^T) + (\Sigma P^T)(P\Sigma P^T)^{-1}(\Omega)(P\Sigma P^T)^{-1}(\Sigma P)$$

(69) $$Var(\epsilon) = \Sigma + (\Sigma P^T)((P\Sigma P^T)^{-1}(\Omega)(P\Sigma P^T)^{-1} - (P\Sigma P^T)^{-1})(\Sigma P^T)$$

Note that formula (69) exactly matches (QG 17).

(QG 17) $$\tilde{\Sigma} = \Sigma + (\Sigma P^T)(\Sigma_v^{-1}\tilde{\Sigma}_v\Sigma_v^{-1} - \Sigma_v^{-1})(P\Sigma)$$

where $\Sigma_v = P\Sigma P^T$ and $\tilde{\Sigma}_v = \Omega$. substituting

$$\tilde{\Sigma} = \Sigma + (\Sigma P^T)((P\Sigma P^T)^{-1}\Omega(P\Sigma P^T)^{-1} - (P\Sigma P^T)^{-1})(P\Sigma)$$

Qian and Gorman demonstrate a conditional covariance which is not derived from either Theil's mixed estimation nor Bayesian updating. It's behavior may increase the variance of the posterior vs the prior in the event that the view variance is larger than the prior variance. They describe this as allowing the investor to have a view on the variance, and they do not suggest that Ω needs to be diagonal. In this model the conditional covariance is proportional to the investor's views on variance, but the blending process is not clear.

Compare this formula to our formula (30) for the variance of the posterior mean estimate.

$$M = \tau\Sigma - \tau\Sigma P^T(P\tau\Sigma P^T + \Omega)^{-1} P\tau\Sigma$$

Intuition tells us that the updated posterior estimate of the mean will be more precise (lower variance)

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than the prior estimate, thus we would like the posterior (conditional) variance to always be less than the prior variance. The Black-Litterman posterior variance achieves this goal and arrives at the well known result of Bayesian analysis in the case of unknown mean and known precision. As result, it seems we should favor these results over Qian and Gorman.

Future Directions

Future directions for this research include reproducing the results from the original papers, either Black and Litterman (1991) or Black and Litterman (1992). These results have the additional complication of including currency returns and partial hedging.

Later versions of this document should include more information on process and a synthesized model containing the best elements from the various authors. A full example from the CAPM equilibrium, through the views to the final optimized weights would be useful, and a worked example of the two factor model from Krishnan and Mains (2005) would also be useful.

Meucci (2006) and Meucci (2008) provide further extensions to the Black-Litterman Model for non-normal views and views on parameters other than return. This allows one to apply the Black-Litterman Model to new areas such as alternative investments or derivatives pricing. His methods are based on simulation and do not provide a closed form solution. Further analysis of his extensions will be provided in a future revision of this document.

An Asset Allocation Process Using the Black-Litterman Model

When used as part of an asset allocation process, the Black-Litterman model provides for estimates which lead to more stable and more diversified portfolios than estimates derived from historical returns when used with unconstrained mean-variance optimization. Because of this property an investor using mean-variance optimization is less likely to require artificial constraints to get a portfolio without extreme weights. Unfortunately using this model requires a broad variety of data, some of which may be hard to find.

First, the investor needs to identify their investable universe and find the market capitalization of each asset. Then, they need to estimate a covariance matrix for the excess returns of the assets. This is most often done using historical data for an appropriate time window. Both Litterman (2003), and Bevan and Winkelmann (1998) provide details on the process used to compute covariance matrices at Goldman Sachs. In the literature, monthly covariance matrices are most commonly estimated from 60 months of historical excess returns. If the actual asset return itself cannot be used, then an appropriate proxy can be used, e.g. S&P 500 Index for US Domestic Large Cap equities. The return on a short term sovereign bond, e.g US 4 or 13-week treasury bill, would suffice for most United States investor's risk free rate.

When applied to the asset allocation problem, finding the market capitalization information for liquid asset classes might be a challenge for an individual investor, but likely presents little obstacle for an institutional investor because of their access to index information from the various providers. Given the limited availability of market capitalization data for illiquid asset classes, e.g. real estate, private equity, commodities, even institutional investors might have a difficult time piecing together adequate market capitalization information. Return data for these same asset classes can also be complicated by

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delays, smoothing and inconsistencies in reporting. Further complicating the problem is the question of how to deal with hedge funds or absolute return managers. The question of whether they should be considered a separate asset class is beyond the scope of this paper.

Next, the investor needs to quantify their views so that they can be applied and new return estimates computed. The views can be derived from quantitative or qualitative processes, and can be complete or incomplete, or even conflicting.

Finally, the outputs from the model need to be fed into a portfolio selection model to generate the efficient frontier, and an efficient portfolio selected. Bevan and Winkelmann (1999) provide a description of their asset allocation process (for international fixed income) and how they use the Black-Litterman model within that process. This includes their approaches to calibrating the model and information on how they compute the covariance matrices.

The standard Black-Litterman model does not provide direct sensitivity of the prior to market factors besides the asset returns. It is fairly simple to extend the Black-Litterman model to use a multi-factor model for the prior distribution. Krishnan and Mains (2005) have provided extensions to the model which allow adding additional cross asset factors which are not priced in the market. Examples of such factors are a recession, or credit, market factor. Their approach is general and could be applied to other factors if desired.

Most of the Black-Litterman literature reports results using the closed form solution for unconstrained mean variance optimization. They also tend to use non-extreme views in their examples. We believe this is done to keep the papers simple, but it is also a testament to the stability of the outputs of the Black-Litterman model that useful results can be generated via this process. As part of a investment process, it is reasonable to conclude that some constraints would be applied at least in terms of restricting short selling and limiting concentration in asset classes. Lack of a budget constraint is also consistent with a Bayesian investor who may not wish to be 100% invested in the market due to uncertainty about their beliefs in the market. portfolio selection is normally considered as part of a two step process, first compute the optimal portfolio, and then determine position along the Capital Market Line.

For the ensuing discussion, we will refer to the CAPM equilibrium distribution as the prior distribution, and the investors views as the conditional distribution. This is consistent with the original Black and Litterman (1992) paper. It also is consistent with our intuition about the outcome in the absence of a conditional distribution (no views in Black-Litterman terminology.) This is the opposite of the way most examples of Bayes Theorem are defined, they start with a non-statistical prior distribution, and then add a sampled (statistical) distribution of new data as the conditional distribution. The mixing model we will use, and our use of normal distributions, will bring us to the same outcome independent of these choices.

An Asset Allocation Process

The Black-Litterman model is just one part of an asset allocation process. Bevan and Winkelmann (1998) document the asset allocation process they use in the Fixed Income Group at Goldman Sachs. At a minimum. a Black-Litterman oriented investment process would have the following steps:

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- Determine which assets constitute the market
- Compute the historical covariance matrix for the assets
- Determine the market capitalization for each asset class.
- Use reverse optimization to compute the CAPM equilibrium returns for the assets
- Specify views on the market
- Blend the CAPM equilibrium returns with the views using the Black-Litterman model
- Feed the estimates (estimated returns, covariances) generated by the Black-Litterman model into a portfolio optimizer.
- Select the efficient portfolio which matches the investors risk preferences

A further discussion of each step is provided below.

The first step is to determine the scope of the market. For an asset allocation exercise this would be identifying the individual asset classes to be considered. For each asset class the weight of the asset class in the market portfolio is required. Then a suitable proxy return series for the excess returns of the asset class is required. Between these two requirements it can be very difficult to integrate illiquid asset classes such as private equity or real estate into the model. Furthermore, separating public real estate holdings from equity holdings (e.g. REITS in the S&P 500 index) may also be required. Idzorek (2006) provides an example of the analysis required to include commodities as an asset class.

Once the proxy return series have been identified, and returns in excess of the risk free rate have been calculated, then a covariance matrix can be calculated. Typically the covariance matrix is calculated from the highest frequency data available, e.g. daily, and then scaled up to the appropriate time frame. Investor's often use an exponential weighting scheme to provide increased weights to more recent data and less to older data. Other filtering (Random Matrix Theory) or shrinkage methods could also be used in an attempt to impart additional stability to the process.

Now we can run a reverse optimization on the market portfolio to compute the equilibrium excess returns for each asset class. Part of this step includes computing a δ value for the market portfolio. This can be calculated from the return and standard deviation of the market portfolio. Bevan and Winkelmann (1998) discuss the use of an expected Sharpe Ratio target for the calibration of δ. For their international fixed income investments they used an expect Sharpe Ratio of 1.0 for the market. The investor then needs to calibrate τ in some manner. This value is usually on the order of 0.025 ~ 0.050.

At this point almost all of the machinery is in place. The investor needs to specify views on the market. These views can impact one or more assets, in any combination. The views can be consistent, or they can conflict. An example of conflicting views would be merging opinions from multiple analysts, where they may not all agree. The investor needs to specify the assets involved in each view, the absolute or relative return of the view, and their uncertainty in the return for the view consistent with their reference model and measured by one of the methods discussed previously.

Appendix D shows the process of cranking through formulas (25), (30) and (27) to compute the new

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posterior estimate of the returns and the covariance of the posterior returns. These values will be the inputs to some type of optimizer, a mean-variance optimizer being the most common. If the user generates the optimal portfolios for a series of returns, then they can plot an efficient frontier.

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Many of these references are available on the Internet. I have placed a Black-Litterman resources page on my website, (www.blacklitterman.org) with links to many of these papers.

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Appendix A

This appendix includes the derivation of the Black-Litterman master formula using Theil's Mixed Estimation approach which is based on Generalized Least Squares.

Theil's Mixed Estimation Approach

This approach is from Theil (1971) and is similar to the reference in the original Black and Litterman, (1992) paper. Koch (2005) also includes a derivation similar to this.

If we start with a prior distribution for the returns. Assume a linear model such as

A.1 $$\pi = x\beta + u$$

Where π is the mean of the prior return distribution, β is the expected return and u is the normally distributed residual with mean 0 and variance Φ.

Next we consider some additional information, the conditional distribution.

A.2 $$q = p\beta + v$$

Where q is the mean of the conditional distribution and v is the normally distributed residual with mean 0 and variance Ω.

Both Ω and Σ are assumed to be non-singular.

We can combine the prior and conditional information by writing:

A.3 $$\begin{bmatrix}\pi\\ q\end{bmatrix} = \begin{bmatrix}x\\ p\end{bmatrix}\beta + \begin{bmatrix}u\\ v\end{bmatrix}$$

Where the expected value of the residual is 0, and the expected value of the variance is

$$E\left(\begin{bmatrix}u\\ v\end{bmatrix}\begin{bmatrix}u' & v'\end{bmatrix}\right) = \begin{bmatrix}\Phi & 0\\ 0 & \Omega\end{bmatrix}$$

We can then apply the generalized least squares procedure, which leads to estimating β as

A.4 $$\hat{\beta} = \left[\begin{bmatrix}x & p\end{bmatrix}\begin{bmatrix}\Phi & 0\\ 0 & \Omega\end{bmatrix}^{-1}\begin{bmatrix}x'\\ p'\end{bmatrix}\right]^{-1}\begin{bmatrix}x' & p'\end{bmatrix}\begin{bmatrix}\Phi & 0\\ 0 & \Omega\end{bmatrix}^{-1}\begin{bmatrix}\pi\\ q\end{bmatrix}$$

This can be rewritten without the matrix notation as

A.5 $$\hat{\beta} = \left[x\Phi^{-1}x' + p\Omega^{-1}p'\right]^{-1}\left[x'\Phi^{-1}\pi + p'\Omega^{-1}q\right]$$

We can derive the expression for the variance using similar logic. Given that the variance is the expectation of $(\hat{\beta} - \beta)^2$, then we can start by substituting formula A.3 into A.5

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A.6 $$\hat{\beta}=\left[x\Phi^{-1}x'+p\Omega^{-1}p'\right]^{-1}\left[x'\Phi^{-1}(x\beta+u)+p'\Omega^{-1}(p\beta+v)\right]$$

This simplifies to

$$\hat{\beta}=\left[x\Phi^{-1}x+p\Omega^{1}p'\right]^{-1}\left[x\beta\Phi^{-1}x+p'\Omega^{-1}p\beta+x\Phi^{-1}u+p\Omega^{-1}v\right]$$

$$\hat{\beta}=\left[x\Phi^{-1}x'+p\Omega^{-1}p'\right]^{-1}\left[x\Phi^{-1}x'\beta+p\Omega^{-1}p'\beta\right]+\left[x\Phi^{-1}x'+p\Omega^{1}p'\right]^{-1}\left[x\Phi^{-1}u+p\Omega^{-1}v\right]$$

$$\hat{\beta}=\beta+\left[x\Phi^{-1}x'+p\Omega^{1}p'\right]^{-1}\left[x\Phi^{-1}u+p\Omega^{-1}v\right]$$

A.7 $$\hat{\beta}-\beta=\left[x\Phi^{-1}x'+p\Omega^{1}p'\right]^{-1}\left[x\Phi^{-1}u+p\Omega^{-1}v\right]$$

The variance is the expectation of formula A.7 squared.

$$E((\hat{\beta}-\beta)^2)=(\left[x\Phi^{-1}x^T+p\Omega^{1}p^T\right]^{-1}\left[x\Phi^{-1}u^T+p\Omega^{-1}v^T\right])^2$$

$$E((\hat{\beta}-\beta)^2)=\left[x\Phi^{-1}x^T+p\Omega^{1}p^T\right]^{-2}$$
$$\left[x\Phi^{-1}u^T u\Phi^{-1}x^T+p\Omega^{-1}v^T v\Omega^{-1}p^T+x\Phi^{-1}u^T v\Omega^{-1}p^T+p\Omega^{-1}v^T u\Phi^{-1}x^T\right]$$

We know from our assumptions above that $E(uu')=\Phi$, $E(vv')=\Omega$ and $E(uv')=0$ because u and v are independent variables, so taking the expectations we see the cross terms are 0

$$E((\hat{\beta}-\beta)^2)=\left[x\Phi^{-1}x^T+p\Omega^{-1}p^T\right]^{-2}\left[(x\Phi^{-1}\Phi\Phi^{-1}x^T)+(p\Omega^{-1}\Omega\Omega^{-1^T})+0+0\right]$$

$$E((\hat{\beta}-\beta)^2)=\left[x\Phi^{-1}x^T+p\Omega^{-1}p^T\right]^{-2}\left[x\Phi^{-1}x^T+p\Omega^{-1}p^T\right]$$

And we know that for the Black-Litterman model, *x* is the identity matrix and $\Phi=\tau\Sigma$ so after we make those substitutions we have

A.8 $$E((\hat{\beta}-\beta)^2)=\left[(\tau\Sigma)^{-1}+p\Omega^{-1}p^T\right]^{-1}$$

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Appendix B

This section provides a quick overview of the relevant portion of Bayes theory in order to create a common vocabulary which can be used in analyzing the Black-Litterman model from a Bayesian point of view.

A Quick Introduction to Bayes Theory

Bayes theory states

(70) $$P(A|B)=\frac{P(B|A)P(A)}{P(B)}$$

P(A|B)	The conditional (or joint) probability of A, given B Also known as the posterior distribution. We will call this the posterior distribution from here on.
P(B|A)	The conditional probability of B given A. Also know as the sampling distribution. We will call this the conditional distribution from here on.
P(A)	The probability of A. Also known as the prior distribution. We will call this the prior distribution from here on.
P(B)	The probability of B. Also known as the normalizing constant.

When actually applying this formula and solving for the posterior distribution, the normalizing constant will disappear into the constants of integration so from this point on we will ignore it..

A general problem in using Bayes theory is to identify an intuitive and tractable prior distribution. One of the core assumptions of the Black-Litterman model (and Mean-Variance optimization) is that asset returns are normally distributed. For that reason we will confine ourselves to the case of normally distributed conditional and prior distributions. Given that the inputs are normal distributions, then it follows that the posterior will also be normally distributed. When the prior distribution and the posterior have the same structure, the prior is known as a conjugate prior. Given interest there is nothing to keep us from building variants of the Black-Litterman model using different distributions, however the normal distribution is generally the most straight forward.

Another core assumption of the Black-Litterman model is that the variance of the prior and the conditional distributions about the actual mean are known, but the actual mean is not known. This case, known as "Unknown Mean and Known Variance" is well documented in the Bayesian literature. This matches the model which Theil uses where we have an uncertain estimate of the mean, but know the variance.

We define the significant distributions below:

The prior distribution

(71) $P(A) \sim N(x,S/n)$

where S is the sample variance of the distribution about the mean, with n samples then S/n is the

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variance of the estimate of x about the mean.

The conditional distribution

(72) $P(B|A) \sim N(\mu,\Omega)$

Ω is the uncertainty in the estimate μ of the mean, it is not the variance of the distribution about the mean.

Then the posterior distribution is specified by

(73) $P(A|B) \sim N([\Omega^{-1}\mu + nS^{-1}x]^T[\Omega^{-1} + nS^{-1}]^{-1},(\Omega^{-1} + nS^{-1})^{-1})$

The variance term in (73) is the variance of the estimated mean about the actual mean.

In Bayesian statistics the inverse of the variance is known as the precision. We can describe the posterior mean as the weighted mean of the prior and conditional means, where the weighting factor is the respective precision. Further, the posterior precision is the sum of the prior and conditional precision. Formula (73) requires that the precisions of the prior and conditional both be non-infinite, and that the sum is non-zero. Infinite precision corresponds to a variance of 0, or absolute confidence. Zero precision corresponds to infinite variance, or total uncertainty.

A full derivation of formula (73).using the PDF based Bayesian approach is shown in Appendix B.

As a first check on the formulas we can test the boundary conditions to see if they agree with our intuition. If we examine formula (73) in the absence of a conditional distribution, it should collapse into the prior distribution.

$\sim N([nS^{-1}x][nS^{-1}]^{-1},(nS^{-1})^{-1})$

(74) $\sim N(x,S/n)$

As we can see in formula (74), it does indeed collapse to the prior distribution. Another important scenario is the case of 100% certainty of the conditional distribution, where S, or some portion of it is 0, and thus S is not invertible. We can transform the returns and variance from formula (73) into a form which is more easy to work with in the 100% certainty case.

(75) $P(A|B) \sim N(x + (S/n)[\Omega + S/n]^{-1}[\mu - x],[(S/n) - (S/n)(\Omega + S/n)^{-1}(S/n)])$

This transformation relies on the result that $(A^{-1} + B^{-1})^{-1} = A - A(A+B)^{-1}A$. It is easy to see that when S is 0 (100% confidence in the views) then the posterior variance will be 0. If Ω is positive infinity (the confidence in the views is 0%) then the posterior variance will be (S/n).

We will revisit equations (73) and (75) later in this paper where we transform these basic equations into the various parts of the Black-Litterman model. Appendix D contains derivations of the alternate Black-Litterman formulas from the standard form, analogous to the transformation from (73) to (75).

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Appendix C

This appendix contains a derivation of the Black-Litterman master formula using the standard Bayesian approach for modeling the posterior of two normal distributions. One additional derivation is in [Mankert, 2006] where she derives the Black-Litterman 'master formula' from Sampling theory, and also shows the detailed transformation between the two forms of this formula.

The PDF Based Approach

The PDF Based Approach follows a Bayesian approach to computing the PDF of the posterior distribution, when the prior and conditional distributions are both normal distributions. This section is based on the proof shown in [DeGroot, 1970]. This is similar to the approach taken in [Satchell and Scowcroft, 2000].

The method of this proof is to examine all the terms in the PDF of each distribution which depend on E(r), neglecting the other terms as they have no dependence on E(r) and thus are constant with respect to E(r).

Starting with our prior distribution, we derive an expression proportional to the value of the PDF.

$P(A) \propto N(x,S/n)$ with n samples from the population.

So $\xi(x)$ the PDF of P(A) satisfies

C.1 $$\xi(x) \propto \exp\left((S/n)^{-1}(E(r)-x)^2\right)$$

Next, we consider the PDF for the conditional distribution.

$P(B|A) \propto N(\mu,\Sigma)$

So $\xi(\mu|x)$ the PDF of P(B|A) satisfies

C.2 $$\xi(\mu|x) \propto \exp\left(\Sigma^{-1}(E(r)-\mu)^2\right)$$

Substituting C.1 and C.2 into formula (1) from the text, we have an expression which the PDF of the posterior distribution will satisfy.

C.3 $$\xi(x|\mu) \propto \exp\left(-\left(\Sigma^{-1}(E(r)-\mu)^2+(S/n)^{-1}(E(r)-x)^2\right)\right)$$,

or $$\xi(x|\mu) \propto \exp(-\Phi)$$

Considering only the quantity in the exponent and simplifying

$$\Phi = \left(\Sigma^{-1}(E(r)-\mu)^2+(S/n)^{-1}(E(r)-x)^2\right)$$

$$\Phi = \left(\Sigma^{-1}\left(E(r)^2-2E(r)\mu+\mu^2\right)+(S/n)^{-1}\left(E(r)^2-2E(r)x+x^2\right)\right)$$

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$$\Phi = E(r)^2\left(\Sigma^{-1}+(S/n)^{-1}\right)-2E(r)(\mu\,\Sigma^{-1}+x(S/n)^{-1})+\Sigma^{-1}\mu^2+(S/n)^{-1}x^2$$

If we introduce a new term y, where

C.4 $$y=\frac{\left(\mu\,\Sigma^{-1}+x(S/n)^{-1}\right)}{\left(\Sigma^{-1}+(S/n)^{-1}\right)}$$

and then substitute in the second term

$$\Phi = E(r)^2\left(\Sigma^{-1}+(S/n)^{-1}\right)-2E(r)\,y(\Sigma^{-1}+(S/n)^{-1})+\Sigma^{-1}\mu^2+(S/n)^{-1}x^2$$

Then add $0 = y^2(\Sigma^{-1}+(S/n)^{-1})-(\mu\,\Sigma^{-1}+x(S/n)^{-1})^2(\Sigma^{-1}+(S/n)^{-1})^{-1}$

$$\begin{aligned}\Phi &= E(r)^2\left(\Sigma^{-1}+(S/n)^{-1}\right)-2E(r)\,y(\Sigma^{-1}+(S/n)^{-1})+\Sigma^{-1}\mu^2+(S/n)^{-1}x^2\\ &\quad +y^2(\Sigma^{-1}+(S/n)^{-1})-(\mu\,\Sigma^{-1}+x(S/n)^{-1})^2(\Sigma^{-1}+(S/n)^{-1})^{-1}\end{aligned}$$

$$\begin{aligned}\Phi &= E(r)^2\left(\Sigma^{-1}+(S/n)^{-1}\right)-2E(r)\,y(\Sigma^{-1}+(S/n)^{-1})+y^2(\Sigma^{-1}+(S/n)^{-1})\\ &\quad +\Sigma^{-1}\mu^2+(S/n)^{-1}x^2-(\mu\,\Sigma^{-1}+x(S/n)^{-1})^2(\Sigma^{-1}+(S/n)^{-1})^{-1}\end{aligned}$$

$$\begin{aligned}\Phi &= \left(\Sigma^{-1}+(S/n)^{-1}\right)\left[E(r)^2-2E(r)\,y+y^2\right]+(\Sigma^{-1}\mu^2+(S/n)^{-1}x^2)\\ &\quad -(\mu\,\Sigma^{-1}+x(S/n)^{-1})^2(\Sigma^{-1}+(S/n)^{-1})^{-1}\end{aligned}$$

$$\begin{aligned}\Phi &= \left(\Sigma^{-1}+(S/n)^{-1}\right)\left[E(r)^2-2E(r)\,y+y^2\right]-(\mu\,\Sigma^{-1}+x(S/n)^{-1})^2(\Sigma^{-1}+(S/n)^{-1})^{-1}\\ &\quad +(\Sigma^{-1}\mu^2+(S/n)^{-1}x^2)(\Sigma^{-1}+(S/n)^{-1})(\Sigma^{-1}+(S/n)^{-1})^{-1}\end{aligned}$$

$$\begin{aligned}\Phi &= \left(\Sigma^{-1}+(S/n)^{-1}\right)\left[E(r)^2-2E(r)\,y+y^2\right]\\ &\quad -(\mu^2\Sigma^{-2}+2\mu\,x\Sigma^{-1}(S/n)^{-1}+x^2(S/n)^{-2})(\Sigma^{-1}+(S/n)^{-1})^{-1}\\ &\quad +(\Sigma^{-2}\mu^2+(S/n)^{-1}\Sigma^{-1}x^2+\mu^2\Sigma^{-1}(S/n)^{-1}+x^2(S/n)^{-2})(\Sigma^{-1}+(S/n)^{-1})^{-1}\end{aligned}$$

$$\begin{aligned}\Phi &= \left(\Sigma^{-1}+(S/n)^{-1}\right)\left[E(r)^2-2E(r)\,y+y^2\right]\\ &\quad +((S/n)^{-1}\Sigma^{-1}x^2-2\mu\,x\Sigma^{-1}(S/n)^{-1}+\mu^2\Sigma^{-1}(S/n)^{-1})(\Sigma^{-1}+(S/n)^{-1})^{-1}\end{aligned}$$

$$\begin{aligned}\Phi &= \left(\Sigma^{-1}+(S/n)^{-1}\right)\left[E(r)^2-2E(r)\,y+y^2\right]\\ &\quad +(\Sigma^{-1}+(S/n)^{-1})^{-1}(x-\mu)(\Sigma^{-1}(S/n)^{-1})\end{aligned}$$

The second term has no dependency on E(r), thus it can be included in the proportionality factor and we are left we

C.5 $$\xi(x|\mu)\propto\exp\left(-\left[\left(\Sigma^{-1}+(S/n)^{-1}\right)^{-1}(E(R)-y)^2\right]\right)$$

Thus the posterior mean is y as defined in formula C.4, and the variance is

C.6 $$\left(\Sigma^{-1}+(S/n)^{-1}\right)^{-1}$$

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Appendix D

This appendix presents a derivation of the alternate formulation of the Black-Litterman master formula for the posterior expected return. Starting from formula (24) we will derive formula (25).

$$E(r)=\left[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi+P^T\Omega^{-1}Q\right]$$

Separate the parts of the second term

$$E(r)=\left[\left[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P\right]^{-1}(\tau\Sigma)^{-1}\Pi\right]+\left[\left[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P\right]^{-1}(P^T\Omega^{-1}Q)\right]$$

Replace the precision term in the first term with the alternate form

$$E(r)=\left[\left[\tau\Sigma-\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\tau\Sigma\right](\tau\Sigma)^{-1}\Pi\right]+\left[\left[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P\right]^{-1}(P^T\Omega^{-1}Q)\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[\left[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P\right]^{-1}(P^T\Omega^{-1}Q)\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[(\tau\Sigma)(\tau\Sigma)^{-1}\left[(\tau\Sigma)^{-1}+P^T\Omega^{-1}P\right]^{-1}(P^T\Omega^{-1}Q)\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[(\tau\Sigma)\left[I_n+P^T\Omega^{-1}P\tau\Sigma\right]^{-1}(P^T\Omega^{-1}Q)\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[\tau\Sigma\left[I_n+P^T\Omega^{-1}P\tau\Sigma\right]^{-1}(P^T\Omega^{-1})Q\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[\tau\Sigma\left[I_n+P^T\Omega^{-1}P\tau\Sigma\right]^{-1}(\Omega(P^T)^{-1})^{-1}Q\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[\tau\Sigma\left[\Omega(P^T)^{-1}+P\tau\Sigma\right]^{-1}Q\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[\tau\Sigma P^T(P^T)^{-1}\left[\Omega(P^T)^{-1}+P\tau\Sigma\right]^{-1}Q\right]$$

$$E(r)=\left[\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}P\Pi\right]\right]+\left[\tau\Sigma P^T\left[\Omega+P\tau\Sigma P^T\right]^{-1}Q\right]$$

Voila, the alternate form of the Black-Litterman formula for expected return.

$$E(r)=\Pi-\left[\tau\Sigma P^T\left[P\tau\Sigma P^T+\Omega\right]^{-1}\right]\left[Q-P\Pi\right]$$

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Appendix E

Derivation of formula (12)

We start with the definition of the views.

(76) $\hat{Q}=Q+\epsilon$

Where

- $\hat{Q}$ is the k x 1 vector of the unknown mean returns to the views.
- Q is the k x 1 vector of the estimated mean returns to the views.
- ϵ is a n x 1 matrix of residuals from the regression where $E(\epsilon)=0; V(\epsilon)=E(\epsilon\,\epsilon^T)=\Omega$ and Ω is non-singular.

We can rewrite formula (1) into a distribution of $\hat{Q}$ as follows:

(77) $\hat{Q}\sim N(Q,\Omega)$

We can also write our definition of the the unknown mean returns of the views based on the unknown mean returns of the assets and the portfolio pick matrix P

(78) $P\,\hat{\Pi}=\hat{Q}$

Where

- P is the k x n vector of weights for the view portfolios.
- $\hat{\Pi}$ is the n x 1 vector of the unknown returns of the assets

Substituting (1) into (2) we get the following

(79) $P\,\hat{\Pi}=Q+\epsilon$

Assuming that P is invertible, which requires it to be of full rank then we can multiply both sides by P^{-1}. This is the projection of the view estimated means into asset space representing the Black-Litterman conditional distribution. If P is not invertible then we would need a slightly different formulation here, adding another term on the right hand side.

(80) $\hat{\Pi}=P^{-1}Q+P^{-1}\epsilon$

We would like to represent formula (5) as a distribution. In order to do this we need to compute the covariance of the random term. The variance of the unknown asset means about the estimated view means projected into asset space is calculated as follows:

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$$
\begin{aligned}
&Variance = E\left[P^{-1}\epsilon\epsilon^{T}\left[P^{-1}\right]^{T}\right] \\
&Variance = P^{-1}E\left[\epsilon\epsilon^{T}\right]\left[P^{-1}\right]^{T} \\
&Variance = P^{-1}\Omega\left[P^{-1}\right]^{T} \\
&\quad using\left[P^{T}\right]^{-1} = \left[P^{-1}\right]^{T} \\
&Variance = P^{-1}\Omega\left[P^{T}\right]^{-1} \\
&\quad using\left[AB\right]^{-1} = B^{-1}A^{-1} \\
&Variance = \left[P^{T}\Omega^{-1}P\right]^{-1}
\end{aligned} \tag{81}
$$

So we arrive at the projection of the views into asset space as

$$\hat{\Pi} \sim N\left(P^{-1}Q, \left[P^{T}\Omega^{-1}P\right]^{-1}\right) \tag{82}$$

The covariance term here is the covariance of the unknown mean returns about the estimated returns from the views, it is not the covariance of expected returns.

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Appendix F

This section of the document summarizes the steps required to implement the Black-Litterman model.

Given the following inputs

- w Equilibrium weights for each asset class. Derived from capitalization weighted CAPM Market portfolio,
- Σ Matrix of covariances between the asset classes. Can be computed from historical data.
- r_f Risk free rate for base currency
- δ The risk aversion coefficient of the market portfolio. This can be assumed, or can be computed if one knows the return and standard deviation of the market portfolio.
- τ A measure of uncertainty of the equilibrium variance. Usually set to a small number of the order of 0.025 – 0.050.

First we use reverse optimization to compute the vector of equilibrium returns, Π using formula (7).

(7) $$\Pi = \delta \Sigma w$$

Then we formulate the investors views, and specify P, Ω, and Q. Given k views and n assets, then P is a k × n matrix where each row sums to 0 (relative view) or 1 (absolute view). Q is a k × 1 vector of the excess returns for each view. Ω is a diagonal k × k matrix of the variance of the views, or the confidence in the views. As a starting point, most authors call for the values of ω_i to be set equal to $p^T \tau \Sigma_i p$ (where p is the row from P for the specific view).

Next assuming we are uncertain in all the views, we apply the Black-Litterman 'master formula' to compute the posterior estimate of the returns using formula (25).

(25) $$\hat{\Pi} = \Pi + \tau \Sigma P^T \left[(P \tau \Sigma P^T) + \Omega \right]^{-1} \left[Q - P \Pi \right]$$

We compute the posterior variance using formula (30).

(30) $$M = \tau \Sigma - \tau \Sigma P^T \left[P \Sigma P^T + \Omega \right]^{-1} P \tau \Sigma$$

Closely followed by the computation of the sample variance from formula (27).

(27) $$\Sigma_p = \Sigma + M$$

And now we can compute the portfolio weights for the optimal portfolio on the unconstrained efficient frontier from formula (10).

(10) $$\hat{w} = \hat{\Pi} (\delta \Sigma_p)^{-1}$$

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Lesson Notes

MODULE 3 | LESSON 1

THE BLACK-LITTERMAN MODEL

Reading Time	90 minutes
Prior Knowledge	Traditional mean-variance optimization, Working knowledge of Bayesian theory, Matrix algebra
Keywords	CAPM, reverse optimization, equilibrium excess returns, market portfolios and global market portfolio, Black-Litterman model.

The previous lesson centered around factor models. Factor models can be seen as an alternative to more traditional mean-variance optimization in the sense that they allow an investor or a portfolio manager to select their desired exposure to specific asset classes.

In this lesson mean-variance optimization is brought back to the center stage, but with a new twist. Investors are allowed both to have no views or to have strong opinions about how assets can perform and their views/opinions are allowed to enter the model and become a part of it. Incidentally, doing so also alleviates some of the issues that are least desirable about traditional mean-variance optimal portfolios (concentrated positions and instability of solutions to changes in inputs).

The next lesson will introduce the critical-line algorithm, which is an important tool both for traditional portfolio optimization and the Black-Litterman model.

1. What is the Black-Litterman Model About?

The Black-Litterman model is not a completely new technique, and it builds upon the same type of mathematical tools used in mean-variance optimization of portfolios. The novelty of the model is that it calculates expected returns using two sources of information:

- By estimating implied expected excess returns in equilibrium, the model is based on the current market's views.
- By allowing a portfolio manager to express one's views about excess returns of stocks, asset classes, or countries, the model incorporates the managers' experience and subjective views.

Putting together the two sources of information, the model produces new estimates of expected excess returns, which become the new inputs of a mean-variance optimization model. Despite the fact that there is a certain amount of pure theory behind it, the Black-Litterman model is very much an applied technique of portfolio theory. In fact, this model was created by experienced practitioners Fischer Black and Robert Litterman of Goldman, Sachs & Company in the very early 1990s.

The model also makes it possible to overcome the particular problem (that we saw in the previous lessons) of highly concentrated, input-sensitive portfolios.

Highly concentrated positions are a problem because financial theory preaches that diversification is a good thing and input sensitivity is bad news for portfolio managers. This is because portfolio managers may find themselves forced to undertake heavy rebalancing and/or end up with portfolios that are counterintuitive and/or containing massive long and short positions and therefore untradable.

Best and Grauer in a 1991 paper have shown that a small increase in the expected return of one of the portfolio's assets can force half of the assets from the portfolio. Some researchers have tried to explore alternative estimates for expected returns (historical, returns, equal-mean returns, risk-adjusted equal mean returns, and more), but no matter what they tried, the end results were portfolios with large long and short positions concentrated on a small number of assets.

The Black-Litterman model uses a Bayesian approach in which the investors' subjective views regarding the expected returns of one more assets are combined with the market equilibrium vector (the prior distribution) of expected returns. The result is a

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new mixed estimate of expected returns (the posterior distribution), which can be described as a weighted average of the investor's views and the market equilibrium.

Academic and applied research confirm that the methodology is largely successful in its ability to avoid the tendency to generate non-tradable, extreme, and oversensitive portfolios every time the parameters of the model are updated.

This lesson is based on the required reading "The Black Litterman Model in Detail".The assigned reading is a fairly detailed survey paper that contains the theory and the technical proofs (discussed in a mathematical appendix).To avoid the risk of "missing the forest for the trees", this lesson aimes at providing a narrative to simplify and/or justify certain aspects of the theory to guide students to read the assigned materials and reflect on a certain amount of details and best practices to implement the model in real case studies.

2. Market Equilibrium Portfolio and Reverse Optimization

2.1 Market Equilibrium

It should be known from previous essons that we can write the linear regression model to test whether the CAPM holds as::

$$r_{it} - r_f = \alpha_i + \beta_i(r_{mt} - r_f) + \epsilon_{it},$$

where

- r_f is the risk-free rate;
- r_{it} is the return of asset i at time t;
- r_{mt} is the return of the equilibrium market portfolio at time t;
- ϵ_{it} represents the idiosyncratic risk for asset i at time t.

If $\alpha_i = 0$ then the CAPM model holds. The equilibrium market portfolio is the result of the decision making of all market participants, and it assumes that all economic agents hold symmetrical information about stock or asset classes returns, variances, and covariances. Furthermore, the model assumes that there is continuous and frictionless trading that allows an adjustment towards a long-term equilibrium. As such, the equilibrium market portfolio represents the shared objective view about excess returns (indirectly, the student should have recognized that we are postulating a semi-strong form of market efficiency). The idiosyncratic risk can be reduced or even made to disappear via diversification, and as such, it is not rewarded; only the systematic risk (measured via β) is.

We are also aware from a previous lesson that the Sharpe index for the market portfolio:

$$\frac{E[r_m] - r_f}{\sigma_m}$$

is also the slope of the capital allocation line (CAL) line, which allows us to identify the optimal (tangent) portfolio.

2.2 Reverse Optimization

The concept of reverse optimization was invented by Sharpe in 1974, and it is nothing too complicated, although the name could lead one to think that we are about to get involved with some type of sophisticated mathematical tool. Reverse optimization is simply used to derive the equilibrium expected excess returns from the CAPM via a certain number of parameters. To this purpose, we notice that the return on the market portfolio in equilibrium can be written in terms of the return of its components as:

$$r_m = \sum_{j=1}^{N} w_{m,j} r_j.$$

Now, let $\pi_i = E[r_i] - r_f$. Then, from the CAPM, we have

$$\pi_i = \beta_i(E[r_m] - r_f).$$

However, recalling that $\beta_i = \frac{\text{cov}(r_i, r_m)}{\sigma_m^2}$, we can rewrite the previous equality as

$$\pi_i = \frac{cov(r_i, r_m)}{\sigma_m^2}(E[r_m] - r_f) = \frac{cov(r_i, r_m)}{\sigma_m}\frac{E[r_m] - r_f}{\sigma_m} = \frac{cov(r_i, r_m)}{\sigma_m} \cdot SR_m, \text{ or}$$

$$\pi_i = \frac{E[r_m] - r_f}{\sigma_m^2} \sum_{j=1}^{N} \text{cov}(r_i, r_j) w_{m,j},$$

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where we just used the definition of r_m and the linearity property of the covariance operator.

Finally, if we let

$$\Sigma = \begin{pmatrix} \operatorname{cov}(r_1, r_1) & \operatorname{cov}(r_1, r_2) & \cdots & \operatorname{cov}(r_1, r_N) \\ \operatorname{cov}(r_2, r_1) & \operatorname{cov}(r_2, r_2) & \cdots & \operatorname{cov}(r_2, r_N) \\ \vdots & \vdots & \ddots & \vdots \\ \operatorname{cov}(r_N, r_1) & \operatorname{cov}(r_N, r_2) & \cdots & \operatorname{cov}(r_N, r_N), \end{pmatrix}$$

then we can write in matrix notation:

$$\pi = d\Sigma w_m$$

with π the $N \times 1$ vector of risk premiums in equilibrium,d is the risk-aversion coefficient, which corresponds to the Sharpe ratio, Σ is the $N \times N$ matrix of excess returns for the N assets, and w_m is the $N \times 1$ vector, which represents the weights of the individual assets in the portfolio based on market capitalization.

Why is it called "reverse optimization"? Following Sharpe in a 2007 paper, we could describe mean-variance portfolio optimization as a process like:

$$\begin{aligned} &\textbf{expected returns} \ + \ \textbf{risk} \ + \ \textbf{correlations} \\ &+ \ \textbf{risk aversion coefficient} \ \mapsto \ \textbf{optimal portfolio.} \end{aligned}$$

Using the market portfolio as the optimal portfolio is a legitimate choice if we assume that all economic agents have the same information and the same coefficient of risk aversion because they would all compute the same optimal portfolio. Then, the previous equation can be rewritten as:

$$\begin{aligned} &\textbf{market portfolio} \ + \ \textbf{risk} \ + \ \textbf{correlations} \\ &+ \ \textbf{risk aversion coefficient} \ \mapsto \ \textbf{expected returns.} \end{aligned}$$

Note that the risk-aversion coefficient,d in our notation, acts like a scaling factor for the reverse optimization estimate of excess returns. In the next section we will show that it corresponds to the Sharpe ratio for the market portfolio.

An individual who has no personal views about the market should hold the market portfolio, which becomes a reference point. In the language of Bayesian statistics, it could also be referred to as an **uninformative uniform prior.**

Arguably, the most tedious part about the Black-Litterman model is the preparation of the inputs as we need to identify the investment universe and estimate excess returns, covariance matrices, and risk-aversion coefficients. A global index (e.g., the MSCI World Index, UBS's GSMI, etc.) can be used as proxy for the market portfolio.

Since the CAPM contains all investable assets, it is very hard to specify.

However, because we are considering a situation of equilibrium, all sub-markets must also be in a situation of equilibrium; therefore, we could choose any investable sub-universe to act as the market portfolio.

Important caveat: If a portfolio manager decides that their universe of possible investments is just a handful of stocks, that would become the market portfolio. Because of our remark in the previous paragraph, that is legitimate. This makes computing the weights of the assets in the market portfolio much simpler. However, we need to be careful that now we need to compute the excess returns and the Sharpe ratio for this new market portfolio we selected and not keep using the MSCI or the GSMI indexes! In other words, now we need to construct the excess returns for our own market portfolio, which acts as our unique benchmark. So we gain in one direction, but we lose in another as now we need to construct the returns for our unique market portfolio weighing the returns of each of the component assets.

Example (Equivalence with approach based on implied betas). An alternative choice is to use the market portfolio capitalization weights to compute expected excess returns—that is, we could use what are called **implied betas**. What are implied betas? Implied betas are the betas of the N assets relative to the market capitalization-weighted portfolio. The formula (for the full $N \times 1$ vector of betas) is given by

$$\beta = (w_m^T \Sigma w_m)^{-1} \Sigma w_m.$$

Once we have the vector of implied betas, we can find the vector of CAPM excess returns. It is possible to prove that the vector of CAPM excess returns coincides with the reverse optimized returns when the CAPM returns are based on implied betas.

Why do we bother so much with market portfolios and the reverse optimization to produce excess equilibrium returns? Well, it turns out that these estimates are less volatile then using historical ones and produce asset weights that are much less extreme

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besides allowing the portfolio manager not to deal with very large long and short positions. As these are also inputs required for the Black-Litterman model, this is important because the methodology leads to cleaner and less extreme portfolios.

Example To show how to compute the excess returns in equilibrium we are going to use an example in which the market portfolio is the S&P 500 Index. As the individual assets we are considering the 11 sectors of the index listed immediately below with their tickers:

S&P 500 SECTORS

^SP500-15 Materials sector
^SP500-20 Industrial sector
^SP500-25 Consumer discretionary sector
^SP500-30 Consumer Staples sector
^SP500-35 Healthcare sector
^SP500-40 Financials sector
^SP500-45 Information Technology sector
^SP500-50 Telecommunication Services sector
^SP500-55 Utilities sector
^SP500-60 Real Estate sector
^GSPE Energy sector.

We are considering the period from January 1, 2022 to August 4, 2023 as our reference period. As of July 12, 2023, the weights of each of the 11 sectors with respect to the S&P 500 were in terms of capitalizations (with minimal roundings so that they sum to 1):

^SP500-15 0.02
^SP500-20 0.08
^SP500-25 0.11
^SP500-30 0.075
^SP500-35 0.135
^SP500-40 0.13
^SP500-45 0.275
^SP500-50 0.09
^SP500-55 0.02
^SP500-60 0.02
^GSPE 0.045

Furthermore during the period $\sigma_m = \sigma_{SP500} = 0.21$ and the Sharpe ratio for the S&P 500 was calculated at 0.62. These numbers are likely to change over different periods, so it is important to specify the time interval used.

At this point we will use some code to compute the excess return in equilibrium, π which will become one of the inputs later when using the Black-Litterman approach.

```
#---------------------------------------------------

import libraries (some are for other applications)

#---------------------------------------------------
import yfinance as yf
import datetime
import math
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
#---------------------------------------------------

uire the data using Yahoo Finance
```

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```
-------------------------------------------

sectors=["^SP500-15", "^SP500-20","^SP500-25","^SP500-30","^SP500-35",
"^SP500-40","^SP500-45","^SP500-50","^SP500-55", "^SP500-60",
"^GSPE" ]
start = datetime.datetime(2022, 1, 1)
end = datetime.datetime(2023, 8, 4)
data = yf.download(sectors, start=start, end=end)['Adj Close']

plot the time series for the individual sectors

data.plot.line(y=sectors, figsize=(10,6),title='S&P500 sectors', ylabel='level' )

calculate the geometric returns

df = np.log(data).diff()
df1= df.dropna()
#-----------------------------------------------------

Compute annualized mean returns and covariance matrix

#-----------------------------------------------------
def expected_return(dataframe):
expected_returns = []
for i in df1.columns:
expected_returns.append(df[i].mean()*252)
return np.array(expected_returns)
returns = expected_return(df1)
df_cov = df1.cov()*252
df1_cor = df1.corr()

-------------------------------------------------------------------------------

weights for each of the sector (manually coded as Yahoo Finance does not

provide this type of data)

#-------------------------------------------------------------------------------
weights_sectors = [0.02, 0.08, 0.11, 0.075, 0.135, 0.13, 0.275, 0.09, 0.02,0.02,0.045
d = 0.62/0.21  # d=2.95
d*(df_cov.dot(weights_sectors))
```

This produces the following excess returns in equilibrium, π **:**

^GSPE 0.083615

^SP500-15 0.115286

^SP500-20 0.104200

^SP500-25 0.145102

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^SP500-30 0.065655

^SP500-35 0.077166

^SP500-40 0.117381

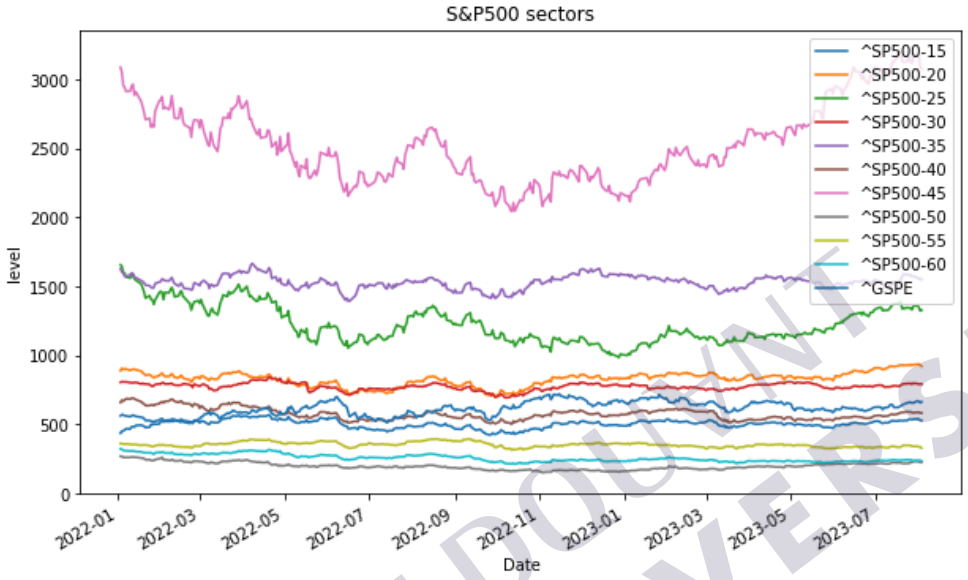
^SP500-45 0.141676

^SP500-50 0.127710

^SP500-55 0.070367

^SP500-60 0.105803

Figure 1: The time series for the 11 sectors that compose the S\&P 500 Index during the period from January 1, 2022 to August 4, 2023.



3. Black-Litterman Model: Implementation

The Black-Litterman model starts from the market equilibrium and introduces in it the portfolio manager's opinions about the returns for each asset. The model does so using a Bayesian approach:

- investor's views play the role of prior distribution;
- market excess returns in equilibrium play the role of additional information;
- Bayes' theorem produces optimal weights like a posterior distribution.

In what follows, we explain the Black-Litterman model using steps to break down the procedure as described in the assigned reading. Some are straightforward while one or two require a bit more thinking.

Take your time to understand each of the steps.

Step 1: The expected returns are assumed to follow a normal distribution, i.e.,

$$E[r] \sim N(\mu, \Sigma).$$

However, μ is not a parameter but a random variable. We express this as:

$$\mu \sim N(\pi, \Delta)$$

in which π is an estimate of the mean μ and Δ is the variance of the estimate μ. The relationship between μ and π can be written as

$$\mu = \pi + \epsilon$$

or

$$\pi = \mu - \epsilon.$$

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If $\epsilon \sim N(0, \Delta)$, then $-\epsilon \sim N(0, \Delta)$ as well, and we can just write

$$\pi = \mu + \epsilon.$$

If we make the further assumption that μ and ϵ are not correlated, then simple properties of the normal distribution yield:

$$E[r] \sim N(\pi, \Psi \equiv \Sigma + \Delta).$$

Before moving on to the next steps, we stress the importance of noticing the difference between the two formulations.:

$$E[r] \sim N(\mu, \Sigma) \quad \text{and} \quad E[r] \sim N(\pi, \Sigma + \Delta).$$

The second form is what is referred to as the **canonical Black-Litterman model**. However, some authors use the first model, and for this reason, it is always very important to be aware of what model is being used.

Step 2: In this step we need to specify π. Black and Litterman do this using the reverse optimization method seen earlier based on the CAPM and the market equilibrium, that is:

$$\pi = d\Sigma w_m$$

where the symbols have been introduced in the previous section.

Step 3: We need to calculate the coefficient of market risk aversion,d. To do that, we notice that if we start from the formula for π obtained in Step 2 and multiply both sides of the formula on the left by w_m^T, we obtain

$$w_m^T \pi = d w_m^T \Sigma w_m$$

(remember that d is a scalar quantity and as such can be moved anywhere in the formula). Now, we pay attention to the fact that

$$w_m^T \Sigma w_m = \sigma_m^2,$$

the variance of the market portfolio. Furthermore, we have also

$$\pi_m = E[r_m] - r_f = \sum_{i=1}^{N} w_{m,i} \pi_i \equiv w_m^T \pi$$

and therefore we have shown that

$$d = \frac{E[r_n] - r_f}{\sigma_m^2} = \frac{SR_m}{\sigma_m}$$

In other words d is the Sharpe ratio of the market portfolio divided by the volatility of the market portfolio which must be obtained from one of the global indices mentioned earlier.

Step 4: This is just a reminder that the weights w_{m_i} of each asset in the market portfolio are determined as

$$w_{m,i} = \frac{c_i}{c_m}$$

with c_i and c_m the capitalized value of asset i and the market portfolio, respectively.

Step 5: We need now to estimate Δ, the variance of μ and π. Black and Litterman make a simplifying assumption, i.e., they assume that

$$\Delta = \tau \Sigma$$

i.e.,Δ is proportional to the variance-covariance matrix of returns, Σ. In other words, they assume that

$$\mu \sim N(\pi, \tau \Sigma).$$

We eliminated one unknown quantity, but at the price of introducing another one. However τ is a scalar while Δ was a matrix. Hence, we have certainly simplified our problem.τ is a number between 0 and 1. The justification for this further assumption is that it measures the uncertainty of a mean. As such, its variability should be smaller than that of the individual excess return components.

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There is also another way to look at it: f an economic agent entertains the view that excess returns are very close to those calculated from the market portfolio via reverse optimization, then the variance of μ should be very close to zero. The value of τ can be calculated via a calibration process, but practitioners often assume that τ is a value in the range between 0.01 and 0.05.

Using MLE, the value of τ can be estimated as

$$\hat{\tau}_{MLE} = \frac{1}{T}$$

with T the number of observations. However, the MLE is not unbiased, and an unbiased version of the estimate is

$$\tilde{\tau} = \frac{1}{T-N}$$

where N was defined earlier as the number of assets in the market portfolio. These are not the only choices, and others have been proposed in the literature.

Step 6: When we work with the Markowitz model, economic agents/investors are allowed to have views regarding the expected excess returns of the assets. However, their views are absolute, meaning that they agree that an asset would achieve a return of, say, 3% more than the year before. In case they had no views, they would be assuming the same excess returns implied by the market portfolio via reverse optimization. In particular, the more a portfolio manager's views differ from the market portfolio's, the more extreme the shift would be in the weights of the individual assets vs. the weights in the market portfolio. No consideration is paid to the fact that views could be subject to errors and thus have a random nature.

The Black-Litterman model does not take this away. However, it allows for relative views/opinions (like when a manager estimates that an asset would produce a return higher (or smaller) than that of another asset. And it introduces random errors in the setup. For instance, let's keep assuming that we have N assets and that we have K opinions. We could express this as:

$$P\mu = q + \epsilon^v$$

where

P is a $K \times N$ matrix whose rows contains the weight of each of the investor's individual views. We have K rows, i.e., as many as the views the investor entertains about the overall universe of views. In case an investor had no view about one or more of the assets, then the corresponding entry along the row will be zero.μ is the $N \times 1$ vector that contains the average of expected excess returns,q is a$K \times 1$ vector that contains the returns forecasted for each of the K views. Finally, ϵ^v is a $K \times 1$ vector that contains the random error associated with the views.

At this point, an example is definitely needed to understand how this works in practice.

Example (Views about assets in Black Litterman). Let's assume that we have $N = 5$ assets and $K = 3$ views. The three views are as follows:

- During the next period, asset A will achieve a return of 8%;
- Again during the next period, asset B will achieve a return of 3%;
- During the next period, asset D will register a return higher than that of A,B,C, and E by 2.5%.

In this particular case P is a 3×5 matrix that looks like

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1/4 & -1/4 & -1/4 & 1 & -1/4 \end{pmatrix};$$

q is a3×1 matrix that looks like

$$q = \begin{pmatrix} 0.08 \\ 0.03 \\ 0.025 \end{pmatrix}.$$

The errors in the views are accounted for with the assumption that\

$$\epsilon^v \sim N(0, \Omega)$$

with Ω a $K \times K$ matrix that incorporates the manager's uncertainty about each view. In particular, the views are independent fr ach other and therefore Ω is a diagonal matrix, that is $\Omega = \text{diag}(\omega_1^2, \ldots, \omega_K^2)$. In turn, this implies that

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$$P\mu \sim N(q, \Omega).$$

Ω is a critical input, and it depends on the individual investor as they are the best economic agent to decide the certainty of their opinions. However, in the literature, a few formulas have been proposed and one often used is the following:

$$\Omega = \text{diag}(P(\tau\Sigma)P^T).$$

A still different approach could start with the individual investors assigning a confidence level to each of their views. For instance, an investor could think that asset i will appreciate by 5% during the next period, and he is 95% confident that the return will be between 1% and 9%. In this case, using simple rules for normal random variables it is easy to realize that $\omega_i = 2\%$.

Example (Views and confidence about assets in Black Litterman). Assume that we have 4 assets (A, B, C, D) and an investor with the following views:

- Asset A will achieve a return of 5% more than asset C. He think that his confidence is $\omega_1 = 3\%$.
- Asset D will achieve a return of 3% more than in the previous period. Confidence level $\omega_2 = 4\%$.

Well, in this case we have

$$P = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad q = \begin{pmatrix} 0.05 \\ 0.03 \end{pmatrix}, \quad \Omega = \begin{pmatrix} .03^2 & 0 \\ 0 & 0.04^2 \end{pmatrix}.$$

Step 7: This is the complex step, but the good news is that while the calculations are hard (we need to compute the mean of a posterior distribution using a multivariate normal with a multivariate normal prior), the operational result is available. The posterior distribution for the expected excess return for the assets that incorporates the portfolio manager's views is given by

$$E[r] = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}q]$$

which can be shown to be equal to

$$\mu_{BL} := E[r] = \pi + \tau\Sigma P^T[(P\tau\Sigma P^T) + \Omega]^{-1}[q - P\pi].$$

This second expression has an advantage in the sense that one does not need to invert the matrix Ω, which can make computations more precise and/or faster. Keep in mind that usually the matrix Ω contains small entries along the main diagonal and, on inverting, there could be some numerical loss of precision.

The posterior covariance matrix is given by

$$\text{cov}(r) \equiv M = ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}.$$

Is important to keep in mind that the posterior covariance matrix just supplied is the covariance matrix of the posterior mean estimate about the actual mean, or, said differently, it measures the uncertainty in the estimated posterior mean. If we wanted to compute the variance of the returns to use in a mean-variance optimization program, we need to compute

$$\Sigma_{BL} := \widetilde{\Sigma} = \Sigma + M = \Sigma + ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}.$$

Step 8: Now we are ready to perform the last step, i.e., we can run the following optimization program.

$$\begin{gathered} \max_w w^T\mu_{BL} - \tfrac{1}{2}w^T\Sigma_{BL}w \\ \textbf{subject to: add constraints, if any.} \end{gathered}$$

4. Further Readings

One could ask what is the outcome if the investor's views are wrong or only partially correct? Clearly, the result obtained using the Black-Litterman model would not be optimal. To be precise, it is optimal in the sense of the model, but it is not optimal from the point of view of practical results. Often, upon noticing that the portfolio is not performing well, the investor will correct their views and obtain a new optimal portfolio. However, to do so means that there will be rebalancing and associated transaction costs. Yanou in 2010 has proposed an alternative to the standard Black-Litterman model that incorporates the possibility of wrong views and opportunity costs. Yanou's model tries to arrive at an optimal solution when the opportunity costs are taken into consideration as part of the model itself.

Yanou's working paper can be found here: The Black-Litterman Model: Wrong Views vs. Opportunity Cost

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5. Python Implementation

As we said several times, nothing has really changed vs. the regular mean-variance optimization problem, except that we need to obtain some additional inputs and create views. A working Python implementation can be found here:

Stuart Jamieson, Black-Litterman Portfolio Allocation Model in Python (2020)

Another couple of useful links are the following, which points to the PyPortfolioOpt package that contains general tools for portfolio optimization and contains the Black-Litterman model, among other things:

- PyPortfolioOpt
- Robert Andrew Martin PyPortfolioOpt: portfolio optimization in Python, Journal of Open Source Software, 6, (61), 2021

6. Conclusion

In this lesson, we learned a different way to do mean-variance optimization to determine optimal portfolios. More specifically, we learned how to estimate an optimal portfolio using the consensus view (the market's view) and how to introduce the portfolio manager's views into the problem. The Black-Litterman model is a relevant tool for portfolio management, and it provides a generalization of previous algorithms we saw in earlier lessons.

In the next lesson, we will cover the critical line algorithm, which is quite useful in solving portfolio optimization problems (both for the traditional problem and Black-Litterman) when inequality constraints are required.

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Lesson Notes

MODULE 3 | LESSON 2

CRITICAL LINE ALGORITHM AND CORNER PORTFOLIOS

Reading Time	90 minutes
Prior Knowledge	Optimization procedures required in obtaining optimal portfolios, Calculus with matrix
Keywords	Critical line algorithm, turning points, corner portfolios

In the previous lesson, we extended the classical mean-variance optimization paradigm to incorporate portfolio managers' views into a model via the Black-Litterman model.

Regardless of what model is used, when inequality constraints are part of the model, we need a reliable tool to compute optimal portfolios. This is the critical line algorithm that will be presented next.

Both in the classical mean-variance optimization model and the Black-Litterman model, one has to resort to some kind of portfolio optimization. In its essence, this amounts to identifying the most suitable portfolio on the efficient frontier given a certain set of constraints.

The constraints change depending on the particular portfolio manager, their tactical portfolio choices, target returns, or threshold volatility assigned to the manager by the clients. In some cases, one may want to hold a position in a specific asset because such an asset is recognized as having strong upside, but its lack of liquidity may suggest to limit the exposure of the portfolio to this particular asset.

When we start to have inequality constraints in our portfolio, we are already aware that there is no closed form solution, unlike the case when we have only equality constraints. While there are packages that can compute numerical solutions for most feasible problems, Markowitz himself developed a method to compute a solution, and his method is called the **critical line algorithm** (CLA). Indeed, CLA has become a favorite among practitioners because it is arguably the only algorithm specifically designed for inequality-constrained portfolio optimization, which guarantees that an exact solution will be identified in a finite number of iterations.

Why can we not rely on the other numerical methods? Well, we can, but gradient-based algorithms depend on a seed vector of initial values and could converge to local maxima or minima. Not so for the CLA. In addition, gradient-based algorithms identify only the optimal solution (assuming it is a

global solution and not a local solution, which would obviously be sub-optimal); on the other hand, the CLA does not find a single portfolio and actually derives the entire efficient frontier.

It is worthwhile to keep in mind that CLA was developed by Markowitz to handle optimizing quadratic functions subject to linear inequality constraints. If, for any reason, one found oneself working with nonlinear inequality constraints, the methodology would not work.

This paper provides a step-by-step implementation of the CLA in the Python language. As the paper intersperses theory and coding, we will cover some important points separately in the notes to make sure that *you will not miss the forest for the trees*.

Before starting the lesson, students should be warned that a mere copy pasting of code in the paper to a Jupyter notebook or any other IDE may not work smoothly as the code in the paper may contain additional spaces which are likely to produce error messages when the code is required to run in Python.

1. Set up and essence of the CLA

We assume that we have n assets, and therefore, we have an $n \times 1$ vector of means (expected returns or, in the parlance of the Black-Litterman model, equilibrium excess returns or posterior excess returns) and an $n \times n$ variance covariance matrix (which could be the posterior variance covariance matrix Σ_{BL} if working in the Black-Litterman model setting). As before, we need to assume that these parameters are time homogeneous, i.e., they are constant over time (at least for the period of time that is of relevance to us). So far, nothing is different from what we did before.

Now, we need to introduce some additional elements. Specifically:

- $N = \{1, 2, \ldots, n\}$ is a set of indices associated with the individual assets in our universe;
- $w_i, i \in N$ denotes the weights of each asset in the portfolio.$w = (w_1, w_2, \ldots, w_n)^T$ is the $n \times 1$ vector of weights;
- $l = (l_1, l_2, \ldots, l_n)^T$ is an $n \times 1$ vector of lower bounds for the weights.$u = (u_1, u_2, \ldots, u_n)^T$ is an $n \times 1$ vector of upper bounds for the weights. Hence, we have that

$$l_i \leq w_i \leq u_i, \quad i \in N;$$

- $F \subset N$ is a subset that collects the free assets, i.e., assets for which $l_i < w_i < u_i$ for $i \in F$;
- $B \subset N$ is a subset that collects the assets whose weights lie on one of the bounds, i.e., either $w_i = l_i$ or $w_i = u_i$;
  \item Clearly $B \cup F = N$, i.e. any asset belongs to either F or B;
- Weights are required to sum up to 1, and the portfolio must achieve the required mean return,r_p, (assuming we solve the problem by minimizing the variance of the portfolio that achieves the minimum required return).

We need to maintain a distinction between free assets' indices and assets whose weights lie on one of their bounds. For this purpose, we will decompose the mean return vector and the variance covariance matrix of the assets as follows:

$$\mu = \begin{pmatrix} \mu_F \\ \mu_B \end{pmatrix}, \quad \text{and} \quad \Sigma = \begin{pmatrix} \Sigma_F & \Sigma_{FB} \\ \Sigma_{BF} & \Sigma_B \end{pmatrix}$$

where Σ_F is a $k \times k$ matrix,k being the number of free assets,Σ_B an $(n-k) \times (n-k)$ matrix, Σ_{FB} is $k \times (n-k)$, and Σ_{BF} is $(n-k) \times k$. The vector of portfolio weights is partitioned accordingly, i.e.,$w^T = (w_F^T, W_B^T)$. Finally, note also that $\Sigma_{FB} = \Sigma_{BF}^T$.

The key technical element for CLA is the concept of **turning point**. The definition of turning point is a practical one: a vector w^* is a turning point if, close to it, there is another solution vector with different free assets, i.e., at least one of the weights previously free becomes bounded and/or one of the previously bounded weights becomes free. This is important. Say that $W \subset R^N$ is the region of the Euclidean space R^N where our weights are allowed to exist. Then, in the regions of W away from turning points, the inequality constraints are for practical purposes irrelevant with respect to the free assets. Since the free weights are, as the name implies, free to move in that part of the region, this is equivalent to solving an unconstrained problem like (3) in the required reading, i.e.:

$$L(w_F, w_B, \gamma, \lambda) = \frac{1}{2} w_F^T \Sigma_F w_F + \frac{1}{2} w_F^T \Sigma_{FB} w_B + \frac{1}{2} w_B^T \Sigma_{BF} w_F + \frac{1}{2} w_B^T \Sigma_B w_B \\ - \gamma (w_F^T \iota_k + w_B^T \iota_{n-k} - 1) - \lambda (w_f^T \mu_F + w_B^T \mu_B - \mu_p)$$

with ι_k a $k \times 1$ vector of ones and ι_{n-k} an $(n-k) \times 1$ vector of ones. The nature of the Lagrangian function should be familiar by now. The thing to keep in mind is that, now,w_B is constrained and thus does not change between turning points. Hence, we need to find only the optimal $w_F^\star$ and since, again in between turning points, the w_F's are not constrained (in the sense that the w_F's are not subject to inequality constraints, more properly), this is an unconstrained optimization problem so to speak.

Markowitz's CLA works in the following way:

- Identify the turning points.
- At each turning point (pay attention since this means that some of the weights will be set equal to some value based on their active constraint), compute the optimal portfolio.
- The efficient frontier is derived as the convex combination between any two neighboring turning points (similar to what we did when using the two fund theorem in a previous lesson).

How do we find the turning points?

The algorithm requires an initial solution on the constrained minimum variance frontier, and it starts by identifying the turning point with the highest expected return. Then, we move to the next lower turning point.

Step 1. Order all assets with respect to their expected return values. Then, the first assets have the highest expected return and the n-th assets the lowest expected return.

Step 2. Set all weights initially at their lower bounds,$w_i = l_i$.

Step 3. Increase the weights of the first asset. If the upper bound is reached and $w^T \iota < 1$, increase the second asset's weight and so forth. This part terminates when $w^T \iota = 1$. Note that this requires that $w^T l \leq 1 \leq w^T u$ or there would be no solution! In particular, we stop for the asset whose index number k is such that

$$\sum_{i=1}^{k} u_i + \sum_{j=k+1}^{n} l_j \leq 1 \quad \text{and}$$

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$$\sum_{i=1}^{k+1} u_i + \sum_{j=k+2}^{n} l_j > 1.$$

Step 4. The solution is a vector of weights where the weights of the first assets are set to the upper bounds, the last assets' weights are set to their lower bounds, and there is an asset in the middle part whose weight is between its lower and upper bounds. Specifically, the weight of the k-th asset (now they are ordered) is going to be such that $l_k < 1 - \sum_{i=1}^{k-1} u_i - \sum_{j=k+1}^{n} l_i$. This particular asset (k) is the only one that is not set to either its upper or lower bound, and it is called the **free asset**. For convenience, we can relabel the index k as i_f. Then

$$w_{i_f} = 1 - \sum_{j \in U} w_j - \sum_{k \in L} w_k = 1 - w_B^T \iota_{n-k}.$$

Pay attention that at this first stage $F = \{i_f\}$ and $B = N \setminus \{i_f\}$.

Step 5. In going from one turning point to the next, it is necessary that one element is either added to the list of free assets, F, or removed from it. This can be done by changing the value of λ in $L(w_F, w_B, \gamma, \lambda)$ as defined above, letting it become smaller and smaller. This is not immediately obvious, and it happens because λ and $w^T \mu$ are linearly and positively related. (This is not really dealt with in the required reading). If we differentiate $L(w_F, w_B, \gamma, \lambda)$ with respect to w_F and set that system of equations to zero, we get

$$\Sigma_F w_F + \Sigma_{FB} w_B - \lambda \mu_F = \gamma \iota_F$$

where ι_F is a vector of 1s of length equal to the number of elements in F. Now, using the condition that the weights must sum to one, we have also

$$w_F^T \iota_F = 1 - w_B^T \iota_B.$$

Solving the last two systems of equations together with respect to γ implies that

$$\gamma = -\lambda \frac{\iota_F^T \Sigma_F^{-1} \mu_F}{\iota_F^T \Sigma_F^{-1} \iota_F} + \frac{1 - \iota_B^T w_B + \iota_F^T \Sigma_F^{-1} \Sigma_{FB} w_B}{\iota_F^T \Sigma_F^{-1} \iota_F}.$$

Also, we have easily determined that

$$w_F = -\Sigma_F^{-1} \Sigma_{FB} w_B + \gamma \Sigma_F^{-1} \iota_F + \lambda \Sigma_F^{-1} \mu_F.$$

Then, we have

$$\mu^T w = \mu_F^T w_F + \mu_B^T w_B = -\mu_F^T \Sigma_F^{-1} \Sigma_{FB} w_B + \gamma H + \lambda K$$

where

$$H = \iota_F^T \Sigma_F^{-1} \mu_F, \;\; K = \mu_F^T \Sigma_F^{-1} \mu_F.$$

Differentiating the last expression with respect to λ and using the expression for γ found above, we get

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$$\frac{\partial \mu^T w}{\partial \lambda} = K - \frac{H^2}{M}$$

with $M = \iota_F^T \Sigma_F^{-1} \iota_F$. At this point, we know that between turning points Σ_F does not change and therefore $\mu^T w$ is a linear function of λ and the slope is given by $K - \frac{H^2}{M}$. Furthermore, one can show that this slope is positive (it is a bit tedious, but it is only algebra).

This linear relationship between λ and $\mu^T w$ implies that every subsequent turning point will have a lower value for λ. In this particular step, our F has only one asset and we have only the option to add another asset to F. However, in general, in the steps that will follow, moving from turning point to turning point by decreasing λ, we can have one of the following two situations:

1. either one free assets moves to one of its bounds or
2. an asset that was previously on one of its bounds becomes free.

These two cases must be considered to compute the λ and w associated with each of the turning points. To this purpose, the required paper deals with two separate cases.

(Case 1: Free asset moves to its bound). In this case, suppose that λ_t is associated with the current turning point and let F be the set of free assets slightly below this turning point, which is the same as selecting a$\lambda_{t+1} < \lambda < \lambda_t$. The index t refers to the successive values of λ associated with a turning point (hence the index "t"). Exploiting the documented linear relationship between λ and $\mu^T w$, if we decrease λ, asset i will hit the lower bound if $dw_i/d\lambda > 0$, and the upper bound whenever $dw_i/d\lambda < 0$.

If $\lambda^{(i)}$ is the λ associated with the turning point where asset i hits one of the bounds, then one could show that its formula is given by

$$\lambda^{(i)} = \frac{1}{C_i}\left[\left(1 - \iota_B^T w_B + \iota_F^T \Sigma_F^{-1} \Sigma_{FB} w_B\right)(\sigma_F^{-1}\iota_F)_i - (\iota_F^T(\Sigma_F^{-1}\iota_F)_i(b_i + (\Sigma_F^{-1}\Sigma_{FB} w_B)_i\right]$$

where

$$C_i = -(\iota_F^T \Sigma_F^{-1} \iota_F)(\Sigma_F^1 \mu_F)_i + (\iota_F^T \Sigma_F^{-1} \mu_F)(\Sigma_F^{-1}\iota_F)_i$$

and

$$b_i = \begin{cases} u_i & \text{if } C_i > 0; \\ l_i & \text{if } C_i < 0. \end{cases}$$

This case should be used only when we have at least two free assets since for $k = 1\, C_i = 0$.

Then, the next $\lambda < \lambda_t$ where an asset leaves the subset F is given by

$$\lambda_{out} = \max_{i \in F}\{\lambda^{(i)}\}.$$

(Case 2. One asset is on its bound and becomes free). When we decrease λ, it is possible that an asset i that was on its bound becomes free. The corresponding portfolio is therefore a turning point, and the subsets F and B must be redefined:\

$$F_i = F \cup \{i\}, \quad B_i = B \setminus \{i\}, \;\; i \in B.$$

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In this case, the formula for λ_i is like the one before, but we have to replace B with B_i and F with F_i just defined. That is:

$$\lambda^{(i)} = \frac{1}{C_i}\left[\left(1 - \iota_{B_i}^T w_{B_i} + \iota_{F_i}^T \Sigma_{F_i}^{-1}\Sigma_{F_i B_i} w_{B_i}\right)(\sigma_{F_i}^{-1}\iota_{F_i})_i - (\iota_{F_i}^T \Sigma_{F_i}^{-1}\iota_{F_i})_i(b_i + (\Sigma_{F_i}^{-1}\Sigma_{F_i B_i} w_{B_i})_i\right]$$

where

$$C_i = -(\iota_{F_i}^T \Sigma_{F_i}^{-1}\iota_{F_i})(\Sigma_{F_i}^1 \mu_{F_i})_i + (\iota_{F_i}^T \Sigma_{F_i}^{-1}\mu_{F_i})(\Sigma_{F_i}^{-1}\iota_{F_i})_i$$

and

$$b_i = \begin{cases} u_i & \text{if asset i was previously on its upper bound} \\ l_i & \text{if asset i was previously on its lower bound.} \end{cases}$$

NOTE: These results for λ_i, C_i, b_i are derived by starting from the equation

$$w_F = -\Sigma_F^{-1}\Sigma_{FB} w_B + \gamma \Sigma_F^{-1}\iota_F + \lambda \Sigma_F^{-1}\mu_F,$$

replacing γ with its values derived earlier, which yields an expression that contains only λ and many constants. Then, one takes the derivatives of the i-th components with respect to λ and perform algebra on the equations setting the bound equal to l_i or b_i as needed. It is not particularly difficult, but it is clearly tedious work.

Define

$$\lambda_{in} = \max_{i \in B}\{\lambda^i : \lambda^i < \lambda_t\}$$

and if there is no $\lambda^i < \lambda_t$, there is no solution for λ_{in}.

Step 6. To identify the next turning point, we use the values found for λ_{in} and λ_{out}. We define

$$\lambda_{new} \equiv \lambda_{t+1} = \max\{\lambda_{in}, \lambda_{out}\}$$

If there is only a solution for λ_{in} or λ_{out}, then λ_{new} is overwritten by the respective value.

Recall that λ_{in} means that for that λ asset i leaves B and enters in F while λ_{out} is the value of λ for which asset i enters B. Hence, we need to keep this in mind when using λ_{new}.

Depending on which case occurs we have to replace F by $F \setminus \{i\}$ or $F_i = F \cup \{i\}$, or $B = B \cup \{i\}$ or $B_i = B \setminus \{i\}$. We must be sure to remember to update the current value for λ.

Step 7. Once we reach a situation where there is no longer a solution for λ_{in} and λ_{out}, the algorithm is completed as we have reached the lower turning point.

At this point, a student should be in a position to read the paper and make sense of the coding done there. Once properly implemented, it will be a working algorithm.

2. Corner portfolios

We will discuss an example where short selling is constrained (i.e. $l_i = 0$ for $i = 1, 2, \ldots, n$). In addition, weights are expected to sum to 1.00, as always.

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We are familiar with the two fund (or two portfolio) theorem; in the case we have constraints, it is not difficult to prove that combining two neighboring turning points linearly also leads to a constrained minimum variance portfolio. This relies on the fact that by writing the function $L(w_F, w_B, \gamma, \lambda)$ the way we did, we are actually solving unconstrained optimization problems (where some of the weights are kept fixed at one of their bounds) and thus a linear combination of two solutions of the unconstrained problem is itself a solution.

The optimal portfolios associated with each turning point are called **corner portfolios**. Then, the efficient frontier can be closely approximated using linear combinations of a small number of corner portfolios.

Given five assets, a portfolio manager who only has no short selling constraints has run the critical line algorithm and has obtained the following output.

The typical output using CLA is presented in Figure 1. Note that in this example 6 corner portfolios are identified and that the portfolio compositions are to be read along the rows.

Figure 1: Example of Output Using CLA .

				Asset Weights				
Corner P.folio	Expected Return	Expected Std. dev.	Sharpe Ratio	Asset 1	Asset 2	Asset 3	Asset 4	Asset 5
1	15%	18.5%	0.81	0.00	0.00	1.00	0.00	0.00
2	13.5%	16.2%	0.83	0.00	0.00	0.85	0.00	0.15
3	13.1%	13.6%	0.96	0.22	0.00	0.55	0.00	0.23
4	12.6%	13.0%	0.97	0.21	0.00	0.50	0.05	0.24
5	10.6%	11.0%	0.91	0.10	0.49	0.21	0.00	0.20
6	9.5%	10.9%	0.87	0.00	0.87	0.04	0.00	0.09

The client, a pension fund, has stated to the portfolio manager that they would need a return to match the current spending rate of 8.5%, inflation (estimated at 2.2%) and cost of managing the pension fund (expected to be 0.50%.) The trustees of the pension fund have stated that they would like to preserve the purchasing power of the fund, and they also explained to the portfolio manager that the standard deviation should be no more than 12.0% a year.

Then, the required rate of return that the portfolio manager needs to achieve is:

$$(1 + 8.5\%)(1 + 2.2\%)(1 + 0.50\%) - 1 = 11.44\%.$$

What should the portfolio manager do based on the table they received from their quantitative analyst?

We notice that the required rate of return lies between Corner Portfolio 4 and Corner Portfolio 5. Hence, we consider a linear combination of these two corner portfolios, i.e.:

$$12.6\%\alpha + 10.6\%(1 - \alpha) = 11.44\%.$$

This means that $2\alpha = 0.84$ or $\alpha = 0.42$. Therefore, the manager will invest 0.42 in CP 4 and 0.58 in CP 5. What does this mean in practice in terms of the weights for each of the five asset classes? It's simple:

$$\text{Asset 1}: 0.42 * (0.21) + 0.58 * (0.10) = 0.146;$$

$$\text{Asset 2}: 0.42 * (0.00) + 0.58 * (0.49) = 0.284;$$

$$\text{Asset 3}: 0.42 \times (0.50) + 0.58 \times (0.21) = 0.332;$$

$$\text{Asset 4}: 0.42 \times (0.05) + 0.58 \times (0.00) = 0.021;$$

$$\text{Asset 5}: 0.42 \times (0.24) + 0.58 \times (0.20) = 0.217.$$

Note that the sum of the assets weights is equal to 1 as we would expect.

The portfolio manager was also assigned a constraint on the max volatility the pension fund is willing to accept. The volatility is computed using the volatilities of the corner portfolios. Since we do not know the correlation between the two corner portfolios, the standard deviation of this portfolio can be closely approximated by the weighted average of the two relevant corner portfolios. In this case, we have

$$\sigma_p = 0.42 \times 13\% + 0.58 \times 11\% = 11.84\%$$

which is less than 12% and hence the portfolio satisfies the risk-requirement as well.

The Sharpe ratio for this portfolio (assuming a risk-free interest rate of zero) is 0.966. The Board of Trustees is quite happy with the proposal they receive from the portfolio manager. Out of curiosity, they inquire what the Sharpe ratio is for the market. This would be the Sharpe ratio of the corner portfolio with the highest Sharpe ratio, i.e., 0.97. The trustee are now really happy with their choice of the portfolio manager.

The last point about identifying the market portfolio, i.e., the corner portfolio with the highest Sharpe ratio is quite important. In fact, if there were a risk-free asset and we are allowed to borrow, we could use the CAL line as we did before. But, for that, we need to be able to identify the market portfolio, which we just did.

3. Conclusion

In this lesson, we learned how to solve portfolio optimization problems (both the Markowitz and the Black-Litterman) when inequality constraints are required. Furthermore, we investigated the notion of corner portfolios and how they can be used in real-life applications.

In the next lesson, we'll consider the Black-Litterman model again. However, this time, we will look at what happens when practitioners are tempted to take certain shortcuts.

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Article

An Open-Source Implementation of the Critical-Line Algorithm for Portfolio Optimization

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Abstract: Portfolio optimization is one of the problems most frequently encountered by financial practitioners. The main goal of this paper is to fill a gap in the literature by providing a well-documented, step-by-step open-source implementation of *Critical Line Algorithm* (CLA) in a scientific language. The code is implemented as a Python class object, which allows it to be imported like any other Python module, and integrated seamlessly with pre-existing code. We discuss the logic behind CLA following the algorithm's decision flow. In addition, we developed several utilities that support finding answers to recurrent practical problems. We believe this publication will offer a better alternative to financial practitioners, many of whom are currently relying on generic-purpose optimizers which often deliver suboptimal solutions. The source code discussed in this paper can be downloaded at the authors' websites (see contact details).

Keywords: Portfolio selection; quadratic programming; portfolio optimization; constrained efficient frontier; turning point; Kuhn-Tucker conditions; risk aversion.

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1. Introduction

Since the work of Markowitz [1], portfolio optimization has become one of the most critical operations performed in investment management. In Modern Portfolio Theory, this operation consists in computing the Efficient Frontier, defined as the set of portfolios that yield the highest achievable mean excess return (in excess of the risk-free rate) for any given level of risk (measured in terms of standard deviation). This portfolio optimization problem receives two equivalent formulations: (i) Minimizing the portfolio's standard deviation (or variance) subject to a targeted excess return or (ii) Maximize the portfolio's excess return subject to a targeted standard deviation (or variance). Because the solution to both formulations in the same, we will focus on (i) for the remainder of the paper.

Every financial firm is constantly faced with the problem of optimizing a portfolio. Wealth managers determine optimal holdings in order to achieve a strategic goal. Mutual funds attempt to beat a benchmark by placing active bets, based on some tactical superior knowledge with regard to the future expected returns or their covariance. Multi-manager hedge funds must allocate capital to portfolio managers, based on their performance expectations. Portfolio managers allocate capital to a variety of bets, incorporating their views about the future state of the economy. Risk managers may compute the portfolio that delivers the best hedge against a position that cannot be liquidated, etc. The current size of the global asset management industry is estimated to be in excess of US$58 trillion at year-end 2011. Thus robust portfolio optimization software is a necessity of the first order [2].

Most practitioners are routinely faced with the problem of optimizing a portfolio subject to inequality conditions (a lower and an upper bound for each portfolio weight) and an equality condition (that the weights add up to one). There is no analytic solution to this problem, and an optimization algorithm must be used. Markowitz [3,4] developed a method for computing such a solution, which he named the *"critical line algorithm"* or CLA. Wolfe [5] developed a Simplex version of CLA to deal with inequality constraints on linear combinations of the optimal weights. These more general constraints make Wolfe's algorithm more flexible. However, the standard portfolio optimization problem does not require them, making CLA the approach favored by most practitioners.

Given the importance of CLA, one would expect a multiplicity of software implementations in a wide range of languages. And yet, we are aware of only one published source-code for CLA, by Markowitz and Todd [6]. This is written in Excel's Microsoft Visual Basic for Applications (VBA-Excel), and it is the descendant of a previous implementation in an experimental programming language called EAS-E [7]. Steuer et al. [8] point out the inconveniences of working with VBA-Excel, most notably its low computational performance and its limitation to covariance matrices with a maximum order of *(256x256)*. Also, the VBA-Excel spreadsheet has to be manually adjusted for different problems, which prevents its industrial use.[1] Hence, it would be highly convenient to have the source code of CLA in a more scientific language, such as C++ or Python.

[1] Kwak [9] explains that VBA-Excel implementations are ubiquitous in the financial world, posing a systemic risk. Citing an internal JP Morgan investigation, he mentions that a faulty Excel implementation of the Value-at-Risk model may have

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A number of authors seem to have implemented CLA in several languages, however their code does not appear to be publicly available. For example, Hirschberger et al. [10] mention their implementation of CLA in Java. These authors state: "For researchers intending to investigate mid- to large-scale portfolio selection, good, inexpensive and understandable quadratic parametric programming software, capable of computing the efficient frontiers of problems with up to two thousand securities without simplifications to the covariance matrix, is hardly known to be available anywhere." This insightful paper discusses the computational efficiency of their CLA implementation, and makes valuable contributions towards a better understanding of CLA's properties. However, it does not provide the source code on which their calculations were based.

Niedermayer et al. [11] implemented CLA in Fortran 90, and concluded that their algorithm was faster than Steuer et al. [8] by a factor of 10,000, on a universe of 2,000 assets. These authors also state that "no publicly available software package exists that computes the entire constrained minimum variance frontier." Like other authors, they discuss the performance of their implementation, without providing the actual source code.

To our knowledge, CLA is the only algorithm specifically designed for inequality-constrained portfolio optimization problems, which guarantees that the exact solution is found after a given number of iterations. Furthermore, CLA does not only compute a single portfolio, but it derives the entire efficient frontier. In contrast, gradient-based algorithms will depend on a seed vector, may converge to a local optimum, are very sensitive to boundary constraints, and require a separate run for each member of the efficient frontier. The Scipy library offers an optimization module called *optimize*, which bears 5 constrained optimization algorithms: The Broyden-Fletcher-Goldfarb-Shanno method (BFGS), the Truncated-Newton method (TNC), the Constrained Optimization by Linear Approximation method (COBYLA), the Sequential Least Squares Programming method (SLSQP) and the Non-Negative Least Squares solver (NNLS). Of those, BFGS and TNC are gradient-based and typically fail because they reach a boundary. COBYLA is extremely inefficient in quadratic problems, and is prone to deliver a solution outside the feasibility region defined by the constraints. NNLS does not cope with inequality constraints, and SLSQP may reach a local optimum close to the original seed provided. Popular commercial optimizers include AMPL, NAG, TOMLAB and Solver. To our knowledge, none of them provide a CLA implementation. Barra offers a product called *Optimizer*, which "incorporates proprietary solvers developed in-house by MSCI's optimization research team".[2] The details of these algorithms are kept confidential, and documentation is only offered to customers, so we have not been able to determine whether this product includes an implementation of CLA. The lack of publicly available CLA software, commercially or open-source, means that most researchers, practitioners and financial firms are resorting to generic linear or quadratic programming algorithms that have not been specifically designed to solve the constrained portfolio optimization problem, and

been partly responsible for the US$6 billion trading loss suffered by JP Morgan in 2012, popularly known as the "London whale" debacle.

[2] http://www.msci.com/resources/factsheets/Barra_Optimizer.pdf

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will often return suboptimal solutions. This is quite astounding, because as we said most financial practitioners face this problem with relatively high frequency.

The main goal of this paper is to fill this gap in the literature by providing a well-documented, step-by-step open-source implementation of CLA in a scientific language.[3] We have chosen Python[4] because it is free, open-source, and it is widely used in the scientific community in general, and has more than 26,000 extension modules currently available.[5] An additional reason is that the Python language stresses readability. Its "pseudocode" appearance makes it a good choice for discussion in an academic paper. Because the procedure published in [11] seems to be the most numerically efficient, our implementation will follow closely their analysis. We have solved a wide range of problems using our code, and ensured that our results exactly match those of the VBA-Excel implementation in [6].

CLA was developed by Harry Markowitz to optimize general quadratic functions subject to linear inequality constraints. CLA solves any portfolio optimization problem that can be represented in such terms, like the standard Efficient Frontier problem. The posterior mean and posterior covariance derived by Black-Litterman [12] also lead to a quadratic programming problem, thus CLA is also a useful tool in that Bayesian framework. However, the reader should be aware of portfolio optimization problems that cannot be represented in quadratic form, and therefore cannot be solved by CLA. For example, the authors of this paper introduced in [13] a Sharpe ratio Efficient Frontier framework that deals with moments higher than 2, and thus has not a quadratic representation. For that particular problem, the authors have derived a specific optimization algorithm which takes skewness and kurtosis into account.

The rest of the paper is organized as follows: Section 2 presents the quadratic programming problem we solve by using the CLA. Readers familiar with this subject can go directly to Section 3, where we will discuss our implementation of CLA in a class object. Section 4 expands the code by adding a few utilities. Section 5 illustrates the use of CLA with a numerical example. Section 6 summarizes our conclusions. Results can be validated using the Python code in the Appendix.

2. The Problem

Consider an investment universe of n assets with observations characterized by a $(nx1)$ vector of means μ and a (nxn) positive definite covariance matrix Σ. The mean vector μ and covariance matrix Σ are computed on time-homogeneous invariants, i.e. phenomena that repeat themselves identically throughout history regardless of the reference time at which an observation is made. Time-homogeneity is a key property that our observations must satisfy, so that we do not need to re-estimate

[3] All code in this paper is provided "as is", and contributed to the academic community for non-business purposes only, under a GNU-GPL license. Users explicitly renounce to any claim against the authors. The authors retain the commercial rights of any for-profit application of this software, which must be pre-authorized in written by the authors.
[4] http://pypi.python.org/pypi
[5] To be precise, our code has been developed using the EPD 7.3 product (Enthought Python Distribution), which efficiently integrates all the necessary scientific libraries. Please visit https://www.enthought.com/products/epd.php for additional details.

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μ, Σ for a sufficiently long period of time. For instance, compounded returns are generally accepted as good time-homogeneous invariants for equity, commodity and foreign-exchange products. In the case of fixed income products, changes in the yield-to-maturity are typically used. In the case of derivatives, changes in the rolling forward at-the-money implied volatility are usually considered a good time-homogeneous invariant (see [14] for a comprehensive discussion of this subject). The numerical example discussed in Section 5 is based on equity products, hence the mean vector μ and covariance matrix Σ are computed on compounded returns.

Following [11], we solve a quadratic programming problem subject to linear constraints in inequalities and one linear constraint in equality, and so we need some nomenclature to cover those inputs:

- $N = \{1,2, \dots, n\}$ is a set of indices that number the investment universe.
- ω is the *(nx1)* vector of asset weights, which is our optimization variable.
- l is the *(nx1)* vector of lower bounds, with $\omega_i \geq l_i, \forall i \in N$.
- u is the *(nx1)* vector of upper bounds, with $\omega_i \leq u_i, \forall i \in N$.
- $F \subseteq N$ is the subset of *free* assets, where $l_i < \omega_i < u_i$. In words, free assets are those that do not lie on their respective boundaries. F has length $1 \leq k \leq n$.
- $B \subset N$ is the subset of weights that lie on one of the bounds. By definition, $B \cup F = N$.

Accordingly, we can construct μ and Σ as a block array and matrix.

$$\Sigma = \begin{bmatrix} \Sigma_F & \Sigma_{FB} \\ \Sigma_{BF} & \Sigma_B \end{bmatrix}, \mu = \begin{bmatrix} \mu_F \\ \mu_B \end{bmatrix}, \omega = \begin{bmatrix} \omega_F \\ \omega_B \end{bmatrix} \tag{1}$$

where Σ_F denotes the *(kxk)* covariance matrix among free assets, Σ_B the *((n-k)x(n-k))* covariance matrix among assets lying on a boundary condition, and Σ_{FB} the *(kx(n-k))* covariance between elements of F and B, which obviously is equal to Σ_{BF}' (the transpose of Σ_{BF}) since Σ is symmetric. Similarly, μ_F is the *(kx1)* vector of means associated with F, μ_B is the *((n-k)x1)* vector of means associated with B, ω_F is the *(kx1)* vector of weights associated with F, ω_B is the *((n-k)x1)* vector of weights associated with B.

The solution to the unconstrained problem[6] consists in minimizing the Lagrange function with respect to the vector of weights ω and the multipliers γ and λ:

$$L[\omega, \gamma, \lambda] = \frac{1}{2}\omega'\Sigma\omega - \gamma(\omega' 1_n - 1) - \lambda\left(\omega'\mu - \mu_p\right) \tag{2}$$

where 1_n is the *(nx1)* vector of ones and μ_p is the targeted excess return. The method of Lagrange multipliers applies first order necessary conditions on each weight and Lagrange multiplier, leading to a linear system of *n+2* conditions. See [14] for an analytical solution to this problem.

[6] "Unconstrained problem" is a bit of a misnomer, because this problem indeed contains two linear equality constraints: Full investment (the weights add up to one) and target portfolio mean. What is meant is to indicate that no specific constraints are imposed on individual weights.

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As the constrained problem involves conditions in inequalities, the method of Lagrange multipliers cannot be used. One option is to apply Karush-Kuhn-Tucker conditions. Alternatively, we can "divide and conquer" the constrained problem, by translating it into a series of unconstrained problems. The key concept is that of *turning point*. A solution vector ω^* is a turning point if in its vicinity there is another solution vector with different free assets. This is important because in those regions of the solution space away from turning points the inequality constraints are effectively irrelevant with respect to the free assets. In other words, between any two turning points, the constrained solution reduces to solving the following unconstrained problem on the free assets.

$$L[\omega,\gamma,\lambda] = \frac{1}{2}\omega_F'\Sigma_F\omega_F + \frac{1}{2}\omega_F'\Sigma_{FB}\omega_B + \frac{1}{2}\omega_B'\Sigma_{BF}\omega_F + \frac{1}{2}\omega_B'\Sigma_B\omega_B - \gamma(\omega_F'1_k + \omega_B'1_{n-k} - 1) - \lambda\left(\omega_F'\mu_F + \omega_B'\mu_B - \mu_p\right) \quad (3)$$

where ω_B is known and does not change between turning points. Markowitz [3] focused his effort in computing the optimal portfolio at each turning point is because *the efficient frontier can be simply derived as a convex combination between any two neighbor turning points*.[7] Hence, the only remaining challenge is the determination of the turning points, to which end we dedicate the following section.

3. The Solution

We have implemented CLA as a class object in Python programming language. The only external library needed for this core functionality is Numpy, which in our code we instantiate with the shorthand *np*. The class is initialized in Snippet 1. The inputs are:

- *mean*: The *(nx1)* vector of means.
- *covar*: The *(nxn)* covariance matrix.
- *lB*: The *(nx1)* vector that sets the lower boundaries for each weight.
- *uB*: The *(nx1)* vector that sets the upper boundaries for each weight.

Implied is the constraint that the weights will add up to one. The class object will contain four lists of outputs:

- *w*: A list with the *(nx1)* vector of weights at each turning point.
- *l*: The value of λ at each turning point.
- *g*: The value of γ at each turning point.
- *f*: For each turning point, a list of elements that constitute *F*.

```
class CLA:
    def __init__(self,mean,covar,lB,uB):
        # Initialize the class
        if (mean==np.ones(mean.shape)*mean.mean()).all():mean[-1,0]+=1e-5
```

[7] In the unconstrained case, this is sometimes referred to as the "two mutual funds theorem". In the constrained case, it still holds between any two neighbor turning points, because as we argued earlier, between them the constrained problem reduces to an unconstrained problem on the free assets.

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```
    self.mean=mean
    self.covar=covar
    self.lB=lB
    self.uB=uB
    self.w=[] # solution
    self.l=[] # lambdas
    self.g=[] # gammas
    self.f=[] # free weights
```

Snippet 1 – CLA initialization

The key insight behind Markowitz's CLA is to find first the turning point associated with the highest expected return, and then compute the sequence of turning points, each with a lower expected return than the previous. That first turning point consists in the smallest subset of assets with highest return such that the sum of their upper boundaries equals or exceeds one. We have implemented this search for the first turning point through a structured array. A structured array is a Numpy object that, among other operations, can be sorted in a way that changes are tracked. We populate the structured array with items from the input *mean*, assigning to each a sequential *id* index. Then we sort the structured array in descending order. This gives us a sequence for searching for the first free asset. All weights are initially set to their lower bounds, and following the sequence from the previous step, we move those weights from the lower to the upper bound until the sum of weights exceeds one. The last iterated weight is then reduced to comply with the constraint that the sum of weights equals one. This last weight is the first free asset, and the resulting vector of weights the first turning point. See Snippet 2 for the actual implementation of this initialization operation.

```
def initAlgo(self):
    # Initialize the algo
    #1) Form structured array
    a=np.zeros((self.mean.shape[0]),dtype=[('id',int),('mu',float)])
    b=[self.mean[i][0] for i in range(self.mean.shape[0])] # dump array into list
    a[:]=zip(range(self.mean.shape[0]),b) # fill structured array
    #2) Sort structured array
    b=np.sort(a,order='mu')
    #3) First free weight
    i,w=b.shape[0],np.copy(self.lB)
    while sum(w)<1:
        i-=1
        w[b[i][0]]=self.uB[b[i][0]]
    w[b[i][0]]+=1-sum(w)
    return [b[i][0]],w
```

Snippet 2 – Algorithm initialization

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The transition from one turning point to the next requires that one element is either added to or removed from the subset of free assets, *F*. Because λ and $\omega'\mu$ are linearly and positively related, this means that each subsequent turning point will lead to a lower value for λ. This recursion of adding or removing one asset from *F* continues until the algorithm determines that the optimal expected return cannot be further reduced. In the first run of this iteration, the choice is simple: *F* has been initialized with one asset, and the only option is to add another one (*F* cannot be an empty set, or there would be no optimization). Snippet 3 performs this task.

```
#2) case b): Free one bounded weight
        l_out=None
        if len(f)<self.mean.shape[0]:
            b=self.getB(f)
            for i in b:
                covarF,covarFB,meanF,wB=self.getMatrices(f+[i])
                covarF_inv=np.linalg.inv(covarF)
                l,bi=self.computeLambda(covarF_inv,covarFB,meanF,wB,meanF.shape[0]-1, \
                    self.w[-1][i])
                if (self.l[-1]==None or l<self.l[-1]) and l>l_out:l_out,i_out=l,i
```

Snippet 3 – Determining which asset could be added to F

In this part of the code, we search within *B* for a candidate asset *i* to be added to *F*. That search only makes sense if *B* is not an empty set, hence the first *if*. Because *F* and *B* are complementary sets, we only need to keep track of one of them. In the code, we always derive *B* from *F*, thanks to the functions *getB* and *diffLists*, detailed in Snippet 4.

```
def getB(self,f):
    return self.diffLists(range(self.mean.shape[0]),f)
#---------------------------------------------------------------
def diffLists(self,list1,list2):
    return list(set(list1)-set(list2))
```

Snippet 4 – Deriving B from F

Snippet 3 invokes a function called *getMatrices*. This function prepares the necessary matrices to determine the value of λ associated with adding each candidate *i* to *F*. In order to do that, it needs to reduce a matrix to a collection of columns and rows, which is accomplished by the function *reduceMatrix*. Snippet 5 details these two functions.

```
def getMatrices(self,f):
    # Slice covarF,covarFB,covarB,meanF,meanB,wF,wB
    covarF=self.reduceMatrix(self.covar,f,f)
    meanF=self.reduceMatrix(self.mean,f,[0])
```

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```
    b=self.getB(f)
    covarFB=self.reduceMatrix(self.covar,f,b)
    wB=self.reduceMatrix(self.w[-1],b,[0])
    return covarF,covarFB,meanF,wB
#-----------------------------------------------------------------
def reduceMatrix(self,matrix,listX,listY):
    # Reduce a matrix to the provided list of rows and columns
    if len(listX)==0 or len(listY)==0:return
    matrix_=matrix[:,listY[0]:listY[0]+1]
    for i in listY[1:]:
      a=matrix[:,i:i+1]
      matrix_=np.append(matrix_,a,1)
    matrix__=matrix_[listX[0]:listX[0]+1,:]
    for i in listX[1:]:
      a=matrix_[i:i+1,:]
      matrix__=np.append(matrix__,a,0)
    return matrix__
```

Snippet 5 – Preparing $\Sigma_F, \Sigma_{FB}, \mu_F, \omega_B$

Using the matrices provided by the function *getMatrices*, λ can be computed as:

$$\lambda = \frac{1}{C}[(1 - 1'_{n-k}\omega_B + 1'_k\Sigma_F^{-1}\Sigma_{FB}\omega_B)(\Sigma_F^{-1}1_k)_i - (1'_k\Sigma_F^{-1}1_k)(b_i + (\Sigma_F^{-1}\Sigma_{FB}\omega_B)_i)]$$

with (4)

$$C = -(1'_k\Sigma_F^{-1}1_k)(\Sigma_F^{-1}\mu_F)_i + (1'_k\Sigma_F^{-1}\mu_k)(\Sigma_F^{-1}1_k)_i$$

$$b_i = \begin{cases} u_i & \text{if } C_i > 0 \\ l_i & \text{if } C_i < 0 \end{cases}$$

A proof of these expressions can be found in [11]. Eq. (4) is implemented in function *computeLambda*, which is shown is Snippet 6. We have computed some intermediate variables, which can be re-used at various points in order to accelerate the calculations. With the value of λ, this function also returns b_i, which we will need in Snippet 8.

```
def computeLambda(self,covarF_inv,covarFB,meanF,wB,i,bi):
    #1) C
    onesF=np.ones(meanF.shape)
    c1=np.dot(np.dot(onesF.T,covarF_inv),onesF)
    c2=np.dot(covarF_inv,meanF)
    c3=np.dot(np.dot(onesF.T,covarF_inv),meanF)
    c4=np.dot(covarF_inv,onesF)
    c=-c1*c2[i]+c3*c4[i]
    if c==0:return None,None
```

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```
    #2) bi
    if type(bi)==list:bi=self.computeBi(c,bi)
    #3) Lambda
    if wB==None:
        # All free assets
        return float((c4[i]-c1*bi)/c),bi
    else:
        onesB=np.ones(wB.shape)
        l1=np.dot(onesB.T,wB)
        l2=np.dot(covarF_inv,covarFB)
        l3=np.dot(l2,wB)
        l2=np.dot(onesF.T,l3)
        return float(((1-l1+l2)*c4[i]-c1*(bi+l3[i]))/c),bi
```

Snippet 6 – Computing λ

In Snippet 3 we saw that the value of λ which results from each candidate i is stored in variable l. Among those values of l, we find the maximum, store it as *l_out*, and denote as *i_out* our candidate to become free. This is only a candidate for addition into F, because before making that decision we need to consider the possibility that one item is removed from F, as follows.

After the first run of this iteration, it is also conceivable that one asset in F moves to one of its boundaries. Should that be the case, Snippet 7 determines which asset would do so. Similarly to the addition case, we search for the candidate that, after removal, maximizes λ (or to be more precise, minimizes the reduction in λ, since we know that λ becomes smaller at each iteration). We store our candidate for removal in the variable *i_in*, and the associated λ in the variable *l_in*.

```
#1) case a): Bound one free weight
        l_in=None
        if len(f)>1:
            covarF,covarFB,meanF,wB=self.getMatrices(f)
            covarF_inv=np.linalg.inv(covarF)
            j=0
            for i in f:
                l,bi=self.computeLambda(covarF_inv,covarFB,meanF,wB,j,[self.lB[i],self.uB[i]])
                if l>l_in:l_in,i_in,bi_in=l,i,bi
                j+=1
```

Snippet 7 – Determining which asset could be removed from F

All auxiliary functions used in Snippet 7 have been discussed earlier. At this point, we must take one of two alternative actions: We can either add one asset to F, or we can remove one asset from F. The answer is whichever gives the greater value of λ, as shown in Snippet 8. Recall that in Snippet 6 the

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function *computeLambda* had also returned b_i. This is the boundary value that we will assign to a formerly free weight (now a member of *B*).

```
#4) decide lambda
        if l_in>l_out:
            self.l.append(l_in)
            f.remove(i_in)
            w[i_in]=bi_in # set value at the correct boundary
        else:
            self.l.append(l_out)
            f.append(i_out)
        covarF,covarFB,meanF,wB=self.getMatrices(f)
        covarF_inv=np.linalg.inv(covarF)
```

Snippet 8 – Deciding between the two alternative actions

We finally know how to modify *F* in order to compute the next turning point, but we still need to compute the actual turning point that results from that action. Given the new value of λ, we can derive the value of γ, which together determine the value of the free weights in the next turning point, ω_F.

$$\gamma = -\lambda \frac{1_k' \Sigma_F^{-1} \mu_F}{1_k' \Sigma_F^{-1} 1_k} + \frac{1 - 1_{n-k}' \omega_B + 1_k' \Sigma_F^{-1} \Sigma_{FB} \omega_B}{1_k' \Sigma_F^{-1} 1_k} \tag{5}$$
$$\omega_F = -\Sigma_F^{-1} \Sigma_{FB} \omega_B + \gamma \Sigma_F^{-1} 1_k + \lambda \Sigma_F^{-1} \mu_F$$

Eq. (5) is evaluated by the function *computeW*, which is detailed in Snippet 9.

```
def computeW(self,covarF_inv,covarFB,meanF,wB):
    #1) compute gamma
    onesF=np.ones(meanF.shape)
    g1=np.dot(np.dot(onesF.T,covarF_inv),meanF)
    g2=np.dot(np.dot(onesF.T,covarF_inv),onesF)
    if wB==None:
        g,w1=float(-self.l[-1]*g1/g2+1/g2),0
    else:
        onesB=np.ones(wB.shape)
        g3=np.dot(onesB.T,wB)
        g4=np.dot(covarF_inv,covarFB)
        w1=np.dot(g4,wB)
        g4=np.dot(onesF.T,w1)
        g=float(-self.l[-1]*g1/g2+(1-g3+g4)/g2)
    #2) compute weights
    w2=np.dot(covarF_inv,onesF)
```

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```
    w3=np.dot(covarF_inv,meanF)
    return -w1+g*w2+self.l[-1]*w3,g
```

Snippet 9 – Computing the turning point associated with the new F

Again, we are computing some intermediate variables for the purpose of re-using them, thus speeding up calculations. We are finally ready to store these results, as shown in Snippet 10. The last line incorporates the exit condition, which is satisfied when $\lambda = 0$, as it cannot be further reduced.

```
#5) compute solution vector
        wF,g=self.computeW(covarF_inv,covarFB,meanF,wB)
        for i in range(len(f)):w[f[i]]=wF[i]
        self.w.append(np.copy(w)) # store solution
        self.g.append(g)
        self.f.append(f[:])
        if self.l[-1]==0:break
```

Snippet 10 – Computing and storing a new solution

The algorithm then loops back through Snippets 3-10, until no candidates are left or the highest lambda we can get is negative.[8] When that occurs, we are ready to compute the last solution, as follows. Each turning point is a minimum variance solution subject to a certain target portfolio mean, but at some point we must also explicitly add the global minimum variance solution, denoted Minimum Variance Portfolio. This possibility is not considered by [11], but it is certainly a necessary portfolio computed by [6]. The analysis would be incomplete without it, because we need the left extreme of the frontier, so that we can combine it with the last computed turning point. When the next λ is negative or it cannot be computed, no additional turning points can be derived. It is at that point that we must compute the Minimum Variance portfolio, which is characterized by $\lambda = 0$ and a vector μ_F of zeroes. Snippet 11 prepares the relevant variables, so that the function *computeW* returns that last solution.

```
if (l_in==None or l_in<0) and (l_out==None or l_out<0):
            #3) compute minimum variance solution
            self.l.append(0)
            covarF,covarFB,meanF,wB=self.getMatrices(f)
            covarF_inv=np.linalg.inv(covarF)
            meanF=np.zeros(meanF.shape)
```

Snippet 11 – Computing the Minimum Variance portfolio

[8] We also depart here from [11], who keep searching for turning points with negative λ.

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4. A Few Utilities

The CLA class discussed in Section 3 computes all turning points plus the global Minimum Variance portfolio. This constitutes the entire set of solutions, and from that perspective, Section 3 presented an integral implementation of Markowitz's CLA algorithm. We think that this functionality can be complemented with a few additional methods designed to address problems typically faced by practitioners.

4.1. Search for the Minimum Variance portfolio

The Minimum Variance portfolio is the leftmost portfolio of the constrained efficient frontier. Even if it did not coincide with a turning point, we appended it to *self.w*, so that we can compute the segment of efficient frontier between the Minimum Variance portfolio and the last computed turning point. Snippet 12 exemplifies a simple procedure to retrieve this portfolio: For each solution stored, it computes the variance associated with it. Among all those variances, it returns the squared root of the minimum (the standard deviation), as well as the portfolio that produced it. This portfolio coincides with the solution computed in Snippet 11.

```
def getMinVar(self):
    # Get the minimum variance solution
    var=[]
    for w in self.w:
        a=np.dot(np.dot(w.T,self.covar),w)
        var.append(a)
    return min(var)**.5,self.w[var.index(min(var))]
```

Snippet 12 – The search for the Minimum Variance portfolio

4.2. Search for the Maximum Sharpe Ratio Portfolio

The turning point with the maximum Sharpe ratio does not necessarily coincide with the maximum Sharpe ratio portfolio. Although we have not explicitly computed the maximum Sharpe ratio portfolio yet, we have the building blocks needed to construct it. Every two neighbor turning points define a segment of the efficient frontier. The weights that form each segment result from the convex combination of the turning points at its edges.

For $\alpha \in [0,1]$, $\omega = \alpha\omega_0 + (1-\alpha)\omega_1$ gives the portfolios associated with the segment of the efficient frontier delimited by ω_0 and ω_1. The Sharpe ratio is a strongly unimodal function of λ (see [15] for a proof). We know that λ is a strictly monotonic function of α, because λ cannot increase as we transition from ω_0 to ω_1. The conclusion is that the Sharpe ratio function is also a strongly unimodal function of α. This means that we can use the Golden Section algorithm to find the maximum Sharpe

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ratio portfolio within the appropriate segment.[9] Snippet 13 shows an implementation of such algorithm. The *goldenSection* function receives as arguments:

- *obj*: The objective function on which the extreme will be found.
- *a*: The leftmost extreme of search.
- *b*: The rightmost extreme of search.
- ***kargs*: Keyworded variable-length argument list.

```
def goldenSection(self,obj,a,b,**kargs):
    # Golden section method. Maximum if kargs['minimum']==False is passed
    from math import log,ceil
    tol,sign,args=1.0e-9,1,None
    if 'minimum' in kargs and kargs['minimum']==False:sign=-1
    if 'args' in kargs:args=kargs['args']
    numIter=int(ceil(-2.078087*log(tol/abs(b-a))))
    r=0.618033989
    c=1.0-r
    # Initialize
    x1=r*a+c*b;x2=c*a+r*b
    f1=sign*obj(x1,*args);f2=sign*obj(x2,*args)
    # Loop
    for i in range(numIter):
      if f1>f2:
        a=x1
        x1=x2;f1=f2
        x2=c*a+r*b;f2=sign*obj(x2,*args)
      else:
        b=x2
        x2=x1;f2=f1
        x1=r*a+c*b;f1=sign*obj(x1,*args)
    if f1<f2:return x1,sign*f1
    else:return x2,sign*f2
```

Snippet 13 – Golden section search algorithm

The extremes of the search are delimited by the neighbor turning points, which we pass as a keyworded variable length argument (*kargs*). *kargs* is a dictionary composed of two optional arguments: *'minimum'* and *'args'*. Our implementation of the Golden Section algorithm searches for a minimum by default, however it will search for a maximum when the user passes the optional argument *'minimum'* with value *False*. *'args'* contains a non-keyworded variable-length argument, which (if present) is passed to the objective function *obj*. This approach allows us to pass as many

[9] The Golden Section search is a numerical algorithm for finding the minimum or maximum of a real-valued and strictly unimodal function. This is accomplished by sequentially narrowing the range of values inside which the minimum or maximum exists. [17] introduced this technique, and [18] proved its optimality.

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arguments as the objective function *obj* may need in other applications. Note that, for this particular utility, we have imported two additional functions from Python's *math* library: *log* and *ceil*.

We ignore which segment contains the portfolio that delivers the global maximum Sharpe ratio. We will compute the local maximum of the Sharpe ratio function for each segment, and store the output. Then, the global maximum will be determined by comparing those local optima. This operation is conducted by the function *getMaxSR* in Snippet 14. *evalSR* is the objective function (*obj*) which we pass to the *goldenSection* routine, in order to evaluate the Sharpe ratio at various steps between ω_0 and ω_1. We are searching for a maximum between those two turning points, and so we set *kargs={'minimum':False,'args':(w0,w1)}*.

```
def getMaxSR(self):
    # Get the max Sharpe ratio portfolio
    #1) Compute the local max SR portfolio between any two neighbor turning points
    w_sr,sr=[],[]
    for i in range(len(self.w)-1):
        w0=np.copy(self.w[i])
        w1=np.copy(self.w[i+1])
        kargs={'minimum':False,'args':(w0,w1)}
        a,b=self.goldenSection(self.evalSR,0,1,**kargs)
        w_sr.append(a*w0+(1-a)*w1)
        sr.append(b)
    return max(sr),w_sr[sr.index(max(sr))]
#---------------------------------------------------------------
def evalSR(self,a,w0,w1):
    # Evaluate SR of the portfolio within the convex combination
    w=a*w0+(1-a)*w1
    b=np.dot(w.T,self.mean)[0,0]
    c=np.dot(np.dot(w.T,self.covar),w)[0,0]**.5
    return b/c
```

Snippet 14 – The search for Maximum Sharpe ratio portfolio

4.3. Computing the Efficient Frontier

As argued earlier, we can compute the various segments of the efficient frontier as a convex combination between any two neighbor turning points. This calculation is carried out by the function *efFrontier* in Snippet 15. We use the Numpy function *linspace* to uniformly partition the unit segment, excluding the value 1. The reason for this exclusion is, the end of one convex combination coincides with the beginning of another, and so the uniform partition should contain that value 0 or 1, but not both. When we reach the final pair of turning points, we include the value 1 in the uniform partition, because this is the last iteration and the resulting portfolio will not be redundant. *efFrontier* outputs

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three lists: Means, Standard Deviations and the associated portfolio weights. The number of items in each of these lists is determined by the argument *points.*

```
def efFrontier(self,points):
    # Get the efficient frontier
    mu,sigma,weights=[],[],[]
    a=np.linspace(0,1,points/len(self.w))[:-1] # remove the 1, to avoid duplications
    b=range(len(self.w)-1)
    for i in b:
        w0,w1=self.w[i],self.w[i+1]
        if i==b[-1]:a=np.linspace(0,1,points/len(self.w)) # include the 1 in the last iteration
        for j in a:
            w=w1*j+(1-j)*w0
            weights.append(np.copy(w))
            mu.append(np.dot(w.T,self.mean)[0,0])
            sigma.append(np.dot(np.dot(w.T,self.covar),w)[0,0]**.5)
    return mu,sigma,weights
```

Snippet 15 – Computing the Efficient Frontier

5. A Numerical Example

We will run our Python implementation of CLA on the same example that accompanies the VBA-Excel implementation by [6]. Table 1 provides the vector of expected returns and the covariance matrix of returns. We set as boundary conditions that $0 \leq \omega_i \leq 1, \forall i = 1, \ldots, n$, and the implicit condition that $\sum_{i=1}^{n} \omega_i = 1$.

Table 1. Covariance Matrix, Vector of Expected Returns and boundary conditions

L Bound	0	0	0	0	0	0	0	0	0	0
U Bound	1	1	1	1	1	1	1	1	1	1
Mean	1.175	1.19	0.396	1.12	0.346	0.679	0.089	0.73	0.481	1.08
Cov	0.4075516									
	0.0317584	0.9063047								
	0.0518392	0.0313639	0.194909							
	0.056639	0.0268726	0.0440849	0.1952847						
	0.0330226	0.0191717	0.0300677	0.0277735	0.3405911					
	0.0082778	0.0093438	0.0132274	0.0052667	0.0077706	0.1598387				
	0.0216594	0.0249504	0.0352597	0.0137581	0.0206784	0.0210558	0.6805671			
	0.0133242	0.0076104	0.0115493	0.0078088	0.0073641	0.0051869	0.0137788	0.9552692		
	0.0343476	0.0287487	0.0427563	0.0291418	0.0254266	0.0172374	0.0462703	0.0106553	0.3168158	
	0.022499	0.0133687	0.020573	0.0164038	0.0128408	0.0072378	0.0192609	0.0076096	0.0185432	0.1107929

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Snippet 16 shows a simple example of how to use the CLA class. We have stored the data in a *csv* file, with the following structure:

- Row 1: Headers
- Row 2: Mean vector
- Row 3: Lower bounds
- Row 4: Upper bounds
- Row 5 and successive: Covariance matrix

```
def main():
    import numpy as np
    import CLA
    #1) Path
    path='H:/PROJECTS/Data/CLA_Data.csv'
    #2) Load data, set seed
    headers=open(path,'r').readline().split(',')[:-1]
    data=np.genfromtxt(path,delimiter=',',skip_header=1) # load as numpy array
    mean=np.array(data[:1]).T
    lB=np.array(data[1:2]).T
    uB=np.array(data[2:3]).T
    covar=np.array(data[3:])
    #3) Invoke object
    cla=CLA.CLA(mean,covar,lB,uB)
    cla.solve()
    print cla.w # print all turning points
    #4) Plot frontier
    mu,sigma,weights=cla.efFrontier(100)
    plot2D(sigma,mu,'Risk','Expected Excess Return','CLA-derived Efficient Frontier')
    #5) Get Maximum Sharpe ratio portfolio
    sr,w_sr=cla.getMaxSR()
    print np.dot(np.dot(w_sr.T,cla.covar),w_sr)[0,0]**.5,sr
    print w_sr
    #6) Get Minimum Variance portfolio
    mv,w_mv=cla.getMinVar()
    print mv
    print w_mv
    return
#---------------------------------------------------------------
# Boilerplate
if __name__=='__main__':main()
```

Snippet 16 – Using the CLA class

The key lines of code are:

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- ***cla=CLA.CLA(mean,covar,lB,uB)***: This creates a CLA object named *cla*, with the input parameters read from the *csv* file.
- ***cla.solve()***: This runs the *solve* method within the CLA class (see the Appendix), which comprises all the Snippets listed in Section 3 (Snippets 1-11).

Once the *cla.solve()* method has been run, results can be accessed easily:

- *cla.w* contains a list of all turning points.
- *cla.l* and *cla.g* respectively contain the values of λ and γ for every turning point.
- *cla.f* contains the composition of F used to compute every turning point.

Table 2 reports these outputs for our particular example. Note that sometimes an asset may become free, and yet the turning point has the weight for that asset resting precisely at the same boundary it became free from. In that case, the solution may seem repeated, when in fact what is happening is that the same portfolio is the result of two different F sets.

Table 2. Return, Risk, λ and composition of the 10 turning points

CP Num	Return	Risk	Lambda	X(1)	X(2)	X(3)	X(4)	X(5)	X(6)	X(7)	X(8)	X(9)	X(10)
1	1.190	0.952	58.303	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	1.180	0.546	4.174	0.649	0.351	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	1.160	0.417	1.946	0.434	0.231	0.000	0.335	0.000	0.000	0.000	0.000	0.000	0.000
4	1.111	0.267	0.165	0.127	0.072	0.000	0.281	0.000	0.000	0.000	0.000	0.000	0.520
5	1.108	0.265	0.147	0.123	0.070	0.000	0.279	0.000	0.000	0.000	0.006	0.000	0.521
6	1.022	0.230	0.056	0.087	0.050	0.000	0.224	0.000	0.174	0.000	0.030	0.000	0.435
7	1.015	0.228	0.052	0.085	0.049	0.000	0.220	0.000	0.180	0.000	0.031	0.006	0.429
8	0.973	0.220	0.037	0.074	0.044	0.000	0.199	0.026	0.198	0.000	0.033	0.028	0.398
9	0.950	0.216	0.031	0.068	0.041	0.015	0.188	0.034	0.202	0.000	0.034	0.034	0.383
10	0.803	0.205	0.000	0.037	0.027	0.095	0.126	0.077	0.219	0.030	0.036	0.061	0.292

We can also access the utilities described in Section 4. For instance, the instruction *mu,sigma,weights=cla.efFrontier(100)* in Snippet 16 computes 100 points of the Efficient Frontier, and plots them using the auxiliary function in Snippet 17. If the optional argument *pathChart* is not used, the chart is plotted, but if a path is provided, the chart is saved as a file in that destination.

```
def plot2D(x,y,xLabel='',yLabel='',title='',pathChart=None):
    import matplotlib.pyplot as mpl
    fig=mpl.figure()
    ax=fig.add_subplot(1,1,1) #one row, one column, first plot
    ax.plot(x,y,color='blue')
    ax.set_xlabel(xLabel)
    ax.set_ylabel(yLabel,rotation=90)
    mpl.xticks(rotation='vertical')
    mpl.title(title)
```

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```
if pathChart==None:
    mpl.show()
else:
    mpl.savefig(pathChart)
mpl.clf() # reset pylab
return
```

Snippet 17 – Plotting the Efficient Frontier

Figure 1. The Efficient Frontier

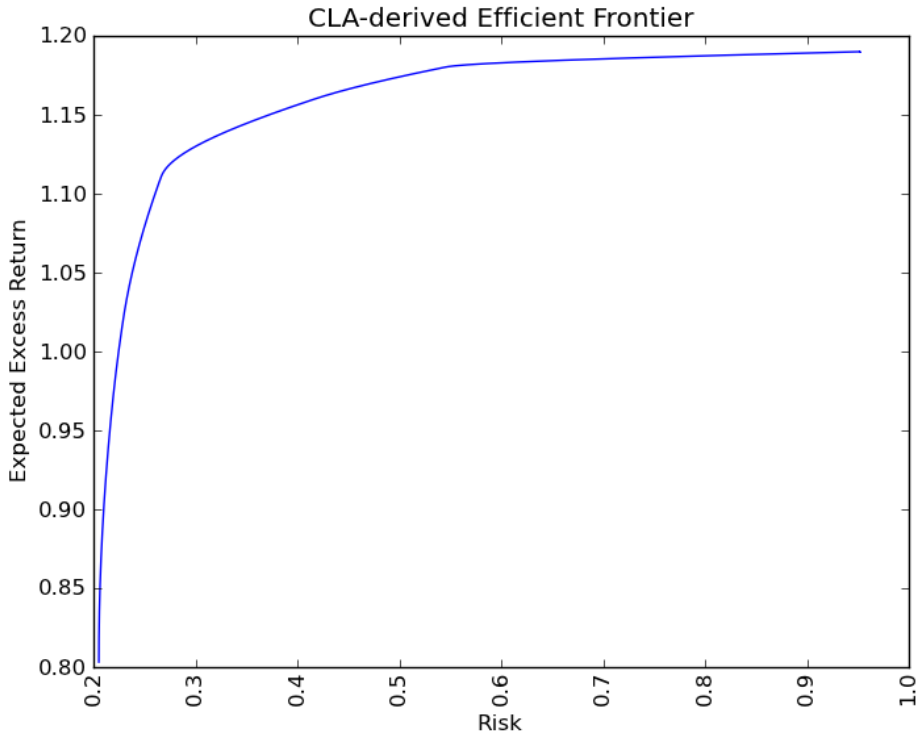
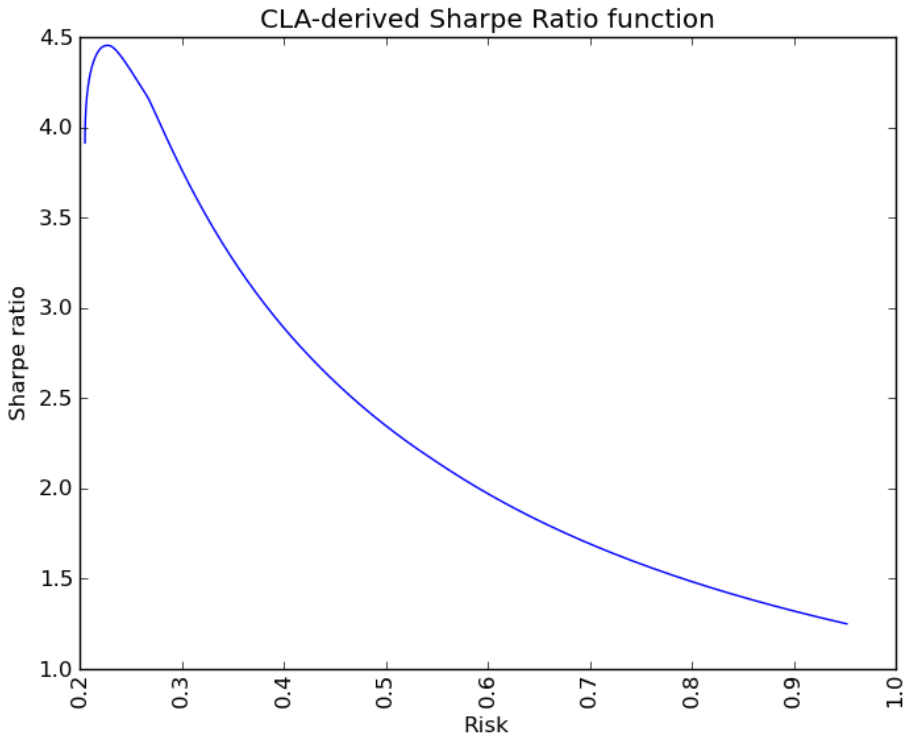


Figure 1 plots the efficient frontier, using the *plot2D* function provided in Snippet 17. Figure 2 plots the Sharpe ratio as a function of risk. Visually, the maximum Sharpe ratio portfolio is located around a risk of 0.2, and reaches a level close to 4.5. This can be easily verified by running the instruction *sr,w_sr=cla.getMaxSR()*, which returns a Sharpe ratio of 4.4535 for a portfolio with risk 0.2274. Similarly, running the instruction *mv,w_mv=cla.getMinVar()* reports a Minimum Variance portfolio with a risk of 0.2052.

VBA-Excel's *double* data type is based on a modified IEEE 754 specification, which offers a precision of 15 significant figures ([16]). Our Python results exactly match the outputs obtained when using the implementation of [6], to the highest accuracy offered by VBA-Excel.

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Figure 2. Sharpe ratio as a function of risk



6. Conclusions

Portfolio optimization is one of the problems most frequently encountered by financial practitioners. Following Markowitz [1], this operation consists in identifying the combination of assets that maximize the expected return subject to a certain risk budget. For a complete range of risk budgets, this gives rise to the concept of Efficient Frontier. This problem has an analytical solution in the absence of inequality constraints, such as lower and upper bounds for portfolio weights.

In order to cope with inequality constraints, [3,4] introduced the Critical Line Algorithm (CLA). To our knowledge, CLA is the only algorithm specifically designed for linear inequality-constrained quadratic portfolio optimization problems, which guarantees that the exact solution is found after a given number of iterations. Furthermore, CLA does not only compute a single portfolio, but it derives the entire efficient frontier. In the context of portfolio optimization problems, this approach is clearly more adequate than generic-purpose quadratic programming algorithms.

Given these two facts that portfolio optimization is a critical task, and that CLA provides an extremely efficient solution method, one would expect a myriad of software implementations to be available. And yet, to our surprise, we are only aware of one open-source, fully documented implementation of CLA, by [6]. Unfortunately, that implementation was done in VBA-Excel. This requires manual adjustments to a spreadsheet, which inevitably narrows its applicability to small-scale problems.

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Consequently, we suspect that most financial practitioners are resorting to generic-purpose quadratic programming software to optimize their portfolios, which often delivers suboptimal solutions.

The main goal of this paper is to fill this gap in the literature by providing a well-documented, step-by-step open-source implementation of CLA in a scientific language. The code is implemented as a Python class object, which allows it to be imported like any other Python module, and integrated seamlessly with pre-existing code. Following the explanation provided in this paper, our class can also be easily translated to other languages, such as C, C++ or Fortran-90. We have discussed the logic behind CLA following the algorithm's decision flow. In addition, we have developed several utilities that facilitate the answering of recurrent practical problems. Our results match the output of [6] at the highest accuracy offered by Excel.

Appendix

A.1. Python implementation of the Critical Line Algorithm

This Python class contains the entirety of the code discussed in Sections 3 and 4 of the paper. Section 4 presents an example of how to generate objects from this class. The following source code incorporates two additional functions:

- *purgeNumErr()*: It removes turning points which violate the inequality conditions, as a result of a near-singular covariance matrix Σ_F.
- *purgeExcess():* It removes turning points that violate the convex hull, as a result of unnecessary drops in λ.

The purpose of these two functions is to deal with potentially ill-conditioned matrices. Should the input matrices and vectors have good numerical properties, these numerical controls would be unnecessary. Since they are not strictly a part of CLA, we did not discuss them in the paper. This source code can be downloaded at http://www.davidhbailey.com/dhbpapers/ or http://www.quantresearch.info/Software.htm

```
#!/usr/bin/env python
# On 20130210, v0.2
# Critical Line Algorithm
# by MLdP <lopezdeprado@lbl.gov>
import numpy as np
#---------------------------------------------------------------
#---------------------------------------------------------------
class CLA:
    def __init__(self,mean,covar,lB,uB):
        # Initialize the class
```

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```
        if (mean==np.ones(mean.shape)*mean.mean()).all():mean[-1,0]+=1e-5
        self.mean=mean
        self.covar=covar
        self.lB=lB
        self.uB=uB
        self.w=[] # solution
        self.l=[] # lambdas
        self.g=[] # gammas
        self.f=[] # free weights
#------------------------------------------------------------------
    def solve(self):
        # Compute the turning points,free sets and weights
        f,w=self.initAlgo()
        self.w.append(np.copy(w)) # store solution
        self.l.append(None)
        self.g.append(None)
        self.f.append(f[:])
        while True:
            #1) case a): Bound one free weight
            l_in=None
            if len(f)>1:
                covarF,covarFB,meanF,wB=self.getMatrices(f)
                covarF_inv=np.linalg.inv(covarF)
                j=0
                for i in f:
                    l,bi=self.computeLambda(covarF_inv,covarFB,meanF,wB,j,[self.lB[i],self.uB[i]])
                    if l>l_in:l_in,i_in,bi_in=l,i,bi
                    j+=1
            #2) case b): Free one bounded weight
            l_out=None
            if len(f)<self.mean.shape[0]:
                b=self.getB(f)
                for i in b:
                    covarF,covarFB,meanF,wB=self.getMatrices(f+[i])
                    covarF_inv=np.linalg.inv(covarF)
                    l,bi=self.computeLambda(covarF_inv,covarFB,meanF,wB,meanF.shape[0]-1, \
                        self.w[-1][i])
                    if (self.l[-1]==None or l<self.l[-1]) and l>l_out:l_out,i_out=l,i
            if (l_in==None or l_in<0) and (l_out==None or l_out<0):
                #3) compute minimum variance solution
                self.l.append(0)
                covarF,covarFB,meanF,wB=self.getMatrices(f)
```

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```
            covarF_inv=np.linalg.inv(covarF)
            meanF=np.zeros(meanF.shape)
         else:
            #4) decide lambda
            if l_in>l_out:
               self.l.append(l_in)
               f.remove(i_in)
               w[i_in]=bi_in # set value at the correct boundary
            else:
               self.l.append(l_out)
               f.append(i_out)
            covarF,covarFB,meanF,wB=self.getMatrices(f)
            covarF_inv=np.linalg.inv(covarF)
         #5) compute solution vector
         wF,g=self.computeW(covarF_inv,covarFB,meanF,wB)
         for i in range(len(f)):w[f[i]]=wF[i]
         self.w.append(np.copy(w)) # store solution
         self.g.append(g)
         self.f.append(f[:])
         if self.l[-1]==0:break
      #6) Purge turning points
      self.purgeNumErr(10e-10)
      self.purgeExcess()
#---------------------------------------------------------------
   def initAlgo(self):
      # Initialize the algo
      #1) Form structured array
      a=np.zeros((self.mean.shape[0]),dtype=[('id',int),('mu',float)])
      b=[self.mean[i][0] for i in range(self.mean.shape[0])] # dump array into list
      a[:]=zip(range(self.mean.shape[0]),b) # fill structured array
      #2) Sort structured array
      b=np.sort(a,order='mu')
      #3) First free weight
      i,w=b.shape[0],np.copy(self.lB)
      while sum(w)<1:
         i-=1
         w[b[i][0]]=self.uB[b[i][0]]
      w[b[i][0]]+=1-sum(w)
      return [b[i][0]],w
#---------------------------------------------------------------
   def computeBi(self,c,bi):
      if c>0:
```

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```
            bi=bi[1][0]
        if c<0:
            bi=bi[0][0]
        return bi
#---------------------------------------------------------------
    def computeW(self,covarF_inv,covarFB,meanF,wB):
        #1) compute gamma
        onesF=np.ones(meanF.shape)
        g1=np.dot(np.dot(onesF.T,covarF_inv),meanF)
        g2=np.dot(np.dot(onesF.T,covarF_inv),onesF)
        if wB==None:
            g,w1=float(-self.l[-1]*g1/g2+1/g2),0
        else:
            onesB=np.ones(wB.shape)
            g3=np.dot(onesB.T,wB)
            g4=np.dot(covarF_inv,covarFB)
            w1=np.dot(g4,wB)
            g4=np.dot(onesF.T,w1)
            g=float(-self.l[-1]*g1/g2+(1-g3+g4)/g2)
        #2) compute weights
        w2=np.dot(covarF_inv,onesF)
        w3=np.dot(covarF_inv,meanF)
        return -w1+g*w2+self.l[-1]*w3,g
#---------------------------------------------------------------
    def computeLambda(self,covarF_inv,covarFB,meanF,wB,i,bi):
        #1) C
        onesF=np.ones(meanF.shape)
        c1=np.dot(np.dot(onesF.T,covarF_inv),onesF)
        c2=np.dot(covarF_inv,meanF)
        c3=np.dot(np.dot(onesF.T,covarF_inv),meanF)
        c4=np.dot(covarF_inv,onesF)
        c=-c1*c2[i]+c3*c4[i]
        if c==0:return None,None
        #2) bi
        if type(bi)==list:bi=self.computeBi(c,bi)
        #3) Lambda
        if wB==None:
            # All free assets
            return float((c4[i]-c1*bi)/c),bi
        else:
            onesB=np.ones(wB.shape)
            l1=np.dot(onesB.T,wB)
```

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```
        l2=np.dot(covarF_inv,covarFB)
        l3=np.dot(l2,wB)
        l2=np.dot(onesF.T,l3)
        return float(((1-l1+l2)*c4[i]-c1*(bi+l3[i]))/c),bi
#---------------------------------------------------------------
    def getMatrices(self,f):
        # Slice covarF,covarFB,covarB,meanF,meanB,wF,wB
        covarF=self.reduceMatrix(self.covar,f,f)
        meanF=self.reduceMatrix(self.mean,f,[0])
        b=self.getB(f)
        covarFB=self.reduceMatrix(self.covar,f,b)
        wB=self.reduceMatrix(self.w[-1],b,[0])
        return covarF,covarFB,meanF,wB
#---------------------------------------------------------------
    def getB(self,f):
        return self.diffLists(range(self.mean.shape[0]),f)
#---------------------------------------------------------------
    def diffLists(self,list1,list2):
        return list(set(list1)-set(list2))
#---------------------------------------------------------------
    def reduceMatrix(self,matrix,listX,listY):
        # Reduce a matrix to the provided list of rows and columns
        if len(listX)==0 or len(listY)==0:return
        matrix_=matrix[:,listY[0]:listY[0]+1]
        for i in listY[1:]:
            a=matrix[:,i:i+1]
            matrix_=np.append(matrix_,a,1)
        matrix__=matrix_[listX[0]:listX[0]+1,:]
        for i in listX[1:]:
            a=matrix_[i:i+1,:]
            matrix__=np.append(matrix__,a,0)
        return matrix__
#---------------------------------------------------------------
    def purgeNumErr(self,tol):
        # Purge violations of inequality constraints (associated with ill-conditioned covar matrix)
        i=0
        while True:
            if i==len(self.w):break
            w=self.w[i]
            for j in range(w.shape[0]):
                if w[j]-self.lB[j]<-tol or w[j]-self.uB[j]>tol:
                    del self.w[i]
```

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```
                del self.l[i]
                del self.g[i]
                del self.f[i]
                break
            i+=1
#---------------------------------------------------------------
    def purgeExcess(self):
        # Remove violations of the convex hull
        i,repeat=0,False
        while True:
            if repeat==False:i+=1
            if i==len(self.w)-1:break
            w=self.w[i]
            mu=np.dot(w.T,self.mean)[0,0]
            j,repeat=i+1,False
            while True:
                if j==len(self.w):break
                w=self.w[j]
                mu_=np.dot(w.T,self.mean)[0,0]
                if mu<mu_:
                    del self.w[i]
                    del self.l[i]
                    del self.g[i]
                    del self.f[i]
                    repeat=True
                    break
                else:
                    j+=1
#---------------------------------------------------------------
    def getMinVar(self):
        # Get the minimum variance solution
        var=[]
        for w in self.w:
            a=np.dot(np.dot(w.T,self.covar),w)
            var.append(a)
        return min(var)**.5,self.w[var.index(min(var))]
#---------------------------------------------------------------
    def getMaxSR(self):
        # Get the max Sharpe ratio portfolio
        #1) Compute the local max SR portfolio between any two neighbor turning points
        w_sr,sr=[],[]
        for i in range(len(self.w)-1):
```

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```
            w0=np.copy(self.w[i])
            w1=np.copy(self.w[i+1])
            kargs={'minimum':False,'args':(w0,w1)}
            a,b=self.goldenSection(self.evalSR,0,1,**kargs)
            w_sr.append(a*w0+(1-a)*w1)
            sr.append(b)
        return max(sr),w_sr[sr.index(max(sr))]
#---------------------------------------------------------------
    def evalSR(self,a,w0,w1):
        # Evaluate SR of the portfolio within the convex combination
        w=a*w0+(1-a)*w1
        b=np.dot(w.T,self.mean)[0,0]
        c=np.dot(np.dot(w.T,self.covar),w)[0,0]**.5
        return b/c
#---------------------------------------------------------------
    def goldenSection(self,obj,a,b,**kargs):
        # Golden section method. Maximum if kargs['minimum']==False is passed
        from math import log,ceil
        tol,sign,args=1.0e-9,1,None
        if 'minimum' in kargs and kargs['minimum']==False:sign=-1
        if 'args' in kargs:args=kargs['args']
        numIter=int(ceil(-2.078087*log(tol/abs(b-a))))
        r=0.618033989
        c=1.0-r
        # Initialize
        x1=r*a+c*b;x2=c*a+r*b
        f1=sign*obj(x1,*args);f2=sign*obj(x2,*args)
        # Loop
        for i in range(numIter):
            if f1>f2:
                a=x1
                x1=x2;f1=f2
                x2=c*a+r*b;f2=sign*obj(x2,*args)
            else:
                b=x2
                x2=x1;f2=f1
                x1=r*a+c*b;f1=sign*obj(x1,*args)
        if f1<f2:return x1,sign*f1
        else:return x2,sign*f2
#---------------------------------------------------------------
    def efFrontier(self,points):
        # Get the efficient frontier
```

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```
    mu,sigma,weights=[],[],[]
    a=np.linspace(0,1,points/len(self.w))[:-1] # remove the 1, to avoid duplications
    b=range(len(self.w)-1)
    for i in b:
        w0,w1=self.w[i],self.w[i+1]
        if i==b[-1]:a=np.linspace(0,1,points/len(self.w)) # include the 1 in the last iteration
        for j in a:
            w=w1*j+(1-j)*w0
            weights.append(np.copy(w))
            mu.append(np.dot(w.T,self.mean)[0,0])
            sigma.append(np.dot(np.dot(w.T,self.covar),w)[0,0]**.5)
    return mu,sigma,weights
#------------------------------------------------------------
#------------------------------------------------------------
```

Snippet 18 – The CLA Python class

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Lesson Notes

MODULE 3 | LESSON 3

SOME ABUSES OF THE BLACK-LITTERMAN MODEL

Reading Time	60 minutes
Prior Knowledge	Black-Litterman model. Some understanding of Bayesian statistics is useful as well.
Keywords	Black-Litterman without equilibrium model, Black-Litterman with data based prior, quadratic utility and loss of utility.

In the previous lesson we studied the critical line algorithm which is an excellent mathematical tool to compute optimal portfolios, both in the traditional setting and in the Black-Litterman approach. In this lesson we will explore some situations where, if the set-up of the Black-Litterman model is not rigorous, one could end up with suboptimal results.

1. Generalities

By now, students are acquainted with the Black-Litterman model and are aware of at least a few techniques that can be used to implement the model in practice. In reality, those techniques apply to Black-Litterman and the classic mean-variance optimization problem alike.

Practitioners are often driven by a certain urgency to avoid complications or look for short-cuts and for such reasons have developed approaches that are not necessarily rigorous.

While these approaches to the Black-Litterman model are good to cut corners, so to speak and tend to work overall, they lead to some loss of precision. This particular topic is the subject of this lesson.

This lesson in based on the mandatory reading, "Uses and misuses of the Black-Litterman model in portfolio construction".The paper describes 3 instances were the implementation of the Black-Litterman model lead to suboptimal solutions.

In the notes for this lesson we will cover some mathematical aspects which are basically assumed in the paper hoping that every student can get as much as possible from the reading itself.

2. Not Using an Equilibrium Model

In Lesson 1 of this Module we covered the Black-Litterman model as intended by its authors: i.e., investors' subjective views are combined with an equilibrium model. Essentially, the portfolio weights are anchored around the implied market capitalization weights and are allowed to move away from those in amounts that depend on the investors uncertainty about their own views. The less convinced an investors or portfolio manager is about his or her views (i.e. higher variance around their mid-point view) the more they will tend to stick close to what we can call the market's view or the equilibrium view.

There is an approach to Black-Litterman in which we dispense with the equilibrium model. To this end, the authors mention a model proposed by Jones, Lim, and Zangari in 2007 and that they call the **JLZ model** and which is a factor-based model. This model requires
the following steps:

Step 1. Identify the factor model by identifying K factors. Then for a portfolio consisting of N assets, we let B be an $N \times K$ matrix that represents factor exposure as we saw in Lesson 4 of Module 2. Hence the portfolio exposure to these factors overall is given by $B^T w$ with w an $N \times 1$ vector of weights.

Step 2. A portfolio manager specifies factor views using a$K \times 1$ vector λ such that $B^T w = \lambda$.

Step 3. The optimal weight that is optimal given the factor view vector λ can now be determined by solving the following problem:

$$\min_{w} w^T \Sigma w$$
$$\text{subject to : } B^T w = \lambda.$$

Now,Σ is the unknown variance covariance matrix of the returns of the assets which is unknown. However this is not a problem as we can estimate it as usual, i.e. we compute $\widehat{\Sigma}$. Then the optimization problem can be run as

$$\min_{w} w^T \widehat{\Sigma} w$$
$$\text{subject to : } B^T w = \lambda.$$

We should also notice that there is no reference to the expected returns μ. If we solve the optimization program, i.e. we form the Lagrangian function

$$L = \frac{1}{2} w \widehat{\Sigma} w + \delta^T (\lambda - B^T w)$$

then the first order conditions, obtained by setting the first derivatives with respect to w and δ equal to zero yield

$$\widehat{\Sigma} w - B\delta = 0 \quad \text{and} \quad B^T w = \lambda.$$

(Pay attention that in our formulation,δ is not a scalar but a $K \times 1$ vector since the constraints are in matrix form and contain actually K constraints).

Now,

$$w = \widehat{\Sigma}^{-1} B \delta$$

$$\delta = (B^T \widehat{\Sigma}^{-1} B)^{-1} \lambda$$

and combining these two sets of equations we find

$$w_{opt} = \widehat{\Sigma}^{-1} B (B^T \widehat{\Sigma}^{-1} B)^{-1} \lambda$$

which is formula (2) in the required reading.

Step 4. At this point we need to compute what the paper calls the Black-Litterman α by solving the problem

$$w_{opt} = \mathbf{argmax}\ w^T \alpha - \frac{1}{2} w^T \widehat{\Sigma} w.$$

If we compute the partial derivatives of the objective function on the right-had side of the equal sign with respect to w and set it equal to zero, then

$$\alpha - \widehat{\Sigma} w = 0$$

yields

$$w = \widehat{\Sigma}^{-1} \alpha$$

and now we use w_{opt} in place of w and we find

$$\alpha = \widehat{\Sigma} w_{opt}$$

which is (4) in the assigned reading. We call this α that was just found α_{BL}.

Step 5. This step is similar to what we have already seen several times already, i.e. we solve the following quadratic problem:

$$\begin{aligned} &\max_w w_{BL}^{\alpha} - \frac{1}{2} w^T \widehat{\Sigma} w \\ &\textbf{subject to}: w^T \iota = 1 \\ &\qquad\qquad l_i \leq w_i \leq u_i, \ \ i = 1, 2, \ldots, N. \end{aligned}$$

which we can solve using gradient based algorithms of the critical line algorithm learned in the lesson before this one and where l_i and u_i represent lower and upper bounds for the weights of asset i for $i = 1, 2, \ldots, N$.

The paper explains that **we are not using an equilibrium model because the data are used to compute $\widehat{\sum}$ but do not use the information about the expected returns μ.** Hence, in the parlance of the Black-Litterman model, subjective views are combined with the model, but only partially as we are ignoring μ.

Then, the authors proceed to analyze what is the impact of such approach when compared to the true Black-Litterman methodology. To do so, they introduce a quadratic utility function

$$U(\mu, \sigma) = \mu - \frac{\delta}{2} \sigma^2$$

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similar to some of those we saw in Module 1 and in Module 2 with μ the mean return of the portfolio and σ the volatility of the portfolio. Then the value of the utility is computed both using the Black-Litterman model and the JLZ model to assess how much utility was lost in the process.

To achieve this, investors' views must be formalized in a probability model and the authors consider:

$$r_{t+1} = B_t f + e_{t+1}, \quad e_{t+1} \sim N(0, (1/h)V_e).$$

In addition, we have also

$$f|h \sim N(m_f, (1/h)V_f)$$

$$h \sim \text{Gamma}(p, q).$$

The choice of using a Gamma distribution is due to the fact that it is a conjugate prior and thus makes the computations much simpler since the posterior distribution would be of the same type as the prior distribution, i.e., multivariate normal.

The paper provides equation (9) as the formula for λ and it also contains a generous note/comment (footnote 13). Formula (9) is a consequence of the use of the hierarchical model involving normal random variables that was selected. In fact, if we start from

$$r_{t+1} = B_t f + e_{t+1}, \quad e_{t+1} \sim N(0, (1/h)V_e), \quad f|h \sim N(m_f, (1/h)V_f)$$

one can prove that the marginal distribution of r_{t+1} is normal.

The easiest way do it is arguably via characteristic functions or moment generating functions paired with the law to total expectation. In fact, in general, if we are given that

$$X|Y = y \sim N(\alpha y, \sigma^2) \quad \text{and} \quad Y \sim N(\mu, \delta^2),$$

then the moment generating function for X can be computed using the definition and the law of total probability, i.e.:

$$\phi_X(t) = E[e^{tX}] = E_Y[E_{X|Y}[e^{tX}|Y]] = E_Y[e^{t\alpha Y + \frac{t^2\sigma^2}{2}}]$$

and now we can compute this second expectation exploiting the fact $Y \sim N(\mu, \delta^2)$. This is not a difficult calculation, albeit a bit tedious since one has to complete the square in the exponent for the exponential, etc. In any case, the calculation yields

$$\phi_X(t) = e^{\alpha\mu t + (\alpha^2\delta^2 + \sigma^2)\frac{t^2}{2}},$$

which is recognizable as the moment generating function of a normal random variable with a certain mean and a certain variance.

The unconditional mean and variance for our variables, which is what we need, are not terribly complicated to compute. In fact, using again the law of total expectation, we have

$$E[r_{T+1}] = E[E[r_{T+1}|f]] = E[B_T f] = B_T E[f] = B_T m_f.$$

For the unconditional variance it is a bit more work but one uses the well-known formula:

$$\mathbf{Var}(X) = E[\mathbf{Var}(X|Y)] + \mathbf{Var}(E]X|Y])$$

which in our case yields

$$\mathbf{Var}(r_{T+1}) = E[\mathbf{Var}(r_{T+1} \mid f)] + \mathbf{Var}(E]r_{T+1}|f])$$

and

$$E[\mathbf{Var}(r_{T+1} \mid f)] = E[B^T(1/h)V_f B_T^T + (1/h)V_e] = B^T(1/h)V_f B_T^T = E[(1/h)V_e] \equiv \Omega.$$

It is not difficult to show that

$$\mathbf{Var}(E]r_{T+1}|f]) = 0.$$

Hence we have shown that r_{t+1} has the following unconditional distribution.

$$r_{T+1} \sim N(B_T f, \Omega).$$

If we now run the optimization program

$$\max_w w^T(B_T f) - \frac{1}{2} w^T \Omega w$$

we find

$$w_{opt}^{\times} = \Omega^{-1} B_T m_f$$

and finally solving for $\lambda = B_T^T w_{opt}^{\times}$ we get

$$\lambda = B_T^T w_{opt}^{\times} = B_T^T \Omega^{-1} B_T m_f$$

which is indeed (9).

Now, one has everything to start predicting. The process is well described in the reading at page 156 (a 7 step process). The authors need to derive the predictive expectation and the predictive variance-covariance matrix (formulae (14), (15), and (16)). These are quite complex and presented without proof. In fairness, the proof is not particularly difficult in terms of mathematics; however it is very long, like 4 pages of algebraic calculations (it is not an unusual occurrence when dealing with Bayesian methods). The proof is contained in the original paper submitted for publication and can be obtained from the authors if students are interested (a reference will be provided at the end of the lesson). Equipped with the predictive mean and variance-covariance, it is shown that the difference in utility between two investors having the same views, but where one uses the information efficiently (RTL) and the other uses the JLZ model, is

$$U_{RTL} - U_{JLZ} = \frac{1}{2}(m_f - \tilde{E}[f])^T B_{T+1} \tilde{V}(r_{T+1})^{-1} B_{T+1}(m_f - \tilde{E}[f])$$

and it is clearly a positive number since the loss is represented by a quadratic form with positive coefficient. The key element is that the loss depends on the magnitude of $m_f - \tilde{E}[f]$, the difference between the prior mean and the the posterior mean of the factor premium.

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3. Data Based Prior

In this case, what the authors refer to the model as the **FFK model**, which utilizes a factor model to represent the portfolio manager's views. That is, we start with a factor model

$$r_{t+1} = B_t f + e_{t+1}, \qquad \epsilon_{t+1} \sim N(0, \Sigma_e)$$

where in this case the unknown are the value of the factor and the variance covariance matrix Σ_e. $Sigma_e$ represents the uncertainty about the view for each factor and, as described in Lesson 1 of this module discussing the Black-Litterman model the views are independent and therefore Σ_e is a diagonal square matrix.

These quantities can be estimated using OLS or, more likely, GLS/WLS techniques as explained in Lesson 4 of Module 2. In this way we will have two estimates $\widehat{f}$ and $\widehat{\Sigma}_e$. Now, the portfolio manager can express his or her prior for the mean return $E[r_{T+1}]$ in the form

$$g \sim N(B_T \widehat{f}, \widehat{\Sigma}_e)$$

Note that this model is *just a belief that a factor model is the underlying data generating process*. However, in this approach one also assumes that π is the estimated equilibrium return obtained via the CAPM and that

$$\pi = g + \epsilon, \;\; \epsilon \sim N(0, \tau\Sigma)$$

where in this case Σ is the variance-covariance matrix of the future returns r_{T+1}

This can be estimated and we can write $\pi = \widehat{\Sigma} w_M$. Then from the definition of π above, we can estimate g and $\mathbf{Var}(g)$; specifically:

$$\hat{g} = \pi = \widehat{\Sigma} w_M, \;\; \widehat{\mathrm{Var}}(\hat{g}) = \tau\widehat{\Sigma}$$

From this, deriving equation (31) in the paper is not exactly complicated, but calculations are a bit tricky.

Using the Bayesian setting for posterior distributions, we have that

$$f_{g|\pi} = \frac{k}{f_\pi} \cdot \exp\left\{ -\frac{1}{2}(\pi - g)^T (\tau\widehat{\Sigma})^{-1} (\pi - g) - \frac{1}{2}(g - B_T\widehat{f})^T (\widehat{\Sigma}_e)^{-1} (g - B_T\widehat{f}) \right\}$$

where k is just a constant of proportionality and f_π is the marginal density for π.

Up to this point the step was relatively straightforward. However, to simplify this expression, one has to introduce some algebraic tricks. Here we follow the process delineated in Satchell and Scowcroft (reference provided at the end of the Lesson). Then, if we let H be the exponent of the exponential in the formula above, i.e.:

$$H \equiv -\frac{1}{2}(\pi - g)^T (\tau\widehat{\Sigma})^{-1} (\pi - g) - \frac{1}{2}(g - B_T\widehat{f})^T (\widehat{\Sigma}_e)^{-1} (g - B_T\widehat{f}),$$

with some simple algebra, we find that

$$H = g^T(\tau\widehat{\Sigma})^{-1}g - 2\pi(\tau\widehat{\Sigma})^{-1}g + \pi^T(\tau\widehat{\Sigma})^{-1}\pi = g^T\widehat{\Sigma}_e^{-1}g + 2(B_T\widehat{f})^T\widehat{\Sigma}_e^{-1}(B_T\widehat{f}).$$

The next step is the somewhat tricky one (but it amounts to just algebra nevertheless) and it is the step described in Satchell and Scowcroft. This step consists in rewriting H as

$$H = g^T M^T M^{-1} M g - 2NM^{-1}Mg + A$$

where

$$M = (\tau\widehat{\Sigma})^{-1} + \widehat{\Sigma}_e^{-1},$$

$$N = (\tau\widehat{\Sigma})^{-1}\pi + \widehat{\Sigma}_e^{-1}B_T\widehat{f},$$

$$A = (BT\widehat{F})^T\widehat{\Sigma}_e^{-1}(B_T\widehat{F}) + \pi^T(\tau\widehat{\Sigma})^{-1}\pi,$$

and we should notice that M is symmetrical as it involves variance covariance matrices. Finally, the exponent can be further simplified to look like

$$H = (g - M^{-1}N)^T M (g - M^{-1}C) + A - N^T M^{-1} N.$$

Now, the terms $A - N^T M^{-1} N$ does not contain g and can be absorbed into the constant of integration. Thus, we have found that

$$f_{g|\pi} = k \exp\{(g - M^{-1}N)^T M (g - M^{-1}C)\}.$$

This is very useful for we can recognize that the posterior is a normal distribution whose mean, $\widehat{E}[g],$ is given by

$$M^{-1}N = [(\tau\widehat{\Sigma})^{-1} + \widehat{\Sigma}_e^{-1}]^{-1}[(\tau\widehat{\Sigma})^{-1}\pi + \widehat{\Sigma}_e^{-1}B_T\widehat{f}]$$

and the variance is found as

$$\widehat{\mathrm{Var}}[g] = [(\tau\widehat{\Sigma})^{-1} + \widehat{\Sigma}_e^{-1}]^{-1},$$

which is (31) as we would have liked to show.

At this point, the relevance of equation (32) is obvious. Now we have a new contradiction: a model where it is assumed that the data is generated by a factor model and a likelihood function that uses a belief according to which the CAPM is the data generating process.

Using the same setup for the previous case, the authors show that the loss in utility between using the true efficient model and the FFK is given by formula (440, p. 162)

$$U_{RTL} - U_{FFK} = -\frac{1}{2}(\widehat{E}[g] - \tilde{E}[r_{T+1}])^T(\widetilde{\mathrm{Var}}[r_{t+1}])^{-1}(\widehat{E}[g] - \tilde{E}[r_{T+1}]).$$

Futhermore, the authors point out that the loss derived from using the model FFK is somewhat smaller than that of using the JLZ model.

4. Reverse Optimization Technique

This part is more straightforward and does not require any particular comment.

5. Conclusion

While the approaches described are simpler and of more immediate application, they are either ignoring part of the information available or using "pieces" from contradictory models (CAPM e factor models). This leads to about a 1% reduction in the estimated mean and, while 1% may not seem like much, we are aware from previous lessons that the estimates for the mean have a big impact on the determination of optimal portfolios.

The third part of the paper explains that in using reverse optimization, practitioners rely on the estimated variance covariance matrix which is estimated with error and such error can often be far from negligible. This should cause practitioners to be very careful and aware of potential misuses of the Black-Litterman model.

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Probabilistic approach for optimal portfolio selection using a hybrid Monte Carlo simulation and Markowitz model



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robustness

Abstract In this paper, a probabilistic form of the portfolio selection problem is established in which the uncertainty of risky assets is considered through a probabilistic optimization problem. To this end, by taking seven portfolios of Shanghai stock as a case study, the mean and standard deviation of daily return values are calculated based on five years of real data. The optimal values corresponding to each random case were then stored as a comprehensive database of system responses. Then, by sorting the resulting optimal values from best to worst, the exceedance probabilities of return and risk rankings were calculated for each portfolio and presented in the form of probabilistic pie charts. The results showed that the portfolio with the highest deterministic rate of return has the highest probability of getting the best return ranking as well. However, since the probability of risk in all cases was calculated, the probability of each portfolio to place in the lower rankings (i.e., ranking 2–7) could also be discussed. Additionally, to check the convergence of the model, the probability values calculated by the Monte Carlo method against the sample size were plotted to ensure the accuracy of the final answer. Eventually, by generating Gaussian random noise and importing it into the model input, probability changes were calculated to assess the robustness of the proposed algorithm.

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1. Introduction

The capital market is one of the market places where the holder of financial resources, on the one hand, and organizations, on the other hand, are actively involved in financial activities [1,2]. The task of the capital market is to provide financial resources to those in need, and the share is known as the most important commodity traded in the capital market [3,4]. Most people make some investment decisions in their lives [5], and capital market is one of the most important places to help optimize asset allocation [6,7]. Individuals in their analyses eventually hope to get an appropriate return with low investment risk [8–11], but one of the essential features of the capital market is uncertainty [12,13]. The price fluctuations make the investors worried about the future of their capitals [14,15]. For investors, achieving the high return is a positive parameter, while the volatility is a negative parameter

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[16,17]. The question here is how investors should allocate their existing capitals to different stocks, that is, what fraction of portfolio should be allocated to existing stocks [18,19]. Therefore, the portfolio optimization is of considerable significance in the finance and investment projects [20–22]. The study and analysis of the basic principles of portfolio management can lead to a better financial plan and increase the wealth of investors [23].

The theory of portfolio selection was first put forward by Harry Markowitz, an American economist and Nobel Prize winner [24]. In his paper in 1952, he postulated that investors behave reasonably, always trying to maximize the expected return and reduce the overall risk of the portfolio, taking into account the possible correlation between assets. The theory he proposed in his research later formed the basis for modern portfolio theory (MPT). The Markowitz's concept has been developed by different researchers. In 1963, William Sharpe [25] developed an effective calculation method to analyze available capital and build a portfolio with expected attributes. Later, Robert Merton [26] published an article in which he showed how to analytically calculate the "efficient Pareto frontier". Pogue [27] later extended the model to include "transaction costs" and short selling. In 1992, Black and Litterman [28] proposed a model based on the Markowitz model where no asset with specific expected returns is required to be considered during the analysis. In 1991, Rom and Ferguson [29] extended the model by removing the limitations of the original model, considering the distribution of returns and taking the variance of returns as a measure of investment risk. The work done after this research is often referred to as the "postmodern portfolio theory". Much new research is still being done on the basis of Markowitz's theory, which utilizes the latest modeling and problem-solving approaches. Woodside-Oriakhi et al. [30] introduced a comprehensive model of portfolio optimization and implemented the model with quadratic programming of complex integers. Mittal and Mehlawat [31] proposed a model involving transaction costs and also solved the optimization problem by adopting the genetic algorithm. A more comprehensive review on deterministic portfolio optimization can be found in [32].

To the best of the authors' knowledge, most existing studies on portfolio optimization have used the deterministic methods for the problem formulation [33]. However, the unknown nature of the uncertainty and market volatility involved in the portfolio project is so large that it could not be ignored [34]. In fact, the deterministic modeling is equivalent to assuming 100% certainty about the status of stocks, which is unacceptable in the reality [35]. Hence, an alternative approach is needed to consider the uncertainty and probabilistically address the problem. In this study, an optimal probability algorithm is used for this purpose. In this way, a deterministic optimization algorithm is first implemented for the problem. Then, the input parameters (including the daily data of seven different portfolios of the Shanghai stock market) are modeled as random variables with the known mean and standard deviation. The resulting random variables are then incorporated into the optimization algorithm to establish a systematic approach to the formulation of the probabilistic optimization. The Monte Carlo sampling method is then used to solve the established problem and to present a probabilistic solution for the individual return and risk values [36,37]. Fig. 1 provides an overview of this process. The convergence of the Monte Carlo method is then evaluated to ensure that the number of samples is sufficient and that the algorithm converges to the solution with a reasonable accuracy. Finally, the input data are polluted with Gaussian random noise to assess the robustness of the proposed algorithm to random fluctuations.

2. Review of Shanghai stock exchange (SSE)

2.1. History of SSE

As mentioned in the introduction, the algorithm proposed in this paper is applied to the SSE. Shanghai is the first city in China that welcomed the stock exchange. SSE opened in the 1860s. In 1891, the Shanghai Share Brokers Association was established in Shanghai. Later, in the 1920s, with the establishment of the Shanghai Security Goods Exchange and the Shanghai Securities Exchange in China, Shanghai became the trading center in the Far East, where the Chinese and foreign investors could trade the stocks.

Since 1980, the China's securities market has grown with the reforms and opening in the country and the development of a socialist market economy. In 1981, the treasury bond issue was resumed. In 1984, the stocks and enterprise bonds were issued in Shanghai and other cities. On November 26, 1990, the SSE was established, and on December 19 of that year, the trading officially started.

2.2. Index introduction

SSE is the largest stock exchange in the mainland China run by the China Securities Regulatory Commission as a non-profit organization. It trades stocks, funds, and bonds. To be listed on the Shanghai exchange, a company must have earned profit for at least three years. Most of the exchange in terms of the total market capitalization is comprised of formally state-run companies, including major banks and insurance companies. The exchange uses the A-shares and B-shares. The A-shares are the shares of companies based on mainland China and are quoted in Yuan, but previously available for the mainland citizens. However, recent changes and regulations allow the foreign investors to buy the shares through the tightly regulated system for the qualified foreign investors. The B-shares are quoted in foreign currencies such as the US Dollar and open to domestic and foreign investors.

In this study, the data of seven different portfolio indices of the Shanghai stock market are used as listed in Table 1. A brief introduction of the portfolio indices is provided in the Appendix A.

3. Modeling and results

As noted in the introduction, this study attempts to express the optimal portfolio selection in the form of a probabilistic problem, because the deterministic models are generally limited to a particular scenario, even for real-life cases that are associated with a high degree of uncertainty [38–40]. However, one must first start with a deterministic model to calculate the optimal solution of the problem for a particular input vector. Then, the deterministic model can be transformed into a probabilistic model by defining the input parameters in the form of random

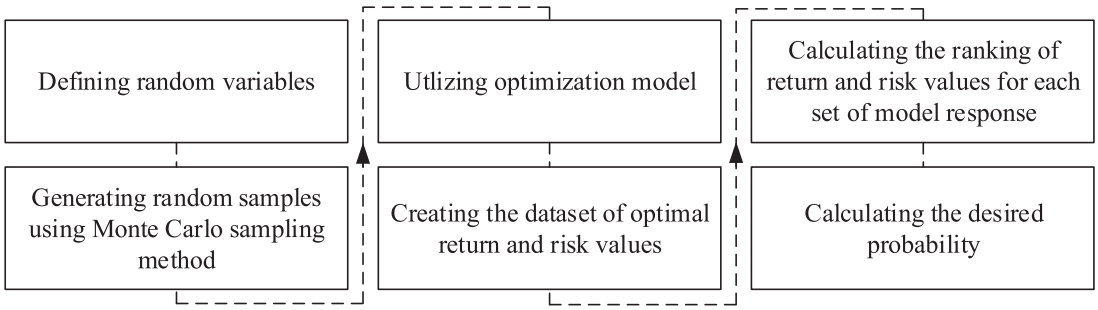


Fig. 1 The process to establish probabilistic model.

Table 1 Seven portfolios of Shanghai stock market used in this research.

No.	Symbol	Associated stock index
1	Portfolio A	SSE Composite Index 000001
2	Portfolio B	SSE 180 index 000010
3	Portfolio C	SSE 50 Index 000016
4	Portfolio D	SSE 380 Index 000009
5	Portfolio E	SSE Fund Index 000011
6	Portfolio F	T-Bond index
7	Portfolio G	SHSE-SZSE300 index

variables [41]. Next, probabilistic analysis methods can be used to solve the recorded problem. Finally, post-processing is required to extract and report the desired response from the randomly generated database. A schematic view of the proposed algorithm is shown in Fig. 2.

3.1. Data collection and deterministic portfolio optimization

For this study, the data on the previously-described seven major SSE indices were collected on a daily basis for five years (from 2014 to 2018). First, given the daily close prices of each index, the daily returns were obtained by the following formula:

$$r_i(t) = \frac{p_i(t+1) - p_i(t)}{p_i(t)} = \frac{p_i(t+1)}{p_i(t)} - 1, \tag{1}$$

where $r_i(t)$ represents the return at time t, $p_i(t)$ denotes the closing price at time t, and $p_i(t+1)$ represents the closing price at time $t+1$. Then, the summation of the daily return values was calculated for each month and considered as monthly return.

As mentioned earlier, the optimization models can provide stocks arrangement with the lowest risk over expected return (or correspondingly highest expected return over risk). For this means, the Markowitz mean–variance model is adopted in this paper as follows:

$$\begin{aligned} Max\{\mu_p\} &= \sum_i w_i \mu_i \\ Min\{\sigma_p^2\} &= \sum_i \sum_j w_i w_j \sigma_{ij} \\ \text{Subject to} &: \begin{cases} \sum_i w_i = 1 \\ w_i \geqslant 0 \end{cases}, \end{aligned} \tag{2}$$

where μ_p represents the portfolio return, σ_p^2 denotes the portfolio variance (risk), σ_{ij} represents the covariance between the two stocks i and j, and w_i is regarded as the share invested in stock i. As shown in Eq. (2), the problem is formulated as a multi-objective optimization. The main purpose of this study is to maximize the return and minimize the risk. Therefore, two objective functions were required to define the problem. Of course, we could formulate the problem by only one of these two objective functions, i.e., either maximize return or minimize risk, which would have made the problem more straightforward, but it would not provide a comprehensive answer. For this reason, it was decided to add complexity to the problem, but cover both objectives at the same time. Additionally, it should be noted that different methodologies, including intelligent algorithms, can be used to solve the established portfolio problem. However, intelligent methods are usually implemented through a population-based approach. That is, the initial population should be generated first, and then updated in each increment to gradually approach the optimal solution. The combination of such methods and Monte Carlo analysis (which itself is based on random sample generation) will impose a high computational cost, and consequently, challenge the ease of analysis. Therefore, the authors decided to use the Markowitz method for deterministic portfolio selection. In the following, the Markowitz mean–variance optimization model was implemented in MATLAB. Then, the implemented algorithm was applied to the data obtained from seven different portfolio indices of the Shanghai stock market, and the desired results were consequently obtained. In this model, the summation of daily return in each month is used to estimate the expected return, and the variance is utilized to determine the risk. One hundred strategies were adopted to run the code and draw the efficient Pareto frontier for the seven stock indices using the monthly return data of each index (each scenario means an optimal response). The results are shown in Fig. 3.

In this graph, the horizontal axis represents the risk amount, and the vertical axis denotes the return of different combinations of seven stock indexes in this study. As can be seen from the figure, the SSE180 index has the highest return but also the highest risk among all indexes.

According to Fig. 3, SSE180 Index, SSE50 Index, Fund Index, SSE300 Index, SSE Composite Index, SSE380 Index, and T-Bond Index have the highest returns, respectively. Additionally, the risk indices ordered from low to high are: T-Bond Index, Fund Index, SSE Composite Index, SHSE300 Index, SSE380 Index, SSE50 Index, and SSE180 Index. The efficient frontier graph represents 100 different combinations of the

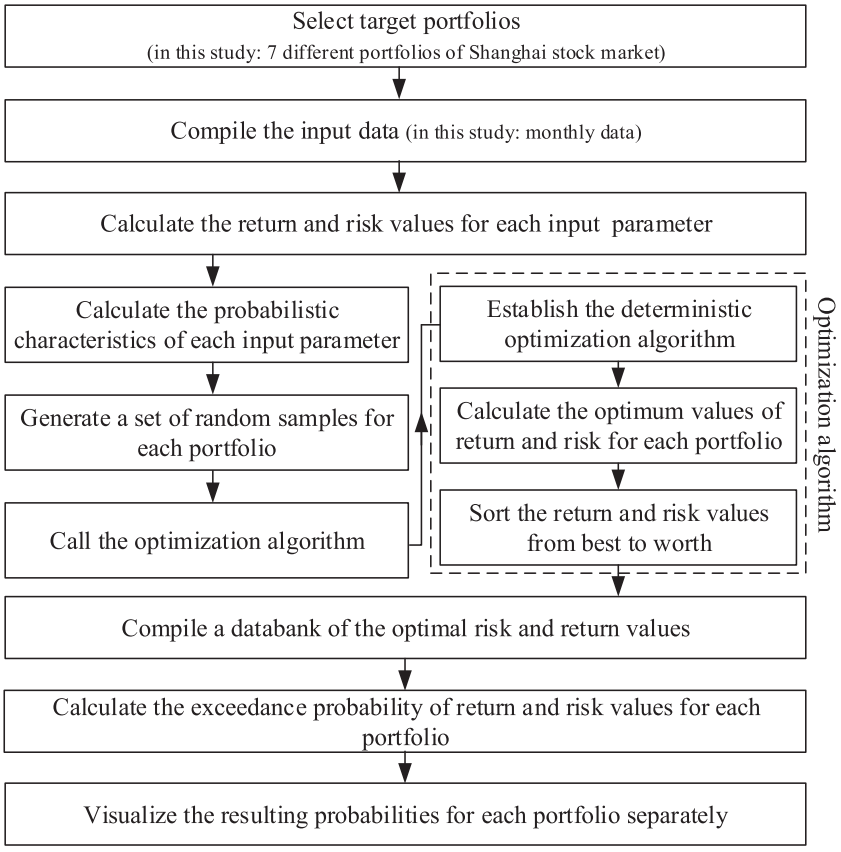


Fig. 2 A schematic view of the proposed algorithm.

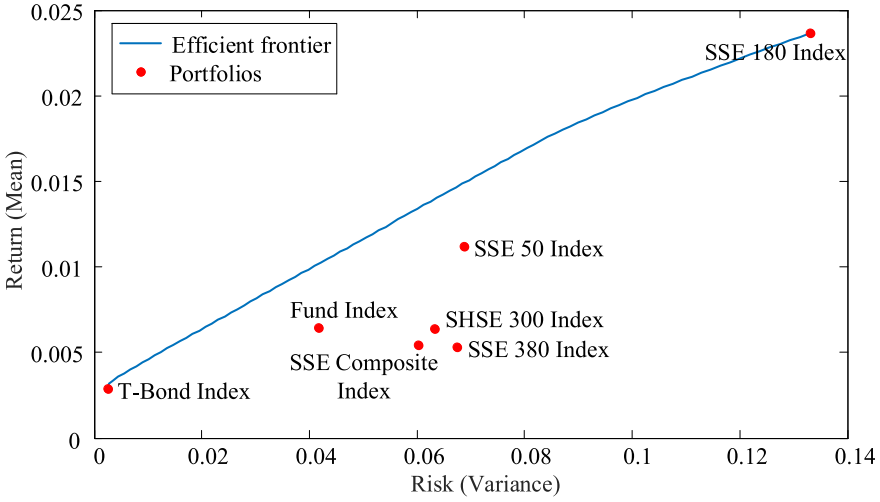


Fig. 3 Pareto frontier for seven different portfolios from Shanghai stock market.

seven indices where each scenario has a risk and a return. The investors can choose the best combination of stock indices for their investment by specifying the maximum risk or minimum expected return.

3.2. Probabilistic modelling of portfolio optimization

As outlined in the introduction, the Monte Carlo sampling method is adopted in this paper to provide the probabilistic analysis of the established problem. To begin implementing the Monte Carlo algorithm, each of the variables involved in the problem (i.e., the monthly values of each portfolio) should be defined as a random variable. In this way, a relatively wide range of possible scenarios would be covered in the analysis process, and the uncertainties in the model parameters would be imported to the problem formulation. The type of probability distribution function can have a significant effect on the resulting probability. Therefore, the distribution function governing the random variables should be selected according to

the actual data. To this end, the cumulative frequency of daily return data is calculated along with different distribution functions, including Normal, Exponential, Uniform, Logistic, and Gumbel. The results for the return values of portfolio A in 2018/12 is shown in Fig. 4. As can be seen, the Normal distribution provides a closer estimation of original data than other distribution functions. As such, the random variables are modeled by Normal distribution function. The mean and standard deviation values of the variables were calculated from the existing daily data. The results are shown in Figs. 5a and b for the whole 60 months.

Then, following the probabilistic characteristics shown in Fig. 5, a normal random vector was generated for each month separately. Each vector contains 30 components corresponding to daily values. A sample of the generated random vectors for the first portfolio is shown in columns 2 to 10 of Table 2. The summation of the random vectors was then calculated for each month and considered as monthly random values. The results are shown in the last column of Table 2.

By repeating this process, 20,000 random vectors were generated for each portfolio. An overview of 200 random vectors for the first portfolio is given in Fig. 6. Only 200 random sam-

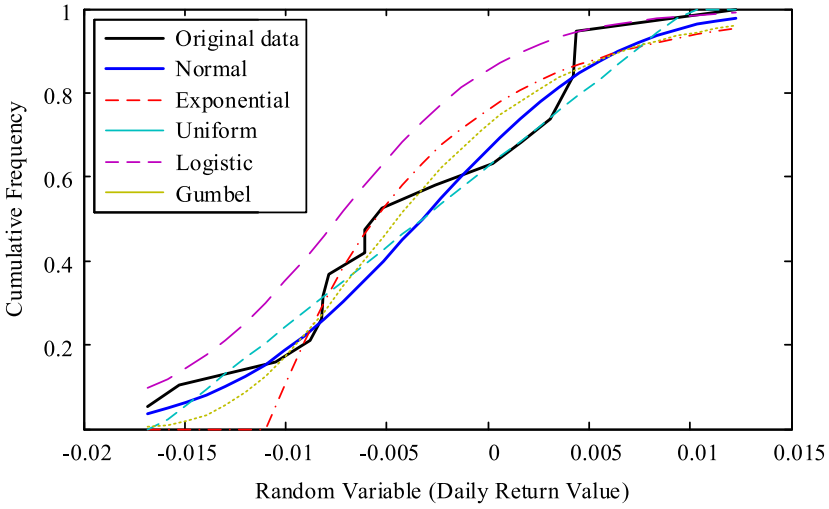


Fig. 4 Cumulative frequency calculated from the return value of Portfolio A and different distribution functions for the year 2018/12.

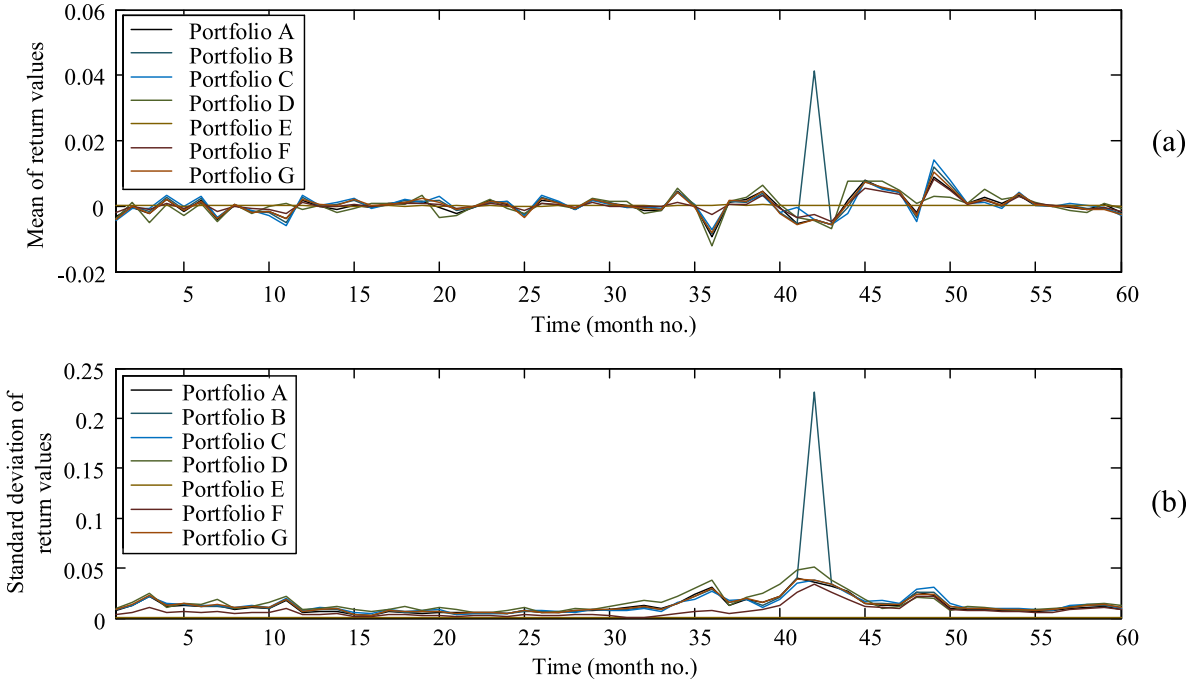


Fig. 5 Mean and standard deviation of return values for each portfolio in a monthly basis.

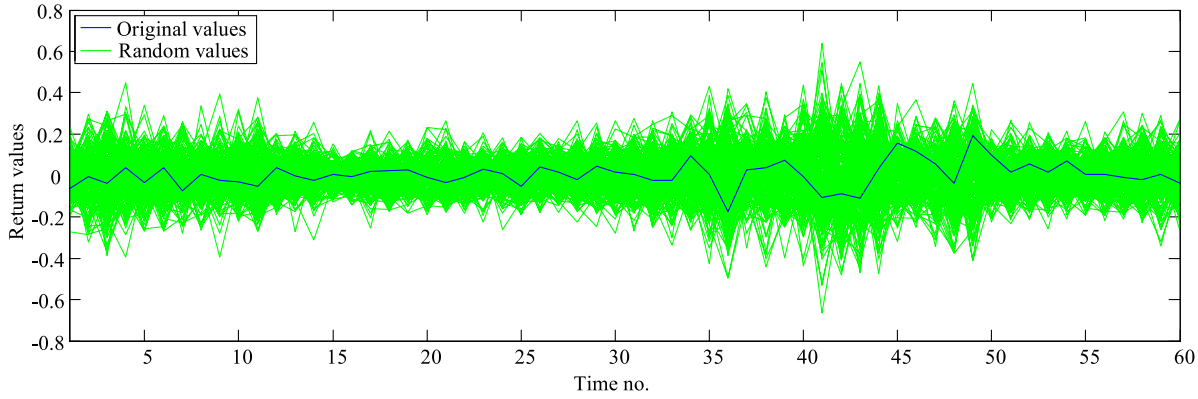


Fig. 6 Overview of 200 random samples generated for the first portfolio.

ples are used to draw this graph, because the fluctuation of random samples and original data can be seen more clearly.

The random vectors generated for each portfolio were then combined to form a matrix in each random scenario. Subsequently, random matrices were imported into the optimization algorithm, and the corresponding optimal return and risk values were calculated and stored in a vector. A schematic view of this process is shown in Fig. 7.

The histogram of the optimal return and risk values for the first portfolio is shown in Fig. 8. As can be seen, the optimal response of each portfolio is not limited to a certain solution and covers a relatively wide range of possible cases.

To process the output data, the optimal return and risk values obtained from each random scenario should first be sorted from best to worst. The best return value corresponds to a sample with the highest possible value, while the best risk value represents a sample with the lowest value. This means that the return data should be arranged from large to small, while the risk data should be sorted from small to large. For example, the optimal return and risk values obtained from the first random sample are shown in columns 1 to 3 of Table 3. Additionally, the return values (sorted from large to small) are presented in columns 4 and 5 of Table 3 beside the corresponding portfolios. As can be observed, portfolio A ranked 1, portfolio D ranked 7, and the rest of portfolios achieved their own rank. Moreover, the risk values (sorted from small to large) and the corresponding portfolios are shown in columns 6 and 7 of Table 3. In this table, ranks 1 and 7 are assigned to portfolios E and G, respectively, and the rest of the portfolios are located in between. Thus, by sorting the optimal results obtained from the random scenarios, each portfolio received a return and risk rank.

The ranking of both return and risk was then listed for all seven portfolios. The result is shown in a matrix form in Tables 4 and 5.

Then, the number of cases where the rank of return or risk in each portfolio equals j ($j = 1, 2, \ldots, 7$) was counted. Dividing this number by the total number of random scenarios (i.e., 20,000), the probability of reaching rank j for each portfolio is calculated as follows (Eq. 3):

$$P_{ij} = \frac{n_{ij}}{N}, \tag{3}$$

where n_{ij} represents the number of cases in which portfolio i ($i = A, B, \cdots, G$) reaches rank j ($j = 1, 2, \ldots, 7$), and P_{ij} indicates the probability of case n_{ij}. The probability for the both risk and return variables is presented in Tables 6 and 7, respectively.

These values were then plotted as a pie chart for each portfolio. The results for the return and risk variables are provided in Fig. 9.

This representation provides a probabilistic statement of ranking for each portfolio that is not limited to a particular

Table 2 A sample of random vectors generated in each month for the first portfolio.

Month	Day in each month									Summation
	1	2	3	4	⋯	27	28	29	30	
1	−0.00091	0.00207	−0.01032	0.00051	⋯	−0.01728	−0.01614	0.00566	−0.00271	−0.11943
2	0.01183	−0.01057	0.00363	−0.01300	⋯	−0.00395	−0.00492	−0.00960	0.00683	−0.01537
3	−0.04688	0.06419	0.02565	−0.02541	⋯	0.00847	−0.03746	−0.01532	0.01916	−0.03435
4	0.00061	−0.00721	−0.00171	−0.00023	⋯	0.00856	0.00223	0.00914	−0.01156	0.07123
5	−0.01843	−0.00989	0.01269	0.00461	⋯	−0.01433	−0.00028	0.01186	−0.00496	−0.04019
⋮	⋮	⋮	⋮	⋮	⋯	⋮	⋮	⋮	⋮	⋮
56	0.00069	0.01015	0.00426	−0.01124	⋯	−0.00532	0.00914	0.00002	−0.00841	0.01506
57	−0.00352	0.01182	−0.00184	0.00425	⋯	0.00229	0.00152	−0.00769	−0.00682	−0.05060
58	−0.00546	0.00398	−0.00937	−0.00764	⋯	−0.00508	−0.00508	0.01082	0.00659	−0.07495
59	−0.00542	−0.00362	0.00750	−0.01254	⋯	0.00531	0.01131	−0.01767	0.00857	0.00897
60	−0.00721	−0.01310	0.00495	0.00201	⋯	−0.01295	−0.00813	−0.00475	−0.01634	−0.05202

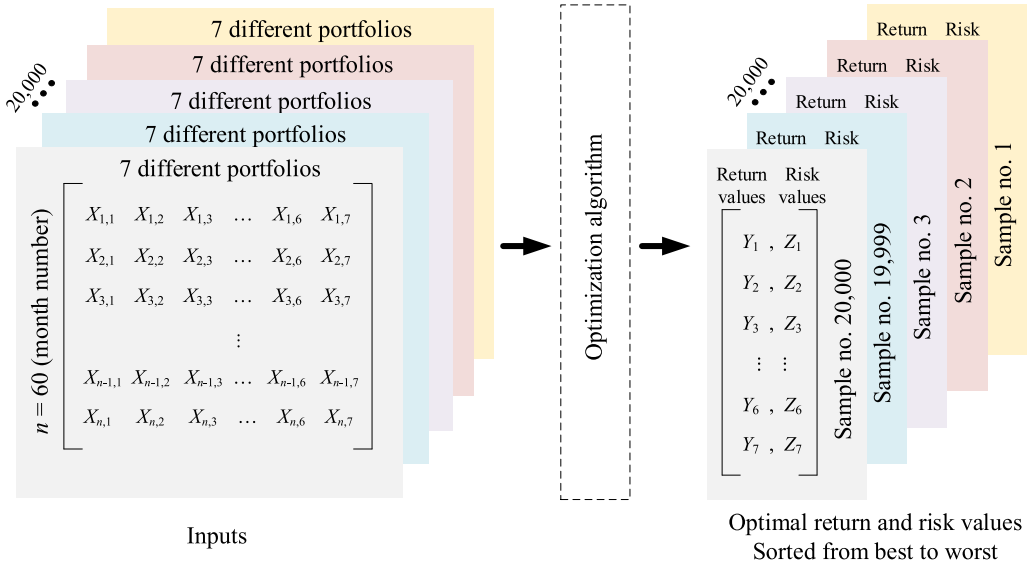


Fig. 7 Schematic view of input random samples and corresponding optimal values.

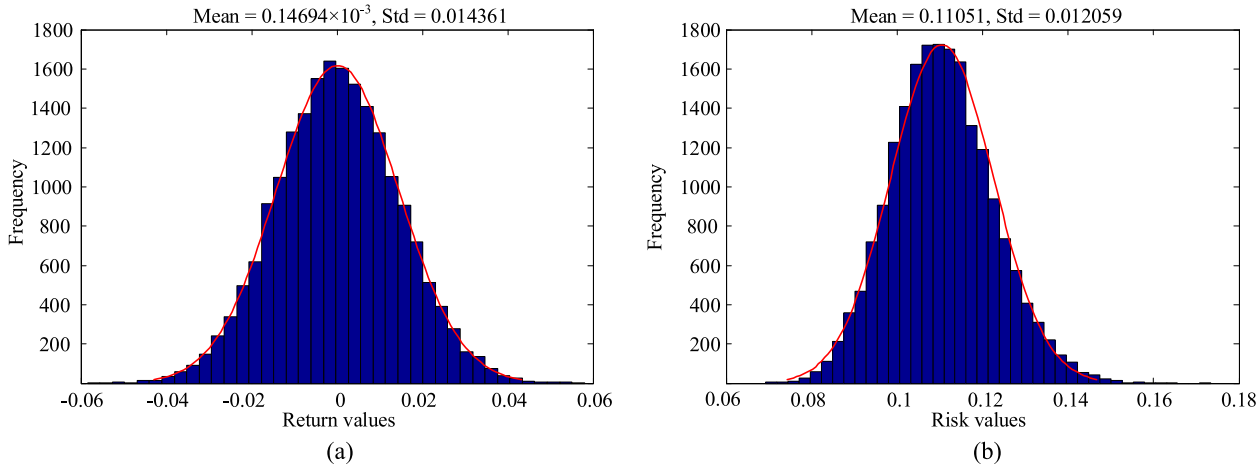


Fig. 8 Histograms of optimal return and risk values for first portfolio.

Table 3 Optimal return and risk values obtained from the first random scenario before and after the sorting process.

Results of optimization algorithm			Sorted return		Sorted risk	
Portfolio	Return	Risk	Portfolio	Return	Portfolio	Risk
A	0.0136	0.1168	A	0.0136	E	0.0168
B	−0.0083	0.1107	G	0.0118	F	0.0893
C	−0.0070	0.1187	E	−0.0013	D	0.1029
D	−0.0120	0.1029	F	−0.0059	B	0.1107
E	−0.0013	0.0168	C	−0.0070	A	0.1168
F	−0.0059	0.0893	B	−0.0083	C	0.1187
G	0.0118	0.1265	D	−0.0120	G	0.1265

case, but covers a wide range of possible scenarios by taking the uncertainties into account in the both return and risk variables. For example, portfolio B (SSE 180 index) was located in the upper right corner of the deterministic optimization chart (Fig. 3), indicating the best return and the worst risk scenario. The probabilistic chart of this portfolio (Fig. 9) also indicates that this portfolio is very likely to achieve the best rank and the worst risk. However, more information about this portfolio is

Table 4 Ranking of return and corresponding portfolio at each random scenario.

Ranking of return	Random scenario										
	1	2	3	4	5	···	19996	19997	19998	19999	20000
Rank 1	A	C	D	A	G	···	G	B	F	C	C
Rank 2	G	E	A	C	F	···	C	F	E	E	G
Rank 3	E	B	F	B	D	···	D	E	G	B	F
Rank 4	F	A	B	G	E	···	E	D	B	A	E
Rank 5	C	G	G	F	C	···	F	G	C	F	B
Rank 6	B	F	E	E	B	···	B	A	D	G	A
Rank 7	D	D	C	D	A	···	A	C	A	D	D

Table 5 Ranking of risk and corresponding portfolio at each random scenario.

Ranking of risk	Random scenario										
	1	2	3	4	5	···	19996	19997	19998	19999	20000
Rank 1	E	E	E	E	E	···	E	E	E	E	E
Rank 2	F	F	F	F	F	···	F	F	F	F	F
Rank 3	D	G	G	G	A	···	A	A	B	C	A
Rank 4	B	D	D	D	C	···	C	C	G	G	C
Rank 5	A	C	A	C	G	···	D	G	D	A	B
Rank 6	C	A	C	A	D	···	G	D	A	B	G
Rank 7	G	B	B	B	B	···	B	B	C	D	D

Table 6 Estimated probability of return for seven major stock indices.

Portfolio	Ranking of return						
	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	Rank 6	Rank 7
A	0.1525	0.1513	0.1331	0.1133	0.1305	0.1542	0.1652
B	0.1970	0.1547	0.1176	0.1031	0.1125	0.1404	0.1750
C	0.1715	0.1539	0.1259	0.1130	0.1260	0.1454	0.1645
D	0.1838	0.1472	0.1150	0.1016	0.1199	0.1464	0.1864
E	0.0167	0.0965	0.2257	0.2989	0.2347	0.1064	0.0212
F	0.1121	0.1496	0.1574	0.1519	0.1518	0.1598	0.1176
G	0.1666	0.1471	0.1254	0.1183	0.1248	0.1476	0.1703

Table 7 Estimated probability of risk for seven major stock indices.

Portfolio	Ranking of risk						
	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	Rank 6	Rank 7
A	0.0	0.0437	0.2995	0.2443	0.1975	0.1425	0.0727
B	0.0	0.0150	0.1281	0.1438	0.1651	0.1988	0.3493
C	0.0	0.0226	0.1957	0.2259	0.2267	0.1997	0.1295
D	0.0	0.0056	0.0703	0.1220	0.1814	0.2822	0.3386
E	1.0	0.0	0.0	0.0	0.0	0.0	0.0
F	0.0	0.8831	0.0824	0.0233	0.0077	0.0031	0.0005
G	0.0	0.0301	0.2242	0.2408	0.2216	0.1739	0.1095

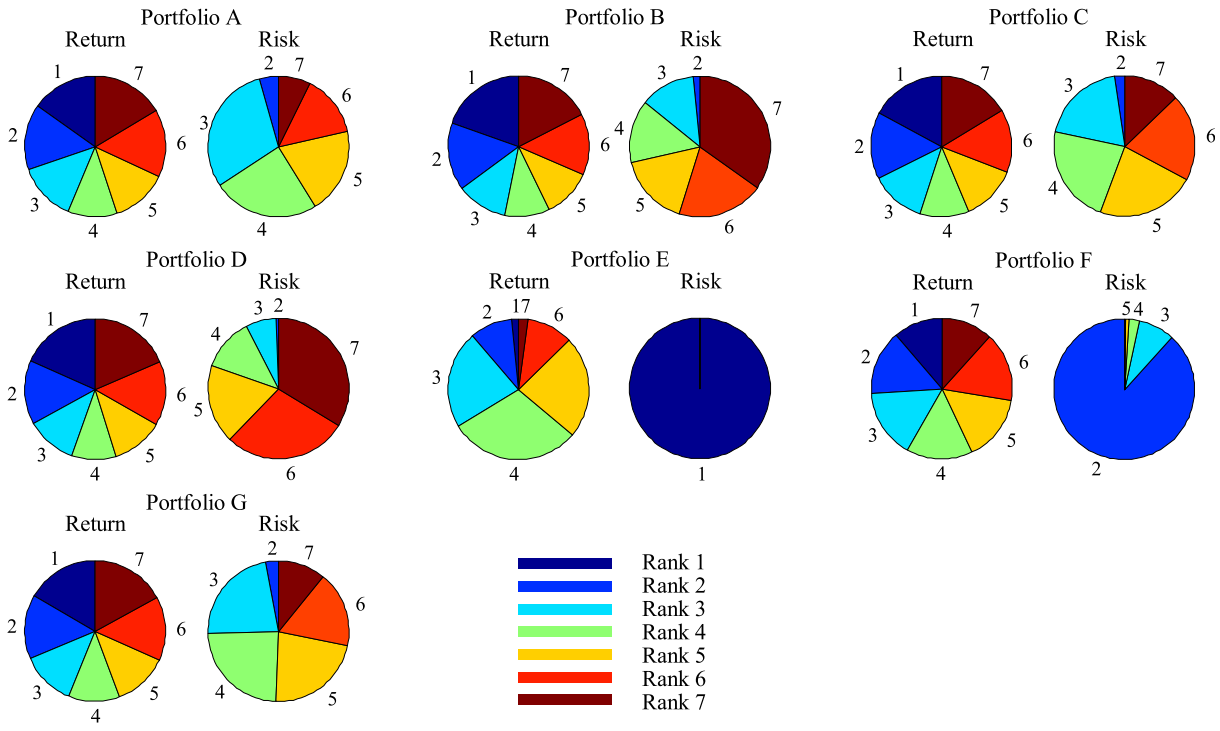


Fig. 9 Pie chart demonstrating probability of return and risk for seven different portfolios.

Table 8 The resulting probabilities for the third rank of portfolio A corresponding to different sample sizes.

Sample size	Return		Risk	
	Number of samples with ranking = 3	Probability	Number of samples with ranking = 3	Probability
100	15	15	21	21
500	77	15.4	112	22.4
1000	153	15.3	224	22.4
2000	311	15.55	443	22.15
5000	760	15.2	1118	22.36
10000	1480	14.8	2303	23.03
15000	2203	14.69	3451	23.01
20000	2916	14.58	4582	22.91

provided in the probabilistic chart, and other probability ranks are also reported. For instance, although the probability of achieving the best return for this portfolio is relatively high ($p = 19.7\%$), the probability of achieving the worst return is also high ($p = 34.93\%$). Such information is also available for all other portfolio ranks.

4. Discussion

4.1. Convergence assessment

One of the most important points to note is to ensure that the number of random samples used in the Monte Carlo method was sufficient enough to achieve an accurate result. In other words, it should be proved that the number of random samples used in the analysis was sufficient to make the probability converge, so that by increasing the number of random samples, the probability would not change significantly [42]. To this end, the algorithm was repeated for different random sample sizes to evaluate the probability convergence process. The results for the third rank of portfolio A corresponding to different values of random sample sizes are shown in Table 8.

By taking a step smaller than that in Table 8 (i.e., the difference between each two sample size), the resulting probability values were plotted against the number of random samples. The result is shown in Fig. 10. As can be seen, in small random sample sizes, there is a large fluctuation in the probability values. However, with the increase in the size of random samples, the fluctuations decrease. Finally, after using about 12000 random samples, the probability converges, and the fluctuations continue to be very low and negligible.

4.2. Robustness of algorithm

In addition to the importance of the accuracy and efficiency of the proposed method, it should also be ensured that if an error

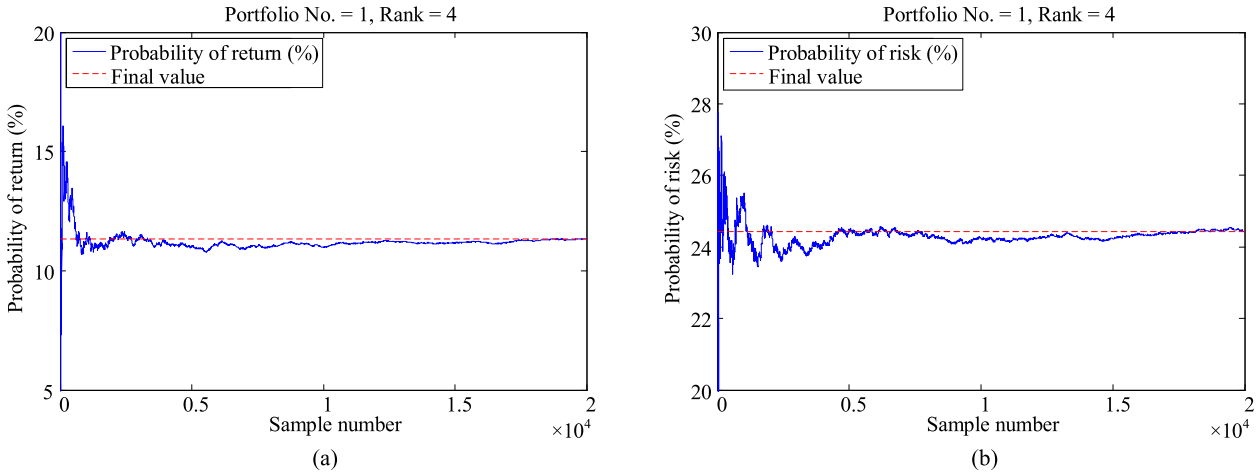


Fig. 10 Sample number versus probability.

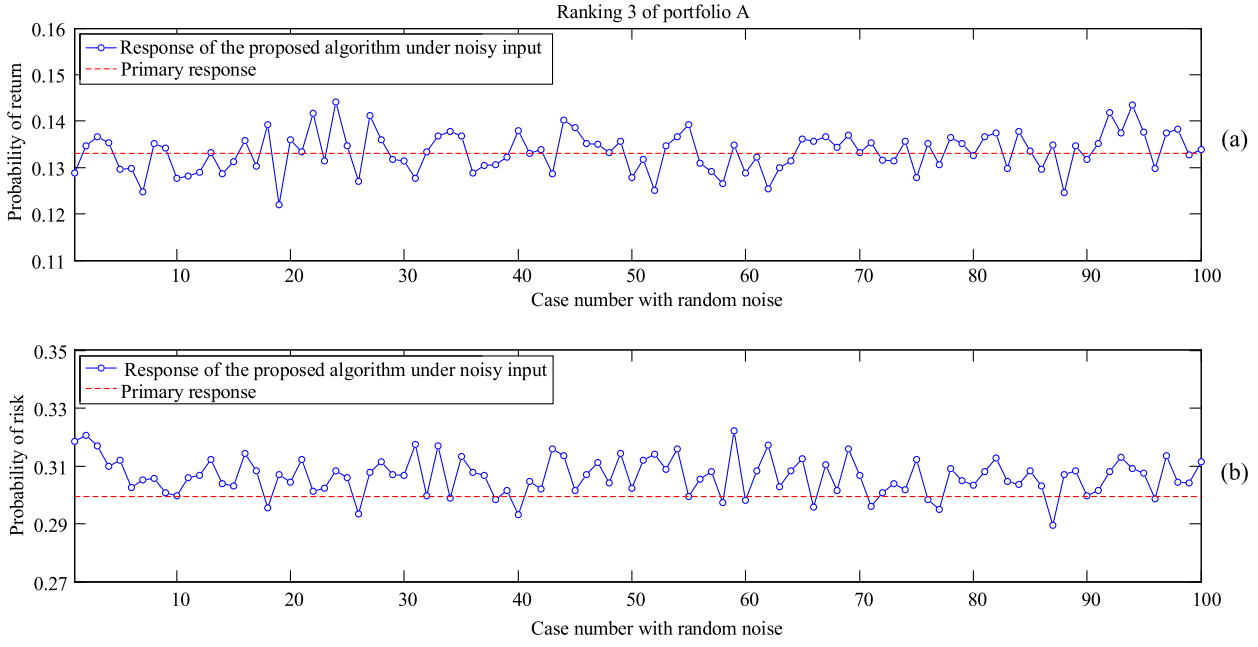


Fig. 11 Response of proposed algorithm to noisy input, reported for rank 3 of portfolio A: (a) return and (b) risk.

occurs in the model inputs for unintended reasons, the answer will not be affected, and the proposed method will still be able to provide an acceptable answer. This indicates the concept of model robustness [43]. This is very important, because when determining the probabilistic characteristics of variables and distribution functions, the inaccurate estimation of uncertainty can easily lead to a margin of error in the system response, resulting in the unreliable solutions. To numerically investigate this issue, a perturbation function was used to generate a random noise and incorporate it into the variables to evaluate the system response. For this purpose, a Gaussian distribution function with a coefficient of variation of 0.01 was adopted. The mean of the Gaussian function was considered a factor of the mean of random variables (Eq. (4)):

$$\mu_{f_i} = \alpha \times \mu_i, \tag{4}$$

where μ_i represents the mean of random variable i, α denotes the noise contribution factor, and μ_{f_i} is regarded as the mean of the Gaussian distribution function for variable i. Assuming $\alpha = 2\%$, a series of 100 random noises were generated and added to the initial data. The proposed algorithm was then utilized to calculate the optimal probabilistic solution for all seven portfolios under the noise-polluted inputs. The results for the rank 3 of return and risk in portfolio A and the histogram of the absolute error are shown in Figs. 11 and 12, respectively. Additionally, the mean and standard deviation of both probability and the absolute error are calculated and reported in Table 9.

As can be seen, the proposed algorithm exhibits a very robust response against the applied changes, so that the obtained results are very close to the actual values in almost all cases, and the resulting error is close to zero. This is because

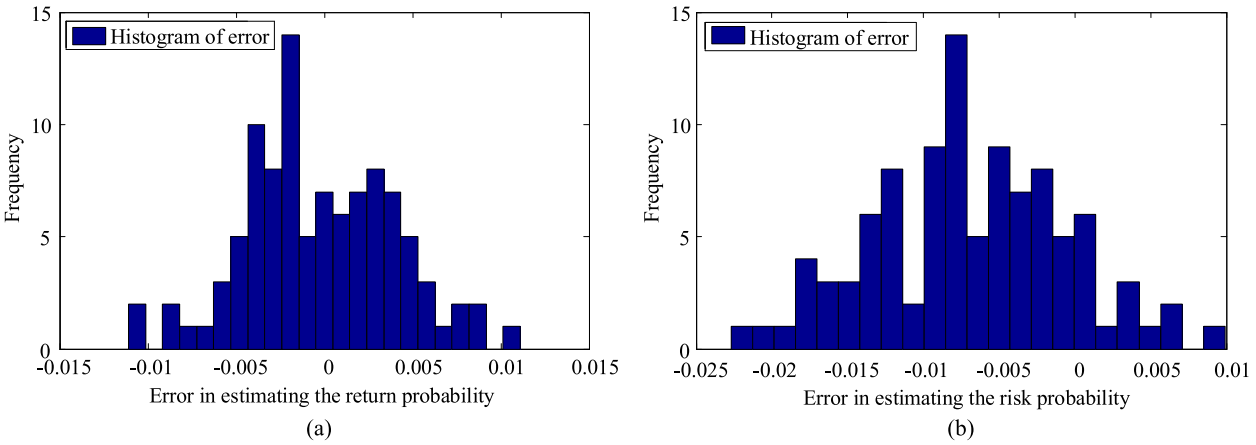


Fig. 12 Histogram of error in estimating the probability of (a) return and (b) risk.

Table 9 Mean and standard deviation of both probability (P) and absolute error (ΔP).

Target parameter	Mean of P	Standard deviation of P	Mean of ΔP	Standard deviation of ΔP
Return	0.13347	0.00431	0.00037	0.00431
Risk	0.30678	0.00646	0.00728	0.00646

the noise applied to the input variables is accommodated by the algorithm, and consequently, no significant change is made in the results.

5. Conclusion

In this paper, a probabilistic form of portfolio optimization was provided for Shanghai stock market considering the uncertainty in the return and risk values. For this purpose, a comprehensive database of input values was collected on a daily basis from January 2014 to December 2018. Then, the mean and standard deviation of the data were calculated for each month and imported to the Monte Carlo algorithm as a vector of random variables. Afterwards, by generating a vector of random numbers for each random variable, 20,000 random scenarios were defined, and the optimal results were separately calculated for each one. Then, by sorting the optimal return and risk values, the rank of each portfolio and the corresponding probabilities were calculated. The results were then presented as a set of two probabilistic pie charts for the return and risk ranks. The most important contributions of this paper can be summarized as follows:

- Instead of direct use of the dataset collected from the stock market, the monthly values corresponding to each portfolio were defined as random variables with the specified means and standard deviations to incorporate the uncertainty into the optimization process.
- By introducing the optimization algorithm into the Monte Carlo sampling method, a systematic approach was presented to compile a database of optimal returns and risks.
- By sorting the obtained artificial data, a simple definition of optimal ranking was presented to calculate the corresponding probability for all seven portfolios.
- By applying a Gaussian random noise to the input data and evaluating the fluctuation of the results, the robustness of the proposed algorithm was investigated.

The results of this paper are essential for evaluating the desired stock indices and the amount of investment in each portfolio, providing a straightforward criterion for the decision-makers and investors. Finally, it should be noted that one of the simplified assumptions in this paper is that the input portfolios are assumed to be independent. In case of having correlations between the inputs, it should be seen when using the Monte Carlo algorithm and generating random samples. If the correlation is linear, it is sufficient to use the Pearson correlation coefficient and the covariance matrix to model the correlation. However, if the correlation is nonlinear, other methods such as Copula functions or joint distribution function governing random input variables can be used to generate random samples. The probabilistic portfolio optimization assuming the correlations between different portfolios is an interesting and challenging topic to be considered by the authors in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. The seven portfolio indices used in the paper are as follows. More information about indices can be found in [44].

SSE Composite Index 000001: SSE composite index consists of all the stocks (including A-shares and B-shares)

listed on SSE. The index aims to reflect the overall performance of the Shanghai stock market. The index is weighted by total market capitalization to reflect the price performance of listed stocks in SSE. The index was launched on July 15, 1991.

SSE 180 Index 000010: SSE 180 index consists of the 180 largest and most liquid A-share stocks listed on SSE. The index aims to reflect the performance of the Shanghai blue chips. SSE 180 indices are composed of the constituents with good performance, without serious financial problems or large price volatility that shows strong evidence of manipulation.

SSE 50 Index 000016: SSE 50 index consists of the 50 largest and most liquid A-share stocks listed on SSE. The index aims to reflect the overall performance of the most influential leading Shanghai stocks. SSE 50 indices are selected from SSE 180 Index constituents and composed of 50 Shanghai stocks with large market cap and good liquidity.

SSE 380 Index 000009: SSE 380 index consists of 380 stocks with mid-size market cap, high growth and good profitability which are selected from the remaining Shanghai listed A-shares after deleting the constituents of SSE 180 index. The index aims to comprehensively reflect the performance of the new Shanghai blue chips. SSE 380 indices are composed of Shanghai stocks from the SSE 180 index universe except for certain stocks. The constituents are selected by ranking of revenue growth, return on net assets, turnover, and total market value.

SSE Fund Index 000011: The constituents for SSE fund index are all the securities investment funds listed on SSE. The base day for SSE fund index is May 8, 2000. The base period is the total market capitalization of all securities investment funds of that day. The base value is 1000. The index was launched on June 9, 2000.

T-Bond index: Treasury bonds, a kind of government bonds issued for raising financial funds, are the certificates for the claims and debts issued to investors by the central government. In the exchange market, there are mainly two kinds of treasury bonds: coupon-bearing treasury bonds and discount treasury bonds.

SHSE-SZSE300 Index: SZSE 300 index and its sub-indices (SZSE 100 Price Index and SZSE 200 Price Index) are the stock market indices of Shenzhen Stock Exchange, representing top 300 companies by free-float adjusted market capitalization. The sub-indices represent the top 100 companies and next 200 (the 101st to 300th) companies, respectively. SZSE 300 index itself is a sub-index of SZSE Component Index, SZSE 1000 Index and SZSE Composite Index.

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Lesson Notes

MODULE 3 | LESSON 4

Probabilistic Scenario Optimization and Probabilistic Approaches Lesson Notes

Reading Time	30 minutes
Prior Knowledge	Students are expected to be familiar with the concepts developed in the previous lessons. In particular a solid grasp of the traditional mean-variance optimization is a necessary requirement.
Keywords	Probability Scenario Optimization (PSO), probabilistic risk measures, algorithms for these situations.

In the previous model we completed the coverage of the Black-Litterman model analysis and a few abuses of that model. Practitioners like to simplify things and, arguably, rightly so, but often actions have consequences. In this lesson, we will cover a completely different topic that, not unlike the Black-Litterman model, can be used to extend the validity and the usefulness of Markowitz's approach.

In the next lesson that starts a new module of the course, we will consider a very important problem: in all models we have visited so far it has been assumed that investors or economic agents have a clear view of what they desire (risk takers, risk averse, etc.) and that they always know exactly how to optimize their choices based on some type of utility functions. Is that always the case? The field of behavioral economics and finance would beg to differ and would say that the fully rational man of classic economics is just a simplification.

1. Probabilistic Scenario Optimization (PSO)

Imagine you are assigned to manage the investment portfolio of a company's pension plan. In particular, let's assume the pension plan is a defined benefit pension plan. In a defined benefit pension plan, the employers promise to pay a defined benefit to employees for life after they retire. In contrast, in a defined contribution plan (401(k), for instance) employees put their money in an employer-sponsored investment program, but the company is not responsible for making periodical payments.

As a portfolio manager for the pension fund, you may still have the long-term goal to maximize the risk-return profile of the portfolio. However, you need more than that. Depending on certain demographic and actuarial characteristics of the company's work-force (past and present), you may have a need to produce specific streams of income to be able to make the payments while insuring the long-term goal of optimizing the portfolio's risk-return profile. Or since the periodic payments may be indexed to inflation, how should you invest to make up for the higher and higher payments?

A similar issue would arise if you were asked to manage the investment portfolio for a property and casualty insurance company. There is no certainty about how much or how often the company has to make payments; however, a certain amount of money must be very liquid in case it had to make large payments to the clients it has insured. Generally speaking, these companies invest with a short horizon as their liabilities have short duration.

The problem does not change much if we consider life insurance companies. Insurance companies sell policies that promise a payment to the policyholders if an event covered under the policy occurs during the life of the policy. The funds received from the policyholders (premiums) are invested until needed for payout. Traditionally, life insurance companies worry the most about a mismatch in the duration of assets and liabilities (as their liabilities are overall long term), cash flow volatility, and credit risk. Thus, to reduce the first type of risk, they focus on a minimum return target (usually set equal to the actuarial assumed rate of growth of policyholder reserves, since it is the minimum return needed to meet payouts), and tend to invest mostly in fixed income assets. Often, they could face legal and regulatory constraints as well.

These issues do exist even if we were managing the investment portfolio for a foundation or an endowment where the goal is to grow the capital after the spending requirements are met but not to risk it in any unreasonable manner. In some cases there could be additional legal and regulatory constraints to consider.

Where are we going with all this? We hope it helps to illustrate the point that the objective function to maximize cannot always be a simple mean-variance representation and that additional elements should be added to it. And, furthermore, we may have a view about several possible scenarios that may develop during the relevant period under consideration and that could affect different securities and/or different classes of assets in different manners.

We need to accept that a generic allocation $w^T = (w_1, w_2, \ldots, w_n)$ is not just a vector that represents the amount of units or the percentage of wealth invested in each of the n securities in a given portfolio. The allocation must be considered as the result of a decision process together with the available information. And, if the available information had been different, we would have ended up with a different allocation vector $\tilde{w}^T$ for our portfolio.

If we wanted to evaluate the pros and cons of a specific allocation, we would need to evaluate the decision process behind it. It is really quite similar to what is done in statistics when we introduce many estimators and then study the properties of such estimators from a probabilistic point of view. This is arguably a good description of PSO.

It must be stressed from the start that PSO is not a simple task and that it is part arts and part science. In its essence, PSO is a risk-based technique aimed at facilitating the implementation of goal-based investing in portfolios managed by or for specific institutions. The decision making extends to long-term investments that investors can optimize and that do not include just the max

return for a given level of risk. Putting it in a more simplistic manner and simplifying to the maximum extent, we could try to decide which of two allocations would be most suitable to us given that we foresee certain potential scenarios and are in a position to assign probabilities to each of the scenarios. We could run many simulations and consider which of the two allocations of investments minimizes the probability to be below a certain threshold or we could mix this probability to reduce the need to minimize the risk, etc.

1.1. How Is It Done?

This is not a simple task, and there is more than one way to implement it, but in the end, it is based on an apparatus of well-conceived simulations. Lots of simulations.

Here, we provide a description of the steps involved.

Step 1: Fix the investment horizon. This could be a period of, say 10 years. Such period can be split in even subintervals or we could use an uneven time step discrete grid with higher frequency in the short terms than in the long terms, say the time period could be 1 week for the first 6 months, 1 month for the next 1.5 years, 6 months after that, etc. These will be the steps at each of which we will perform a Monte Carlo simulation.

Step 2: Select the investment universe, which is obviously no surprise, and indicate the allocation constraints (no short selling, max currency exposure, max single asset exposure, etc.). Furthermore, we must specify the investment risk-return profile for our client. For instance:

- We could have a risk-averse profile and set the target return similar to risk-free rate. Say we want to achieve a return of 1% per year. In this profile, we could set a risk adequacy so that our capital never loses value starting from a specific period, say the third year or the fifth.
- Alternatively, we could have a more accommodating risk-return profile and set a target return of, say LIBOR + 200 bps. We could also allow for some capital risk over the short term and enforce tighter capital insurance in the medium to long terms.
- Or, we could adopt a risk-return profile more tolerant of risk in which the return goal could be set to, say, 5% a year at least and no control on risk over the full time horizon of the investment.

Step 3: Decide on the type of optimization: multi-period (usually) or single-period. In the case of a single period optimization, say 5 years, one would not model any intermediate risk. In the multi-period, one can set liquidity constraints for a certain number of years, i.e., during this period, we periodically check the adequacy constraints at the end of every year and after that period, we no longer check for portfolio weights' adequacy.

Step 4: We need to specify an objective function, which could be the probability or weighted probability of a total return higher than our minimum goal while complying with allocation constraints and risk limits. Regarding the risk limits, the risk measure to be used could be implemented via a one-tail VaR.

Step 5: Before running the Monte Carlo simulations, we have to set up the scenarios in terms of probability distributions. A multivariate normal is a common choice, but it is not the only possible choice. Furthermore, one could also think of introducing a few ad-hoc views that will serve to stress test our results. We could imagine a booming economy for the first 5 years, which could be followed by a sharp decline or, vice versa, an economy that starts with a sharp decline in the equity markets followed by a protracted expansion until another threshold time and then contracts again. Notice

that we need to generate Monte Carlo scenarios on a variety of risk factors, which could include market indices, individual securities, currency exchange rates, and credit curves for government and credit issuers. This requires that we collect time series data over a pre-specified period and then compute daily returns, correlations, etc.

Step 6: Generate the entire quasi-random space of admissible asset allocations that satisfy the asset allocation constraints given above. This must be understood in the sense that, even if we start with asset allocations that fulfill certain requirement of no short selling, or limit on the concentration, under certain scenarios, as the value of the assets change, we could break through those constraints, and therefore, we need to re-check them. Recall that the weights are in market capitalization percentage of the total portfolio.

Step 7. Reduce the admissible asset allocations to the set of risk-adequate portfolios with respect to the investor's risk-profile. In other words, we filter out those portfolios that do not comply with our ambitions.

While this step could have been performed on the space of all admissible asset allocations, working on the sample allocations that are compliant with the allocation constraints is advantageous from a computational point of view.

This step could take the form of assigning boundary conditions, like that we do not want to lose more than a certain amount with a certain probability, i.e., we want the α quantile profile of the investor to be higher than such threshold. Or it would be a constraint where the right quantile of the optimal portfolio at any selected time point t has a probability $1 - \alpha$ to be greater than or equal to the investor's return required return. In this way, if we choose a low α level, we are imposing a risk cap. By choosing a medium to high α-level, we can impose a probabilistic return floor.

Step 8: Measure the probability required and optimize the objective function. This can be quite different than what we are accustomed to. In fact, we would not just look for the maximum return, which could be achieved by different portfolios under different scenarios in general. But, we could maximize the probability of achieving a minimum time dependent target return level, say l_t^A where $t \in T$ and T is the universe of time steps. Indicating with Θ the set of portfolios that made it to this point, we are trying to determine the portfolio(s) such that:

$$\max_{\theta \in \Theta} \left\{ \sum_{t \in T} k_t P[R_\theta > l_t^A] \right\}$$

where R_θ represents the return on portfolio θ, $k_t \in K$ is a multi-period weighting scheme that allocates a positive weight to each $t \in T$. If we assigned full weight to only a discrete set of points $t \in T$, then the other points would be irrelevant. In an extreme case, we could attach the entire weight to one point only, say the last point.

Step 9: Measure the performance and compare portfolios, i.e., track the investment performance over time and compute the probability to achieve or beat a target return across time or the probability to fall short of it.

1.2 Low Discrepancy of Quasi-Random Numbers

When running Monte Carlo simulations, admissible asset allocations can be a significantly large set, running into the millions or billions. As if this were not enough, the underlying space depends on

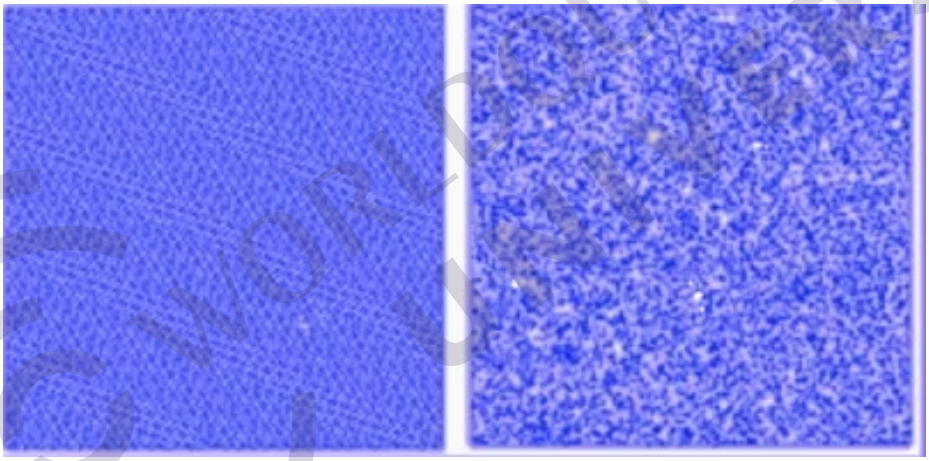
multidimensional sets of variables. For instance, if we model 100 assets and need to compute a probability (an area), this is like doing integration on a high dimensional space. Evaluation of high-dimensional integrals is performed by stochastic Monte Carlo simulations. Simulations that use **quasi-random numbers** can achieve faster convergence and better accuracy than simulations via conventional pseudo-random numbers.

The purpose of **low discrepancy methods** is to build a series of simulations by sampling from the interval $[0, 1]$ in such a way that every draw is further away from the previous one, and the draws are as far apart from each other as possible to guarantee better overall coverage and avoid situations in which parts of the domain contain many points and others are nearly deserted.

Monte Carlo integration uses a finite quantity of pseudo-random numbers. A critical issue with these numbers is that they may not be perfectly equi-distributed, i.e., the finite quantity of pseudo-random numbers used are not spread uniformly throughout the domain of the integrand function, which leads to poorer results. Equi-distribution, and therefore accuracy, improve as more random numbers are used, but this requires longer run-times. Low-discrepancy sequences that are well-distributed even for small quantities were introduced.

Figure 1 compares 10,000 simulations on the square $[0, 1) \times [0, 1)$ using traditional methods (right) and quasi-random numbers (left).

Figure 1: Comparing Simulations using Quasi-Random Numbers and the Traditional Technique



To understand the concept of discrepancy, let's consider a hypercube like $I_s = [0, 1)^s$ for some s an integer greater or equal to 1. Consider now M points in $I_s : z_1, z_2, \ldots, z_M$. Furthermore, consider a subset J or I_s, i.e.$I_s \subset J$ and let $A(J)$ be the number of points $z_1, \ldots, z_M$ that are in J. Finally, let $V(J)$ denote the volume of J. Then, the discrepancy $D(J, M)$ is defined as

$$D(J, M) = |A(J)/M - V(J)|.$$

Intuitively, the discrepancy is the difference between the proportion of points in J compared to the volume of the box J.

We can also identify a worst-case discrepancy, i.e., the worst-case distribution of a sequence of points $z_1, z_2, \ldots, z_M$, such that

$$D^{\times}(M) \equiv \max_J D(J, M).$$

The goal of a low-discrepancy sequence is to minimize $D^{\times}(M)$.

One important method to generate low-discrepancy sequences is provided by Halton sequences. In this method, each dimension is represented in a different prime base $(e.g., 2, 3, 5, 7, \ldots)$. To generate the n-th point in a sequence, consider the b-ary expansion of a number n $\sum_{i=0}^{\infty} a_i b^i$ where $a_i \in \{0, 1, 2, \ldots, b-1\}$. The n-th Halton point $H(n)$ is the radical-inverse function in base b :

$$H(n) = \sum_{i=0}^{\infty} a_i b^{-1-i}.$$

Example. Consider the base $b = 2$. And let's draw the first 7 numbers $H(1), \ldots, H(7)$. To do this, we need to write each number in its $b = 2$-ary expansion:

$$1 = 1, 2 = 10, 3 = 11, 4 = 100, 5 = 101, 6 = 110, 7 = 111.$$

Now, pay attention to the fact that in representing 2 for instance, we have $a_0 = 0$ and $a_1 = 1$. In representing 4, $a_0 = 0, a_1 = 0, a_2 = 1$, etc. Then

$$H(1) = 1 \times 2^{-1} = 0.1$$

which, in decimal notation, is $1/2$. Then

$$H(2) = 0 \times 2^{-1} + 1 \times 2^{-2} = 0.01,$$

or, in decimal notation,$H(2) = 1/4$. Moving along, we find also:

$$H(3) = 1 \times 2^{-1} + 1 \times 2^{-2} = 0.11,$$

i.e.,$3/4$ in decimal notation.

$$\ldots\ldots$$

$$H(7) = 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 0.111$$

or $7/8$ in decimal notation. As one can see, we are slowly populating the interval between 0 and 1.

How do we generate sequences in $(0, 1) \times (0, 1)$ or higher dimensions? We partition the first $(0, 1)$ interval using a 2-ary expansion as we did above. For the second space $(0, 1)$, we use instead a 3-ary expansion. Let's call these numbers using $\tilde{H}$. Hence, we start with

$$1_{10} = 1_3, 2_{10} = 2_3, 3_{10} = 10_3, 4_{10} = 11_3, 5_{10} = 12_3, \ldots,$$

where the symbols a_{10} and a_3 indicate in which base the number a is written. Then, we use the same approach as before, i.e.,

$$\tilde{H}(1) = 0.1 = 1 \times 3^{-1} = 1/3, \tilde{H}(2) = 2 \times 3^{-1} = 2/3, \tilde{H}(3) = 0 \times 3^{-1} + 1 \times 3^{-2} = 0.01 = 1/9, \tilde{H}(4) = 4/9,$$

and so on. Finally, we consider the pairs of numbers:

$$(1/2, 1/3), (1/4, 2/3), (3/4, 1/9), \ldots$$

Students interested in this topic could read the paper Monte Carlo integration with quasi random numbers which contains also relevant Python code.

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1.3 Ordering, Lexicographical Ordering

Admissible portfolios must be ordered so that we can apply to each its Halton sequence. There is a bit of a problem in doing this because say that the universe of investments consists of only four assets, A, B, C, and D, and the tick sizes (weight of the assets in the portfolio) can only be one of the following values: $\{0, 0.25, 0.50, 0.75, 1\}$. Which portfolio comes first among $(0.25, 0.25, 0.5, 0)$ and $(0.25, 0.50, 0, 0.25)$ if we were to list them? To do this we need to introduce some suitable ordering. In this case, **lexicographical ordering** can be used (but it is not the only one possible). To this purpose, we rewrite the list $\{0, 0.25, 0.5, 0.75, 1\}$ as $\{a, b, c, d, e\}$. Now, we make further assumptions that we are given minimum and maximum allocations for each asset, say

$$A : [0.25, 0.50], B : [0.25, 0.75], C : [0, 0.50], D : [0, 0.75]$$

and we also have the usual constraint that the sum of the weights is equal to 1.

We can represent these constraints based on our identification above also in the format:

$$A : [1, 2], B : [1, 3], C : [0, 2], D : [0, 3].$$

We can build an ordered set of allocations by starting to allocate the minimum amount allowed for each asset in the alphabetical order where $0 < 1 < 2 < 3 < 4$ but *thought of as letters and not numbers*!

So:

$$\text{Allocation } 1 : 0.25, 0.25, 0, 0.5 = \{b, b, a, c\};$$

$$\text{Allocation } 2 : 0.25, 0.25, 0.25, 0.25 = \{b, b, b, b\};$$

$$\text{Allocation } 3 : 0.25, 0.25, 0.5, 0.0 = \{b, b, c, a\};$$

and so on. You should be able to see the lexicographical order of the letters starting from the first to the last as if looking for "words" in a dictionary. The words are "bbac," "bbbb," "bbca," etc. How many allocations are there? It is possible to find that number, but the task requires a complex recursive formula that depends on several inputs and as this is just an introductory lesson to the topic, we will not pursue it.

However, once the notion of lexicographical order is understood, we can use the portfolios in a sequence that can be programmed.

2. Portfolio Optimization Using a Probabilistic Risk Measure

Earlier, we mentioned that an optimal portfolio cannot always be expressed in terms of mean and variance. There are instances where an investor cannot measure the risk of their portfolio only based on volatility alone. In Module 1, Lesson 1, we mentioned that semi-variance can also be used, but other risk measures have been suggested as well, like the mean absolute deviation for instance. Further, in Module 2, we looked at how to include skewness and kurtosis in an optimization scheme. We should not forget VaR as another possible choice, despite the fact that it suffers from certain known limitations. Minimax measures have also been suggested: We compute the max of certain risks that are relevant for our situation and then choose the portfolio that minimize the maximum risk.

This is less general than working with probabilistic scenarios, but it is still a pace in a direction that is outside the classical optimization problems in portfolio theory. The aim of this section is to introduce students to a different way to state the problem.

Let's look a little more at how this could be done. To this purpose, we assume that W_0 is the investor's initial wealth, which they would like to invest in a universe of N possible risky assets, $S_1, \ldots, S_N$. To avoid confusion with wealth, we use the symbols $x_1, \ldots, x_N$ to denote the percentage of W_0 invested in each asset. The usual constraints apply, i.e., the space of viable portfolios is defined as:

$$\mathcal{X} = \{(x_1, \ldots, x_N) : \sum_{i=1}^{N} x_i = 1 \quad \text{and} \quad x_i \geq 0\}$$

Now, let R_i be the return on asset i for $i = 1, 2, \ldots, N$. Furthermore, we assume that the R_i's are stochastic, and, for convenience, that

$$R_i \sim N(\mu_i \sigma_i^2).$$

We do not know μ_i, but it can be estimated if we have observations for T periods, that is

$$\hat{\mu}_i \equiv r_i = \frac{1}{T} \sum_{t=1}^{T} R_{it}$$

where R_{it} simply denotes the return for the i-th asset during the t-th time period. Then, we can estimate the return on the portfolio as

$$r(x_1, x_2, \ldots, x_N) \sum_{i=1}^{N} r_i x_i.$$

If the σ_is are not known, they can be estimated as well.

So far, nothing is really different from what we have done many times already. However, now we introduce what we call a **probabilistic risk measure**:

$$\omega_p(x_1, \ldots, x_N) = \min_{1 \leq i \leq N} \{P[|R_i - r_i| x_i \leq \theta \cdot \epsilon]\}.$$

We should notice that $|R_i - r_i|$ represents the deviation from the average return. $|R_i - r_i| x_i$ informs us how much that deviation impacts our portfolio via the weight of asset i in the portfolio, x_i. ϵ is a measure of risk if we think that larger deviations are less desirable than fewer deviations. For instance, we could link ϵ to a measure of average risk for the entire portfolio:

$$\epsilon = \frac{1}{N} \sum_{i=1}^{N} \sigma_i,$$

but this is just one possible way.

Note: The risk measure that we introduced penalizes assets in the portfolio whose returns exceeds the average return by more than a certain amount of market volatility. We could modify it to penalize

only assets that underperform. The only caveat is that, as we will see briefly, we need to be able to derive certain analytical results in order to create a suitable algorithm.

θ is a constant used to adjust the risk aversion. The smaller θ, the more conservative the investor.

We assume that the investor is still risk averse and rational, and therefore, their portfolio selection can be described by a bi-criteria optimization problem:

$$\max_{x_1,\ldots,x_N} \{\omega_p, \sum_{i=1}^{N} r_i x_i\}.$$

To make the notation more compelling, let

$$p(x_i) \equiv P[|R_i - r_i| x_i \leq \theta\epsilon], \;\; i = 1, \ldots, N.$$

Then, the maximization problem can be written as:

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$$\begin{aligned} &\max_{x_1,\ldots,x_N} \{\min p(x_i), \sum_{i=1}^{N} r_i x_i\} \\ &\text{s.t. } (x_1, \ldots, x_N) \in \mathcal{X}. \end{aligned}$$

which is the bi-criteria optimization problem. In this problem, we are still concerned about the volatility of returns, but it is quite different from the optimal problem we considered before, i.e.:

$$\begin{aligned} &\max_{x_1,\ldots,X_N} \sum_{i=1}^{N} r_i x_i - \frac{\theta}{2} \sum_{i=1}^{n} \sum_{j=1}^{N} \sigma_i \sigma_j x_i x_j \\ &\text{s.t. } (x_1, \ldots, x_N) \in \mathcal{X}. \end{aligned}$$

In the new setup, we are concerned about the probability of certain outcomes and not the overall variability.

This problem can be solved by introducing a new variable and N additional constraints:

$$\begin{aligned} &\max_{x_1,\ldots,x_N, y} \{y, \sum_{i=1}^{N} r_i x_i\} \\ &\text{s.t. } y \leq f(x_i)), \;\; i = 1, \ldots, N, \\ &(x_1, \ldots, x_N) \in \mathcal{X}. \end{aligned}$$

Using the assumption of normality, with some calculus, we find that

$$p(x_i) = 2 \int_0^{\frac{\theta\epsilon}{x_i}} \frac{1}{\sqrt{2\pi}\sigma_i} e^{-(x_i^2/2\sigma_i^2)} \, dz_i$$

where $z_i = R_i - r_i$. We are trying to minimize $f(x_i)$ and it should be easy to realize that $f(x_i)$ is monotonically decreasing with respect to x_i. Hence, since we need to have $y \leq f(x_i)$ and that to maximize our objective function above we need to push y to reach $\min_{i=1,\ldots,N} p(x_i)$, then for any number $y \in (0, 1)$ (recall that y is a probability!), the fact that $p(x_i)$ is monotonic implies that the upper bound for x_i is given by $p^{-1}(y)$. More specifically, we use the notation $\tilde{x}_i = p^{-1}(y)$ and

define $U_i = \min\{1, \tilde{x}_i\}$. Thus, for any fixed value of y, the optimization problem can be further re-written as a linear programming (LP) problem:

$$\begin{aligned} &\max_{x_1,\ldots,x_N} \sum_{i=1}^{N} r_i x_i \\ &\text{s.t.} \sum_{i=1}^{N} x_i = 1 \\ &\quad 0 \le x_i \le U_i, \;\; i = 1, \ldots, N. \end{aligned}$$

Now, assuming that none of the assets have the same returns and standard deviation (in continuous setting this probability is zero!), _Sun et al. _(see References) have shown that the problem can be solved as follows:

- Order the asset in order of return, such that $r_1 \ge r_2 \ge \ldots \ge r_N$.
- Find the integer $k \le N$ such that $\sum_{i=1}^{k-1} U_i < 1$, and $\sum_{i=1}^{k} U_i \ge 1$.
- Let

$$x_i^\star = \begin{cases} U_i & \text{if } i = 1, 2, \ldots, k-1; \\ 1 - \sum_{i=1}^{k-1} U_i & \text{if } j = k, \\ 0 & \text{otherwise.} \end{cases}$$

and $x_1^\star, x_2^\star, \ldots, x_N^\star$ is the optimal solution for our problem.

Note that in the previous solution, we fixed y. This is not different than what we did when computing the efficient frontier in the sense of Markowitz's mean-variance optimization where we can fix σ and then find the optimum portfolio and its return, and we keep repeating the process.

Another interesting type of risk function is the group known as ℓ functions. In practice, the ones used are ℓ_1, ℓ_2, and ℓ_∞. The name comes from their respective metrics in analysis and functional analysis. The ℓ_2 risk measure is just the classic M-V formulation and it amounts to quadratic programming, and therefore, there is not much to add. The ℓ_1 risk measure can be introduced like this:

$$\begin{aligned} &\min_{x_1,\ldots,x_N} L(x_1, \ldots, x_N) \\ &\text{s.t.} \sum_{i=1}^{N} r_i x_i \ge \rho, \\ &\quad \sum_{i=1}^{N} x_i = 1, \\ &\quad 0 \le x_i \le u_i, \;\; i = 1, 2, \ldots, N \end{aligned}$$

where (see the paper by Konno and Suzuki listed among the references)

$$L(x_1, \ldots, x_N) \equiv E\left[\left|\sum_{i=1}^{N} R_i x_i - E[\sum_{i=1}^{N} R_i x_i]\right|\right] = \sqrt{\frac{2}{\pi}}\sigma(x_1, \ldots, x_N)$$

with

$$\sigma(x_1,\ldots,x_N) \equiv \sqrt{\sum_{i=1}^{N}\sum_{j=1}^{N}\sigma_{ij}x_i x_j}.$$

The interesting part to notice is that this is now an LP problem! Then, the risk function can be approximated like this:

$$L(x_1,\ldots,x_N) = \frac{1}{T}\sum_{t=1}^{T}\left|\sum_{i=1}^{N}(r_i t - r_i)x_i\right|$$

and letting $a_{it} = r_{it} - r_i$ for each $i = 1, 2, \ldots, N$ and $t = 1, 2, \ldots, T,$ we can write a new LP problem as:

$$\begin{aligned}
&\min_{x_1,\ldots,x_N} \frac{1}{T}\sum_{i=1}^{T} y_t \\
&\text{s. t. } y_t + \sum_{i=1}^{N} a_{it}x_i \geq 0, \;\; t_1,\ldots,T, \\
&\quad y_t - \sum_{i=1}^{N} a_{it}x_i \geq 0, \;\; t_1,\ldots,T, \\
&\quad \sum_{i=1}^{N} r_i x_i \geq \rho, \\
&\quad \sum_{i=1}^{N} x_i = 1, \\
&\quad 0 \leq x_i \leq u_i, \;\; i = 1,\ldots,N.
\end{aligned}$$

In this setup,u_i is the maximum percentage of capital that can be invested in the i-th asset. In addition, while the original problem has N variables, the new one has $N + T.$

The ℓ_∞ formulation of the problem looks like this:

$$\begin{aligned}
&\min\left(\max_{x_1,\ldots,x_N} q_i r_i, -\sum_{i=1}^{N} r_i x_i\right) \\
&\text{s.t. } (x_1,\ldots,x_N) \in \mathcal{X}
\end{aligned}$$

where The problem can then be transformed to:

$$\begin{aligned}
&\min_{x_1,\ldots,X_N,y} \lambda y + (1-\lambda)\left(-\sum_{i=1}^{N} r_i x_i\right) \\
&\text{s.t. } q_i x_i \leq y, \;\; i = 1,\ldots,N \\
&\quad (x_1,\ldots,x_N) \in \mathcal{X}
\end{aligned}$$

for a fixed $\lambda \in (0,1).$ The equivalence in this case is not straightforward, and interested students can find some references at the end of the lesson. Using the latest program, it is possible to show that for any $\lambda \in (0,1),$

$$x_i^\star = \begin{cases} \frac{W_0}{q_i}\left(\sum_{j\in H(\lambda)} \frac{1}{q_j}\right)^{-1} & \text{if } i \in H(\lambda); \\ 0 & \text{otherwise}, \end{cases} \quad y^\star = W_0\left(\sum_{j\in H(\lambda)} \frac{1}{q_j}\right)^{-1}$$

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with $H(\lambda)$ defining the set of assets to be invested and that is determined according to a complicated rule provided in the Cai et al.'s paper.
The different approaches, as expected, return different efficient frontiers, and it is up to the practitioner to decide which one is the most suitable for specific applications.

3. An Application of the Probabilistic Approach for Optimal Portfolio Selection

So far we have covered the main ideas of PSO. PSO is not a small feat to pull off, and we discussed some of the main issues. Then, we considered probabilistic measures to determine optimal portfolios because the mean-variance setup is not ideal in all cases.

The following **required** reading:

Probabilistic approach for optimal portfolio selection using a hybrid Monte Carlo simulation and Markowitz model

is reasonably simple and illustrates well some of the nuances implicit in probabilistic approaches to optimal portfolio determination and, because of the simulations, can also be linked to a simple instance of PSO.

4. Conclusion

In this lesson, we have covered the basics of probability scenario optimization and looked at alternative ways to deal with risk in portfolio problems. Variance is just one way to do it. Finally, the application contained in the required reading illustrates some the practical nuances that practitioners need to deal with in solving real problems.

The next lesson will introduce a totally different topic as we move to consider behavioral finance, behavioral portfolio theory, and how accounting for individuals' emotional and cognitive biases changes the way we look at the problem of portfolio optimization.

Pyportfolioopt Package

We are at the end of Module 3 and by now the key traditional approaches to portfolio optimization have been considered. Furthermore, at this point in time, students should have learned several algorithms that are necessary to implement those techniques. Thus, it is the right time to bring up that there exist several large packages that allow practitioners to implement the methodologies that were considered and more. One such package is **pyportfolioopt**.

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Uses and Misuses of the Black-Litterman Model in Portfolio Construction*

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ABSTRACT

The Black-Litterman model has gained popularity in applications in the area of quantitative equity portfolio management. Unfortunately, many recent applications of the Black-Litterman to novel aspects of quantitative portfolio management have neglected the rigor of the original Black-Litterman modelling. In this article, we critically examine some of these applications from a Bayesian perspective. We identify three reasons why these applications may create losses to investors. These three reasons are: 1) Using a prior without "anchoring" the prior to an equilibrium model; 2) Using a prior and an equilibrium model that conflict with one another; and 3) Ignoring the implications of the estimation error of the variance-covariance matrix. We also quantify the loss first analytically and also numerically based on historical data on 10 major world stock market indices. Our conservative estimate of the loss is around a 1% reduction in the annualized return of the portfolio.

Keywords: Black-Litterman; Bayes Rule; Portfolio Construction; Econometric Techniques; Simulations

1. Introduction

The Black-Litterman model is a powerful tool in the portfolio construction process. It has gained popularity among practitioners for the past two decades, and its applications to various aspects of the portfolio construction process have been discussed in the literature. We believe, however, that some of the applications adopted by practitioners and discussed in the literature deserve more critical examination. In this paper, we will highlight the uses of the Black Litterman model that we find problematic, or at least lacking the vigor of the original formulation of the Black-Litterman model. We will present numerical examples to make our case stronger and, where appropriate, will propose alternative approaches.

Black and Litterman [1,2] saw two strengths of their approach:

1) The subjective views of the investors can be easily incorporated in the portfolio construction process.

2) The Black-Litterman mean-variance optimization does not produce unreasonable solutions, as the standard mean-variance framework does.

The first of these two comes from the feature of the model that investors' subjective views are expressed as linear combinations of expected returns of assets, rather than as expected returns of individual assets. That is, the subjective view need not be an exact value of the expected return of an individual asset, but rather can be expressed as the expected return of two assets or more in relation to each other. This type of formulation is easier for investors to apply. The second strength comes from the model's feature that the investors' subjective views are combined with an equilibrium model that tilts the portfolio weights away from the market capitalization weights based on the relative uncertainty in the investor's views. This anchors the portfolio weights towards the implied market capitalization weights, thus not allowing for extreme weights due to differences in expected returns.

Out of these two features, the first feature is less essential to the Black-Litterman model. It is relatively easy to come up with an alternative way of specifying investors' subjective views.[1] The spirit of the Black Litterman model can be retained from a variety of subjective prior specifications. The second feature, however, is much more significant. An attempt to modify this feature of the

*We would like to especially thank Josh Ruben for research assistance. We also thank Frank Fabozzi for comments.

[1]The Black Litterman model allows an expression of subjective views such as "the sum of asset A return and asset B return will be positive." That is, it allows the investor to make a statement on a linear combination of many asset returns. If this type of statement is not allowed, one may have to be more explicit. An example is: "asset a return is likely to be around 10% while asset B return is likely to be around 5%."

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model is likely to introduce inconsistencies. The reason is that there is really only one right way to combine investors' subjective views with a given model, and there is no other correct alternative. One may choose different views or different models, but once the views and a model are chosen, there is only *one* way to combine them according to Bayes' rule.

Some applications of the Black-Litterman model by other authors have attempted to modify the second feature of the model, and, in doing so, they have lost the mathematical rigor of the original Black-Litterman model. Other applications have simply ignored the nature of the subjective view and mixed it up with the model in an inconsistent manner.

In this paper we discuss three representative (mis-) uses of the Black-Litterman model in portfolio construction. We will attempt to quantify the possible losses created by the misuse of the model and, where appropriate, propose an alternative.[2] The first application we consider was discussed in Jones, Lim, and Zangari [4].[3] We will argue that their approach could create a loss to investors because the prior is not "anchored" to an equilibrium model. More specifically, the estimate of the mean return from the model is not included in the process, making the prediction of the mean return less than optimal. The second application we consider was discussed in Fabbozi, Forcardi, and Kolm [5]. We will argue that their approach could create losses to investors if the prior and the equilibrium model conflict each other. The third application we consider is the so called reverse optimization, which is quite popular among practitioners. We will argue that this reverse optimization could have surprisingly large errors in the resulting mean estimates.

In this article, we take a Bayesian perspective. A Bayesian perspective allows us to quantify the losses of investors without too many complications in the analysis. We borrow the framework of Satchell and Scowcroft [6], and extend it so that investors' loss can be discussed.

The paper is organized as follows: Section 2 discusses the use of the Black-Litterman technique without an equilibrium model, and the potential losses associated with that methodology; Section 3 discusses the use of the Black-Litterman model with data based priors that conflict with the model, and the loss associated with that methodology; Section 4 discusses the use of the Black-Litterman approach as a reverse optimization and the implication of using an estimated variance-covariance matrix; and Section 5 concludes the paper.

[2]Chincarini and Kim [3] examined other situations in which improper portfolio construction led to utility losses to the investor.

[3]We are not implying that the researchers we cite were unaware of the problems we are raising. In fact, the applications we discuss were not the main point of their articles, which explains why the problems we discuss now did not receive full attention in their article.

2. Using the Black Litterman Approach without an Equilibrium Model

2.1. How It Is Usually Done

Jones, Lim, and Zangari (2007; henceafter JLZ) presented a way in which the Black Litterman model can be used to incorporate a factor-based view in a structured equity portfolio.

Let $\boldsymbol{w}$ be an N-dimensional portfolio weight vector, where the i-th element w_i is the portfolio weight of asset i. Let $\boldsymbol{B}$ be an N-by-K factor exposure matrix, where (i, j) element $b_{i,j}$ is the exposure of stock i to factor j.[4] Then the factor exposure of the portfolio is $\boldsymbol{B}'\boldsymbol{w}$. A portfolio manager's factor view can be expressed as a K-dimensional vector $\boldsymbol{\lambda}$, which is the portfolio manager's desired value of $\boldsymbol{B}'\boldsymbol{w}$.[5]

JLZ suggest the following steps for the portfolio construction:

1) Specify the factor view vector $\boldsymbol{\lambda}$.

2) Calculate the optimal tilt portfolio weight, *i.e.* the weight of the portfolio that is optimal given the factor view vector. That is, solve:

$$\min_{\boldsymbol{w}} \quad \boldsymbol{w}'\hat{\boldsymbol{\Sigma}}\boldsymbol{w} \quad \text{s.t.} \quad \boldsymbol{B}'\boldsymbol{w} = \boldsymbol{\lambda} \tag{1}$$

where $\hat{\boldsymbol{\Sigma}}$ is an N-by-N variance-covariance matrix estimate of asset returns. The solution to this problem is

$$\boldsymbol{w}_{OTP} = \hat{\boldsymbol{\Sigma}}^{-1}\boldsymbol{B}\left(\boldsymbol{B}'\hat{\boldsymbol{\Sigma}}^{-1}\boldsymbol{B}\right)^{-1}\boldsymbol{\lambda} \tag{2}$$

where OTP is an acronym for optimal tilt portfolio.[6]

3) Compute the Black Litterman alpha, *i.e.* the expected return that produces the optimal tilt portfolio identified in the previous step. That is, solve the following equation for $\boldsymbol{\alpha}$:

$$\boldsymbol{w}_{OTP} = \text{argmax} \quad \boldsymbol{w}'\boldsymbol{\alpha} - \frac{1}{2}\boldsymbol{w}'\hat{\boldsymbol{\Sigma}}\boldsymbol{w} \tag{3}$$

The solution to this problem is[7]

$$\boldsymbol{\alpha}_{BL} = \hat{\boldsymbol{\Sigma}}\boldsymbol{w}_{OTP} \tag{4}$$

4) Find an optimal portfolio based on the Black Litterman alpha, $\boldsymbol{\alpha}_{BL}$, and any other constraints using a standard mean variance optimization. For example, an optimal portfolio can be found by solving the following quadratic programming problem:

$$\begin{aligned} &\max_{\boldsymbol{w}} \quad \boldsymbol{w}'\boldsymbol{\alpha}_{BL} - \frac{1}{2}\boldsymbol{w}'\hat{\boldsymbol{\Sigma}}\boldsymbol{w} \\ &\text{s.t.} \quad \boldsymbol{w}'\boldsymbol{\iota} = 1 \quad l_i \le w_i \le u_i, \quad i = 1, \cdots, N \end{aligned} \tag{5}$$

[4]In many applications, $\boldsymbol{B}$ is a matrix of observable characteristics.

[5]Portfolio managers may determine the value of $\boldsymbol{\lambda}$ based on their expectation of "factor returns." For example, if they expect the size of return to be high, they may choose a large value for the size exposure.

[6]The derivation of this result and other results is included in an appendix available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id = 2191278.

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where l_i and u_i are lower and upper bounds on asset i weights.[8]

In the four step procedure described above, no explicit reference is made to an equilibrium model, which is an integral component of the original Black-Litterman model. In fact, there is no explicit reference to *any* model of stock returns.

Implicitly, though, the above procedure requires *a* model to produce an estimate $\hat{\boldsymbol{\Sigma}}$. Even when one decides to use a historical variance-covariance matrix for $\hat{\boldsymbol{\Sigma}}$, one is using a model. The implicit model is that the historical variance-covariance matrix is a good estimator of the true variance-covariance matrix.

Using a model to produce an estimate $\hat{\boldsymbol{\Sigma}}$ and using the estimate in calculating the Black Litterman alpha, $\boldsymbol{\alpha}_{BL}$, however, is not equivalent to "combining a subjective view with an equilibrium model" as is done in the original Black Litterman model. The problem with this approach is that the model is used only to produce an estimate of the variance-covariance matrix. *This approach ignores the information in the data about the mean return*.[9] The consequence is that a subjective view is combined with a model, but only partially and incompletely. The incompleteness does create a problem, as will be shown in the next sub-section.

2.2. The Cost of Ignoring Mean Estimates

The JLZ procedure calculates the Black Litterman alpha, $\boldsymbol{\alpha}_{BL}$, based on a view and a model. The view is summarized in $\boldsymbol{\lambda}$ and the model is summarized in an estimate of $\boldsymbol{\Sigma}$. As mentioned above, however, the view and the model are not combined in an efficient way. In particular, the mean component of the model is ignored. This creates some loss, the magnitude of which we will quantify in this subsection.

We first discuss our measures of loss. In order to measure an investor's loss, we need to develop a formal model of returns and investor views. In the subsequent text, we discuss a measure of loss based upon a formal model of returns and views.

2.2.1. A Measure of Loss

One can quantify the loss generated in a portfolio construction process in terms of the Sharpe ratio, the information ratio, or an investor's utility. For the purpose at hand, the most natural choice is to use the investor's utility implicit in the Black Litterman approach, which is

$$U(\mu,\sigma) = \mu - \frac{\delta}{2}\sigma^2 \tag{6}$$

where μ is the mean return of the portfolio and σ is the standard deviation of the portfolio return. In particular, we will set the value of δ to 1, as was done in the JLZ procedure.[10] Given the utility function, we can calculate the highest utility one can achieve given the model and the view, and we can also calculate the utility that one actually achieves. The difference is our measure of the loss.

To compute the utility, we will use the *predictive* mean and variance based on the model and the view. This is justified as our primary interest is in the investor's portfolio construction strategy, not the validity of the investor's belief. In addition to this, the predictive mean and variance are the only mean and variance that the investor can actually calculate. It would not make sense to define the investor's utility in terms of the quantities that are not known to the investor.[11]

If a portfolio construction process combines the model and the view efficiently, then by construction the loss is zero. This is true in the original Black Litterman approach. It is not possible to improve the utility *given the model and the view*. In the JLZ procedure, however, it is possible to improve the utility *given the model and the view*. It is possible to do so by adding the mean component of the model, as we will show below.

2.2.2. The Model and the View

Consider the following model:

$$\boldsymbol{r}_{t+1} = \boldsymbol{B}_t \boldsymbol{f} + \boldsymbol{e}_{t+1}, \quad \boldsymbol{e}_{t+1} \sim N\left(0, \frac{1}{h}\boldsymbol{V}_e\right) \tag{7}$$

where $\boldsymbol{r}_{t+1}$ represents the vector of stock returns, $\boldsymbol{B}_t$ represents the $N \times K$ matrix of factor exposures for K factors and N stocks, and $\boldsymbol{f}$ is a $K \times 1$ vector of factor premiums, $\boldsymbol{e}_{t+1}$ is the error vector, and $\frac{1}{h}\boldsymbol{V}_e$ is the covariance matrix of the error. We include a scalar h that affects the overall volatility of the market.

While this model is not explicitly stated in the four step procedure outlined in the previous section, this is probably the most natural model for the JLZ investor, if he were forced to adopt one. (Recall that the lack of an explicit model is one weakness of the JLZ procedure.) Without a model of this kind, it would be hard to mo-

[8]Without the constraints, the solution to this optimization would be the optimal tilt weights, $\boldsymbol{w}_{OTP}$.

[9]One might argue that means are typically not estimated accurately, and that ignoring the mean estimate is not a big deal. However, a variance estimate does not have the same meaning if one ignores the mean estimate. Recall that the variance measures the deviation *from the mean*. Without specifying a mean, a variance is an incomplete concept. One might end up making a decision as if the true mean is zero, which can be worse than using an imprecisely estimated mean.

[10]Our main arguments are not affected by the value of δ, nor by the choice of how loss is measured.

[11]See the discussion preceeding Equation (14) for an example of alternatives that we did not take.

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tivate a factor tilted portfolio using the JLZ procedure.[12]

The variables $\boldsymbol{r}_{t+1}$ and $\boldsymbol{B}_t$ are observable, while $\boldsymbol{f}$ and $\boldsymbol{e}_{t+1}$ are not observable. The error variance $\frac{1}{h}\boldsymbol{V}_e$ is not known either. Before creating a portfolio for $\boldsymbol{T}+1$, the investor observes $(\boldsymbol{r}_1,\cdots,\boldsymbol{r}_T)$ and $(\boldsymbol{B}_0,\cdots,\boldsymbol{B}_{T-1})$ and also $\boldsymbol{B}_T$. With these data, the investor must determine the distribution of $\boldsymbol{r}_{T+1}$, by estimating $\boldsymbol{f}$, h, and $\boldsymbol{V}_e$, or by updating his belief about these quantities.

The JLZ procedure introduced $\boldsymbol{\lambda}$ to capture the investor's belief. $\boldsymbol{\lambda}$ is the desired tilt to specific factors, which depends on the investor's belief about factor returns. Rather than consider the prior of (*i.e.* distribution of possible values of) $\boldsymbol{\lambda}$, we consider the prior of factor returns. While this approach may seem like an unnecessary detour, it makes the analysis much more tractable.

We interpret $\boldsymbol{\lambda}$ in the context of the following prior of $\boldsymbol{f}$ and h:

$$\begin{aligned} \boldsymbol{f}\,|h &\sim N\left(\boldsymbol{m}_f,\frac{1}{h}\boldsymbol{V}_f\right) \\ h &\sim G(p,q) \end{aligned} \tag{8}$$

That is, $\boldsymbol{f}$ has a multivariate normal distribution conditional on h, and h has a Gamma distribution.[13] By including $\frac{1}{h}$ in the variance of $\boldsymbol{f}$, we are assuming that the overall uncertainty of the factor premium is proportional to the overall market volatility.[14] This is the simplest prior one can adopt, and is general enough for our purpose. Satchell and Scowcroft (2000) also use this prior when they interpret the Black Litterman model. Note also that this prior allows for the noninformative prior as a special case, as will be shown in the next section.

It is easy to show that $\boldsymbol{\lambda}$ is related to the parameters in Equations (7) and (8) in the following way:

$$\boldsymbol{\lambda} = \boldsymbol{B}'\boldsymbol{\Sigma}^{-1}\boldsymbol{B}\boldsymbol{m}_f \tag{9}$$

where $\boldsymbol{\Sigma}$ is the variance-covariance matrix of $\boldsymbol{r}_{T+1}$, *i.e.*, $\boldsymbol{B}_T\frac{1}{h}\boldsymbol{V}_f\boldsymbol{B}_T' + E\left(\frac{1}{h}\boldsymbol{V}_e\right)$.[15]

Equation (7) represents the model, and Equation (8) represents the investor's view. And together, they describe the belief system of the JLZ investor completely.

Let us be clear about our distinction between the model and the view. The *model* specifies the relationship among variables that we observe. The *view* is about the quantities that we do not observe. In particular, the view specifies the relationship among *parameters* of the model. The view has *hyper-parameters*, whose values investors choose on their own. In Equation (7), there are three parameters: $\boldsymbol{f},\boldsymbol{V}_e,h$. In Equation (8), there are four hyper-parameters: $\boldsymbol{m}_f,\boldsymbol{V}_f,p,q$. The distinction between the parameters and the hyper-parameters can be made clearly: parameters are a part of the model, while hyper-parameters are a part of the view.

These concepts might be clearer if we describe them in a series of steps.

1) All the equations are known to everyone;

2) At the beginning of our portfolio formulation stage, God decides the value of the hyper-parameters $\boldsymbol{m}_f,\boldsymbol{V}_f,p,q$. God announces this decision to every investor;

3) On day 1, God draws parameters $\boldsymbol{f},\boldsymbol{V}_e,h$ using Equation (8). These quantities are *never known* to the investor;

4) On day 2, God draws data $(\boldsymbol{r}_1,\cdots,\boldsymbol{r}_T)$, $(\boldsymbol{B}_0,\cdots,\boldsymbol{B}_{T-1})$, and also $\boldsymbol{B}_T$, and shows them to the investor;[16]

5) After observing data, the investor estimates the

[12]JLZ do mention this model and even suggest to use it to determine the view. We will discuss the problem of using a data based view in the next section. For now, we interpret the view as completely independent from the data.

[13]The Gamma distribution makes the analysis tractable. The Gamma distribution, when used with normal distribution, has the property that the posterior will be of the same type of distribution as the prior. Due to this property, the Gamma-Normal prior is called a conjugate prior, and is widely used in Bayesian analysis. We follow the parameterization of the Gamma distribution used by Koop (2003). The pdf of the above distribution is

$$\left(\frac{2p}{q}\right)^{\frac{q}{2}}\Gamma\left(\frac{q}{2}\right)^{-1}h^{\frac{q}{2}-1}e^{-\frac{hq}{2p}}$$

[14]This aspect of the prior is required for a conjugate prior, *i.e.* a prior that leads to a posterior of the same distribution. Using a conjugate prior makes our analysis simple, but the conclusion of our analysis does not depend on a conjugate prior.

[15] $\boldsymbol{\lambda}$ is the factor tilt implied by the investor's view and the model. Thus, we can find an optimal portfolio given the investor's view and the model, and determine the factor tilt of that portfolio, which should be $\boldsymbol{\lambda}$. Given the investor's view and the model, the unconditional distribution of $\boldsymbol{r}_{T+1}$ is

$$\boldsymbol{r}_{T+1} \sim N\left(\boldsymbol{B}_T\boldsymbol{m}_f,\boldsymbol{\Sigma}\right) \tag{10}$$

where $\boldsymbol{\Sigma}$ is the variance-covariance matrix of $\boldsymbol{r}_{T+1}$, *i.e.*, $\boldsymbol{B}_T\frac{1}{h}\boldsymbol{V}_f\boldsymbol{B}_T' + E\left(\frac{1}{h}\boldsymbol{V}_e\right)$. If the investor has the utility function specified in Equation (6) with δ equal to **1**, then the optimal portfolio, given the above distribution, can be found from the following optimization:

$$\max_{\boldsymbol{w}} \quad \boldsymbol{w}'\boldsymbol{B}_T\boldsymbol{m}_f - \frac{1}{2}\left(\boldsymbol{w}'\boldsymbol{\Sigma}\boldsymbol{w}\right) \tag{11}$$

The solution to the above problem is

$$\boldsymbol{w}^* = \boldsymbol{\Sigma}^{-1}\boldsymbol{B}_T\boldsymbol{m}_f \tag{12}$$

From the factor tilt of this portfolio, we can obtain Equation (9). This comes from the second equation in the maximization problem, that $\boldsymbol{B}'\boldsymbol{w} - \boldsymbol{\lambda} = 0$. Thus, substitution of Equation 12 gives $\boldsymbol{\lambda} = \boldsymbol{B}'\boldsymbol{w} = \boldsymbol{B}'\boldsymbol{\Sigma}^{-1}\boldsymbol{B}_T\boldsymbol{m}_f$.

[16]Note that, in our analysis, $\boldsymbol{B}$'s are matrices of observable characteristics, not estimates of factor loadings. Estimation error does not arise as far as $\boldsymbol{B}$'s are concerned.

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parameters, makes a prediction of $\boldsymbol{r}_{T+1}$, and creates a portfolio;

6) On day 3, God draws $\boldsymbol{r}_{T+1}$ using Equation (7);

7) After observing $\boldsymbol{r}_{T+1}$, the investor calculates his portfolio return.

In this scheme, one could think of a number of different expectations and variances of the portfolio return. Let us consider three of them. First, we can think of "God's expectation," which is conditional on the true value of parameters $\boldsymbol{f}$, h, and $\boldsymbol{V}_e$. Since only God knows the value of $\boldsymbol{f}$, h, and $\boldsymbol{V}_e$, this expectation is probably not very interesting to our hypothetical investor. Second, we can think of the "predictive expectation," which is conditional on the data as well as on the hyper-parameters, but not on the true value of the parameters. That is, it is the expectation based on the investor's best prediction given the information available. This is something rational investors can calculate. Third, we can think of "an economist's expectation," which is conditional only on the hyper-parameters $\boldsymbol{m}_f, \boldsymbol{V}_f, p, q$. This is the expectation one obtains by imagining many different worlds where different parameters could have been chosen from the same hyper-parameters. To investors who live only once, this concept is not particularly interesting. Their utility does not depend on what could have happened in other worlds. This concept, however, could be interesting to economists who want to make general statements about certain decision making schemes. Note also that "an economist's expectation" is obtained if we integrate the data out of the "predictive expectation."[17]

Our analysis is based on the "predictive expectation" and similar concepts, as we are working with a utility-based measure of loss. Predictive expectation is conditional on the value of the hyper-parameters. Thus, an analysis of predictive expectation can be influenced by the choice of hyper-parameter values. In our numerical analysis, we vary the values of the hyper-parameters to make sure that our results are robust, and that they are not peculiar to the chosen values of the hyper-parameters.

We provide the formula for the predictive mean and the predictive variance-covariance matrix of the returns here without proof.[18] The predictive mean and the predictive variance-covariance matrix of returns are obtained by applying statistical operators to the model and then using the Bayesian updating formula (see Zellner [7] and Koop [8]). Let us denote the predictive mean of the return as $\tilde{E}(\boldsymbol{r}_{T+1})$ and the predictive variance-covariance matrix of the return as $\tilde{V}(\boldsymbol{r}_{T+1})$. Then

$$\begin{aligned}\tilde{E}(\boldsymbol{r}_{T+1}) &= \boldsymbol{B}_T \tilde{E}(\boldsymbol{f}) \\ \tilde{V}(\boldsymbol{r}_{T+1}) &= \boldsymbol{B}_T \tilde{V}(\boldsymbol{f}) \boldsymbol{B}_T' + \tilde{V}(\boldsymbol{e}_{T+1})\end{aligned} \tag{14}$$

The predictive mean of the factor premium $\tilde{E}(\boldsymbol{f})$, the predictive variance-covariance matrix of the factor premium $\tilde{V}(\boldsymbol{f})$, and the predictive variance-covariance matrix of the error $\tilde{V}(\boldsymbol{e}_{T+1})$ are determined by the Bayesian updating formula:

$$\begin{aligned}\tilde{E}(\boldsymbol{f}) &= \left(\boldsymbol{V}_f^{-1} + \boldsymbol{S}_B\right)^{-1}\left(\boldsymbol{V}_f^{-1}\boldsymbol{m}_f + \boldsymbol{S}_B\hat{\hat{\boldsymbol{f}}}\right) \\ \tilde{V}(\boldsymbol{f}) &= \frac{q+NT}{q+NT-2}\frac{1}{\tilde{E}(h)}\left(\boldsymbol{V}_f^{-1} + \boldsymbol{S}_B\right)^{-1} \\ \tilde{V}(\boldsymbol{e}_{T+1}) &= \frac{1}{\tilde{E}(h)}\tilde{\boldsymbol{V}}_e\end{aligned} \tag{15}$$

The first equation simply says that the predictive mean of the factor premium is a "precision-weighted" average of the prior mean $\boldsymbol{m}_f$ and the GLS estimator $\hat{\boldsymbol{f}}$. The second and the third equation simply break down the components of the variance-covariance matrix. The predictive mean of the error precision parameter $\tilde{E}(h)$ can be expressed as follows:[19]

$$\begin{aligned}&\frac{1}{\tilde{E}(h)} \\ &= \frac{1}{q+NT}\left[\frac{q}{p} + \boldsymbol{S}_e + \left(\boldsymbol{m}_f - \hat{\hat{\boldsymbol{f}}}\right)'\left(\boldsymbol{V}_f + \boldsymbol{S}_B^{-1}\right)^{-1}\left(\boldsymbol{m}_f - \hat{\hat{\boldsymbol{f}}}\right)\right]\end{aligned} \tag{16}$$

2.2.3. An Analytic Formula for Loss

Now we are ready to compare the utility of an investor who follows the JLZ procedure (the JLZ investor) to the utility of a rational investor who combines all information efficiently (the RTL investor), both having the same view and the same model. We first discuss the utility of the two investors in general form. Then we examine a special case where we can provide an analytic solution for loss.

Let us first characterize the utility of the RTL investor. As noted before, the investor would not consider $\boldsymbol{B}\boldsymbol{m}_f$

[17] That is,

$$\begin{aligned}&E(\text{portfolioreturn}\,|\,\text{hyperparameters}) \\ &= \int E(\text{portfolioreturn}\,|\,\text{hyperparameters}, \text{data})\, p(\text{data})\end{aligned} \tag{13}$$

where $p(\text{data})$ is the probability density function of the data. The integration is taken over the data. In other words, if we average the predictive expectation over all the possible values of the data, then we get an economist's expectation.

[18] The derivation of this result is included in an appendix available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2191278.

[19] The GLS estimator of the factor premium $\hat{\hat{\boldsymbol{f}}}$ is $\left(\sum_{t=0}^{T-1}\boldsymbol{B}_t'\hat{\boldsymbol{V}}_e^{-1}\boldsymbol{B}_t\right)^{-1}\sum_{t=0}^{T-1}\boldsymbol{B}_t'\hat{\boldsymbol{V}}_e^{-1}\boldsymbol{r}_{t+1}$, its precision matrix $\boldsymbol{S}_B$ is $\sum_{t=0}^{T-1}\boldsymbol{B}_t'\hat{\boldsymbol{V}}_e^{-1}\boldsymbol{B}_t$, and the error sum of square matrix $\boldsymbol{S}_e$ is $\sum_{t=0}^{T-1}\left(\boldsymbol{r}_{t+1} - \boldsymbol{B}_t\hat{\hat{\boldsymbol{f}}}\right)'\hat{\boldsymbol{V}}_e^{-1}\left(\boldsymbol{r}_{t+1} - \boldsymbol{B}_t\hat{\hat{\boldsymbol{f}}}\right)$, where $\hat{\boldsymbol{f}}$ is $\left(\sum_{t=0}^{T-1}\boldsymbol{B}_t'\boldsymbol{B}_t\right)^{-1}\sum_{t=0}^{T-1}\boldsymbol{B}_t'\boldsymbol{r}_{t+1}$, and $\hat{\boldsymbol{V}}_e$ is $\frac{p}{T}\sum_{t=0}^{T-1}\left(\boldsymbol{r}_{t+1} - \boldsymbol{B}_t\hat{\boldsymbol{f}}\right)\left(\boldsymbol{r}_{t+1} - \boldsymbol{B}_t\hat{\boldsymbol{f}}\right)'$.

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to be the best estimate of the mean return, nor would he consider the Black Litterman alpha implied by the optimal tilted portfolio to be the best estimate of the mean return. Using the Black Litterman alpha would be equivalent to not using any information from the estimation about the mean return. Instead, the RTL investor would estimate the mean and the variance-covariance matrix of returns using all the information available, and then choose a portfolio based on these estimates.

The best estimates of the mean and of the variance-covariance matrix of the returns are the predictive mean and the predictive variance-covariance matrix described above. Given the predictive mean and the predictive variance-covariance matrix of the return, the rational investor will choose the portfolio by solving the optimization problem comparable to Equation (5). Then his utility is determined as:[20]

$$U_{RTL} = \boldsymbol{w}_{RTL}\tilde{E}\left(\boldsymbol{r}_{T+1}\right) - \frac{1}{2}\boldsymbol{w}_{RTL}\tilde{V}\left(\boldsymbol{r}_{T+1}\right)\boldsymbol{w}_{RTL} \tag{17}$$

where

$$\begin{aligned} &\boldsymbol{w}_{RTL} = \operatorname{argmax}_{\boldsymbol{w}} \quad \boldsymbol{w}'\tilde{E}\left(\boldsymbol{r}_{T+1}\right) - \frac{1}{2}\boldsymbol{w}'\tilde{V}\left(\boldsymbol{r}_{T+1}\right)\boldsymbol{w} \\ &\text{s.t.} \quad \boldsymbol{w}'\boldsymbol{\iota} = 1, \quad l_i \le w_i \le u_i, \quad i = 1, \cdots, N \end{aligned} \tag{18}$$

Now let us consider the utility of the JLZ investor. The JLZ investor would follow the four steps specified in the previous section. Thus, her portfolio is the solution to Equation (5), and her utility is determined from the mean and the variance of her portfolio. That is,

$$U_{JLZ} = \boldsymbol{w}_{JLZ}\tilde{E}\left(\boldsymbol{r}_{T+1}\right) - \frac{1}{2}\boldsymbol{w}_{JLZ}\tilde{V}\left(\boldsymbol{r}_{T+1}\right)\boldsymbol{w}_{JLZ} \tag{19}$$

where

$$\begin{aligned} &\boldsymbol{w}_{JLZ} = \operatorname{argmax}_{\boldsymbol{w}} \\ &\boldsymbol{w}'\boldsymbol{B}_T\left(\boldsymbol{B}_T\tilde{V}\left(\boldsymbol{r}_{T+1}\right)^{-1}\boldsymbol{B}_T\right)^{-1}\boldsymbol{\lambda} - \frac{1}{2}\boldsymbol{w}'\tilde{V}\left(\boldsymbol{r}_{T+1}\right)\boldsymbol{w} \\ &\quad \text{s.t.} \quad \boldsymbol{w}'\boldsymbol{\iota} = 1, \quad l_i \le w_i \le u_i, \quad i = 1, \cdots, N \end{aligned} \tag{20}$$

Computation of the loss based on the above utility can be done numerically, which we explain in the next subsection. For now, we will present an analytic solution for the special case where there are no constraints in the optimization. Although, this is not a realistic way of constructing an optimal portfolio, it will help illustrate some key concepts.

Let us start with the rational investor. When there are no constraints, the solution to the optimization problem in Equation (18) is simply

$$\boldsymbol{w}_{RTL} = \tilde{V}\left(\boldsymbol{r}_{T+1}\right)^{-1}\tilde{E}\left(\boldsymbol{r}_{T+1}\right) \tag{21}$$

[20]Recall that we evaluate the utility in terms of predictive mean and variance.

Then the utility is given by

$$U_{RTL} = \frac{1}{2}\tilde{E}\left(\boldsymbol{r}'_{T+1}\right)\tilde{V}\left(\boldsymbol{r}_{T+1}\right)^{-1}\tilde{E}\left(\boldsymbol{r}_{T+1}\right) \tag{22}$$

Now let us calculate the utility of the JLZ investor. When there are no constraints, the portfolio of the JLZ investor is simply the optimal tilted portfolio of Equation (2). Combining Equation (2) and Equation (9), we get

$$\begin{aligned} \boldsymbol{w}_{JLZ} &= \tilde{V}\left(\boldsymbol{r}_{T+1}\right)^{-1}\boldsymbol{B}_{T+1}\boldsymbol{m}_f \\ &= \boldsymbol{w}_{RTL} + \tilde{V}\left(\boldsymbol{r}_{T+1}\right)^{-1}\boldsymbol{B}_{T+1}\left(\boldsymbol{m}_f - \tilde{E}\left(\boldsymbol{f}\right)\right) \end{aligned} \tag{23}$$

Note that we made one simplifying assumption to get the above formula. Firstly, we assumed that the JLZ investor's covariance estimate of returns $\left(\hat{\boldsymbol{\Sigma}}\right)$ is identical to the rational investor's covariance estimate of returns $\left(\tilde{V}\left(\boldsymbol{r}_{T+1}\right)\right)$. Since we did not specify how the JLZ investor obtains $\hat{\boldsymbol{\Sigma}}$, we chose the one that was the best and most advantageous to the JLZ investor. In other words, by doing this we eliminated one mistake that the JLZ investor made, *i.e.* using an inferior estimate of the error covariance.

Now we can express the utility of the JLZ investor in the following way:

$$\begin{aligned} U_{JLZ} &= \boldsymbol{w}'_{JLZ}\tilde{E}\left(\boldsymbol{r}_{T+1}\right) - \frac{1}{2}\boldsymbol{w}'_{JLZ}\tilde{V}\left(\boldsymbol{r}_{T+1}\right)\boldsymbol{w}_{JLZ} \\ &= U_{RTL} - \frac{1}{2}\left[\boldsymbol{m}_f - \tilde{E}\left(\boldsymbol{f}\right)\right]'\boldsymbol{B}_{T+1}\tilde{V}\left(\boldsymbol{r}_{T+1}\right)^{-1} \\ &\quad \cdot \boldsymbol{B}_{T+1}\left[\boldsymbol{m}_f - \tilde{E}\left(\boldsymbol{f}\right)\right] \end{aligned} \tag{24}$$

The above formula shows that the difference in utility between the JLZ investor and the RTL investor, *i.e.* the loss, is quadratic in the difference between the prior mean and the posterior mean of the factor premium. This can be understood intuitively. When one ignores the mean estimates, the portfolio weights deviate from the optimal weights and the deviation is linear in the difference between the prior mean and the posterior mean. The utility, however, depends on the second moments of the portfolio returns, so the loss is quadratic in the difference between the prior mean and the posterior mean.

2.2.4. Investor Simulations

In order to quantify the losses from an incorrect procedure, we perform a numerical simulation.

The simulation is based on the monthly returns and other characteristics of the MSCI country indices.[21] Specifically, we select the top 10 countries by market capitalization. We used the price-to-earnings ratio, 6 months momentum return, and the market capitalization of the indices as factors in our factor model. Our data set

[21]We could have carried out the simulations at the individual stock level. All our formulas would work at the individual stock level as well.

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spans the 7-year period from September 2001 to August 2008. The summary statistics of the variables for each country are presented in **Table 1**.

For the simulation, we use the Chi-square distribution for the error precision parameter h. The Chi-square distribution is a special case of the Gamma distribution, and all the formulas presented above are still applicable. The model and the prior now look as follows:

$$\boldsymbol{r}_{t+1} = \boldsymbol{B}_t \boldsymbol{f} + \boldsymbol{e}_{t+1}, \quad \boldsymbol{e}_{t+1} \sim N\left(0, \frac{1}{h}\boldsymbol{V}_e\right) \tag{25}$$

and

$$\begin{aligned} &\boldsymbol{f}\,|h \sim N\left(\boldsymbol{m}_f, \frac{1}{h}\boldsymbol{V}_f\right) \\ &h \sim \chi(p) \end{aligned} \tag{26}$$

The simulation proceeds according to the following steps.[22]

1) We choose a reasonable value for the hyper parameters $\boldsymbol{m}_f, \boldsymbol{V}_f, p$;

2) We draw a single value of parameters $\boldsymbol{f}$ and h using the hyper parameters above. We use the OLS estimator for the remaining parameter $\boldsymbol{V}_e$;

3) We draw $(\boldsymbol{r}_1, \cdots, \boldsymbol{r}_T)$ from the parameters determined above. For the factor exposures $(\boldsymbol{B}_0, \cdots, \boldsymbol{B}_{T-1})$ and $\boldsymbol{B}_T$, we use historical values;

4) Given the values above, we determine the portfolio of the JLZ investor and the RTL investor;

5) We calculate the loss;

6) We repeat Steps 2 to 5 a 1000 times, without changing the hyper parameters. Then we obtain a distribution of losses, which will allow us to make general statements about the JLZ investor's decision making process;

7) We repeat the above six steps by making small perturbations to the hyper parameter values.

The outcome of the simulations is summarized in **Tables 2** and **3**. A figure of the actual losses in 1000 simulations is shown in **Figure 1**. The magnitude of the loss is fairly stable across various sets of hyper parameter values. The mean loss is between 0.10 and 0.15, while the median loss is between 0.06 and 0.11. As the loss is the difference in the quadratic utility, we could interpret the loss as a reduction in the expected monthly return. That is, the loss of the JLZ investor amounts to about a 0.1% reduction in the expected monthly return. This would imply a fairly conservative estimate of the return loss as 1.2% annually.[23]

[22]Further details about the simulation procedure are provided in an appendix available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2191278.

[23]This estimate is fairly conservative, as we allow the JLZ investor to use the "correct" variance-covariance matrix of errors.

Table 1. Summary Statistics of Returns and Factor Exposures.

Country	Variable	Nobs	Mean	SD	Min	Max
Australia	Return	84	1.50	4.95	−12.57	14.84
	Market Cap	84	5.04	2.27	2.02	10.13
	P/E	84	18.01	2.17	13.51	24.02
	Momentum	84	11.39	11.46	−14.33	33.07
Canada	Return	84	1.46	4.88	−11.26	9.78
	Market Cap	84	6.74	3.16	2.69	12.68
	P/E	84	19.95	4.04	16.09	32.53
	Momentum	84	10.12	12.76	−19.41	33.30
France	Return	84	0.83	5.11	−15.34	15.63
	Market Cap	84	9.21	3.25	4.28	15.05
	P/E	84	23.28	13.66	10.70	70.26
	Momentum	84	5.96	13.64	−29.15	33.91
Germany	Return	84	1.06	6.66	−24.35	23.42
	Market Cap	84	6.93	2.94	2.69	13.09
	P/E	84	19.50	27.84	−211.30	68.96
	Momentum	84	7.45	18.67	−39.44	50.42
Italy	Return	84	0.76	5.00	−13.78	13.62
	Market Cap	84	3.60	1.12	1.78	5.52
	P/E	84	17.92	4.89	9.83	38.65
	Momentum	84	5.87	13.16	−26.41	28.19
Japan	Return	84	0.46	4.62	−9.34	13.43
	Market Cap	84	20.50	6.70	10.19	30.31
	P/E	84	15.09	55.45	−175.67	109.71
	Momentum	84	3.03	16.01	−32.99	36.54
Spain	Return	84	1.31	5.57	−16.19	14.79
	Market Cap	84	3.56	1.43	1.44	6.18
	P/E	84	15.28	1.83	9.90	18.79
	Momentum	84	9.69	14.24	−26.17	36.12
Switzerland	Return	84	0.85	3.76	−12.02	11.35
	Market Cap	84	6.55	1.84	3.88	9.55
	P/E	84	20.33	4.49	12.92	32.88
	Momentum	84	6.04	11.30	−19.37	26.98
UK	Return	84	0.69	3.88	−10.35	10.40
	Market Cap	84	22.42	6.00	13.29	33.61
	P/E	84	14.47	2.23	9.88	19.71
	Momentum	84	5.22	11.59	−19.04	27.67
USA	Return	84	0.33	3.71	−11.33	9.04
	Market Cap	84	105.25	21.13	66.22	140.48
	P/E	84	20.79	4.77	15.69	34.60
	Momentum	84	1.98	10.49	−29.37	22.93

Note: Return is the monthly percentage return. Market cap is the market capitalization in hundred of billions of US dollars. P/E is the price-to-earnings ratio as calculated by MSCI. Momentum is latest 6 month return.

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Table 2. Utility of the JLZ, FFK, and Rational Investor from Various Values of m_f and p.

H.P. Set	Mean	Median	S.D.	Max	Min
1.00	−0.99	−0.79	1.32	1.64	−10.58
	−1.10	−0.92	1.27	1.44	−10.63
	−1.09	−0.89	1.30	1.43	−11.97
2.00	−0.78	−0.53	1.39	1.83	−9.05
	−0.90	−0.63	1.35	1.57	−9.05
	−0.89	−0.60	1.38	1.57	−10.15
3.00	−1.15	−0.89	1.42	1.48	−10.73
	−1.28	−1.00	1.38	1.22	−10.79
	−1.26	−0.98	1.40	1.46	−11.40
4.00	−1.11	−0.85	1.43	1.71	−11.59
	−1.23	−0.97	1.38	1.29	−11.62
	−1.23	−0.97	1.40	1.27	−11.61
5.00	−1.09	−0.79	1.57	1.55	−15.13
	−1.19	−0.90	1.53	1.28	−15.14
	−1.18	−0.87	1.57	1.29	−15.97
6.00	−0.94	−0.66	1.41	2.30	−15.90
	−1.07	−0.80	1.37	2.03	−15.92
	−1.05	−0.77	1.39	2.02	−16.14
7.00	−0.78	−0.51	1.33	1.84	−9.68
	−0.90	−0.64	1.29	1.62	−9.70
	−0.89	−0.60	1.31	1.52	−10.42
8.00	−1.27	−1.02	1.46	2.21	−8.94
	−1.38	−1.09	1.42	1.81	−8.94
	−1.37	−1.08	1.46	1.82	−10.18
9.00	−0.97	−0.69	1.41	1.95	−13.60
	−1.08	−0.82	1.37	1.45	−13.60
	−1.07	−0.80	1.38	1.57	−13.93
10.00	−1.01	−0.73	1.31	1.82	−8.61
	−1.12	−0.81	1.28	1.46	−8.61
	−1.11	−0.79	1.30	1.50	−8.61
11.00	−0.71	−0.46	1.39	1.72	−10.85
	−0.86	−0.57	1.33	1.40	−10.89
	−0.83	−0.56	1.35	1.40	−11.08

Note: The H.P. is for Hyper-Parameter set. Sets 1 - 11 alter the value of vector m_f and p and as follows: [1 1 1 1 1; 1.1 1 1 1 1; 0.9 1 1 1 1; 1 1.1 1 1 1; 1 0.9 1 1 1; 1 1 1.1 1 1; 1 1 0.9 1 1; 1 1 1 1.1 1; 1 1 1 0.9 1; 1 1 1 1 0.9; 1 1 1 1 1.1]. Each set is separated by a semi-colon. For example, set 1 is the baseline case where m_f is not altered. Set 2 alters m_f by multiplying the 2nd element of m_f by 1.1 and so on and so forth. Each row in the table within each set represents the utility for a particular investor. The first row is Rational, the second row is the JLZ, and the third row is the FFK investor.

Table 3. Utility Losses to the JLZ and FFK Investors from Various Values of m_f and p.

H.P. Set	Mean	Median	S.D.	Max	Min
1.00	0.11	0.08	0.12	1.21	0.00
	0.09	0.06	0.10	1.39	0.00
2.00	0.12	0.08	0.13	1.24	0.00
	0.11	0.08	0.12	1.10	0.00
3.00	0.13	0.09	0.14	1.37	0.00
	0.11	0.08	0.11	1.07	0.00
4.00	0.13	0.09	0.12	0.84	0.00
	0.12	0.08	0.11	0.88	0.00
5.00	0.10	0.06	0.12	0.82	0.00
	0.09	0.06	0.11	1.46	0.00
6.00	0.13	0.09	0.13	1.14	0.00
	0.11	0.08	0.11	0.80	0.00
7.00	0.12	0.09	0.13	1.00	0.00
	0.11	0.08	0.11	1.38	0.00
8.00	0.12	0.07	0.14	1.19	0.00
	0.11	0.07	0.13	1.58	-0.00
9.00	0.12	0.08	0.12	0.85	0.00
	0.10	0.07	0.10	1.30	0.00
10.00	0.11	0.08	0.13	0.91	0.00
	0.10	0.07	0.11	1.03	0.00
11.00	0.15	0.11	0.15	1.10	0.00
	0.12	0.10	0.11	0.79	0.00

Note: The H.P. is for Hyper-Parameter set. Sets 1 - 11 alter the value of vector m_f and p and as follows: [1 1 1 1 1; 1.1 1 1 1 1; 0.9 1 1 1 1; 1 1.1 1 1 1; 1 0.9 1 1 1; 1 1 1.1 1 1; 1 1 0.9 1 1; 1 1 1 1.1 1; 1 1 1 0.9 1; 1 1 1 1 0.9; 1 1 1 1 1.1]. Each set is separated by a semi-colon. For example, set 1 is the baseline case where m_f is not altered. Set 2 alters m_f by multiplying the 2nd element of m_f by 1.1 and so on and so forth. The first row represents the utility loss to the JLZ investor and the second row represents the utility loss to the FFK investor for each hyper parameter set.

3. Using the Black-Litterman Approach with a Data Based Prior

3.1. How It Is Usually Done

Fabozzi, Forcardi, and Kolm (2006; henceafter FFK) suggested that one could incorporate a trading strategy, possibly based on a factor model, as a prior in the Black Litterman approach. The portfolio construction process might look as follows.[24]

1) Estimate the factor model that represents the portfolio manager's belief, and specify the "prior" from the estimates of the model. Suppose that the factor model can be written in the following form

[24]See p. 32 of Fabbozi, Forcardi, and Kolm (2006).

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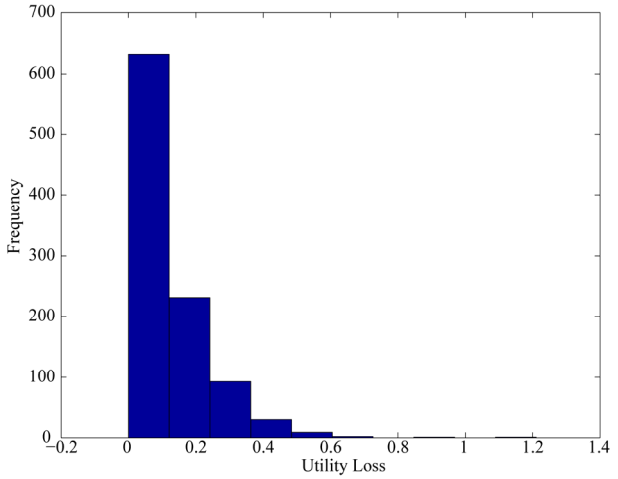


Figure 1. Utility Losses from 1000 Simulations for the JLZ Investor.

$$\boldsymbol{r}_{t+1} = \boldsymbol{B}_t \boldsymbol{f} + \boldsymbol{e}_{t+1}, \quad \boldsymbol{e}_{t+1} \sim N\left(0, \boldsymbol{\Sigma}_e\right) \tag{27}$$

Given the estimate $\hat{\boldsymbol{f}}$ and $\hat{\boldsymbol{\Sigma}}_e$, the "prior" for the mean of the future return, $\boldsymbol{g} \equiv E\left(\boldsymbol{r}_{T+1}\right)$, can be specified as:[25]

$$\boldsymbol{g} \sim N\left(\boldsymbol{B}_T \hat{\boldsymbol{f}}, \hat{\boldsymbol{\Sigma}}_e\right) \tag{28}$$

2) Let $\boldsymbol{\pi}$ be the estimate of the "equilibrium return" implied by the Capital Asset Pricing Model. Assume the following "model" for $\boldsymbol{\pi}$:

$$\boldsymbol{\pi} = \boldsymbol{g} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N\left(0, \tau\boldsymbol{\Sigma}\right) \tag{29}$$

where $\boldsymbol{\Sigma}$ is the variance-covariance matrix of future returns, *i.e.*, $\boldsymbol{\Sigma} \equiv V\left(\boldsymbol{r}_{T+1}\right)$, and τ is a known constant. Note that $\boldsymbol{\pi} = \hat{\boldsymbol{\Sigma}}\boldsymbol{w}_M$, where $\hat{\boldsymbol{\Sigma}}$ is an estimate of $\boldsymbol{\Sigma}$.[26] From this value of $\boldsymbol{\pi}$, one can get an estimate of $\boldsymbol{g}$ and also get a variance-covariance matrix estimate of the estimator:

$$\hat{\boldsymbol{g}} \equiv \boldsymbol{\pi}, \quad \hat{V}\left(\hat{\boldsymbol{g}}\right) \equiv \tau\hat{\boldsymbol{\Sigma}} \tag{30}$$

3) Combine the "prior" with the likelihood of the "model" according to Bayes rule. The posterior of the mean returns $\boldsymbol{g}$ is specified by[27]

$$\begin{aligned} \hat{E}\left(\boldsymbol{g}\right) &\equiv \left(\hat{\boldsymbol{\Sigma}}_e^{-1} + \frac{1}{\tau}\hat{\boldsymbol{\Sigma}}^{-1}\right)^{-1}\left(\hat{\boldsymbol{\Sigma}}_e^{-1}\boldsymbol{B}_T\hat{\boldsymbol{f}} + \frac{1}{\tau}\hat{\boldsymbol{\Sigma}}^{-1}\boldsymbol{\pi}\right) \\ \hat{V}\left(\boldsymbol{g}\right) &\equiv \left(\hat{\boldsymbol{\Sigma}}_e^{-1} + \frac{1}{\tau}\hat{\boldsymbol{\Sigma}}^{-1}\right)^{-1} \end{aligned} \tag{31}$$

The posterior mean $\hat{E}\left(\boldsymbol{g}\right)$ is sometimes referred to as the "Black Litterman alpha".

4) Find the optimal portfolio from a mean variance optimization using the posterior mean $\hat{E}\left(\boldsymbol{g}\right)$ and the posterior variance-covariance matrix of stock returns $\hat{\boldsymbol{\Sigma}}$.[28] That is, solve

$$\begin{aligned} &\max_{\boldsymbol{w}} \quad \boldsymbol{w}'\hat{E}\left(\boldsymbol{g}\right) - \frac{1}{2}\boldsymbol{w}'\hat{\boldsymbol{\Sigma}}\boldsymbol{w} \\ &\text{s.t.} \quad \boldsymbol{w}'\boldsymbol{\iota} = 1 \qquad l_i \le w_i \le u_i, \quad i = 1, \cdots, N \end{aligned} \tag{32}$$

where l_i and u_i are lower and upper bounds on asset i weight.

This procedure takes care of the problem identified in the previous section. The Black Litterman alpha is derived from an equilibrium model as well as from a prior. Information on the mean returns is not discarded, removing the potential for the type of loss identified in the previous section.

This procedure, however, introduces a new type of problem. It is based on two models that are contradictory to each other. The prior is built on the belief that a factor model is the underlying data generating process, while the likelihood is based on the belief that a CAPM is the underlying data generating process. Both beliefs cannot be true at the same time. As the procedure is based on two beliefs, the procedure cannot be optimal regardless of which belief is accurate.[29] In the following subsection, we will show the magnitude of the loss arising from adopting this procedure.

3.2. The Cost of Using Contradictory Models

3.2.1. The Model and the View

We will assume that the factor model in Equation (27) is the true data generating process, and we will show that using the FFK procedure creates a loss. Our argument is independent of what the true model is.

In the previous section, we already discussed the factor model in Equation (27), a prior for this model, and the resulting predictive mean and the variance. The only difference is that in Equation (27) we have $\boldsymbol{\Sigma}_e$ instead of $\frac{1}{h}\boldsymbol{V}_e$. We will also make one modification to the prior

[25] $\hat{\boldsymbol{\Sigma}}_e$ is an estimate of the error variance, not an estimate of the uncertainty of the mean estimate. In some special cases, however, $\hat{\boldsymbol{\Sigma}}_e$ can be interpreted as as estimate of the uncertainty of the mean estimate, for example, if there is only one variable on the right hand side of the model.

[26] See the section entitled "The Derivation of the CAPM Equilibrium Return $\boldsymbol{\pi}$" in the appendix available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2191278 for a detailed explanation.

[27] See the appendix available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2191278 for the proof.

[28] $\hat{E}\left(\boldsymbol{g}\right)$ is the posterior mean of the stock return, as it is the posterior mean of the mean returns.

[29] Meucci's [9] extension of BL has a similar feature. That is, it is based on two inconsistent models. Meucci, however, provides an alternative, more appealing justification of his approach based on the idea of relative entropy. Thus, our criticism has limited relevance to Meucci's approach.

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in this section. We will assume that the investor *does not have any prior opinion*. This modification not only makes the calculation much simpler, but it also makes our comparison more relevant. If we suppose that the investor who follows the FFK procedure has a certain prior opinion, then this investor is making twice as many mistakes. The first mistake is ignoring his prior opinion, and the second mistake is not using the factor model exclusively. Thus, by assuming that the investor does not have any prior opinion, we are eliminating one possible mistake that the investor could commit by following the FFK procedure.

Formally, the non-informative prior can be expressed as

$$\begin{aligned} &\boldsymbol{f}\,|h \sim N\left(\boldsymbol{m}_f, \frac{1}{h}\boldsymbol{V}_f\right), \quad \boldsymbol{V}_f \to \infty \\ &h \sim G(p,q), \quad q \to 0 \end{aligned} \tag{33}$$

This is rather intuitive as it says that the non-informative prior is just the prior distribution of the factor premium as the variance of the factor premium converges to infinity. The formula for the posterior must be modified as well. The new formulae are:[30]

$$\begin{aligned} \tilde{E}(\boldsymbol{f}) &= \hat{\hat{\boldsymbol{f}}} \\ \tilde{V}(\boldsymbol{f}) &= \frac{\boldsymbol{S}_e}{NT-2}\boldsymbol{S}_B^{-1} \\ \tilde{V}(\boldsymbol{e}_{T+1}) &= \frac{\boldsymbol{S}_e}{NT}\hat{\boldsymbol{\Sigma}}_e \end{aligned} \tag{34}$$

The predictive mean and the predictive variance-covariance matrix of the returns are then:

$$\begin{aligned} \tilde{E}(\boldsymbol{r}_{T+1}) &= \boldsymbol{B}_T\hat{\hat{\boldsymbol{f}}} \\ \tilde{V}(\boldsymbol{r}_{T+1}) &= \frac{\boldsymbol{S}_e}{NT-2}\boldsymbol{B}_T\boldsymbol{S}_B^{-1}\boldsymbol{B}_T' + \frac{\boldsymbol{S}_e}{NT}\hat{\boldsymbol{\Sigma}}_e \end{aligned} \tag{35}$$

3.2.2. An Analytic Formula for Loss

Consider two investors. One investor, called the FFK investor, follows the FFK procedure as described in the previous subsection. The other investor, called the RTL investor, makes rational decisions based on the factor model.

The RTL investor's portfolio can be specified as

$$\begin{aligned} &\boldsymbol{w}_{RTL} = \text{argmax}_{\boldsymbol{w}} \quad \boldsymbol{w}'\tilde{E}(\boldsymbol{r}_{T+1}) - \frac{1}{2}\boldsymbol{w}'\tilde{V}(\boldsymbol{r}_{T+1})\boldsymbol{w} \\ &\text{s.t.} \quad \boldsymbol{w}'\boldsymbol{\iota} = 1 \quad l_i \le w_i \le u_i, \; i = 1,\cdots,N \end{aligned} \tag{36}$$

and the RTL investor's utility can be written as

$$U_{RTL} = \boldsymbol{w}_{RTL}'\tilde{E}(\boldsymbol{r}_{T+1}) - \frac{1}{2}\boldsymbol{w}_{RTL}'\tilde{V}(\boldsymbol{r}_{T+1})\boldsymbol{w}_{RTL} \tag{37}$$

The FFK investor's portfolio can be specified as

$$\begin{aligned} &\boldsymbol{w}_{FFK} = \text{argmax}_{\boldsymbol{w}} \quad \boldsymbol{w}'\hat{E}(\boldsymbol{g}) - \frac{1}{2}\boldsymbol{w}'\hat{\boldsymbol{\Sigma}}\boldsymbol{w} \\ &\text{s.t.} \quad \boldsymbol{w}'\boldsymbol{\iota} = 1 \quad l_i \le w_i \le u_i, \; i = 1,\cdots,N \end{aligned} \tag{38}$$

and the FFK investor's utility can be written as

$$U_{FFK} = \boldsymbol{w}_{FFK}'\tilde{E}(\boldsymbol{r}_{T+1}) - \frac{1}{2}\boldsymbol{w}_{FFK}'\tilde{V}(\boldsymbol{r}_{T+1})\boldsymbol{w}_{FFK} \tag{39}$$

As in the previous section, we calculate the loss analytically for the simplest case, *i.e.* the case where there are no constraints in the optimization. If there are no constraints in the optimization, the RTL investor's portfolio weight and his utility are identical to those presented in the previous section, *i.e.*

$$\boldsymbol{w}_{RTL} = \tilde{V}(\boldsymbol{r}_{T+1})^{-1}\tilde{E}(\boldsymbol{r}_{T+1}) \tag{40}$$

$$U_{RTL} = \frac{1}{2}\tilde{E}(\boldsymbol{r}_{T+1}')\tilde{V}(\boldsymbol{r}_{T+1})^{-1}\tilde{E}(\boldsymbol{r}_{T+1}) \tag{41}$$

The FFK investor's portfolio weight, on the other hand, is:

$$\boldsymbol{w}_{FKK} = \hat{\boldsymbol{\Sigma}}^{-1}\hat{E}(\boldsymbol{g}) \tag{42}$$

For simplicity, let us assume that the FFK investor's estimate of the variance-covariance matrix $\tilde{V}(\boldsymbol{r}_{T+1})$ is not very different from the predictive variance-covariance matrix $\tilde{V}(\boldsymbol{r}_{T+1})$ of the RTL investor.[31] Then we could write

$$\boldsymbol{w}_{FKK} \approx \boldsymbol{w}_{RTL} + \tilde{V}(\boldsymbol{r}_{T+1})^{-1}\left[\hat{E}(\boldsymbol{g}) - \tilde{E}(\boldsymbol{r}_{T+1})\right] \tag{43}$$

and

$$\begin{aligned} &U_{FFK} \\ &\approx U_{RTL} - \frac{1}{2}\left[\hat{E}(\boldsymbol{g}) - \tilde{E}(\boldsymbol{r}_{T+1})\right]'\tilde{V}(\boldsymbol{r}_{T+1})^{-1}\left[\hat{E}(\boldsymbol{g}) - \tilde{E}(\boldsymbol{r}_{T+1})\right] \end{aligned} \tag{44}$$

Thus, the loss is approximately quadratic in the difference between the FFK investor's posterior mean and the RTL investor's posterior mean.

Let us consider the difference between the FFK investor's posterior mean and the RTL investor's posterior mean further. Assuming that the difference between the OLS estimator $\hat{\boldsymbol{f}}$ and the GLS estimator $\hat{\hat{\boldsymbol{f}}}$ is not large[32], one can write

$$\hat{E}(\boldsymbol{g}) - \tilde{E}(\boldsymbol{r}_{T+1}) \approx \left(\hat{\boldsymbol{\Sigma}}_e^{-1} + \frac{1}{\tau}\hat{\boldsymbol{\Sigma}}^{-1}\right)^{-1}\frac{1}{\tau}\hat{\boldsymbol{\Sigma}}^{-1}\left(\boldsymbol{\pi} - \boldsymbol{B}_T\hat{\hat{\boldsymbol{f}}}\right) \tag{45}$$

Thus, the loss depends on the difference between the

[30] $\boldsymbol{S}_e$, $\boldsymbol{S}_B$, $\hat{\hat{\boldsymbol{f}}}$, $\hat{\boldsymbol{\Sigma}}_e$ need to be modified as well.

[31] That is, we are assuming that the FFK investor is using the best estimator of the variance-covariance matrix. This assumption is of course more favorable to the FFK investor.

[32] This happens when the error covariance is close to an identity matrix, *i.e.* homeskedatic and no serial correlation.

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multifactor model estimates and the CAPM estimates. This is a rather obvious result, considering that the very source of the problem was the use of the CAPM, which is not the correct model in our example.

3.2.3. Investor Simulations

We follow the procedure identical to that of the previous section to numerically calculate the magnitude of the loss.[33] **Tables 2** and **3** summarize the simulation results. The actual distribution of losses for the FFK investor for 1000 simulations is shown in **Figure 2**. The mean loss for the FFK investor is between 0.09 and 0.12, while the median loss is between 0.06 and 0.10. As before, we might interpret this loss as a reduction in the expected return of the investor. A conservative estimate would be a loss of around 0.09% in expected return per month or 1.08% on an annualized basis.

One may notice that the loss of the FFK investor is somewhat smaller than the loss of the JLZ investor. This is what we could have expected. The loss of the JLZ investor is a quadratic function of the error in the mean estimate, while the loss of the FFK investor is a linear function of the difference in mean estimates of the two models.

4. Using the Black-Litterman Approach as a Reverse Optimization Technique

Practitioners often use the Black-Litterman approach as a tool to extract implied expected returns for a given portfolio weight vector. Step 3 of the JLZ procedure discussed in Section 2 of this paper is an example of this.

Given a portfolio weight vector $\boldsymbol{w}_0$, one can derive the implied expected return by assuming that the portfolio weight vector is a solution to the following optimization problem:

$$\boldsymbol{w}_0 = \operatorname{argmax}_{\boldsymbol{w}} \quad \boldsymbol{w}'\boldsymbol{\alpha} - \frac{1}{2}\boldsymbol{w}'\hat{\boldsymbol{\Sigma}}\boldsymbol{w} \tag{46}$$

As we have shown in Section 2, the solution is given as

$$\boldsymbol{\alpha} = \boldsymbol{\Sigma}\boldsymbol{w}_0 \tag{47}$$

where $\boldsymbol{\Sigma}$ is the variance-covariance matrix of returns.

One aspect that is often ignored in the "reverse optimization" is that the validity of the implied expected return depends on the validity of the variance-covariance matrix used.[34] To the extent that the variance-covariance matrix includes an error, the implied expected return also includes the error.

Let $\boldsymbol{\Sigma}_0$ be the true variance-covariance matrix. Let $\boldsymbol{\Sigma}_1$ be our estimate of the variance-covariance matrix. Then the error in the implied expected return is $\left(\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_0\right)\boldsymbol{w}_0$.

To have a more intuitive sense of the magnitude of the error, we focus on the implied market return, which is $\boldsymbol{w}_M'\left(\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_0\right)\boldsymbol{w}_0$, where $\boldsymbol{w}_M$ is the market capitalization weight. We present typical values of this bias in **Table 4**, which were calculated from a simulation.[35] When $\boldsymbol{w}_0$ reflects a value weight portfolio , the 80% range of the error is from −2.6 to 2.75. That is, excluding the 20% worst cases, the error in the expected monthly market return can be as high as 2.75%. When $\boldsymbol{w}_0$ reflects an equal-weighted portfolio, the same error can be as high as 3.52%. This magnitude of the error can be compared to the long-term average return of the market, which is around 2%. This is a large error in relative terms.

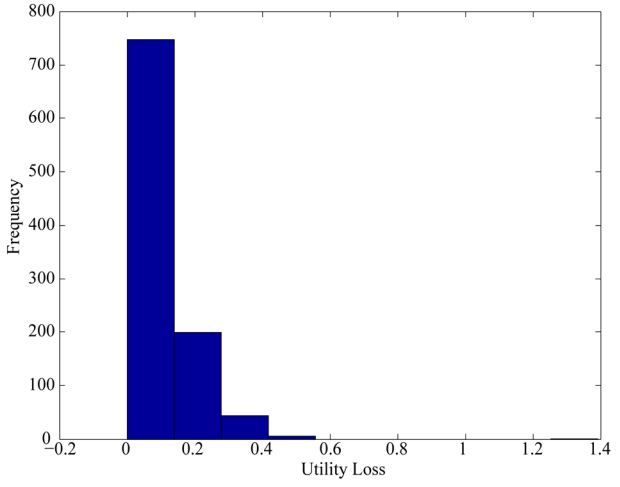


Figure 2. Utility Losses from 1000 Simulations for the FFK Investor.

[33]See the appendix available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2191278 for more details.

5. Conclusions

The Black-Litterman model has been used extensively in

Table 4. Bias in the Implied Expected Monthly Market Returns.

Type	Mean	SD	10th Pct	Med	90th Pct
VW	0.001	2.09	−2.61	0.11	2.75
EW	0.010	2.68	−3.33	−0.13	3.52

Note: VW represents the reverse optimization bias for value-weighted portfolio, while EV represents the reverse optimization bias for an equal-weighted portfolio in percentage terms.

[34]That one recovers the expected return vector rather than the variance-covariance matrix from the reverse optimization may reflect the belief that estimating the variance-covariance matrix is easier than estimating the expected return vector. Even if such a belief is reasonable, one shouldn't forget that the implied expected return is an estimate.

[35]See the appendix available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2191278 for details of the simulation.

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asset allocation and in recent years, a number of people have proposed applications of it to areas such as trading and quantitative portfolio management. Some of these methods have necessitated a transformation or adaptation of the Black-Litterman model to the specific application. Unfortunately, these transformations could be associated with unintended consequences. In this paper, we discuss three applications of the Black Litterman model that result in unnecessary costs to the investor.

The first type of application creates a portfolio out of a prior and an estimate of the variance-covariance matrix, but fails to utilize the mean estimate. Not using the mean estimate amounts to ignoring a valuable piece of information present in the data. Our conservative estimate of the loss from neglecting the mean is about a 1% reduction in expected annual returns. Although it is well known that means are not estimated as reliably as variance and covariances, ignoring the mean estimate cannot be an optimal solution. For even if the mean estimate is not extremely precise, it is still better to use this estimate than to entirely ignore it.

The second type of application creates a portfolio out of two conflicting models of security returns, e.g. the CAPM and a multi-factor model. Assuming that the multi-factor model is the true model of stock returns, we estimate the loss to an investor that chooses to combine both a CAPM model and a multi-factor model in his portfolio construction. We find the magnitude of this loss to be around a 1% reduction in expected annual returns. One might justify the use of two models if the portfolio manager has no idea regarding which model is more likely to be true. In most practical situations, it would be unlikely that the portfolio manager or analyst did not have a clear idea about which model they believed represented the behavior of stock returns. Thus, it makes it hard to justify using two contradictory models of stock returns.

The third application is the so-called reverse optimization technique. That is, practitioners often use the weights of an index and reverse optimize to obtain the implied expected returns of the market. Since the variance-covariance matrix is estimated with error, the implied expected returns of such a procedure will also be estimated with error. We quantify the magnitude of the errors associated with this technique. We found this error to be quite high, in some cases as high as 3.5% per month and much higher if the original benchmark was an equal-weighted benchmark. This brings into question how useful this reverse optimization technique really is.

No doubt the Black-Litterman model brought a new tool to aid asset allocators, portfolio managers, and traders with the construction of optimal portfolios. In particular, it provided a theoretical framework upon which an investor's prior views about asset markets or stocks could be combined with the actual historical data in order to construct optimal investments. The main application of the Black-Litterman model has been to asset allocation across broad asset classes. Recent research has tried to integrate the Black-Litterman model in the trading framework and the quantitative equity portfolio framework. These advances are inspiring as they improve the tool set for quantitative managers, unfortunately some of the applications have side effects which we must be aware of. In this paper, we have discussed some of the potential side effects of these applications of the Black-Litterman model. Our estimates of the losses associated with these side effects are large from a portfolio management perspective. We also suggest a straightforward way to apply the Black-Litterman model without this bias in the portfolio construction process.

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